



Intelligent Security API (Thermal)

Developer Guide

Legal Information

TO THE MAXIMUM EXTENT PERMITTED BY APPLICABLE LAW, THE DOCUMENT IS PROVIDED "AS IS" AND "WITH ALL FAULTS AND ERRORS". OUR COMPANY MAKES NO REPRESENTATIONS OR WARRANTIES, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO, WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE OR NON-INFRINGEMENT. IN NO EVENT WILL OUR COMPANY BE LIABLE FOR ANY SPECIAL, CONSEQUENTIAL, INCIDENTAL, OR INDIRECT DAMAGES, INCLUDING, AMONG OTHERS, DAMAGES FOR LOSS OF BUSINESS PROFITS, BUSINESS INTERRUPTION OR LOSS OF DATA, CORRUPTION OF SYSTEMS, OR LOSS OF DOCUMENTATION, WHETHER BASED ON BREACH OF CONTRACT, TORT (INCLUDING NEGLIGENCE), OR OTHERWISE, IN CONNECTION WITH THE USE OF THE DOCUMENT, EVEN IF OUR COMPANY HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES OR LOSS.

Contents

Chapter 1 Overview	1
1.1 Introduction	1
1.2 Update History	1
Chapter 2 Upload Real-Time Thermal Stream via RTSP	10
Chapter 3 Alarm or Event	15
3.1 Alarm or Event Settings	15
3.1.1 Configure Temperature Screening Alarm	15
3.1.2 Configure Temperature Measurement Alarm in Normal Mode	18
3.1.3 Configure Temperature Measurement Alarm in Expert Mode	21
3.1.4 Configure Fire and Smoke Alarm	25
3.1.5 Configure Ship Flow Detection Alarm	26
3.1.6 Configure Dredger Detection Alarm	30
3.2 Receive Alarm/Event in Arming Mode	34
3.3 Receive Alarm/Event in Listening Mode	36
3.4 Subscribe Alarm/Event in Arming Mode	38
Chapter 4 Other Thermal Applications	41
Chapter 5 Request URIs	47
5.1 /ISAPI/Event/notification/subscribeEventCap	47
5.2 /ISAPI/Event/notification/subscribeEvent	47
5.3 /ISAPI/Event/notification/subscribeEvent/<ID>	48
5.4 /ISAPI/Event/notification/unSubscribeEvent	48
5.5 /ISAPI/Event/schedules/faceThermometrys	49
5.6 /ISAPI/Event/schedules/faceThermometrys/<ID>	49
5.7 /ISAPI/Event/schedules/dredgerDetection/<ID>/scene/<SID>	50
5.8 /ISAPI/Event/schedules/shipsFlowDetection/<ID>/scene/<SID>	51
5.9 /ISAPI/Event/schedules/Thermometry	51

5.10 /ISAPI/Event/schedules/thermometry/<ID>	52
5.11 /ISAPI/Event/schedules/TMPA/<ID>	53
5.12 /ISAPI/Event/schedules/TMPA	54
5.13 /ISAPI/Event/triggers	54
5.14 /ISAPI/Event/triggers/dredgerDetection/<ID>/scene/<SID>	55
5.15 /ISAPI/Event/triggers/shipsFlowDetection/<ID>/scene/<SID>	55
5.16 /ISAPI/Event/triggers/<eventType>-<ID>	56
5.17 /ISAPI/Event/triggersCap	57
5.18 /ISAPI/Image/channels/<ID>/DPC/correction	57
5.19 /ISAPI/Image/channels/<ID>/DPC/correction/capabilities	58
5.20 /ISAPI/Intelligent/capabilities	58
5.21 /ISAPI/Intelligent/faceTemperature/capabilities?format=json	59
5.22 /ISAPI/Intelligent/faceTemperature/dataStatistics?format=json	59
5.23 /ISAPI/PTZCtrl/channels/<ID>/presets	59
5.24 /ISAPI/System/capabilities	61
5.25 /ISAPI/Thermal/capabilities	61
5.26 /ISAPI/Thermal/channels/<ID>/blackBody	61
5.27 /ISAPI/Thermal/channels/<ID>/blackBody/capabilities	62
5.28 /ISAPI/Thermal/channels/<ID>/bodyTemperatureCompensation	62
5.29 /ISAPI/Thermal/channels/<ID>/bodyTemperatureCompensation/capabilities	63
5.30 /ISAPI/Thermal/channels/<ID>/Correction	64
5.31 /ISAPI/Thermal/channels/<ID>/CorrectionParam	64
5.32 /ISAPI/Thermal/channels/<ID>/CorrectionParam/capabilities	65
5.33 /ISAPI/Thermal /channels/<ID>/CorrectionParam/import	65
5.34 /ISAPI/Thermal/channels/<ID>/faceThermometry	66
5.35 /ISAPI/Thermal/channels/<ID>/faceThermometry/capabilities	66
5.36 /ISAPI/Thermal/channels/<ID>/faceThermometry/regions	67
5.37 /ISAPI/Thermal/channels/<ID>/faceThermometry/regions/<ID>	67

5.38 /ISAPI/Thermal/channels/<ID>/faceThermometry/regions/<ID>/detectionInfo	68
5.39 /ISAPI/Thermal/channels/<ID>/fireDetection	68
5.40 /ISAPI/Thermal/channels/<ID>/fireDetection/advanceParam/capabilities?format=json	69
5.41 /ISAPI/Thermal/channels/<ID>/fireDetection/advanceParam?format=json	70
5.42 /ISAPI/Thermal/channels/<ID>/fireDetection/capabilities	70
5.43 /ISAPI/Thermal/channels/<ID>/historyTemperature	71
5.44 /ISAPI/Thermal/channels/<ID>/historyTemperature/capabilities	71
5.45 /ISAPI/Thermal/channels/<ID>/pip	71
5.46 /ISAPI/Thermal/channels/<ID>/pip/capabilities	72
5.47 /ISAPI/Thermal/channels/<ID>/shipsDetection/advanceParam/capabilities?format=json	72
5.48 /ISAPI/Thermal/channels/<ID>/shipsDetection/advanceParam?format=json	73
5.49 /ISAPI/Thermal/channels/<ID>/shipsDetection/basicParam/capabilities?format=json	73
5.50 /ISAPI/Thermal/channels/<ID>/shipsDetection/basicParam?format=json	74
5.51 /ISAPI/Thermal/channels/<ID>/shipsDetection/capabilities?format=json	75
5.52 /ISAPI/Thermal/channels/<ID>/shipsDetection/sceneTrace/capabilities?format=json	75
5.53 /ISAPI/Thermal/channels/<ID>/shipsDetection/sceneTrace?format=json	76
5.54 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/capabilities?format=json	76
5.55 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/captureRatio/capabilities? format=json	77
5.56 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/captureRatio?format=json	78
5.57 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/dredgerDetection/regions/ capabilities?format=json	78
5.58 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/dredgerDetection/regions/<ID>? format=json	79
5.59 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/dredgerDetection/regions? format=json	80
5.60 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>?format=json	80
5.61 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/goto?format=json	81

5.62 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipsDetectionCount?format=json	82
5.63 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipFlowDetection/regions/capabilities?format=json	82
5.64 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipFlowDetection/regions/<ID>?format=json	83
5.65 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipFlowDetection/regions?format=json	84
5.66 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipsFunctionMode/capabilities?format=json	85
5.67 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipsFunctionMode?format=json	85
5.68 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/resetCount?format=json	86
5.69 /ISAPI/Thermal/channels/<ID>/shipsDetection?format=json	86
5.70 /ISAPI/Thermal/channels/<ID>/streamParam	87
5.71 /ISAPI/Thermal/channels/<ID>/streamParam/capabilities	88
5.72 /ISAPI/Thermal/channels/<ID>/thermIntell	88
5.73 /ISAPI/Thermal/channels/<ID>/thermIntell/capabilities	89
5.74 /ISAPI/Thermal/channels/<ID>/thermometry/<SID>	90
5.75 /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/alarmRules	90
5.76 /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/alarmRules/capabilities	91
5.77 /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/capabilities	91
5.78 /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/regions	92
5.79 /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/regions/<ID>	93
5.80 /ISAPI/Thermal/channels/<ID>/thermometry/basicParam	93
5.81 /ISAPI/Thermal/channels/<ID>/thermometry/basicParam/capabilities	94
5.82 /ISAPI/Thermal/channels/<ID>/thermometry/jpegPicWithAppendData?format=json	94
5.83 /ISAPI/Thermal/channels/<ID>/thermometry/pixelToPixelParam	96
5.84 /ISAPI/Thermal/channels/<ID>/thermometry/pixelToPixelParam/capabilities	97

5.85 /ISAPI/Thermal/channels/<ID>/thermometry/realTimethermometry/rules/<RuleID>?format=json	97
5.86 /ISAPI/Thermal/channels/<ID>/thermometry/realTimethermometry/rules?format=json	98
5.87 /ISAPI/Thermal/channels/<ID>/thermometryMode	98
5.88 /ISAPI/Thermal/channels/<ID>/thermometryMode/capabilities	99
5.89 /ISAPI/Thermal/channels/<ID>/burningPrevention	99
5.90 /ISAPI/Thermal/channels/<ID>/burningPrevention/capabilities	100
5.91 /ISAPI/Thermal/channels/<ID>/clickToThermometry/initialization	100
5.92 /ISAPI/Thermal/channels/<ID>/clickToThermometry/rules/<ID>?format=json	101
5.93 /ISAPI/Thermal/channels/<ID>/temperatureCorrect/capabilities?format=json	101
5.94 /ISAPI/Thermal/channels/<ID>/temperatureCorrect?format=json	102
5.95 /ISAPI/Thermal/channels/<ID>/thermalDataConversion	102
5.96 /ISAPI/Thermal/channels/<ID>/thermalDataConversion/capabilities	103
5.97 /ISAPI/Thermal/channels/<ID>/thermometry/OffLineCapture?format=json	103
5.98 /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/rulesTemperatureInfo?format=json	104
5.99 /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/rulesTemperatureInfo/<ID>?format=json	105
5.100 /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/rulesTemperatureInfo/capabilities?format=json	105
5.101 /ISAPI/Thermal/channels/<ID>/thermometry/basicParam/calibrationCoefficient	106
5.102 /ISAPI/Thermal/Power	106
5.103 /ISAPI/Thermal/Power/capabilities	106
5.104 rtsp://<ipaddress>[:port]/ISAPI/Streaming/thermal/channels/<ID>/streamType/<Type>/trackID=	107
5.105 rtsp://<ipaddress>[:port]/ISAPI/Streaming/thermal/channels/<ID>/streamType/<Type>	107
5.106 /ISAPI/Event/notification/alertStream	108
5.107 /ISAPI/Event/notification/httpHosts	109

5.108 /ISAPI/Event/notification/httpHosts/<ID>	110
5.109 /ISAPI/Event/notification/httpHosts/<ID>/test	112
5.110 /ISAPI/Event/notification/httpHosts/capabilities	112
5.111 http://ipAddress:portNo/url	113
Appendix A. Request and Response Messages	114
A.1 JSON_advanceParam	114
A.2 JSON_advanceParamCap	114
A.3 JSON_basicParam	114
A.4 JSON_basicParamCap	115
A.5 JSON_Cap_functionMode	115
A.6 JSON_ClickToThermometryRule	116
A.7 JSON_DredgerDetectionRule	116
A.8 JSON_DredgerDetectionRuleList	117
A.9 JSON_DredgerDetectionRuleListCap	118
A.10 JSON_EpidemicControlCap	119
A.11 JSON_EventNotificationAlert_dredgerDetection	119
A.12 JSON_EventNotificationAlert_TDAMsg	121
A.13 JSON_FaceThermReportSearchCond	123
A.14 JSON_FaceThermReportSearchResult	123
A.15 JSON_fireDetection_AdvanceParam	124
A.16 JSON_fireDetection_AdvanceParamCap	124
A.17 JSON_functionMode	124
A.18 JSON_ResponseStatus	124
A.19 JSON_ShipDetectionCount	125
A.20 JSON_shipsDetectionCap	125
A.21 JSON_ShipsDetectionScene	126
A.22 JSON_ShipsDetectionSceneCap	127
A.23 JSON_ShipsDetectionSceneList	128

A.24 JSON_ShipsDetectionCaptureRatio	128
A.25 JSON_ShipsDetectionCaptureRatioCap	129
A.26 JSON_ShipsDetectionTraceListCap	129
A.27 JSON_ShipsDetectionSceneTraceList	130
A.28 JSON_ShipFlowDetectionRuleListCap	131
A.29 JSON_ShipFlowDetectionRuleList	132
A.30 JSON_ShipFlowDetectionRule	133
A.31 JSON_thermometryOffLineCapture	134
A.32 JSON_ThermometryRulesTemperatureInfo	135
A.33 JSON_ThermometryRulesTemperatureInfoList	135
A.34 JSON_ThermometryRulesTemperatureInfoListCap	136
A.35 JSON_ThermometryUpload	137
A.36 JSON_ThermometryUploadList	139
A.37 JSON_TemperatureCorrect	140
A.38 JSON_TemperatureCorrectCap	141
A.39 XML_BodyTemperatureCompensation	142
A.40 XML_BodyTemperatureCompensationCap	142
A.41 XML_BurningPrevention	143
A.42 XML_BurningPreventionCap	144
A.43 XML_CalibrationCoefficientDescription	144
A.44 XML_CalibrationCoefficientResult	144
A.45 XML_Cap_DPC	145
A.46 XML_Cap_FaceThermometry	145
A.47 XML_Cap_FireDetection	148
A.48 XML_Cap_Power	149
A.49 XML_Cap_ThermalBlackBody	150
A.50 XML_Cap_ThermIntell	151
A.51 XML_Cap_ThermalPip	151

A.52 XML_Cap_ThermometryAlarmRule	152
A.53 XML_Cap_ThermometryBasicParam	154
A.54 XML_Cap_ThermometryMode	156
A.55 XML_Cap_ThermometryScene	157
A.56 XML_CorrectionParam	159
A.57 XML_CorrectionParamCap	160
A.58 XML_DeviceCap	160
A.59 XML_DPC	169
A.60 XML_EventNotificationAlert_FireSmokeAlarmEg	169
A.61 XML_EventNotificationAlert_IntrusionAlarmMsg	172
A.62 XML_EventNotificationAlert_LineCrossingAlarmMsg	174
A.63 XML_EventNotificationAlert_RegionEntranceAlarmMsg	177
A.64 XML_EventNotificationAlert_RegionExitingAlarmMsg	179
A.65 XML_EventNotificationAlert_shipFlowDetection	181
A.66 XML_EventNotificationAlert_SubscriptionHeartbeat	182
A.67 XML_EventNotificationAlert_TemperatureAlarmMsg	182
A.68 XML_EventNotificationAlert_TMPAMsg	184
A.69 XML_EventTrigger	185
A.70 XML_EventTriggerList	186
A.71 XML_EventTriggerNotification	190
A.72 XML_EventTriggerNotificationList	191
A.73 XML_EventTriggersCap	191
A.74 XML_EventTriggerCapType	194
A.75 XML_FaceThermDetectionInfo	196
A.76 XML_FaceThermometry	197
A.77 XML_FaceThermometryRegionList	198
A.78 XML_FaceThermometryScheduleList	198
A.79 XML_FireDetection	199

A.80 XML_HistoryTemperatureCap	200
A.81 XML_HistoryTemperatureDescription	200
A.82 XML_HistoryTemperatureResult	201
A.83 XML_IntelliCap	201
A.84 XML_PixelToPixelParam	204
A.85 XML_PixelToPixelParamCap	205
A.86 XML_Power	206
A.87 XML_PTZPreset	206
A.88 XML_PTZPresetList	206
A.89 XML_RegionalExposure	206
A.90 XML_ResponseStatus	207
A.91 XML_Schedule	207
A.92 XML_SubscribeEvent	208
A.93 XML_SubscribeEventCap	209
A.94 XML_SubscribeEventResponse	211
A.95 XML_ThermalBlackBody	211
A.96 XML_ThermalCap	212
A.97 XML_ThermalDataConversionCap	214
A.98 XML_ThermalDataConversionDescription	214
A.99 XML_ThermalDataConversionResult	215
A.100 XML_ThermalPip	215
A.101 XML_ThermalStreamParam	216
A.102 XML_ThermalStreamParamCap	216
A.103 XML_ThermIntell	216
A.104 XML_ThermometryAlarmRule	217
A.105 XML_ThermometryBasicParam	218
A.106 XML_ThermometryMode	220
A.107 XML_ThermometryRegion	220

A.108 XML_ThermometryRegionList	222
A.109 XML_ThermometryScene	222
A.110 XML_ThermometryScheduleList	222
A.111 XML_TMPAScheduleList	223
A.112 JSON_EventNotificationAlert_Alarm/EventInfo	223
A.113 XML_EventNotificationAlert_AlarmEventInfo	224
A.114 XML_EventNotificationAlert_HeartbeatInfo	224
A.115 XML_HttpHostNotification	225
A.116 XML_HttpHostNotificationCap	226
A.117 XML_HttpHostNotificationList	227
A.118 XML_HttpHostTestResult	228
Appendix B. Appendixes	229
B.1 RTP Packet Format	229
B.2 ISAPI Response Codes	230

Chapter 1 Overview

1.1 Introduction

Thermal application developed by ISAPI protocol contains temperature screening and temperature measurement, which is mainly for thermal products. You can configure the temperature screening alarm and thermography alarm (including temperature alarm and temperature difference alarm) and receive the alarms if triggered.

1.2 Update History

The update history shows the summary of changes in ISAPI protocol according to different products or projects, and the updated time.

Summary of Changes in Version 2.6_June, 2020

Related Product: DS-2TD1217B series and DS-2TD2636B series Thermal & Optical Bi-spectrum Network Camera

1. Extended the configuration capability of temperature measurement basic parameters ***XML_Cap_ThermometryBasicParam*** (related URI: GET ***/ISAPI/Thermal/channels/<ID>/thermometry/basicParam/capabilities***) and temperature measurement basic parameters ***XML_ThermometryBasicParam*** (related URI: ***/ISAPI/Thermal/channels/<ID>/thermometry/basicParam***):
added two nodes **<distanceMode>** (distance mode) and **<faceTemperatureInfoUploadEnabled>** (whether to enable uploading face temperature information).
2. Extended the message about real-time temperature measurement data ***JSON_ThermometryUpload*** (related URI: GET ***/ISAPI/Thermal/channels/<ID>/thermometry/realTimethermometry/rules/<RuleID>?format=json***):
added two nodes **faceSnapThermometryEnabled** (whether to enable uploading captured face picture with temperature information) and **faceRect** (face thumbnail rectangle).
3. Extended the temperature screening configuration capability ***XML_Cap_FaceThermometry*** (related URI: GET ***/ISAPI/Thermal/channels/<ID>/faceThermometry/capabilities***):
added one sub node **<MaxPupilParam>** (maximum pupil distance) to **<ThermometryRegion>** of **<FaceThermometryRegionList>**;
added four nodes: **<faceSnapUploadEnabled>** (whether to enable uploading captured face picture), **<maxTemperatureCoordinatesEnabled>** (whether to enable displaying the maximum temperature position), **<faceRectShowEnabled>** (whether to enable displaying a frame on the target person), and **<faceTemperatureShowEnabled>** (whether to enable displaying face temperature).
4. Extended the temperature screening parameters ***XML_FaceThermometry*** (related URI: ***/ISAPI/Thermal/channels/<ID>/faceThermometry***):

- added four nodes: **<faceSnapUploadEnabled>** (whether to enable uploading captured face picture), **<maxTemperatureCoordinatesEnabled>** (whether to enable displaying the maximum temperature position), **<faceRectShowEnabled>** (whether to enable displaying a frame on the target person), and **<faceTemperatureShowEnabled>** (whether to enable displaying face temperature).
5. Extended the alarm linkage capabilities ***XML_EventTriggersCap*** (related URI: GET ***/ISAPI/Event/triggersCap***):
added two nodes **<NoMaskDetectionTriggerCap>** (alarm linkage capability of no wearing mask detection) and **<TMPATriggerCap>** (alarm linkage capability of temperature measurement pre-alarm).
 6. Extended the linkage parameter message ***XML_EventTrigger*** (related URI: ***/ISAPI/Event/triggers/<eventType>-<ID>***):
added two event type "noMaskDetection" (no wearing mask detection) and "TMPA" (temperature measurement pre-alarm) to the node **<eventType>**.
 7. Extended the body temperature compensation capability ***XML_BodyTemperatureCompensationCap*** (related URI: GET ***/ISAPI/Thermal/channels/<ID>/bodyTemperatureCompensation/capabilities***) and body temperature compensation parameters ***XML_BodyTemperatureCompensation*** (related URI: ***/ISAPI/Thermal/channels/<ID>/bodyTemperatureCompensation***):
added four sub nodes to **<AutoParam>**: **<smartCorrection>** (smart calibration), **<environmentalTemperature>** (environment temperature), **<temperatureCompensation>** (whether to enable body temperature compensation), and **<environmentalTemperatureMode>** (environment temperature mode).
 8. Extended the black body configuration capability ***XML_Cap_ThermalBlackBody*** (related URI: GET ***/ISAPI/Thermal/channels/<ID>/blackBody/capabilities***) and black body configurations ***XML_ThermalBlackBody*** (related URI: ***/ISAPI/Thermal/channels/<ID>/blackBody***):
added two nodes **<enabled>** (whether to enable black body) and **<BlackBodyReigon>** (black body detection area).
 9. Added two URIs of configuring the arming schedule for temperature measurement pre-alarm:
Get or set the arming schedule of temperature measurement pre-alarm for a specified channel:
GET or PUT ***/ISAPI/Event/schedules/TMPA/<ID>*** ;
Get or set arming schedules of temperature measurement pre-alarm for all channels in a batch:
GET or PUT ***/ISAPI/Event/schedules/TMPA*** .
 10. Extended the temperature screening rule configurations of a specific detection region ***XML_ThermometryRegion*** (related URI: ***/ISAPI/Thermal/channels/<ID>/faceThermometry/regions/<ID>***):
added one node **<MaxPupilParam>** (maximum pupil distance).
 11. Extended the message about temperature measurement precautionary alarm ***XML_EventNotificationAlert_TMPAMsg*** :
added one sub node **<alarmRuleTemperature>** (pre-alarm rule temperature) to **<TMPA>**.
 12. Extended the message about temperature difference alarm details ***JSON_EventNotificationAlert_TDAMsg*** :

added one sub node **<presetNo>** (preset No.) to **TDA**.

13. Extended the thermal capability ***XML_ThermalCap*** (related URI: GET ***/ISAPI/Thermal/capabilities***):
added one node **<isSupportFaceSnapThermometry>** (whether device supports uploading captured face picture with temperature information).

Summary of Changes in Version 2.6_May, 2020

Related Product: TB-4117-3-S Temperature Screening Thermal Camera Module with Software Version 5.5.2

1. Extended the thermal capability ***XML_ThermalCap*** (related URI: GET ***/ISAPI/Thermal/capabilities***):
added one node **<isSupportThermalTemperatureCorrect>** (whether device supports temperature calibration).
2. Added two URIs for temperature calibration:
Get temperature calibration configuration capability: GET ***/ISAPI/Thermal/channels/<ID>/temperatureCorrect/capabilities?format=json*** ;
Get or set temperature calibration configuration parameters: GET or POST ***/ISAPI/Thermal/channels/<ID>/temperatureCorrect?format=json*** .

Summary of Changes in Version 2.6_Feb., 2020

Related Product: Thermal & Optical Bi-spectrum Network Speed Dome; Thermal & Optical Bi-spectrum Positioning System; Thermal & Optical Bi-spectrum Network Stable PTZ Camera in Version 5.5.25

Added URIs of configuring fire detection advanced parameters:

Get capability: GET ***/ISAPI/Thermal/channels/<ID>/fireDetection/advanceParam/capabilities?format=json*** ;

Get or set advanced parameters of fire detection: GET or PUT ***/ISAPI/Thermal/channels/<ID>/fireDetection/advanceParam?format=json*** .

Summary of Changes in Version 2.0_Nov., 2019

Related Product: DS-2TD11 Series, DS-2TD12 Series, DS-2TD21 Series, and DS-2TD16 Series Thermal Network Bullet Camera in Version 5.5.24

1. Extended the preset (scene) configuration capability ***XML_Cap_ThermometryScene*** (related URI: ***/ISAPI/Thermal/channels/<ID>/thermometry/<SID>/capabilities***):
added a sub node **<RegionBoundary>** (rule region boundary) to the node **<ThermometryRegion>** of **<ThermometryRegionList>**.
2. Added two nodes **<visibleLightURL>** (optical picture URI) and **<thermalURL>** (thermal picture URI) to the four messages: ***XML_EventNotificationAlert_LineCrossingAlarmMsg*** , ***XML_EventNotificationAlert_IntrusionAlarmMsg*** , ***XML_EventNotificationAlert_RegionEntranceAlarmMsg*** and ***XML_EventNotificationAlert_RegionExitingAlarmMsg*** .
3. Added two sub status codes to status code 6 (Invalid Message Content) in ***ISAPI Response Codes*** :

ruleAndShieldingMaskConflict (the line-rule region overlaps with the shielded area) and wholeRuleInShieldingMask (there are complete temperature measurement rules in the shielded area).

Summary of Changes in Version 2.0_July, 2019

Related Product Type: DS-2TA03-7AVI, DS-2TA03-15VI, DS-2TA03-15SVI, DS-2TA06-25SVI, DS-2TA06-25VI, and DS-2TA03-25SVI Thermographic Automation Thermal Camera in Version 2.2

1. Edited the burning prevention capability message ***XML_BurningPreventionCap*** (related URI: ***GET /ISAPI/Thermal/channels/<ID>/burningPrevention/capabilities***):
changed the default value of node ***<closedDuration>*** (closed status duration) to "10".

Summary of Changes in Version 2.0_July, 2019

Related Product: Thermographic Cube Camera in Version 1.0

1. Extended the picture-in-picture configuration capability ***XML_Cap_ThermalPip*** and the picture-in-picture configuration parameters ***XML_ThermalPip*** :
added one node ***<distance>*** (fusion distance).
2. Extended the basic configuration capability of temperature measurement ***XML_Cap_ThermometryBasicParam*** and the basic temperature measurement parameters ***XML_ThermometryBasicParam*** :
added two temperature ranges "20-350" and "20-45" to the node ***<temperatureRange>***.

Summary of Changes in Version 2.0_June, 2019

Related Product Type: Thermal Network Bullet Camera; Thermal Bi-spectrum Network Bullet Camera; Thermal & Optical Network Turret Camera; Thermal Box Camera; Thermal Network Turret Camera in Version 5.5.20

1. Added one alarm type "TDA" (temperature difference alarm) to temperature measurement alarm. Refer to ***Configure Temperature Measurement Alarm in Expert Mode*** and ***Configure Temperature Measurement Alarm in Normal Mode*** for details.

Summary of Changes in Version 2.0_July, 2019

Related Product: Thermal & Optical Bi-spectrum Network Speed Dome; Thermal & Optical Bi-spectrum Positioning System; Thermal & Optical Bi-spectrum Network Stable PTZ Camera in Version 5.5.19

1. Extended the thermal product capability ***XML_ThermalCap*** (related URI: ***/ISAPI/Thermal/capabilities***):
added one node: ***<isSupportShipsDetectionWithScene>*** (whether supports ship detection by scene).
2. Added the function of configuring ship flow detection alarm, see details in ***Configure Ship Flow Detection Alarm*** .
3. Added the function of configuring dredger detection alarm, see details in ***Configure Dredger Detection Alarm*** .
4. Extended the configuration capability of fire and smoke detection ***XML_Cap_FireDetection*** (related URI: ***/ISAPI/Thermal/channels/<ID>/fireDetection/capabilities***):

added five nodes: **<verificationSensitivity>** (sensitivity of double verification), **<fireAlgorithmModel>** (fire detection algorithm mode), **<agriculturalMachineryFilterEnabled>** (enable agricultural machinery filter), **<waterReflectionEnabled>** (enable water reflection), and **<patrolSensitivity>** (patrol sensitivity).

5. Extended the fire and smoke detection parameters ***XML_FireDetection*** (related URI: ***/ISAPI/Thermal/channels/<ID>/fireDetection***):
added five nodes: **<verificationSensitivity>** (sensitivity of double verification), **<fireAlgorithmModel>** (fire detection algorithm mode), **<agriculturalMachineryFilterEnabled>** (enable agricultural machinery filter), **<waterReflectionEnabled>** (enable water reflection), and **<patrolSensitivity>** (patrol sensitivity).
6. Extended the burning prevention capabilities ***XML_BurningPreventionCap*** (related URI: ***/ISAPI/Thermal/channels/<ID>/burningPrevention/capabilities***):
added three nodes: **<protectionMode>** (protection mode), **<burningRecoveryEnabled>** (enable burning recovery or not), **<movementDuration>** (duration of lens movement for burning protection).
7. Extended the burning prevention parameters ***XML_BurningPrevention*** (related URI: ***/ISAPI/Thermal/channels/<ID>/burningPrevention***):
added three nodes: **<protectionMode>** (protection mode), **<burningRecoveryEnabled>** (enable burning recovery or not), **<movementDuration>** (duration of lens movement for burning protection).

Summary of Changes in Version 2.0_May, 2019

Related Product Type: Thermal Network Bullet Camera; Thermal Bi-spectrum Network Bullet Camera; Thermal & Optical Network Turret Camera; Thermal Box Camera V5.5.18

1. Extended the alarm linkage action ***XML_EventTriggerNotification*** :
added one linkage action "audio" to node **<notificationMethod>**.
2. Extended the preset (scene) configuration capability ***XML_Cap_ThermometryScene*** (request URI: GET ***/ISAPI/Thermal/channels/<ID>/thermometry/<SID>/capabilities***):
added one node **<emissivityMode>** (emissivity type).
3. Extended the capability of temperature measurement basic configuration ***XML_Cap_ThermometryBasicParam*** (request URI: GET ***/ISAPI/Thermal/channels/<ID>/thermometry/basicParam/capabilities***):
added two nodes **<emissivityMode>** (emissivity type) and **<displayTemperatureInOpticalChannelEnabled>** (display the temperature information of optical channel).
4. Extended the temperature measurement basic parameters ***XML_ThermometryBasicParam*** (request URI: ***/ISAPI/Thermal/channels/<ID>/thermometry/basicParam***):
added two nodes **<emissivityMode>** (emissivity type) and **<displayTemperatureInOpticalChannelEnabled>** (display the temperature information of optical channel).
5. Extended the temperature measurement region parameters ***XML_ThermometryRegion*** (request URI: ***/ISAPI/Thermal/channels/<ID>/thermometry/<SID>/regions/<ID>***):
added one node **<emissivityMode>** (emissivity type).

Summary of Changes in Version 2.0_April, 2019

Related Product Type: Handheld Thermography Camera in Version 1.1

1. Extended the message of camera power capability ***XML_Cap_Power*** and message of camera power parameters ***XML_Power*** :
added one node **<batteryPower>** (battery percentage).
2. Extended the thermal capability message ***XML_ThermalCap*** (related URI: GET ***/ISAPI/Thermal/capabilities***):
added one node **<isSupporttthermometryOffLineCapture>** (whether device supports offline capture).
3. Added one URI of capturing one picture offline:
GET ***/ISAPI/Thermal/channels/<ID>/thermometry/OffLineCapture?format=json***

Summary of Changes in Version 2.0_02/2019

Related Product Type: Thermographic Automation Thermal Camera V2.1

Related Product Model: DS-2TA03-7AVI, DS-2TA03-15VI, DS-2TA03-15SVI, DS-2TA06-25SVI, DS-2TA06-25VI

1. Extended the thermal product capabilities ***XML_ThermalCap*** :
added two nodes **<isSupportJpegPicWithAppendData>** (whether supports getting JPEG picture with pixel-to-pixel temperature measurement data) and **<isSupportRealTimethermometryForHTTP>** (whether supports real-time temperature measurement).
2. Added the function of getting JPEG picture with pixel-to-pixel temperature measurement data:
GET ***/ISAPI/Thermal/channels/<ID>/thermometry/jpegPicWithAppendData?format=json***
3. Added the function of getting real-time temperature measurement information:
Get real-time temperature measurement information of all rules: GET ***/ISAPI/Thermal/channels/<ID>/thermometry/realTimethermometry/rules?format=json*** ;
Get real-time temperature measurement information of a single rule: GET ***/ISAPI/Thermal/channels/<ID>/thermometry/realTimethermometry/rules/<RuleID>?format=json*** .
4. Added the function of uploading real-time thermal stream via RTSP, see details in ***Upload Real-Time Thermal Stream via RTSP***
5. Extended the temperature information list message of temperature measurement rules ***JSON_ThermometryRulesTemperatureInfoList*** , temperature information message of temperature measurement rules ***JSON_ThermometryRulesTemperatureInfo*** , and capability of manually getting temperature information of temperature measurement rules ***JSON_ThermometryRulesTemperatureInfoListCap*** :
added one parameter **isFreezedata** (whether freezes data).

Summary of Changes in Version 2.0_01/2019

Related Product Type: Network Thermographic Automation Camera

Related Product Model: DS-2TA03-4AUM, DS-2TA03-7AUF, DS-2TA03-7HUF, DS-2TA03-10AUF, DS-2TA03-15AUF, DS-2TA03-15HUF, DS-2TA06-7AXF, DS-2TA06-15AXF

1. Extended the pixel-to-pixel temperature measurement configuration capability ***XML_PixelToPixelParamCap*** and pixel-to-pixel temperature measurement configuration parameters ***XML_PixelToPixelParam*** :
added one node **<JpegPictureWithAppendData>** (configuration of getting JPEG picture with pixel-to-pixel temperature measurement data).

Summary of Changes in Version 2.0_01/2019

Related Product Type: Thermal & Optical Bi-spectrum Network Speed Dome, Thermal & Optical Bi-spectrum Positioning System, Thermal & Optical Bi-spectrum Network Stable PTZ Camera

Related Product Model: DS-2TD4237-xx/V2, DS-2TD4136-xx/V2, DS-2TD6236-xx/V2, DS-2TD6236-xx/V2, DS-2TD8166xx/V2, DS-2TD8136xx/V2

1. Extended the fire detection capability message ***XML_Cap_FireDetection*** and fire detection message ***XML_FireDetection*** :
added one node **<smokeAuxiliaryDetectionEnabled>** (enable fire and smoke detection or not).
2. Added an integration flow and description of fire and smoke alarm configuration, refer to ***Configure Fire and Smoke Alarm*** .

Summary of Changes in Version 2.0_12/2018

Related Product Type: Thermal Network Bullet Camera; Thermal Bi-spectrum Network Bullet Camera; Thermal & Optical Network Dome Camera V5.5.16

Related Product Model: DS-2TD21xx/V1 series, DS-2TD21xx/VP series, DS-2TD26xx series, DS-2TD26xx/V1 series, DS-2TD28xx series, DS-2TD28xx/V1 series, DS-2TD12xx/V1 series

1. Extended the capability of picture-in-picture configuration ***XML_Cap_ThermalPip*** and picture-in-picture configuration parameters ***XML_ThermalPip*** :
added two nodes: **<imageFusionRatio>** (image fusion ratio), **<borderFusionRatio>** (border fusion ratio).

Summary of Changes in Version 2.0_10/2018

Related Product Type: Thermographic Automation Thermal Camera

Related Product Model: DS-2TA03-15SVI, DS2TA06-25SVI

1. Extended the capability of temperature measurement basic parameters configuration ***XML_Cap_ThermometryBasicParam*** :
added ten nodes: **<isSupportAlertOutputIOPortList>**, **<isSupportAlarmOutputIOPortList>**, **<alertFilteringTime>** (temperature pre-alarm dwell time), **<alarmFilteringTime>** (temperature alarm dwell time), **<displayMaxTemperatureEnabled>** (whether displays the maximum temperature), **<displayMinTemperatureEnabled>** (whether displays the minimum temperature), **<displayAverageTemperatureEnabled>** (whether displays the average temperature), **<thermometryInfoDisplayposition>** (position of temperature measurement

information overlay), **<calibrationCoefficientEnabled>** (whether enables calibration coefficient), **<calibrationCoefficient>** (calibration coefficient).

2. Extended the temperature measurement basic configuration parameters

XML_ThermometryBasicParam :

added ten nodes: **<AlertOutputIOPortList>** (list of temperature pre-alarm output ports), **<AlarmOutputIOPortList>** (list of temperature alarm output ports), **<alertFilteringTime>** (temperature pre-alarm dwell time), **<alarmFilteringTime>** (temperature alarm dwell time), **<displayMaxTemperatureEnabled>** (whether displays the maximum temperature), **<displayMinTemperatureEnabled>** (whether displays the minimum temperature), **<displayAverageTemperatureEnabled>** (whether displays the average temperature), **<thermometryInfoDisplayposition>** (position of temperature measurement information overlay), **<calibrationCoefficientEnabled>** (whether enables calibration coefficient), **<calibrationCoefficient>** (calibration coefficient).

3. Added the function of getting calibration coefficient:

Get the calibration coefficient: POST **/ISAPI/Thermal/channels/<ID>/thermometry/basicParam/calibrationCoefficient** .

4. Extended the capability of temperature measurement preset (scene) configuration

XML_Cap_ThermometryScene :

added one sub node **<distanceUnit>** to node **<ThermometryRegion>**.

5. Extended the temperature measurement parameters of a specific detection region

XML_ThermometryRegion :

added one node **<distanceUnit>**.

6. Added the function of manually getting the temperature information of temperature measurement rules:

Get capability of manually getting temperature information of temperature measurement rules: GET **/ISAPI/Thermal/channels/<ID>/thermometry/<SID>/rulesTemperatureInfo/capabilities?format=json** ;

Manually get the temperature information of temperature measurement rules in batch: GET **/ISAPI/Thermal/channels/<ID>/thermometry/<SID>/rulesTemperatureInfo?format=json** ;

Manually get the single temperature information of temperature measurement rules: GET **/ISAPI/Thermal/channels/<ID>/thermometry/<SID>/rulesTemperatureInfo/<ID>?format=json** .

7. Extended the burning prevention capability **XML_BurningPreventionCap** and burning prevention parameters **XML_BurningPrevention** :

added three nodes: **<mode>**, **<closedDuration>** (close duration), **<shutterStatus>** (shutter status).

Summary of Changes in Version 2.0_09/2018

Related Product Type: Network Thermographic Automation Camera DS-2TA03-4AUM

1. Extended the capability of configuring temperature measurement basic parameters

XML_Cap_ThermometryBasicParam and basic configuration parameters of temperature measurement **XML_ThermometryBasicParam** :

- added two nodes: **<thermalOpticalTransmittance>** (optical transmissivity) and **<externalOpticsWindowCorrection>** (external optical temperature).
- Extended the capability of temperature measurement preset (scene) configuration ***XML_Cap_ThermometryScene*** :
added three nodes: **<maxPointNum>** (maximum number of points that can be configured), **<maxLineNum>** (maximum number of lines that can be configured), **<maxRegionNum>** (maximum number of frames that can be configured).
 - Extended the configuration capability of temperature measurement mode ***XML_Cap_ThermometryMode*** and temperature measurement mode configurations ***XML_ThermometryMode*** :
added one node **<thermometryROIEnabled>** (whether to enable ROI temperature measurement).
 - Extended the thermal product capability set ***XML_ThermalCap*** : added one node **<isSupportTemperatureCorrection>** (whether supports temperature measurement correction).
 - Added the function of configuring temperature correction parameters:
Get the capability set of temperature correction parameters: GET ***/ISAPI/Thermal/channels/<ID>/CorrectionParam/capabilities*** ;
Get or set temperature correction parameters: GET/PUT ***/ISAPI/Thermal/channels/<ID>/CorrectionParam*** ;
Import the temperature correction parameters: PUT ***/ISAPI/Thermal/channels/<ID>/CorrectionParam/import*** ;
Start correction: PUT ***/ISAPI/Thermal/channels/<ID>/Correction*** .
 - Extended the configuration capability of pixel-to-pixel temperature measurement parameters ***XML_PixelToPixelParamCap*** and pixel-to-pixel temperature measurement parameters ***XML_PixelToPixelParam*** :
added three parameters: **<distance>**, **<distanceUnit>** (distance unit), and **<temperatureDataLength>** (temperature data length).

Summary of Changes in Version 2.0_07/2018

- Extended the temperature screening capability ***XML_Cap_FaceThermometry*** and temperature screening configuration parameters ***XML_FaceThermometry*** :
added one node **<mode>** (temperature screening mode).
- Extended the basic configuration capability of temperature measurement ***XML_Cap_ThermometryBasicParam*** and basic configuration parameters of temperature measurement ***XML_ThermometryBasicParam*** :
added one temperature range "-20 to 120" to node **<temperatureRange>**, added two nodes **<distanceUnit>** (distance unit) and **<Distance>** (distance).

Summary of Changes in Version 1.0_01/2018

- All contents are newly added (the first version).

Chapter 2 Upload Real-Time Thermal Stream via RTSP

You can upload the real-time thermal stream via RTSP directly without any plug-in. The Real Time Streaming Protocol (RTSP) is a network control protocol designed for use in entertainment and communications systems to control streaming media servers.

Before You Start

Make sure you have logged in to the device.

Steps

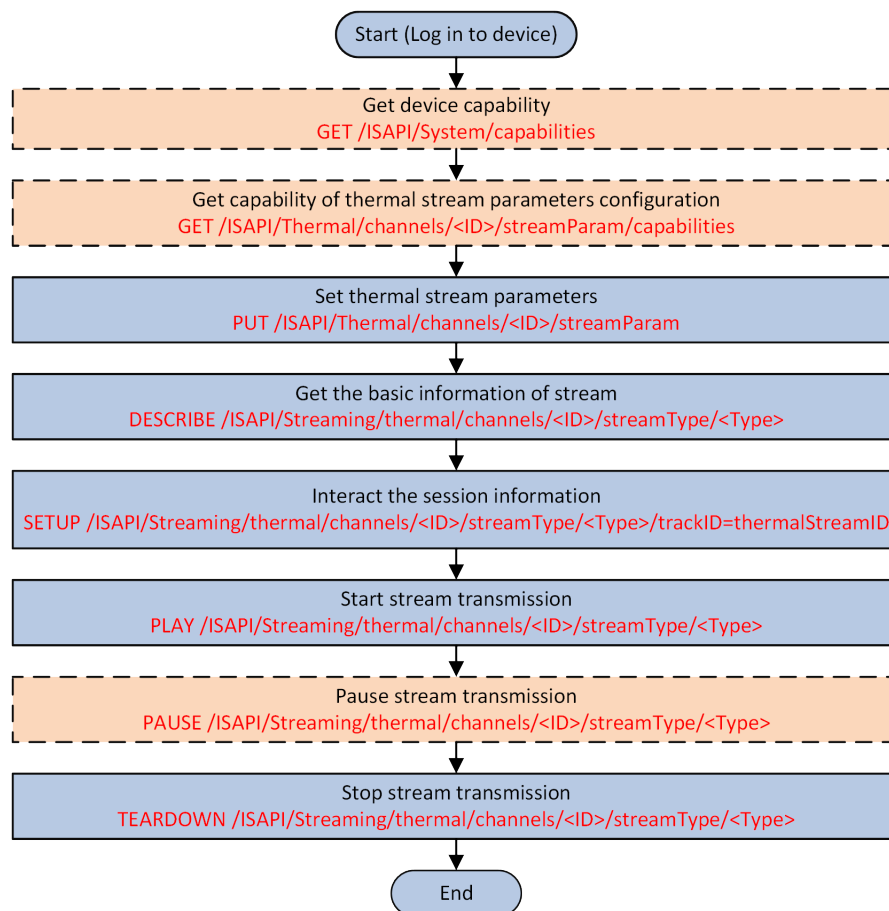


Figure 2-1 Programming Flow of Uploading Real-Time Thermal Stream via RTSP

1. **Optional:** Call `/ISAPI/System/capabilities` by GET method to get the device capability to check whether the device supports uploading real-time thermal stream via node `<isSupportThermalStreamData>`.

The device capability is returned in *XML_DeviceCap*

2. **Optional:** Call `/ISAPI/Thermal/channels/<ID>/streamParam/capabilities` by GET method to get the video coding type.

The thermal stream parameters are returned in *XML_ThermalStreamParamCap* .

3. Call `/ISAPI/Thermal/channels/<ID>/streamParam` by PUT method to set the thermal stream parameters.



Note

Before setting the thermal stream parameters, you can get the thermal stream parameters by the URI: GET `/ISAPI/Thermal/channels/<ID>/streamParam` .

4. Call `rtsp://<ipaddress>[:port]/ISAPI/Streaming/thermal/channels/<ID>/streamType/<Type>` by DESCRIBE method to get the basic information of stream via RTSP.

Example

Interaction Sample of DESCRIBE Method

DESCRIBE `rtsp://10.10.36.212:554/ISAPI/Streaming/thermal/channels/1/streamType/thermal_raw_data RTSP/1.0`

```
DESCRIBE
rtsp://10.10.36.212:554/ISAPI/Streaming/thermal/channels/1/streamType/thermal_raw_data RTSP/1.0
CSeq:0
Accept:application/sdp
Authorization:
User-Agent:NKPlayer-VSPlayer1.0
```

```
RTSP/1.0 401 Unauthorized
CSeq: 0
WWW-Authenticate: Digest realm="IP Camera(51289)",
nonce="9fe43f1322976c0aa218b0f23f554b44", stale="FALSE"
Date: Mon, Dec 18 2017 04:37:30 GMT
```

```
DESCRIBE
rtsp://10.10.36.212:554/ISAPI/Streaming/thermal/channels/1/streamType/thermal_raw_data RTSP/1.0
CSeq:0
Accept:application/sdp
Authorization:Digest username="admin", realm="IP Camera(51289)",
nonce="9fe43f1322976c0aa218b0f23f554b44",
uri="rtsp://10.10.36.212:554/ISAPI/Streaming/thermal/channels/1/streamType/thermal_raw_data",
response="99b683462210bd819c8d6bafb39dd262"
User-Agent:NKPlayer-VSPlayer1.0
```

```
RTSP/1.0 200 OK
CSeq: 0
Content-Type: application/sdp
Content-Length: 737
v=0 o=- 1513571850040093 1513571850040093 IN IP4 10.10.36.212
s=Media Presentation
e=NONE
b=AS:5100 t=0 0
```

```
a=control:*
m=video 0 RTP/AVP 96
c=IN IP4 0.0.0.0
b=AS:5000
a=recvonly
a=x-dimensions:1920,1080
a=control:trackID=1
a=rtpmap:96 H264/90000
a=fmtp:96 profile-level-id=420029; packetization-mode=1; sprop-parameter-sets=Z00AKp2oHgCJ
+WbglCAoAAADAAGAAAMBICA=,aO48gA==
m=audio 0 RTP/AVP 0
c=IN IP4 0.0.0.0
b=AS:50
a=recvonly
a=control:trackID=2
a=rtpmap:0 PCMU/8000
m=application 0 RTP/AVP 107
c=IN IP4 0.0.0.0 b=AS:50
a=recvonly
a=control:trackID=3
a=rtpmap:107 isapi.metadata/90000
m=application 0 RTP/AVP 109
c=IN IP4 0.0.0.0
b=AS:50
a=recvonly
a=control:trackID=5
a=rtpmap:109 thermalStream/90000
a=Media_header:MEDIAINFO=494D4B48010200000400000110710110401F000000FA0000000000000000000000
00000000000000;
a=appversion:1.0
```

The parsed stream information is returned by the device, and for thermal stream data, the trackID=5, that is the information followed trackID=5 is the thermal stream information.

5. Call **rtsp://<ipaddress>[:port]/ISAPI/Streaming/thermal/channels/<ID>/streamType/<Type>/trackID=** by SETUP method to interact the session information.

Example

Interaction Sample of SETUP Method

SETUP rtsp://10.10.36.212:554/ISAPI/Streaming/thermal/channels/1/streamType/
thermal_raw_data/trackID=5

```
SETUP
rtsp://10.10.36.212:554/ISAPI/Streaming/thermal/channels/1/streamType/thermal_raw_data/trackID=5 RTSP/1.0
Seq:3
Authorization:Digest username="admin", realm="IP Camera(51289)",
nonce="9fe43f1322976c0aa218b0f23f554b44",
uri="rtsp://10.10.36.212:554/ISAPI/Streaming/thermal/channels/1/streamType/thermal_raw_data",
response="46b39a563fd9977eff83067adc4e587b"
Session: 408703626
Transport: RTP/AVP/UDP;unicast;
User-Agent:NKPlayer-VSPlayer1.0
```

```
RTSP/1.0 200 OK
CSeq:0
Session: 408703626;timeout=60
Transport: RTP/AVP/UDP;unicast;interleaved=6-7;ssrc=14b37590;mode="play"
Date: Mon, Dec 18 2017 04:37:30 GMT
```

6. Call ***rtsp://<ipaddress>[:port]/ISAPI/Streaming/thermal/channels/<ID>/streamType/<Type>*** by PLAY method to start stream transmission.

Example

Interaction Sample of PLAY Method

PLAY ***rtsp://10.10.36.212:554/ISAPI/Streaming/thermal/channels/1/streamType/thermal_raw_data/1.0***

```
PLAY
rtsp://10.10.36.212:554/ISAPI/Streaming/thermal/channels/1/streamType/thermal_raw_data RTSP/1.0
CSeq:4
Authorization: Digest username="admi/n", realm="IP Camera(C7968)",
nonce="e554bd56d935913abec7cc6d8f37a175",
uri=" rtsp://10.10.36.212:554/ISAPI/Streaming/thermal/channels/1/streamType/thermal_raw_data",
response="a568b39cb69f993462c9d03f0c3e69fd"
Session:1061719290;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112

RTSP/1.0 200 OK
CSeq: 4
Session: 1061719290
RTP-Info: url=trackID=5;seq=29477
Date: Fri, Dec 28 2018 15:36:28 GMT
```

The device starts to upload stream data in the format of RTP packet.



Note

The RTP header payload type of thermal stream is 0x6D, the upper layer can check whether the data is thermal stream data via the payload type, refer to for details.

7. **Optional:** Call ***rtsp://<ipaddress>[:port]/ISAPI/Streaming/thermal/channels/<ID>/streamType/<Type>*** by PAUSE method to pause stream transmission.
8. Call ***rtsp://<ipaddress>[:port]/ISAPI/Streaming/thermal/channels/<ID>/streamType/<Type>*** by TEARDOWN method to stop stream transmission.

Example

Interaction Sample of TEARDOWN Method

TEARDOWN ***rtsp://10.10.36.212:554/ISAPI/Streaming/thermal/channels/1/streamType/thermal_raw_data RTSP/1.0***

```
TEARDOWN
rtsp://10.10.36.212:554/ISAPI/Streaming/thermal/channels/1/streamType/thermal_raw_data RTSP/1.0
```

```
CSeq:4
Authorization: Digest username="admin", realm="IP Camera(51289)",
nonce="9fe43f1322976c0aa218b0f23f554b44",
uri="rtsp://10.10.36.212:554/ISAPI/Streaming/thermal/channels /1/streamType/thermal_raw_data",
response="2822238cec4028c34749cf3a48ce7519"
Session: 408703626;timeout=60

User-Agent:NKPlayer-VSPlayer1.0
```

Chapter 3 Alarm or Event

When the alarm is triggered or the event occurred, if you have configured alarm/event uploading parameters, you can receive and process the alarm/event information in the third-party platform or system. Two modes are available for receiving alarms, including arming mode and listening mode.

Arming Mode

When the alarm is triggered or event occurred, the third-party platform or system can send the request URL to the device for getting the alarm/event stream, and then the device uploads the response message with alarm/event information.

Listening Mode

When alarm is triggered or event occurred, the device uploads the alarm information automatically, and then the third-party platform or system can receives the alarm/event by configuring listening port of HTTP host server.



Note

Currently, for traffic camera or capture camera, receiving alarm or event in arming mode is not supported.

3.1 Alarm or Event Settings

Before the alarms can be triggered or the events can be detected, you must configure parameters, such as detection rule, alarm threshold, arming schedule, linkage action, and so on, for different alarms or events. This chapter shows the configuration processes of alarms or events with multiple types.

3.1.1 Configure Temperature Screening Alarm

Temperature screening can detect the temperature of human face or body at large-scale area in the public places (e.g., customs, airports, schools, hospitals) with high people density, and help to fast find and track the person with high temperature. Then, the alarm will be remotely uploaded to the monitor center via the network for real-time analysis and handling.

Before You Start

Make sure you have logged in to the device.

Steps

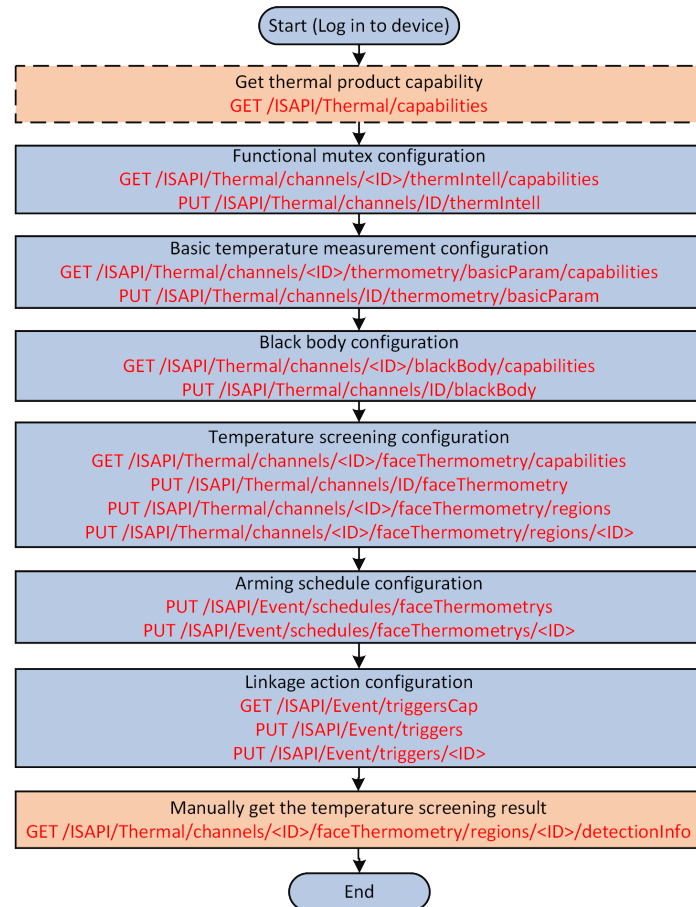


Figure 3-1 Programming Flow of Configuring Temperature Screening Alarm

1. **Optional:** Call **/ISAPI/Thermal/capabilities** by GET method to get the thermal capability to check if the following functions are supported.
The thermal product capability **XML_ThermalCap** is returned.
2. Configure functional mutex parameters.
 - 1) **Optional:** Call **/ISAPI/Thermal/channels/<ID>/thermIntell/capabilities** by GET method to get the mutex capability to check the supported configuration options.
 - 2) Call **/ISAPI/Thermal/channels/<ID>/thermIntell** by PUT method to set functional mutex parameters.

Note

- To switch to each exclusive functions, you should reboot the device to take effect.
- Before setting the functional mutex parameters, you'd better perform GET on the configuration URL to get the existing or configured parameters for reference.

3. Configure basic temperature measurement parameters.

- 1) **Optional:** Call **`/ISAPI/Thermal/channels/<ID>/thermometry/basicParam/capabilities`** by GET method to get the basic configuration capability to check the supported configuration options.
- 2) Call **`/ISAPI/Thermal/channels/<ID>/thermometry/basicParam`** by PUT method to set the basic temperature measurement parameters.



Note

Before setting the basic temperature measurement parameters, you'd better perform GET on the configuration URL to get the existing or configured parameters for reference.

4. Configure black body parameters.

- 1) **Optional:** Call **`/ISAPI/Thermal/channels/<ID>/blackBody/capabilities`** by GET method to get black body configuration capability to check the supported configuration options.
- 2) Call **`/ISAPI/Thermal/channels/<ID>/blackBody`** by PUT method to set the black body parameters.



Note

- The black body is a heating element with adjustable temperature and regular shape, which is used as the reference temperature for the calibration of temperature screening. It can improve the accuracy of temperature measurement.
 - Before setting the black body parameters, you'd better perform GET on the configuration URL to get the existing or configured parameters for reference.
-

5. Configure temperature screening parameters.

- 1) **Optional:** Call **`/ISAPI/Thermal/channels/<ID>/faceThermometry/capabilities`** by GET method to get the temperature screening configuration capability to check the supported configuration options.
- 2) Call **`/ISAPI/Thermal/channels/<ID>/faceThermometry`** by PUT method to set the temperature screening parameters.
- 3) Call **`/ISAPI/Thermal/channels/<ID>/faceThermometry/regions`** or **`/ISAPI/Thermal/channels/<ID>/faceThermometry/regions/<ID>`** by PUT method to set the temperature measurement rules of all detection regions or a specific detection region.



Note

Before setting temperature screening parameters or rules, you'd better perform GET on the configuration URL to get the existing or configured parameters for reference.

6. Call **`/ISAPI/Event/schedules/faceThermometrys`** or **`/ISAPI/Event/schedules/faceThermometrys/<ID>`** by PUT method to configure arming schedule for all temperature screening alarms or a specific alarm.



Note

Before setting the arming schedule, you'd better perform GET on the configuration URLs to get the existing or configured parameters for reference.

7. Configure linkage actions.

- 1) **Optional:** Call `/ISAPI/Event/triggersCap` by GET method to get the linkage action configuration capability to check the supported configuration options.
- 2) Call `/ISAPI/Event/triggers` or `/ISAPI/Event/triggers/<eventType>-<ID>` by PUT method to set linkage actions for all events/alarms or a specific event/alarm.



Note

- The linkage action must be set to "center" (upload to alarm center) if you want to receive and view the alarm information.
- Before setting linkage actions, you'd better perform GET on the configuration URLs to get the existing or configured parameters for reference.

-
8. **Optional:** Call `/ISAPI/Thermal/channels/<ID>/faceThermometry/regions/<ID>/detectionInfo` by GET method to manually get the temperature screening results.

3.1.2 Configure Temperature Measurement Alarm in Normal Mode

Temperature measurement is mainly applied to the electric and industrial automation fields, which provides accurate and stable temperature measurement for the mid-to-high end manufacturers (e.g., intelligent robot) to realize the process control, automatic detection, and status monitoring of electromechanical devices. You can set the temperature measurement in normal mode to measure the global temperature during the industrial production. When the measured temperature does not meet the configured temperature threshold, the alarm will be triggered and uploaded to monitoring center if configured. And then, you can receive the alarm.

Before You Start

Make sure you have logged in to the device.

Steps

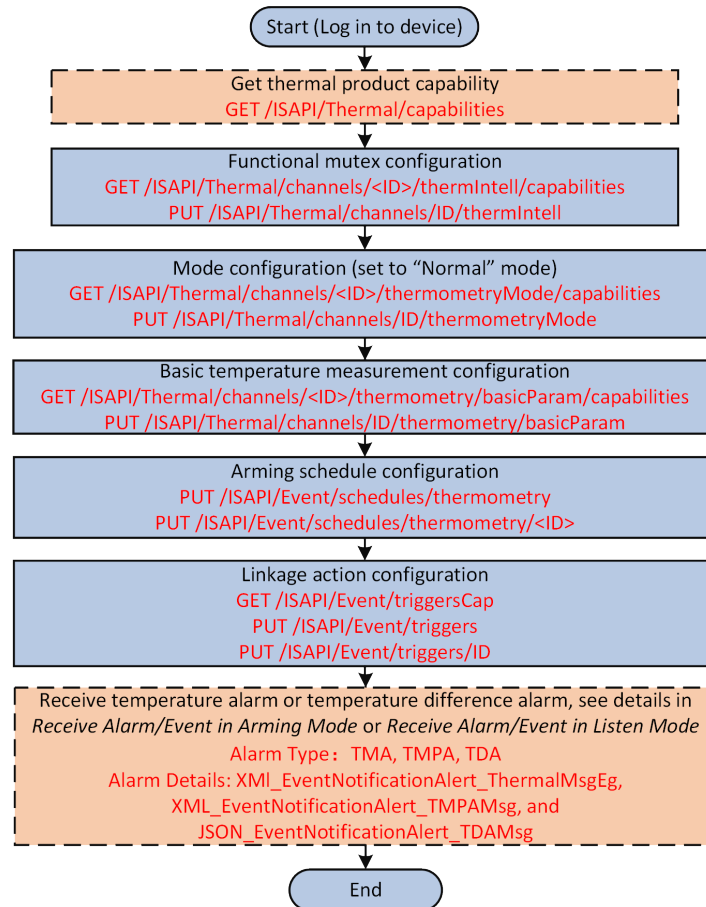


Figure 3-2 Programming Flow of Configuring Temperature Measurement Alarm in Normal Mode

Note

The temperature measurement parameters, temperature measurement schedule, rules, and linkage actions can be configured directly by logging in to the device via web page.

1. **Optional:** Call `/ISAPI/Thermal/capabilities` by GET method to get the thermal capability to check if the following functions are supported.
The thermal product capability `XML_ThermalCap` is returned.
2. Configure functional mutex parameters.
 - 1) **Optional:** Call `/ISAPI/Thermal/channels/<ID>/thermIntell/capabilities` by GET method to get the mutex capability to check the supported configuration options.
 - 2) Call `/ISAPI/Thermal/channels/<ID>/thermIntell` by PUT method to set functional mutex parameters.

Note

- To switch to each exclusive functions, you should reboot the device to take effect.
 - Before setting the functional mutex parameters, you'd better perform GET on the configuration URL to get the existing or configured parameters for reference.
-

3. Configure temperature measurement mode.

- 1) **Optional:** Call ***/ISAPI/Thermal/channels/<ID>/thermometryMode/capabilities*** by GET method to get the configuration capability of temperature measurement mode to check the supported configuration options.
 - 2) Call ***/ISAPI/Thermal/channels/<ID>/thermometryMode*** by PUT method to set the temperature measurement mode to "normal".
-

Note

Before setting the mode, you can perform GET operation on the URL to get the existing or configured mode for reference.

4. Configure basic temperature measurement parameters.

- 1) **Optional:** Call ***/ISAPI/Thermal/channels/<ID>/thermometry/basicParam/capabilities*** by GET method to get the basic configuration capability to check the supported configuration options.
 - 2) Call ***/ISAPI/Thermal/channels/<ID>/thermometry/basicParam*** by PUT to set the basic temperature measurement parameters.
-

Note

- Before setting the basic temperature measurement parameters, you'd better perform GET on the configuration URL to get the existing or configured parameters for reference.
 - You can get the calibration coefficient via the URL: POST ***/ISAPI/Thermal/channels/<ID>/thermometry/basicParam/calibrationCoefficient*** .
-

5. Call ***/ISAPI/Event/schedules/thermometry/<ID>*** or ***/ISAPI/Event/schedules/Thermometry*** by PUT method to configure arming schedule for all temperature measurement alarms or a specific alarm.

Note

Before setting the arming schedule, you'd better perform GET on the configuration URLs to get the existing or configured parameters for reference.

6. Configure linkage actions.

- 1) **Optional:** Call ***/ISAPI/Event/triggersCap*** by GET method to get the linkage action configuration capability to check the supported configuration options.
- 2) Call ***/ISAPI/Event/triggers*** or ***/ISAPI/Event/triggers/<eventType>-<ID>*** by PUT method to set linkage actions for all events/alarms or a specific event/alarm.

Note

- The linkage action must be set to "center" (upload to alarm center) if you want to receive and view the alarm information.
- Before setting linkage actions, you'd better perform GET on the configuration URLs to get the existing or configured parameters for reference.

-
7. **Optional:** Receive temperature alarm or temperature difference alarm in arming mode (see *Receive Alarm/Event in Arming Mode* for details) or listening mode (see *Receive Alarm/Event in Listening Mode* for details).

Note

The alarm type (**eventType**) refers to "TMA" (temperature measurement alarm), "TMPA" (temperature measurement pre-alarm), and "TDA" (temperature difference alarm), and for the alarm details, refer to the message *XML_EventNotificationAlert_TemperatureAlarmMsg* , *XML_EventNotificationAlert_TMPAMsg* , and *JSON_EventNotificationAlert_TDAMsg* .

3.1.3 Configure Temperature Measurement Alarm in Expert Mode

Temperature measurement is mainly applied to the electric and industrial automation fields, which provides accurate and stable temperature measurement for the mid-to-high end manufacturers (e.g., intelligent robot) to realize the process control, automatic detection, and status monitoring of electromechanical devices. You can set the temperature measurement in expert mode to measure the temperature according to areas or presets during the industrial production. When the measured temperature does not meet the configured temperature threshold, the alarm will be triggered and uploaded to monitoring center if configured. And then, you can receive the alarm.

Before You Start

Make sure you have logged in to the device.

Steps

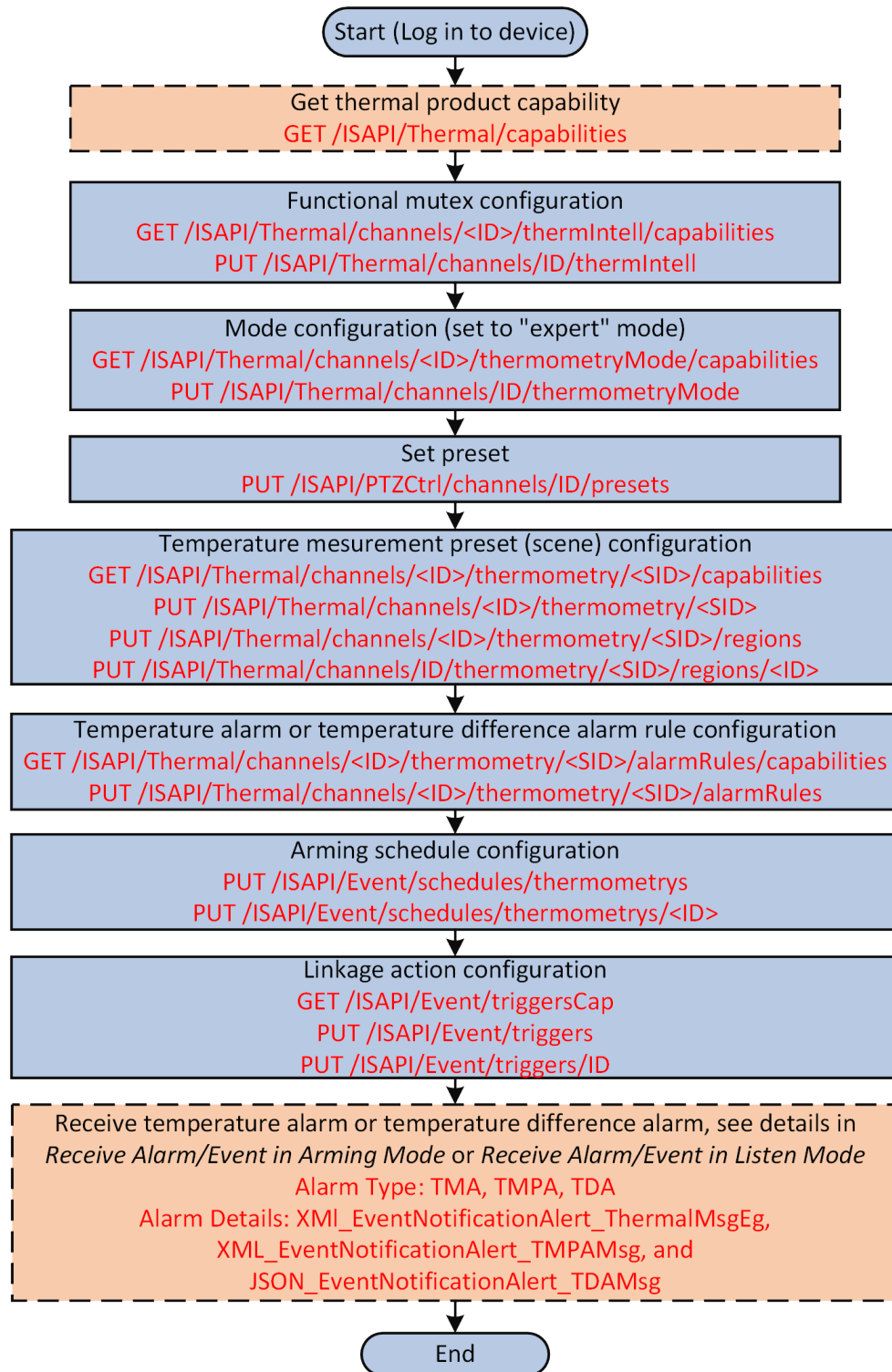


Figure 3-3 Programming Flow of Configuring Temperature Measurement Alarm in Expert Mode

Note

The temperature measurement parameters, temperature measurement schedule, rules, and linkage actions can be configured directly by logging in to the device via web page.

1. **Optional:** Call ***/ISAPI/Thermal/capabilities*** by GET method to get the thermal capability to check if the following functions are supported.

The thermal product capability ***XML_ThermalCap*** is returned.

2. Configure functional mutex parameters.

- 1) **Optional:** Call ***/ISAPI/Thermal/channels/<ID>/thermIntell/capabilities*** by GET method to get the mutex capability to check the supported configuration options.
 - 2) Call ***/ISAPI/Thermal/channels/<ID>/thermIntell*** by PUT method to set functional mutex parameters.
-

Note

- To switch to each exclusive functions, you should reboot the device to take effect.
 - Before setting the functional mutex parameters, you'd better perform GET on the configuration URL to get the existing or configured parameters for reference.
-

3. Configure temperature measurement mode.

- 1) **Optional:** Call ***/ISAPI/Thermal/channels/<ID>/thermometryMode/capabilities*** by GET method to get the temperature measurement mode configuration capability to check if it is supported.
 - 2) Call ***/ISAPI/Thermal/channels/<ID>/thermometryMode*** by PUT method to set the temperature measurement mode to "expert".
-

Note

Before setting the mode, you can perform GET operation on the configuration URL to get the existing or configured mode for reference.

4. Call ***/ISAPI/PTZCtrl/channels/<ID>/presets*** by PUT to set the preset for measuring temperature.
 5. Configure parameters (including temperature measurement points, lines, and frames) for the temperature measurement preset (scene).
 - 1) **Optional:** Call ***/ISAPI/Thermal/channels/<ID>/thermometry/<SID>/capabilities*** by GET method to get the temperature measurement preset (scene) configuration capability check the supported configuration options.
 - 2) Call ***/ISAPI/Thermal/channels/<ID>/thermometry/<SID>*** by PUT method to set the preset (scene) parameter.
 - 3) Call ***/ISAPI/Thermal/channels/<ID>/thermometry/<SID>/regions*** or ***/ISAPI/Thermal/channels/<ID>/thermometry/<SID>/regions/<ID>*** by PUT method to set the temperature measurement points, lines, and frames for all detection regions or a specific region in a preset (scene).
-

Note

Before setting the temperature measurement preset (scene) parameters, you'd perform GET operation on the configuration URLs to get the existing or configured parameters for reference.

6. Configure rules for temperature alarm or temperature difference alarm of a preset (scene).
 - 1) **Optional:** Call `/ISAPI/Thermal/channels/<ID>/thermometry/<SID>/alarmRules/capabilities` by GET method to get the rule configuration capability to check the supported configuration options.
 - 2) Call `/ISAPI/Thermal/channels/<ID>/thermometry/<SID>/alarmRules` by PUT method to set the rule parameters of a preset (scene).
-

Note

Before setting the rule parameters, you'd perform GET operation on the configuration URL to get the existing or configured parameters for reference.

7. Call `/ISAPI/Event/schedules/Thermometry` or `/ISAPI/Event/schedules/thermometry/<ID>` by PUT method to configure arming schedule for all temperature measurement alarms or a specific alarm.
-

Note

Before setting the arming schedule, you'd better perform GET on the configuration URLs to get the existing or configured parameters for reference.

8. Configure linkage actions.
 - 1) **Optional:** Call `/ISAPI/Event/triggersCap` by GET method to get the linkage action configuration capability to check the supported configuration options.
 - 2) Call `/ISAPI/Event/triggers` or `/ISAPI/Event/triggers/<eventType>-<ID>` by PUT method to set linkage actions for all events/alarms or a specific event/alarm.
-

Note

- The linkage action must be set to "center" (upload to alarm center) if you want to receive and view the alarm information.
 - Before setting linkage actions, you'd better perform GET on the configuration URLs to get the existing or configured parameters for reference.
-

9. **Optional:** Receive temperature alarm or temperature difference alarm in arming mode (see *Receive Alarm/Event in Arming Mode* for details) or listening mode (see *Receive Alarm/Event in Listening Mode* for details).
-

Note

The alarm type (**eventType**) refers to "TMA" (temperature measurement alarm), "TMPA" (temperature measurement pre-alarm), and "TDA" (temperature difference alarm), and for the alarm details, refer to the message *XML_EventNotificationAlert_TemperatureAlarmMsg* , *XML_EventNotificationAlert_TMPAMsg* , and *JSON_EventNotificationAlert_TDAMsg* .

3.1.4 Configure Fire and Smoke Alarm

In the actual fire detection, the fire source may be concealed or the distance is too far to be detected, so the fire alarm may failed to be triggered sometimes. To avoid the above situation, you can configure the fire and smoke alarm, which includes fire source detection and smoke detection, once the fire source or smoke is detected, the alarm will be triggered.

Before You Start

Make sure you have logged in to device.

Steps

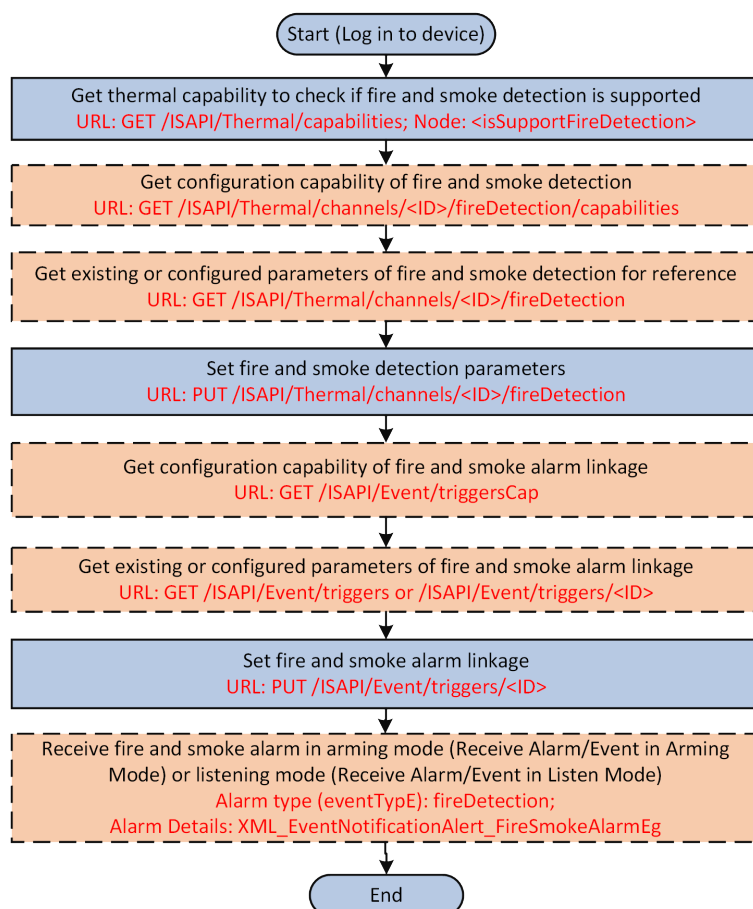


Figure 3-4 Programming Flow of Configuring Fire and Smoke Alarm

1. Call `/ISAPI/Thermal/capabilities` by GET method to get thermal capability to check if fire and smoke detection is supported.

The thermal capability is returned in the message ***XML_ThermalCap***.

If the node **`<isSupportFireDetection>`** exists in the thermal capability message and its value is "true", it indicates that fire and smoke detection is supported by the device, and you can perform the following steps.

Otherwise, the fire and smoke detection supported, you should end the task.

2. **Optional:** Call `/ISAPI/Thermal/channels/<ID>/fireDetection/capabilities` by GET method to get configuration capability of fire and smoke detection.
The configuration capability of fire and smoke detection is returned in the message ***XML_Cap_FireDetection***.
3. **Optional:** Call `/ISAPI/Thermal/channels/<ID>/fireDetection` by GET method to get existing or configured parameters of fire and smoke detection for reference.
4. Call `/ISAPI/Thermal/channels/<ID>/fireDetection` by PUT method to set fire and smoke detection parameters by the request URL: .
5. **Optional:** Call `/ISAPI/Event/triggersCap` by GET to get configuration capability of fire and smoke alarm linkage.
The configuration capability of fire and smoke alarm linkage is returned in the message
6. **Optional:** Call `/ISAPI/Event/triggers` or by GET method to get existing or configured parameters of fire and smoke alarm linkage for reference.
7. Call by PUT method to set fire and smoke alarm linkage.



Note

- To receive alarm in platform, the linkage action must be set to "center" (upload to center).
- The fire and smoke detection parameters and alarm linkage can also be configured by logging in to device via web browser.

-
8. Receive fire and smoke alarm in arming mode (see ***Receive Alarm/Event in Arming Mode*** for details) or listening mode (see for details).



Note

The alarm type (**eventType**) refers to "fireDetection", and for the alarm details, refer to the message ***XML_EventNotificationAlert_FireSmokeAlarmEg***.

3.1.5 Configure Ship Flow Detection Alarm

Ship flow detection can calculate the amount of ships passing through a predefined area, and triggers an alarm when the ship crosses the predefined line. It is widely applied to river monitoring under severe weather conditions, such as heavy rain, heavy fog, and so on.

Before You Start

Make sure you have logged in to the device.

Steps

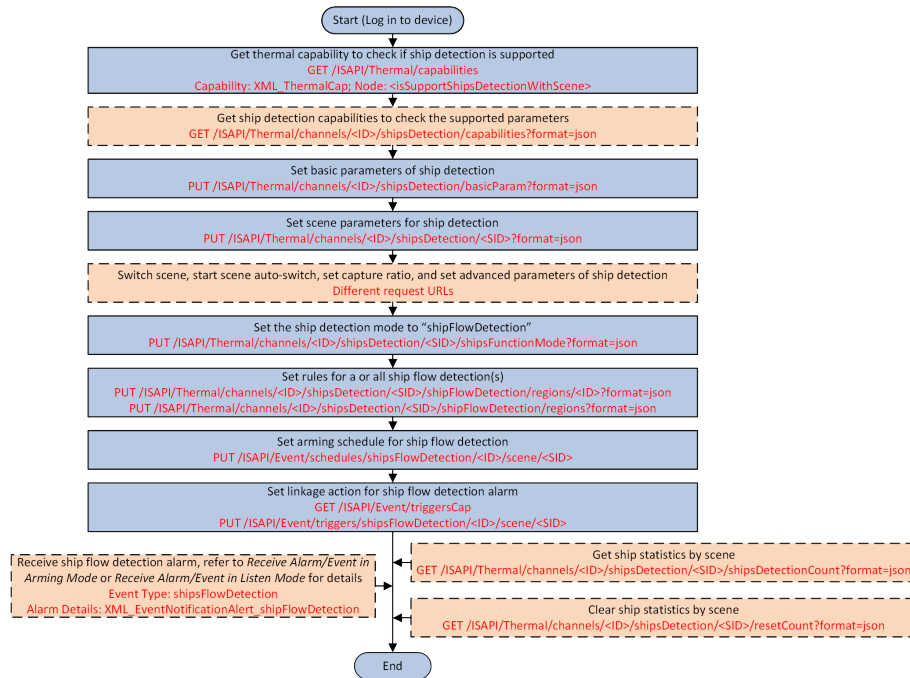


Figure 3-5 Programming Flow of Configuring Ship Flow Detection Alarm

1. Call **/ISAPI/Thermal/capabilities** by GET method for getting the thermal capability to check if the device supports ship detection by scene.
The thermal capability is returned in the message **XML_ThermalCap** .
If supports, the node **<isSupportShipsDetectionWithScene>** is returned in the message and its value is "true", and then you can perform the following steps.
Otherwise, ship detection by scene is not supported by device, please end this task.
2. **Optional:** Call **/ISAPI/Thermal/channels/<ID>/shipsDetection/capabilities?format=json** by GET method for getting the ship detection capabilities to check the supported parameters.
The ship detection capability is returned in the message **JSON_shipsDetectionCap** .
3. Set basic parameters of ship detection.
 - 1) **Optional:** Call **/ISAPI/Thermal/channels/<ID>/shipsDetection/basicParam/capabilities?format=json** by GET method for getting the basic parameters configuration capability of ship detection to check the supported parameters.
The basic configuration capability of ship detection is returned in the message **JSON_basicParamCap** .
 - 2) **Optional:** Call **/ISAPI/Thermal/channels/<ID>/shipsDetection/basicParam?format=json** by PUT method for getting the default or configured basic parameters of ship detection for reference.
The basic parameters of ship detection are returned in the message **JSON_basicParam** .
 - 3) Call **/ISAPI/Thermal/channels/<ID>/shipsDetection/basicParam?format=json** by PUT method for setting basic parameters for ship detection.

4. Set scene parameters for ship detection.

- 1) **Optional:** Call */ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/capabilities?format=json* by GET method for getting scene configuration capability.

The scene configuration capability is returned in the message

JSON_ShipsDetectionSceneCap .

- 2) **Optional:** Call */ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>?format=json* or */ISAPI/Thermal/channels/<ID>/shipsDetection?format=json* by GET method for getting default or configured parameters of a or all scene(s) for reference.

The parameters of a or all scene(s) are returned in the message *JSON_ShipsDetectionScene* or *JSON_ShipsDetectionSceneList* .

- 3) Call */ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>?format=json* or */ISAPI/Thermal/channels/<ID>/shipsDetection?format=json* by PUT method for creating a or all scenes(s).
- 4) Call */ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>?format=json* or */ISAPI/Thermal/channels/<ID>/shipsDetection?format=json* by PUT method for setting parameters of a or all scenes.
- 5) **Optional:** Call */ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>?format=json* or */ISAPI/Thermal/channels/<ID>/shipsDetection?format=json* by DELETE method for deleting a or all scenes(s).

5. **Optional:** Perform the following operation(s) to switch scene, start scene auto-switch, set capture ratio for scene, and set advanced parameters for ship detection.

Switch Scene	PUT <i>/ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/goto?format=json</i>
---------------------	---

Scene Auto-Switch	a. GET <i>/ISAPI/Thermal/channels/<ID>/shipsDetection/sceneTrace/capabilities?format=json</i> for getting scene auto-switch capability. b. GET <i>/ISAPI/Thermal/channels/<ID>/shipsDetection/sceneTrace?format=json</i> for getting default or configured scene auto-switch parameters for reference. c. PUT <i>/ISAPI/Thermal/channels/<ID>/shipsDetection/sceneTrace?format=json</i> for starting scene auto-switch.
--------------------------	---

Capture Ratio	a. GET <i>/ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/captureRatio/capabilities?format=json</i> for getting configuration capability of capture ratio. b. GET <i>/ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/captureRatio?format=json</i> for getting default or configured capture ratio for reference. c. PUT <i>/ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/captureRatio?format=json</i> for setting capture ratio.
----------------------	--

Advanced Configuration

- a. GET ***/ISAPI/Thermal/channels/<ID>/shipsDetection/advanceParam/capabilities?format=json*** for getting advanced configuration capability of ship detection to check the supported parameters.
 - b. GET ***/ISAPI/Thermal/channels/<ID>/shipsDetection/advanceParam?format=json*** for getting default or configured advanced parameters of ship detection for reference.
 - c. PUT ***/ISAPI/Thermal/channels/<ID>/shipsDetection/advanceParam?format=json*** for setting advanced parameters of ship detection.
6. Set function mode to "shipFlowDetection" to detect the ship flow.
- 1) **Optional:** Call ***/ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipsFunctionMode/capabilities?format=json*** by GET method for getting function mode capability to check the supported modes.
The function mode capability is returned in the message ***JSON_Cap_functionMode***.
 - 2) **Optional:** Call ***/ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipsFunctionMode?format=json*** by GET method for getting default or configured function mode for reference.
The function mode is returned in the message ***JSON_functionMode***.
 - 3) Call ***/ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipsFunctionMode?format=json*** by PUT method for setting the function mode to "shipFlowDetection".
7. Set ship flow detection rules.
- 1) **Optional:** Call ***/ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipFlowDetection/regions/capabilities?format=json*** by GET method for getting rule configuration capability of ship flow detection to check the supported rule parameters.
The rule configuration capability of ship flow detection is returned in the message ***JSON_ShipFlowDetectionRuleListCap***.
 - 2) **Optional:** Call ***/ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipFlowDetection/regions/<ID>?format=json*** or ***/ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipFlowDetection/regions?format=json*** by GET method for getting default or configured rules of a or all ship flow detection(s) for reference.
The rules of a or all ship flow detection(s) are returned in the message ***JSON_ShipFlowDetectionRule*** or ***JSON_ShipFlowDetectionRuleList***.
 - 3) Call ***/ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipFlowDetection/regions/<ID>?format=json*** or ***/ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipFlowDetection/regions?format=json*** by PUT method for setting rules of a or all ship flow detection(s).
8. Call ***/ISAPI/Event/schedules/shipsFlowDetection/<ID>/scene/<SID>*** by PUT method for setting the arming schedule of ship flow detection.



Note

Before setting arming schedule, you can transmission this URL by GET method to get default or configured arming schedule for reference.

9. Call **`/ISAPI/Event/triggers/shipsFlowDetection/<ID>/scene/<SID>`** by PUT method for setting linkage action of ship flow detection alarm.



Note

Before setting linkage action, you can transmission this URL by GET method to get default or configured linkage actions for reference.

10. **Optional:** Perform the following operation(s) after configuring ship flow detection alarm.

Get Ship Statistics by Scene

GET **`/ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipsDetectionCount?format=json`**

Clear Ship Statistics by Scene

PUT **`/ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/resetCount?format=json`**

Receive Alarm in Arming or Listening Mode

Perform the flow of ***Receive Alarm/Event in Arming Mode*** or ***Receive Alarm/Event in Listening Mode*** .



Note

The alarm type (**`eventType`**) is "shipsFlowDetection", and for the alarm details, refer to the message **`XML_EventNotificationAlert_shipFlowDetection`** .

3.1.6 Configure Dredger Detection Alarm

Dredger detection can detect the ship staying time to distinguish if the ship is dredger, when the ship staying time exceeds the configured threshold, the dredger detection alarm will be triggered. It is widely applied to river monitoring under severe weather conditions, such as heavy rain, heavy fog, and so on.

Before You Start

Make sure you have logged in to the device.

Steps

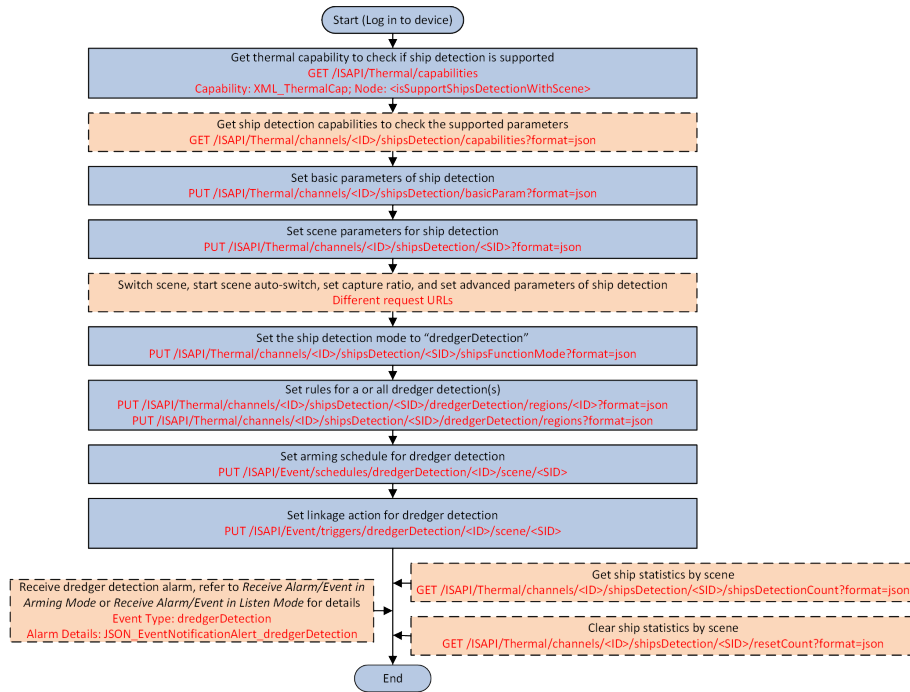


Figure 3-6 Programming Flow of Configuring Dredger Detection Alarm

1. Call **`/ISAPI/Thermal/capabilities`** by GET method for getting the thermal capability to check if the device supports ship detection by scene.
The thermal capability is returned in the message **`XML_ThermalCap`** .
If supports, the node **`<isSupportShipsDetectionWithScene>`** is returned in the message and its value is "true", and then you can perform the following steps.
Otherwise, ship detection by scene is not supported by device, please end this task.
2. **Optional:** Call **`/ISAPI/Thermal/channels/<ID>/shipsDetection/capabilities?format=json`** by GET method for getting the ship detection capability to check the supported parameters.
The ship detection capabilities are returned in the message **`JSON_shipsDetectionCap`** .
3. Set basic parameters of ship detection.
 - 1) Call **`/ISAPI/Thermal/channels/<ID>/shipsDetection/basicParam/capabilities?format=json`** by GET method for getting basic configuration capability of ship detection to check the supported parameters.
The basic configuration capability of ship detection is returned in the message **`JSON_basicParamCap`** .
 - 2) Call **`/ISAPI/Thermal/channels/<ID>/shipsDetection/basicParam?format=json`** by GET method for getting default or configured basic parameters of ship detection for reference.
The basic parameters of ship detection are returned in the message **`JSON_basicParam`** .
 - 3) Call **`/ISAPI/Thermal/channels/<ID>/shipsDetection/basicParam?format=json`** by PUT method for setting basic parameters of ship detection.
4. Set scene parameters for ship detection.

- 1) **Optional:** Call **`/ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/capabilities?format=json`** by GET method for getting scene configuration capability.
The scene configuration capability is returned in the **`JSON_ShipsDetectionSceneCap`**.
 - 2) **Optional:** Call **`/ISAPI/Thermal/channels/<ID>/shipsDetection/basicParam?format=json`** or **`/ISAPI/Thermal/channels/<ID>/shipsDetection?format=json`** by GET method for getting default or configured scene parameters of a or all scene(s) for reference.
The parameters of a or all scene(s) are returned in the message **`JSON_ShipsDetectionScene`** or **`JSON_ShipsDetectionSceneList`**.
 - 3) Call **`/ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>?format=json`** or **`/ISAPI/Thermal/channels/<ID>/shipsDetection?format=json`** by PUT method for creating a or all scenes(s).
 - 4) Call **`/ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>?format=json`** or **`/ISAPI/Thermal/channels/<ID>/shipsDetection?format=json`** by PUT method for setting parameters of a or all scenes.
 - 5) **Optional:** Call **`/ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>?format=json`** or **`/ISAPI/Thermal/channels/<ID>/shipsDetection?format=json`** by DELETE method for deleting a or all scenes(s).
5. **Optional:** Perform the following operation(s) to switch scene, start auto-switch, set capture ratio for scene, and set advanced parameters for ship detection.

Switch Scene PUT **`/ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/goto?format=json`**

Scene Auto-Switch a. GET **`/ISAPI/Thermal/channels/<ID>/shipsDetection/sceneTrace/capabilities?format=json`** for getting scene auto-switch capability.
b. GET **`/ISAPI/Thermal/channels/<ID>/shipsDetection/sceneTrace?format=json`** for getting default or configured scene auto-switch parameters for reference.
c. PUT **`/ISAPI/Thermal/channels/<ID>/shipsDetection/sceneTrace?format=json`** for starting scene auto-switch.

Capture Ratio a. GET **`/ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/captureRatio/capabilities?format=json`** for getting configuration capability of capture ratio.
b. GET **`/ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/captureRatio?format=json`** for getting default or configured capture ratio for reference.
c. PUT **`/ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/captureRatio?format=json`** for setting capture ratio.

Advanced parameters

- a. GET */ISAPI/Thermal/channels/<ID>/shipsDetection/advanceParam/capabilities?format=json* for getting advanced configuration capability of ship detection to check the supported parameters.
 - b. GET */ISAPI/Thermal/channels/<ID>/shipsDetection/advanceParam?format=json* for getting default or configured advanced parameters of ship detection for reference.
 - c. PUT */ISAPI/Thermal/channels/<ID>/shipsDetection/advanceParam?format=json* for setting advanced parameters of ship detection.
6. Set the function mode to "dredgerDetection" to detect the dredger.
- 1) **Optional:** Call */ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipsFunctionMode/capabilities?format=json* by GET method for getting function mode capability to check the supported modes.
The function mode capability is returned in the message *JSON_Cap_functionMode*.
 - 2) **Optional:** Call */ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipsFunctionMode?format=json* by GET method for getting default or configured function mode for reference.
The function mode is returned in the message *JSON_functionMode*.
 - 3) Call */ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipsFunctionMode?format=json* by PUT method for setting the function mode to "dredgerDetection".
7. Set dredger detection rules.
- 1) **Optional:** Call */ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/dredgerDetection/regions/capabilities?format=json* by GET method for getting configuration capability of dredger detection rules to check the supported rule parameters.
The rule configuration capability of dredger detection is returned in the message *JSON_DredgerDetectionRuleListCap*.
 - 2) **Optional:** Call */ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/dredgerDetection/regions/<ID>?format=json* or */ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/dredgerDetection/regions?format=json* by GET method for getting default or configured rules of a or all dredger detection(s) for reference.
The rules of a or all dredger detection(s) are returned in the message *JSON_DredgerDetectionRule* or *JSON_DredgerDetectionRuleList*.
 - 3) Call */ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/dredgerDetection/regions/<ID>?format=json* or */ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/dredgerDetection/regions?format=json* by PUT method for setting rules of a or all dredger detection(s).
8. Call */ISAPI/Event/schedules/dredgerDetection/<ID>/scene/<SID>* by PUT method for setting the arming schedule of dredger detection alarm.

Note

Before setting arming schedule, you can transmission this URL by GET method to get default or configured arming schedule for reference.

9. Call `/ISAPI/Event/triggers/dredgerDetection/<ID>/scene/<SID>` by PUT method for setting linkage action of dredger detection alarm.



Note

Before setting linkage action, you can transmission this URL by GET method to get default or configured linkage actions for reference.

10. **Optional:** Perform the following operation(s) after configuring the dredger detection alarm.

Get Ship Statistics by Scene

GET `/ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipsDetectionCount?format=json`

Clear Ship Statistics by Scene

PUT `/ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/resetCount?format=json`

Receive Alarm in Arming or Listening Mode

Perform the flow of *Receive Alarm/Event in Arming Mode* or *Receive Alarm/Event in Listening Mode*.



Note

The alarm type (**eventType**) is "dredgerDetection", and for the alarm details, refer to the message *JSON_EventNotificationAlert_dredgerDetection*.

3.2 Receive Alarm/Event in Arming Mode

When alarm is triggered or event occurred, and the alarm/event linkage is configured, you can send request message to device for getting the alarm/event stream, and then the device uploads the corresponding response message, which contains alarm/event information.

Before You Start

Make sure you have configured alarm/event and triggered the alarm/event. For configuring alarm/event parameters, refer to the some typical applications of alarm/event configuration.

Steps

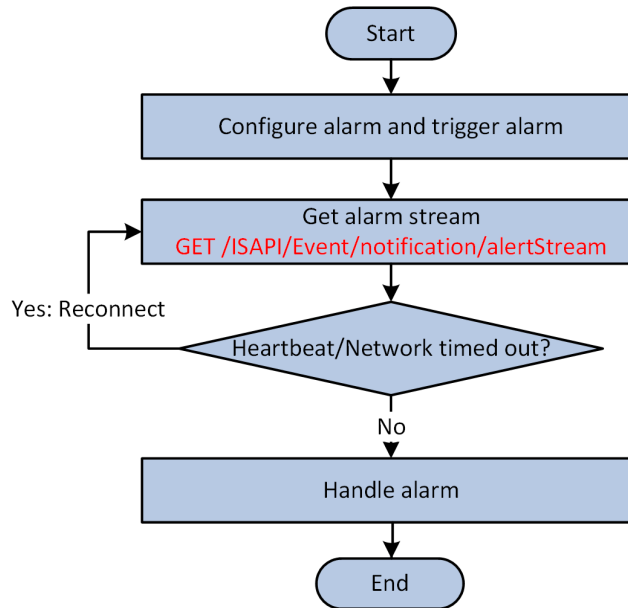


Figure 3-7 API Calling Flow of Receiving Alarm/Event in Arming Mode

1. Call **`/ISAPI/Event/notification/alertStream`** by GET to get the alarm/event stream.
2. Check if the heartbeat receiving timed out or network disconnected.
 - If the heartbeat keeps alive and the network still connected, perform the following step to continue.
 - If the heartbeat receiving timed out or network disconnected, perform the above step repeatedly until reconnected.
3. Receive and process the alarm/event information.

Example

Sample Code of Receiving Alarm/Event in Arming Mode (without Binary Picture Data)

```

GET /ISAPI/Event/notification/alertStream HTTP/1.1
Host: data_gateway_ip
Connection: Keep-Alive

HTTP/1.1 401 Unauthorized
Date: Sun, 01 Apr 2018 18:58:53 GMT
Server:
Content-Length: 178
Content-Type: text/html
Connection: keep-alive
Keep-Alive: timeout=10, max=99
WWW-Authenticate: Digest qop="auth",
realm="IP Camera(C2183)",
nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",
stale="FALSE"

GET /ISAPI/Event/notification/alertStream HTTP/1.1
Authorization: Digest username="admin",
  
```

```
realm="IP Camera(C2183)",
nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",
uri="/ISAPI/Event/notification/alertStream",
cnonce="3d183a245b8729121ae4ca3d41b90f18",
nc=00000001,
qop="auth",
response="f2e0728991bb031f83df557a8f185178"
Host: 10.6.165.192
```

```
HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=<frontier>
```

```
--<frontier>
Content-Type: application/xml; charset="UTF-8"
Content-Length: text_length
```

```
<EventNotificationAlert/>
--<frontier>
```



Note

Some alarm data is in JSON format, so the **Content-Type** may be "application/xml" or "application/json".

3.3 Receive Alarm/Event in Listening Mode

When alarm is triggered or event occurred, and the alarm/event linkage is configured, the device uploads the alarm/event information automatically, you can receive the alarm/event by configuring the listening port of HTTP host server.

Before You Start

Make sure you have configured alarm/event and triggered the alarm/event. For configuring alarm/event parameters, refer to the some typical applications of alarm/event configuration.

Steps

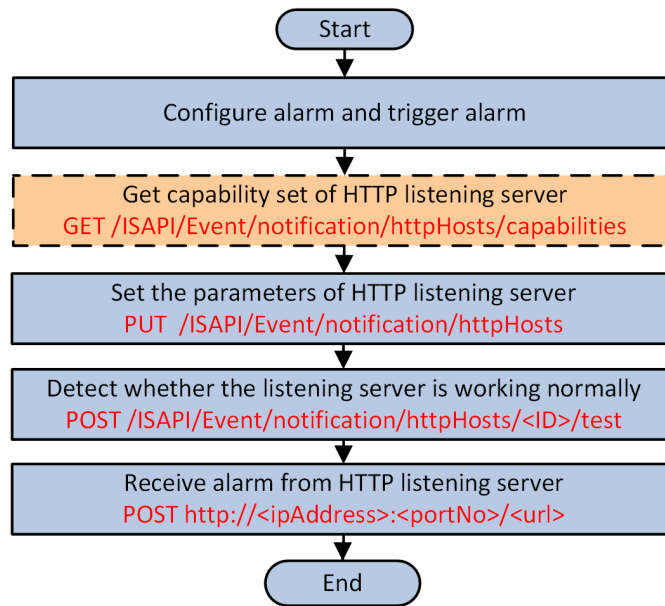


Figure 3-8 API Calling Flow of Receiving Alarm/Event in Listening Mode

1. **Optional:** Call `/ISAPI/Event/notification/httpHosts/capabilities` by GET method to get the capability of HTTP listening server.
2. Call `/ISAPI/Event/notification/httpHosts` by PUT method to set the parameters (including listening address and listening port) of HTTP listening server.



Note

Before setting the listening server, you'd better call the URI by GET method to get the default or configured parameters for reference.

3. Call `/ISAPI/Event/notification/httpHosts/<ID>/test` by POST method to check if the listening server is working normally.
4. Call `http://ipAddress:portNo/url` by POST method to receive the alarm/event information from the listening server.

Example

Sample Code of Receiving Alarm/Event in Listening Mode

- with Binary Picture Data

```

//Request
POST requestUrl HTTP/1.1
Host: data_gateway_ip:port
Accept-Language: en-US
Date: YourDate
Content-Type: multipart/form-data;boundary=<frontier>
Content-Length: body_length
Connection: keep-alive

--<frontier>
  
```

```
Content-Disposition: form-data; name="Event_Type"
Content-Type: text/xml
Content-Length: xml_length

<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Length: image_length
Content-Type: image/jpeg

[binary picture data]
--<frontier>--

//Response
HTTP/1.1 HTTP statusCode
Date: YourDate
Connection: close
```

- without Binary Picture Data

```
//Request
POST requestUrl HTTP/1.1
Host: data_gateway_ip:port
Accept-Language: en-US
Date: YourDate
Content-Type: text/xml;
Content-Length: text_length
Connection: Close

<EventNotificationAlert/>

//Response
HTTP/1.1 HTTP statusCode
Date: YourDate
Connection: close
```

Note

- The **Host** is the HTTP server domain name or IP address and port No.
 - Some alarm data is in JSON format, so the **Content-Type** may be "text/xml" or "text/json".
-

3.4 Subscribe Alarm/Event in Arming Mode

For arming mode, the platform will connect to the devices automatically and send commands to the devices for uploading alarm/event information when the alarm is triggered or event occurred. To reduce the CPU and bandwidth usage of platform, and improve the device processing performance, the platform can subscribe alarm/event types to receive alarm/event information as required.

Before You Start

Configure alarm/event and trigger the alarm/event. For configuring alarm/event parameters, refer to the some typical applications of alarm/event configuration.

Steps

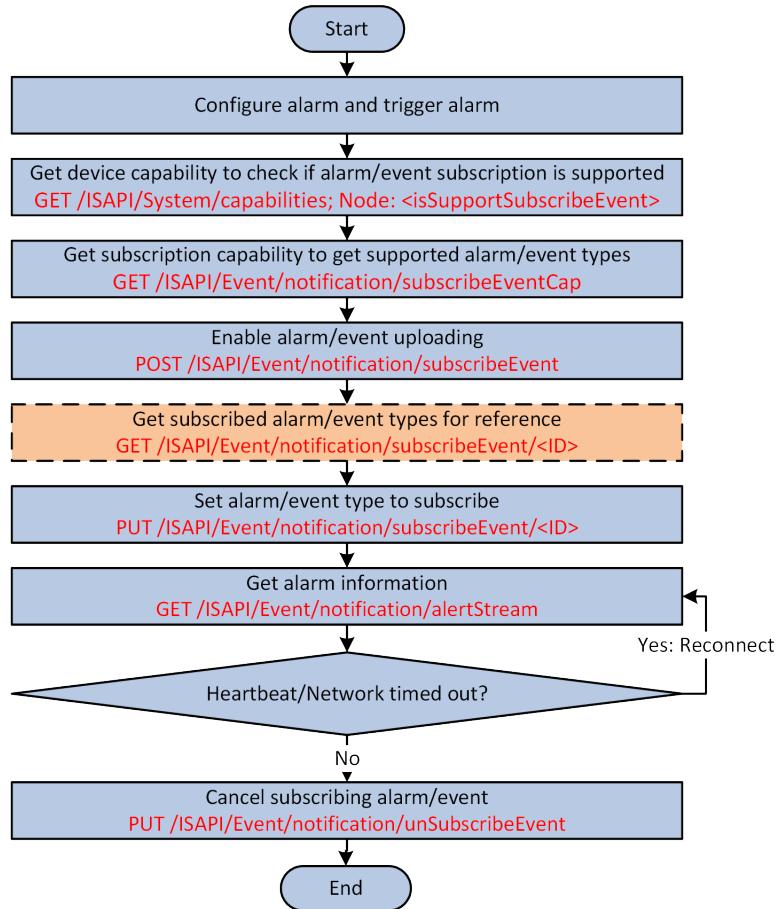


Figure 3-9 Programming Flow of Subscribing Alarm/Event in Arming Mode

1. Call **/ISAPI/System/capabilities** by GET method to get device capability to check whether the alarm/event subscription is supported.

The device capability is returned in the message **XML_DeviceCap**.

If the node **<isSupportSubscribeEvent>** is also returned in the message and its value is "true", it indicates that alarm/event subscription is supported by device, and you can continue to perform the following steps;

Otherwise, alarm/event subscription is not supported, please end the task.

2. Call **/ISAPI/Event/notification/subscribeEventCap** by GET method to get subscription capability to get supported alarm/event types.

The alarm/event subscription capability is returned in the message **XML_SubscribeEventCap**.

3. Call **/ISAPI/Event/notification/subscribeEvent** by POST method to enable alarm/event uploading.



Note

The sub node **<type>** of event type list (**EventList**) in the request message **XML_SubscribeEvent** can be set to different values (refer to *Supported Alarm/Event Types* for details) to subscribe different types of alarm/event information.

4. **Optional:** Call **/ISAPI/Event/notification/subscribeEvent/<ID>** by GET method to get subscribed alarm/event types for reference.
 5. Call **/ISAPI/Event/notification/subscribeEvent/<ID>** by PUT to set alarm/event type to subscribe.
-



Note

The sub node **<type>** of event type list (**EventList**) in the request message **XML_SubscribeEvent** can be set to different values (refer to *Supported Alarm/Event Types* for details) to subscribe different types of alarm/event information.

6. Call **/ISAPI/Event/notification/alertStream** by GET method to receive the alarm/event information.
7. Check if the heartbeat receiving timed out or network disconnected.
 - If the heartbeat keeps alive and the network still connected, perform the following step to continue.
 - If the heartbeat receiving timed out or network disconnected, perform the above step repeatedly until reconnected.
8. Call **/ISAPI/Event/notification/unSubscribeEvent** by PUT method to cancel subscribing alarm/event.

Chapter 4 Other Thermal Applications

Get JPEG Picture

Get JPEG picture with pixel-to-pixel temperature measurement data	GET <i>/ISAPI/Thermal/channels/<ID>/thermometry/jpegPicWithAppendData?format=json</i>  Note Before getting JPEG picture with pixel-to-pixel temperature measurement data, you should check whether the device supports this function via node isSupportJpegPicWithAppendData in <i>XML_ThermalCap</i> .
---	---

Measure Real-Time Temperature

Measure real-time temperature of all rules	GET <i>/ISAPI/Thermal/channels/<ID>/thermometry/realTimethermometry/rules?format=json</i>  Note - Before getting real-time temperature measurement information, you should check whether the device supports this function via node <isSupportRealTimethermometryForHTTP> in <i>XML_ThermalCap</i> . - It is valid in both normal mode and expert mode.
Measure real-time temperature of one single rule	GET <i>/ISAPI/Thermal/channels/<ID>/thermometry/realTimethermometry/rules/<RuleID>?format=json</i>  Note - Before getting real-time temperature measurement information, you should check whether the device supports this function via node <isSupportRealTimethermometryForHTTP> in <i>XML_ThermalCap</i> . - It is valid only in expert mode.

Thermal Stream Configuration

Get thermal stream configuration capability	GET <i>/ISAPI/Thermal/channels/<ID>/streamParam/capabilities</i>
Get thermal stream parameters	GET <i>/ISAPI/Thermal/channels/<ID>/streamParam</i>
Set thermal stream parameters	SET <i>/ISAPI/Thermal/channels/<ID>/streamParam</i>

Pixel-to-Pixel Temperature Measurement Configuration

Get configuration capability of pixel-to-pixel temperature measurement	GET <i>/ISAPI/Thermal/channels/<ID>/thermometry/pixelToPixelParam/capabilities</i>
Get pixel-to-pixel temperature measurement parameters	GET <i>/ISAPI/Thermal/channels/<ID>/thermometry/pixelToPixelParam</i>
Set pixel-to-pixel temperature measurement parameters	PUT <i>/ISAPI/Thermal/channels/<ID>/thermometry/pixelToPixelParam</i>

Temperature Correction Configuration

Get temperature correction configuration capability	GET <i>/ISAPI/Thermal/channels/<ID>/CorrectionParam/capabilities</i>  Note Check whether the device supports the function of temperature correction via the node <isSupportTemperatureCorrection> in <i>XML_ThermalCap</i> .
Get temperature correction parameters	GET <i>/ISAPI/Thermal/channels/<ID>/CorrectionParam</i>
Set temperature correction parameters	PUT <i>/ISAPI/Thermal/channels/<ID>/CorrectionParam</i>
Import temperature correction parameters	PUT <i>/ISAPI/Thermal /channels/<ID>/CorrectionParam/import</i>
Start temperature correction	PUT <i>/ISAPI/Thermal/channels/<ID>/Correction</i>

Picture-in-Picture Configuration

Get picture-in-picture configuration capability	GET <i>/ISAPI/Thermal/channels/<ID>/pip/capabilities</i>
Get picture-in-picture parameters	GET <i>/ISAPI/Thermal/channels/<ID>/pip</i>
Set picture-in-picture parameters	PUT <i>/ISAPI/Thermal/channels/<ID>/pip</i>

Manually Get Temperature Information of Temperature Measurement Rules

Get capability of manually getting temperature information of temperature measurement rules	GET <i>/ISAPI/Thermal/channels/<ID>/thermometry/<SID>/rulesTemperatureInfo/capabilities?format=json</i>
Manually get temperature information of temperature measurement rules in batch	GET <i>/ISAPI/Thermal/channels/<ID>/thermometry/<SID>/rulesTemperatureInfo?format=json</i>
Manually get single temperature information of temperature measurement rules	GET <i>/ISAPI/Thermal/channels/<ID>/thermometry/<SID>/rulesTemperatureInfo/<ID>?format=json</i>

Click to Detect Temperature

Initialize click to detect temperature	PUT <i>/ISAPI/Thermal/channels/<ID>/clickToThermometry/initialization</i>  Note Check whether the device supports the function of click to detect temperature via node <code><isSupportClickToThermometry></code> in <i>XML_ThermalCap</i> .
Set rules of temperature measurement by point	PUT <i>/ISAPI/Thermal/channels/<ID>/clickToThermometry/rules/<ID>?format=json</i>

Burning Prevention

Get burning prevention capability	GET <i>/ISAPI/Thermal/channels/<ID>/burningPrevention/capabilities</i>
-----------------------------------	--

	 Note Check whether the device supports the burning prevention function via node <code><isSupportBurningPrevention></code> in <i>XML_ThermalCap</i> .
Get burning prevention parameters	GET <i>/ISAPI/Thermal/channels/<ID>/burningPrevention</i>
Set burning prevention parameters	PUT <i>/ISAPI/Thermal/channels/<ID>/burningPrevention</i>

Thermal Data Conversion

Get capability of thermal data conversion	GET <i>/ISAPI/Thermal/channels/<ID>/thermalDataConversion/capabilities</i> The capability is returned in the message of <i>XML_ThermalDataConversionCap</i> .
Convert thermal data	POST <i>/ISAPI/Thermal/channels/<ID>/thermalDataConversion</i>

Get History Temperature Data

Get history temperature capability	GET <i>/ISAPI/Thermal/channels/<ID>/historyTemperature/capabilities</i> The capability is returned in <i>XML_HistoryTemperatureCap</i> .
Get history temperature data	POST <i>/ISAPI/Thermal/channels/<ID>/historyTemperature</i>

Advanced Parameters Configuration of Fire Detection

Get capability of configuring advanced parameters of fire detection	GET <i>/ISAPI/Thermal/channels/<ID>/fireDetection/advanceParam/capabilities?format=json</i>
Get advanced parameters of fire detection	GET <i>/ISAPI/Thermal/channels/<ID>/fireDetection/advanceParam?format=json</i>
Set advanced parameters of fire detection	PUT <i>/ISAPI/Thermal/channels/<ID>/fireDetection/advanceParam?format=json</i>

Body Temperature Compensation

Get capability of body temperature compensation	GET <i>/ISAPI/Thermal/channels/<ID>/bodyTemperatureCompensation/capabilities</i>
Get parameters of body temperature compensation	GET <i>/ISAPI/Thermal/channels/<ID>/bodyTemperatureCompensation</i>
Set parameters of body temperature compensation	PUT <i>/ISAPI/Thermal/channels/<ID>/bodyTemperatureCompensation</i>

Temperature Measurement Pre-Alarm

Function	Request URI
Get or set arming schedule of temperature measurement pre-alarm for a specified channel	GET or PUT <i>/ISAPI/Event/schedules/TMPA/<ID></i>
Get or set arming schedule of temperature measurement pre-alarm for all channels in a batch	GET or PUT <i>/ISAPI/Event/schedules/TMPA</i>

Face Thermography

Function	Request URI
Get face thermography capability	GET <i>/ISAPI/Intelligent/faceTemperature/capabilities?format=json</i>
Search for face thermography report	POST <i>/ISAPI/Intelligent/faceTemperature/dataStatistics?format=json</i>



Note

To check whether the device supports face thermography, you can call */ISAPI/Intelligent/capabilities* by GET method to get the intelligent capability (*XML_IntelliCap*). If supports, the node **<isSupportFaceTemperature>** will be returned and its value is true.

Temperature Calibration

Function	Request URI
Get temperature calibration capability	GET <i>/ISAPI/Thermal/channels/<ID>/temperatureCorrect/capabilities?format=json</i>
Get temperature calibration configuration parameters	GET <i>/ISAPI/Thermal/channels/<ID>/temperatureCorrect?format=json</i>
Set temperature calibration parameters	POST <i>/ISAPI/Thermal/channels/<ID>/temperatureCorrect?format=json</i>

Chapter 5 Request URIs

5.1 /ISAPI/Event/notification/subscribeEventCap

Get event/alarm subscription capability.

Request URI Definition

Table 5-1 GET /ISAPI/Event/notification/subscribeEventCap

Method	GET
Description	Get event/alarm subscription capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SubscribeEventCap</i> Failed: <i>XML_ResponseStatus</i>

5.2 /ISAPI/Event/notification/subscribeEvent

Subscribe events/alarms in arming mode.

Request URI Definition

Table 5-2 POST /ISAPI/Event/notification/subscribeEvent

Method	POST
Description	Subscribe events/alarms in arming mode.
Query	None.
Request	<i>XML_SubscribeEvent</i>
Response	Succeeded: <i>XML_SubscribeEventResponse</i> or <i>XML_EventNotificationAlert_SubscriptionHeartbeat</i> and alarm/event details message Failed: <i>XML_ResponseStatus</i>

Remarks

The *XML_EventNotificationAlert_SubscriptionHeartbeat* and alarm/event details message is uploaded repeatedly. The default time interval of uploading heartbeat information is 30s.

5.3 /ISAPI/Event/notification/subscribeEvent/<ID>

Get or set alarm/event subscription parameters.

Request URI Definition

Table 5-3 GET /ISAPI/Event/notification/subscribeEvent/<ID>

Method	GET
Description	Get alarm/event subscription parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SubscribeEvent</i> Failed: <i>XML_ResponseStatus</i>

Table 5-4 PUT /ISAPI/Event/notification/subscribeEvent/<ID>

Method	PUT
Description	Set alarm/event subscription parameters.
Query	None.
Request	<i>XML_SubscribeEvent</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the subscription No. which is returned by the device. After the persistent connection for receiving events or alarms in arming mode is closed, the device will release the resource used by the <ID>.

5.4 /ISAPI/Event/notification/unSubscribeEvent

Cancel subscribing alarm/event.

Request URI Definition

Table 5-5 PUT /ISAPI/Event/notification/unSubscribeEvent

Method	PUT
Description	Cancel subscribing alarm/event.
Query	None.

Request	None.
Response	<i>XML_ResponseStatus</i>

5.5 /ISAPI/Event/schedules/faceThermometrys

Operations about arming schedule configuration of all temperature screening alarms.

Request URI Definition

Table 5-6 GET /ISAPI/Event/schedules/faceThermometrys

Method	GET
Description	Get the arming schedule configurations of all temperature screening alarms.
Query	None.
Request	None.
Response	<i>XML_FaceThermometryScheduleList</i>

Table 5-7 PUT /ISAPI/Event/schedules/faceThermometrys

Method	PUT
Description	Set the arming schedule of all temperature screening alarms.
Query	None.
Request	<i>XML_FaceThermometryScheduleList</i>
Response	<i>XML_ResponseStatus</i>

5.6 /ISAPI/Event/schedules/faceThermometrys/<ID>

Operations about arming schedule configurations of a specified temperature screening alarm.

Request URI Definition

Table 5-8 GET /ISAPI/Event/schedules/faceThermometrys/<ID>

Method	GET
Description	Get the temperature screening arming schedule of a specific camera.
Query	None.

Request	None.
Response	<i>XML_Schedule</i>

Table 5-9 PUT /ISAPI/Event/schedules/faceThermometrys/<ID>

Method	PUT
Description	Set the temperature screening arming schedule of a specific camera.
Query	None.
Request	<i>XML_Schedule</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to temperature screening alarm ID, which consists of the alarm type name and the corresponding video input channel No., e.g., "faceThermometry-1".

5.7 /ISAPI/Event/schedules/dredgerDetection/<ID>/scene/<SID>

Operations about arming schedule of dredger detection in specified scene.

Request URL Definition**Table 5-10 GET /ISAPI/Event/schedules/dredgerDetection/<ID>/scene/<SID>**

Method	GET
Description	Get the arming schedule of dredger detection in specified scene.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Schedule</i> Failed: <i>XML_ResponseStatus</i>

Table 5-11 PUT /ISAPI/Event/schedules/dredgerDetection/<ID>/scene/<SID>

Method	PUT
Description	Set the arming schedule of dredger detection in specified scene.
Query	None.
Request	<i>XML_Schedule</i>
Response	<i>XML_ResponseStatus</i>

Remarks

- The <ID> in the URL refers to the channel No.
- The <SID> in the URL refers to the scene ID.

5.8 /ISAPI/Event/schedules/shipsFlowDetection/<ID>/scene/<SID>

Operations about arming schedule of ship flow detection in specified scene.

Request URL Definition

Table 5-12 GET /ISAPI/Event/schedules/shipsFlowDetection/<ID>/scene/<SID>

Method	GET
Description	Get the arming schedule of ship flow detection in specified scene.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Schedule</i> Failed: <i>XML_ResponseStatus</i>

Table 5-13 PUT /ISAPI/Event/schedules/shipsFlowDetection/<ID>/scene/<SID>

Method	PUT
Description	Set the arming schedule of ship flow detection in specified scene.
Query	None.
Request	<i>XML_Schedule</i>
Response	<i>XML_ResponseStatus</i>

Remarks

- The <ID> in the URL refers to the channel No.
- The <SID> in the URL refers to the scene ID.

5.9 /ISAPI/Event/schedules/Thermometry

Operations about the arming schedule configurations of all temperature measurement alarms.

Request URI Definition

Table 5-14 GET /ISAPI/Event/schedules/Thermometry

Method	GET
Description	Get the arming schedule configurations of all temperature measurement alarms.
Query	None.
Request	None.
Response	<i>XML_ThermometryScheduleList</i>

Table 5-15 PUT /ISAPI/Event/schedules/Thermometry

Method	PUT
Description	Set the arming schedule of all temperature measurement alarms.
Query	None.
Request	<i>XML_ThermometryScheduleList</i>
Response	<i>XML_ResponseStatus</i>

5.10 /ISAPI/Event/schedules/thermometry/<ID>

Operations about the arming schedule configurations of a specified temperature measurement alarm.

Request URI Definition

Table 5-16 GET /ISAPI/Event/schedules/thermometry/<ID>

Method	GET
Description	Get the arming schedule of a specified temperature measurement alarm.
Query	None.
Request	None.
Response	<i>XML_Schedule</i>

Table 5-17 PUT /ISAPI/Event/schedules/thermometry/<ID>

Method	PUT
Description	Set the arming schedule of a specified temperature measurement alarm.

Query	None.
Request	<i>XML_Schedule</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI is temperature measurement alarm ID, which consists of alarm type name and video input channel No., e.g., thermometry-1.

5.11 /ISAPI/Event/schedules/TMPA/<ID>

Get or set the arming schedule of temperature measurement pre-alarm for a specified channel.

Request URI Definition

Table 5-18 GET /ISAPI/Event/schedules/TMPA/<ID>

Method	GET
Description	Get the arming schedule of temperature measurement pre-alarm for a specified channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Schedule</i> Failed: <i>XML_ResponseStatus</i>

Table 5-19 PUT /ISAPI/Event/schedules/TMPA/<ID>

Method	PUT
Description	Set the arming schedule of temperature measurement pre-alarm for a specified channel.
Query	None.
Request	<i>XML_Schedule</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to temperature measurement pre-alarm ID, which consists of event type and video input channel No., e.g., TMPA-1.

5.12 /ISAPI/Event/schedules/TMPA

Get or set arming schedules of temperature measurement pre-alarm for all channels in a batch.

Request URI Definition

Table 5-20 GET /ISAPI/Event/schedules/TMPA

Method	GET
Description	Get arming schedules of temperature measurement pre-alarm for all channels in a batch.
Query	None.
Request	None.
Response	Succeeded: <i>XML_TMPAScheduleList</i> Failed: <i>XML_ResponseStatus</i>

Table 5-21 PUT /ISAPI/Event/schedules/TMPA

Method	PUT
Description	Set arming schedules of temperature measurement pre-alarm for all channels in a batch.
Query	None.
Request	<i>XML_TMPAScheduleList</i>
Response	<i>XML_ResponseStatus</i>

5.13 /ISAPI/Event/triggers

Get linkage actions of multiple alarms.

Request URI Definition

Table 5-22 GET /ISAPI/Event/triggers

Method	GET
Description	Get linkage actions of multiple alarms.
Query	None.
Request	None.
Response	Succeeded: <i>XML_EventTriggerList</i> Failed: <i>XML_ResponseStatus</i>

5.14 /ISAPI/Event/triggers/dredgerDetection/<ID>/scene/<SID>

Operations about linkage configurations of dredger detection in specified scene.

Request URL Definition

Table 5-23 GET /ISAPI/Event/triggers/dredgerDetection/<ID>/scene/<SID>

Method	GET
Description	Get the linkage configurations of dredger detection in specified scene.
Query	None.
Request	None.
Response	Succeeded: <i>XML_EventTrigger</i> Failed: <i>XML_ResponseStatus</i>

Table 5-24 PUT /ISAPI/Event/triggers/dredgerDetection/<ID>/scene/<SID>

Method	PUT
Description	Set the linkage parameters of dredger detection in specified scene.
Query	None.
Request	<i>XML_EventTrigger</i>
Response	<i>XML_ResponseStatus</i>

Remarks

- The <ID> in the URL refers to the channel No.
- The <SID> in the URL refers to the scene ID.

5.15 /ISAPI/Event/triggers/shipsFlowDetection/<ID>/scene/<SID>

Get or set the linkage configurations of ship flow detection in specified scene.

Request URL Definition

Table 5-25 GET /ISAPI/Event/triggers/shipsFlowDetection/<ID>/scene/<SID>

Method	GET
Description	Get the linkage configurations of ship detection in specified scene.
Query	None.

Request	None.
Response	Succeeded: <i>XML_EventTrigger</i> Failed: <i>XML_ResponseStatus</i>

Table 5-26 PUT /ISAPI/Event/triggers/shipsFlowDetection/<ID>/scene/<SID>

Method	PUT
Description	Set the linkage parameters of ship detection in specified scene.
Query	None.
Request	<i>XML_EventTrigger</i>
Response	<i>XML_ResponseStatus</i>

Remarks

- The <ID> in the URL refers to the channel No.
- The <SID> in the URL refers to the scene ID.

5.16 /ISAPI/Event/triggers/<eventType>-<ID>

Get, set, or delete the alarm linkage action by channel.

Request URI Definition**Table 5-27 GET /ISAPI/Event/triggers/<eventType>-<ID>**

Method	GET
Description	Get the alarm linkage action by channel.
Query	None
Request	None
Response	Succeeded: <i>XML_EventTrigger</i> Failed: <i>XML_ResponseStatus</i>

Table 5-28 PUT /ISAPI/Event/triggers/<eventType>-<ID>

Method	PUT
Description	Set the alarm linkage action by channel.
Query	None
Request	<i>XML_EventTrigger</i>
Response	<i>XML_ResponseStatus</i>

Table 5-29 DELETE /ISAPI/Event/triggers/<eventType>-<ID>

Method	DELETE
Description	Delete the alarm linkage action by channel.
Query	None
Request	None
Response	<i>XML_ResponseStatus</i>

Remarks

The <eventType> in the request URI refers to the predefined event or alarm type name, and the <ID> is the No. of the event detection channel. For example, if the No. of the face capture channel is 101, the "<eventType>-<ID>" is "faceSnap-101".

5.17 /ISAPI/Event/triggersCap

Get alarm linkage capability.

Request URI Definition**Table 5-30 GET /ISAPI/Event/triggersCap**

Method	GET
Description	Get alarm linkage capability.
Query	None
Request	None
Response	Succeeded: <i>XML_EventTriggersCap</i> Failed: <i>XML_ResponseStatus</i>

5.18 /ISAPI/Image/channels/<ID>/DPC/correction

Configure defective pixel correction according to channel ID.

Request URL Definition**Table 5-31 PUT /ISAPI/Image/channels/<ID>/DPC/correction**

Method	PUT
Description	Configure defective pixel correction according to channel ID.
Query	None.

Request	<i>XML_DPC</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URL is defined as channel ID.

5.19 /ISAPI/Image/channels/<ID>/DPC/correction/capabilities

Get the capability of defective pixel correction.

Request URL Definition

Table 5-32 GET /ISAPI/Image/channels/<ID>/DPC/correction/capabilities

Method	GET
Description	Get the capability of defective pixel correction.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_DPC</i> Failed: <i>XML_ResponseStatus</i>

5.20 /ISAPI/Intelligent/capabilities

Get the intelligent capability set.

Request URI Definition

Table 5-33 GET /ISAPI/Intelligent/capabilities

Method	GET
Description	Get the intelligent capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IntelliCap</i> Failed: <i>XML_ResponseStatus</i>

5.21 /ISAPI/Intelligent/faceTemperature/capabilities?format=json

Get the face thermography capability.

Request URI Definition

Table 5-34 GET /ISAPI/Intelligent/faceTemperature/capabilities?format=json

Method	GET
Description	Get the face thermography capability.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_EpidemicControlCap</i> Failed: <i>JSON_ResponseStatus</i>

5.22 /ISAPI/Intelligent/faceTemperature/dataStatistics?format=json

Search for face thermography statistical report.

Request URI Definition

Table 5-35 POST /ISAPI/Intelligent/faceTemperature/dataStatistics?format=json

Method	POST
Description	Search for face thermography statistical report.
Query	format : determine the format of request or response message.
Request	<i>JSON_FaceThermReportSearchCond</i>
Response	Succeeded: <i>JSON_FaceThermReportSearchResult</i> Failed: <i>JSON_ResponseStatus</i>

5.23 /ISAPI/PTZCtrl/channels/<ID>/presets

Operations about all presets' configurations.

Request URI Definition

Table 5-36 GET /ISAPI/PTZCtrl/channels/<ID>/presets

Method	GET
Description	Get all presets' parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_PTZPresetList</i> Failed: <i>XML_ResponseStatus</i>

Table 5-37 PUT /ISAPI/PTZCtrl/channels/<ID>/presets

Method	PUT
Description	Set all presets' parameters.
Query	None.
Request	<i>XML_PTZPresetList</i>
Response	<i>XML_ResponseStatus</i>

Table 5-38 POST /ISAPI/PTZCtrl/channels/<ID>/presets

Method	POST
Description	Add a preset.
Query	None.
Request	<i>XML_PTZPreset</i>
Response	<i>XML_ResponseStatus</i>

Table 5-39 DELETE /ISAPI/PTZCtrl/channels/<ID>/presets

Method	DELETE
Description	Delete all presets.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI is the camera ID.

5.24 /ISAPI/System/capabilities

Get device capability.

Request URI Definition

Table 5-40 GET /ISAPI/System/capabilities

Method	GET
Description	Get device capability.
Query	None
Request	None.
Response	Succeeded: <i>XML_DeviceCap</i> Failed: <i>XML_ResponseStatus</i>

5.25 /ISAPI/Thermal/capabilities

Get thermal capability.

Request URI Definition

Table 5-41 GET /ISAPI/Thermal/capabilities

Method	GET
Description	Get the thermal capability.
Query	None.
Request	None.
Response	<i>XML_ThermalCap</i>

5.26 /ISAPI/Thermal/channels/<ID>/blackBody

Operations about black body configuration.

Request URL Definition

Table 5-42 GET /ISAPI/Thermal/channels/<ID>/blackBody

Method	GET
Description	Get the black body parameters of a specific camera.

Query	None.
Request	None.
Response	<i>XML_ThermalBlackBody</i>

Table 5-43 PUT /ISAPI/Thermal/channels/<ID>/blackBody

Method	PUT
Description	Set the black body parameters of a specific camera.
Query	None.
Request	<i>XML_ThermalBlackBody</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URL is the channel ID.

5.27 /ISAPI/Thermal/channels/<ID>/blackBody/capabilities

Get black body capability.

Request URL Definition**Table 5-44 GET /ISAPI/Thermal/channels/<ID>/blackBody/capabilities**

Method	GET
Description	Get the black body capability.
Query	None.
Request	None.
Response	<i>XML_Cap_ThermalBlackBody</i>

Remarks

The <ID> in the URL is the channel ID.

5.28 /ISAPI/Thermal/channels/<ID>/bodyTemperatureCompensation

Get or set configuration parameters of body temperature compensation.

Request URI Definition**Table 5-45 GET /ISAPI/Thermal/channels/<ID>/bodyTemperatureCompensation**

Method	GET
Description	Get the configuration parameters of body temperature compensation.
Query	None
Request	None
Response	Succeeded: <i>XML_BodyTemperatureCompensation</i> Failed: <i>XML_ResponseStatus</i>

Table 5-46 PUT /ISAPI/Thermal/channels/<ID>/bodyTemperatureCompensation

Method	PUT
Description	Set body temperature compensation.
Query	None
Request	<i>XML_BodyTemperatureCompensation</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the channel ID.

5.29 /ISAPI/Thermal/channels/<ID>/bodyTemperatureCompensation/capabilities

Get the capability of body temperature compensation.

Request URI Definition**Table 5-47 GET /ISAPI/Thermal/channels/<ID>/bodyTemperatureCompensation/capabilities**

Method	GET
Description	Get the capability of body temperature compensation.
Query	None
Request	None
Response	Succeeded: <i>XML_BodyTemperatureCompensationCap</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the channel ID.

5.30 /ISAPI/Thermal/channels/<ID>/Correction

Start correction.

Request URL Definition

Table 5-48 PUT /ISAPI/Thermal/channels/<ID>/Correction

Method	PUT
Description	Start correction.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URL is the live view channel ID, by default it is the first channel.

5.31 /ISAPI/Thermal/channels/<ID>/CorrectionParam

Get or set temperature correction parameters.

Request URL Definition

Table 5-49 GET /ISAPI/Thermal/channels/<ID>/CorrectionParam

Method	GET
Description	Get the temperature correction parameters.
Query	None.
Request	None.
Response	<i>XML_CorrectionParam</i>

Table 5-50 PUT /ISAPI/Thermal/channels/<ID>/CorrectionParam

Method	PUT
Description	Set the temperature correction parameters.
Query	None.

Request	<i>XML_CorrectionParam</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URL is the live view channel ID, by default it is the first channel.

5.32 /ISAPI/Thermal/channels/<ID>/CorrectionParam/capabilities

Get the capability set of temperature correction parameters.

Request URL Definition

Table 5-51 GET /ISAPI/Thermal/channels/<ID>/CorrectionParam/capabilities

Method	GET
Description	Get the capability set of temperature correction parameters.
Query	None.
Request	None.
Response	<i>XML_CorrectionParamCap</i>

Remarks

The <ID> in the URL is the live view channel ID, by default it is the first channel.

5.33 /ISAPI/Thermal /channels/<ID>/CorrectionParam/import

Import the temperature measurement correction parameters in batch.

Request URI Definition

Table 5-52 PUT /ISAPI/Thermal /channels/<ID>/CorrectionParam/import

Method	PUT
Description	Import the temperature measurement correction parameters in batch.
Query	None.
Request	Opaque data.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to video input channel ID, and it starts from 1.

5.34 /ISAPI/Thermal/channels/<ID>/faceThermometry

Operations about temperature screening configuration.

Request URI Definition

Table 5-53 GET /ISAPI/Thermal/channels/<ID>/faceThermometry

Method	GET
Description	Get the configuration parameters of temperature screening of a specific camera.
Query	None.
Request	None.
Response	<i>XML_FaceThermometry</i>

Table 5-54 PUT /ISAPI/Thermal/channels/<ID>/faceThermometry

Method	PUT
Description	Set the parameters of temperature screening of a specific camera.
Query	None.
Request	<i>XML_FaceThermometry</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to channel ID.

5.35 /ISAPI/Thermal/channels/<ID>/faceThermometry/capabilities

Get the temperature screening capability by channel.

Request URI Definition

Table 5-55 GET /ISAPI/Thermal/channels/<ID>/faceThermometry/capabilities

Method	GET
Description	Get the temperature screening capability by channel.
Query	None.
Request	None.
Response	<i>XML_Cap_FaceThermometry</i>

Remarks

The <ID> in the URI refers to channel ID.

5.36 /ISAPI/Thermal/channels/<ID>/faceThermometry/regions

Operations about the temperature screening rule configurations of all detection regions.

Request URI Definition

Table 5-56 GET /ISAPI/Thermal/channels/<ID>/faceThermometry/regions

Method	GET
Description	Get the temperature screening rule parameters of all detection regions.
Query	None.
Request	None.
Response	<i>XML_FaceThermometryRegionList</i>

Table 5-57 PUT /ISAPI/Thermal/channels/<ID>/faceThermometry/regions

Method	PUT
Description	Set the temperature screening rules for all detection regions.
Query	None.
Request	<i>XML_FaceThermometryRegionList</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URL refers to the channel ID.

5.37 /ISAPI/Thermal/channels/<ID>/faceThermometry/regions/<ID>

Operations about the temperature screening rule configurations of a specific detection region.

Request URI Definition

Table 5-58 GET /ISAPI/Thermal/channels/<ID>/faceThermometry/regions/<ID>

Method	GET
Description	Get the temperature screening rule parameters of a specific detection region.

Query	None.
Request	None.
Response	<i>XML_ThermometryRegion</i>

Table 5-59 PUT /ISAPI/Thermal/channels/<ID>/faceThermometry/regions/<ID>

Method	PUT
Description	Set the temperature screening rules of a specific detection region.
Query	None.
Request	<i>XML_ThermometryRegion</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The first <ID> in the URI is the channel ID, and the second <ID> is the detection region ID.

5.38 /ISAPI/Thermal/channels/<ID>/faceThermometry/regions/<ID>/detectionInfo

Get the temperature screening results of a specific detection region.

Request URI Definition**Table 5-60 GET /ISAPI/Thermal/channels/<ID>/faceThermometry/regions/<ID>/detectionInfo**

Method	GET
Description	Get the temperature screening results of a specific detection region.
Query	None.
Request	None.
Response	<i>XML_FaceThermDetectionInfo</i>

Remarks

The first <ID> in the URI is channel ID, and second <ID> is the detection region ID.

5.39 /ISAPI/Thermal/channels/<ID>/fireDetection

Operations about the parameter information of fire and smoke detection.

Request URL Definition

Table 5-61 GET /ISAPI/Thermal/channels/<ID>/fireDetection

Method	GET
Description	Get the parameter information of fire and smoke detection.
Query	None.
Request	None.
Response	<i>XML_FireDetection</i>

Table 5-62 PUT /ISAPI/Thermal/channels/<ID>/fireDetection

Method	PUT
Description	Set the parameter information of fire and smoke detection.
Query	None.
Request	<i>XML_FireDetection</i>
Response	<i>XML_ResponseStatus</i>

5.40 /ISAPI/Thermal/channels/<ID>/fireDetection/advanceParam/capabilities?format=json

Get the capability of configuring advanced parameters of fire detection.

Request URI Definition

Table 5-63 GET /ISAPI/Thermal/channels/<ID>/fireDetection/advanceParam/capabilities?format=json

Method	GET
Description	Get the capability of configuring advanced parameters of fire detection.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_fireDetection_AdvanceParamCap</i> Failed: <i>JSON_ResponseStatus</i>

5.41 /ISAPI/Thermal/channels/<ID>/fireDetection/advanceParam?format=json

Get and set the advanced parameters of fire detection.

Request URI Definition

Table 5-64 GET /ISAPI/Thermal/channels/<ID>/fireDetection/advanceParam?format=json

Method	GET
Description	Get the advanced parameters of fire detection.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_fireDetection_AdvanceParam</i> Failed: <i>JSON_ResponseStatus</i>

Table 5-65 PUT /ISAPI/Thermal/channels/<ID>/fireDetection/advanceParam?format=json

Method	PUT
Description	Set the advanced parameters of fire detection.
Query	format : determine the format of request or response message.
Request	<i>JSON_fireDetection_AdvanceParam</i>
Response	<i>JSON_ResponseStatus</i>

5.42 /ISAPI/Thermal/channels/<ID>/fireDetection/capabilities

Get the configuration capability of fire and smoke detection.

Request URL Definition

Table 5-66 GET /ISAPI/Thermal/channels/<ID>/fireDetection/capabilities

Method	GET
Description	Get the configuration capability of fire and smoke detection.
Query	None.
Request	None.
Response	<i>XML_Cap_FireDetection</i>

5.43 /ISAPI/Thermal/channels/<ID>/historyTemperature

Get history temperature.

Request URL Definition

Table 5-67 POST /ISAPI/Thermal/channels/<ID>/historyTemperature

Method	POST
Description	Get history temperature.
Query	None.
Request	<i>XML_HistoryTemperatureDescription</i>
Response	<i>XML_HistoryTemperatureResult</i>

5.44 /ISAPI/Thermal/channels/<ID>/historyTemperature/capabilities

Get the history temperature capabilities.

Request URL Definition

Table 5-68 GET /ISAPI/Thermal/channels/<ID>/historyTemperature/capabilities

Method	GET
Description	Get the history temperature capabilities.
Query	None.
Request	None.
Response	<i>XML_HistoryTemperatureCap</i>

5.45 /ISAPI/Thermal/channels/<ID>/pip

Get or set the picture-in-picture parameters.

Request URL Definition

Table 5-69 GET /ISAPI/Thermal/channels/<ID>/pip

Method	GET
Description	Get the picture-in-picture configuration parameters.
Query	None.

Request	None.
Response	<i>XML_ThermalPip</i>

Table 5-70 PUT /ISAPI/Thermal/channels/<ID>/pip

Method	PUT
Description	Set the picture-in-picture parameters.
Query	None.
Request	<i>XML_ThermalPip</i>
Response	<i>XML_ResponseStatus</i>

5.46 /ISAPI/Thermal/channels/<ID>/pip/capabilities

Get the capability of picture-in-picture configuration.

Request URL Definition

Table 5-71 GET /ISAPI/Thermal/channels/<ID>/pip/capabilities

Method	GET
Description	Get the capability of picture-in-picture configuration.
Query	None.
Request	None.
Response	<i>XML_Cap_ThermalPip</i>

5.47 /ISAPI/Thermal/channels/<ID>/shipsDetection/advanceParam/capabilities?format=json

Get the capabilities of advanced parameters configuration for ship detection.

Request URL Definition

Table 5-72 GET /ISAPI/Thermal/channels/<ID>/shipsDetection/advanceParam/capabilities?format=json

Method	GET
Description	Get the capabilities of advanced parameters configuration for ship detection.
Query	format: determine the format of request or response message.

Request	None.
Response	Succeeded: <i>JSON_advanceParamCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URL refers to the channel No.

5.48 /ISAPI/Thermal/channels/<ID>/shipsDetection/advanceParam?format=json

Get or set the advanced parameters of ship detection.

Request URL Definition

Table 5-73 GET /ISAPI/Thermal/channels/<ID>/shipsDetection/advanceParam?format=json

Method	GET
Description	Get the advanced parameters of ship detection.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_advanceParam</i> Failed: <i>JSON_ResponseStatus</i>

Table 5-74 PUT /ISAPI/Thermal/channels/<ID>/shipsDetection/advanceParam?format=json

Method	PUT
Description	Set the advanced parameters of ship detection.
Query	format: determine the format of request or response message.
Request	<i>JSON_advanceParam</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URL refers to the channel No.

5.49 /ISAPI/Thermal/channels/<ID>/shipsDetection/basicParam/capabilities?format=json

Get the basic parameters configuration capability of ship detection.

Request URL Definition

Table 5-75 GET /ISAPI/Thermal/channels/<ID>/shipsDetection/basicParam/capabilities?format=json

Method	GET
Description	Get the basic parameters configuration capability of ship detection.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_basicParamCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URL refers to the channel No.

5.50 /ISAPI/Thermal/channels/<ID>/shipsDetection/basicParam?format=json

Get or set the ship detection basic parameters.

Request URL Definition

Table 5-76 GET /ISAPI/Thermal/channels/<ID>/shipsDetection/basicParam?format=json

Method	GET
Description	Get the ship detection basic parameters.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_basicParam</i> Failed: <i>JSON_ResponseStatus</i>

Table 5-77 PUT /ISAPI/Thermal/channels/<ID>/shipsDetection/basicParam?format=json

Method	PUT
Description	Set the ship detection basic parameters.
Query	format: determine the format of request or response message.
Request	<i>JSON_basicParam</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URL refers to the channel No.

5.51 /ISAPI/Thermal/channels/<ID>/shipsDetection/capabilities?format=json

Get the ship detection capabilities.

Request URL Definition

Table 5-78 GET /ISAPI/Thermal/channels/<ID>/shipsDetection/capabilities?format=json

Method	GET
Description	Get the ship detection capabilities.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_shipsDetectionCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URL refers to the channel No.

5.52 /ISAPI/Thermal/channels/<ID>/shipsDetection/sceneTrace/capabilities?format=json

Get the scene auto-switch capabilities.

Request URL Definition

Table 5-79 GET /ISAPI/Thermal/channels/<ID>/shipsDetection/sceneTrace/capabilities?format=json

Method	GET
Description	Get the scene auto-switch capabilities.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ShipsDetectionTraceListCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URL refers to the channel No.

5.53 /ISAPI/Thermal/channels/<ID>/shipsDetection/sceneTrace?format=json

Get all scenes information or start the scene auto-switch.

Request URL Definition

Table 5-80 GET /ISAPI/Thermal/channels/<ID>/shipsDetection/sceneTrace?format=json

Method	GET
Description	Get all scenes information.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: Failed: <i>JSON_ResponseStatus</i>

Table 5-81 PUT /ISAPI/Thermal/channels/<ID>/shipsDetection/sceneTrace?format=json

Method	PUT
Description	Start the scene auto-switch.
Query	format: determine the format of request or response message.
Request	
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URL refers to the channel No.

5.54 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/capabilities?format=json

Get the scene parameters configuration capabilities of ship detection.

Request URL Definition**Table 5-82 GET /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/capabilities?format=json**

Method	GET
Description	Get the scene parameters configuration capabilities of ship detection.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ShipsDetectionSceneCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

- The <ID> in the URL refers to the channel No.
- The <SID> in the URL refers to the scene ID.

5.55 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/captureRatio/capabilities?format=json

Get the capture ratio configuration capabilities.

Request URL Definition**Table 5-83 GET /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/captureRatio/capabilities?format=json**

Method	GET
Description	Get the capture ratio configuration capabilities.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ShipsDetectionCaptureRatioCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

- The <ID> in the URL refers to the channel No.
- The <SID> in the URL refers to the scene ID.

5.56 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/captureRatio?format=json

Get or set the capture ratio.

Request URL Definition

Table 5-84 GET /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/captureRatio?format=json

Method	GET
Description	Get the capture ratio.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ShipsDetectionCaptureRatio</i> Failed: <i>JSON_ResponseStatus</i>

Table 5-85 PUT /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/captureRatio?format=json

Method	PUT
Description	Set the capture ratio.
Query	format : determine the format of request or response message.
Request	<i>JSON_ShipsDetectionCaptureRatio</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

- The <ID> in the URL refers to the channel No.
- The <SID> in the URL refers to the scene ID.

5.57 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/dredgerDetection/regions/capabilities?format=json

Get the configuration capabilities of dredger detection rules.

Request URL Definition

Table 5-86 GET /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/dredgerDetection/regions/capabilities?format=json

Method	GET
Description	Get the configuration capabilities of dredger detection rules.

Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_DredgerDetectionRuleListCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

- The <ID> in the URL refers to the channel No.
- The <SID> in the URL refers to the scene ID.

5.58 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/dredgerDetection/regions/<ID>?format=json

Get or set a specified rule of dredger detection.

Request URL Definition

Table 5-87 GET /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/dredgerDetection/regions/<ID>?format=json

Method	GET
Description	Get a specified rule of dredger detection.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_DredgerDetectionRule</i> Failed: <i>JSON_ResponseStatus</i>

Table 5-88 PUT /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/dredgerDetection/regions/<ID>?format=json

Method	GET
Description	Get a specified rule of dredger detection.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

- The first <ID> in the URL refers to the channel No., the second <ID> in the URL refers to the rule ID
- The <SID> in the URL refers to the scene ID.

5.59 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/dredgerDetection/regions?format=json

Get or set all rules of dredger detection.

Request URL Definition

Table 5-89 GET /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/dredgerDetection/regions?format=json

Method	GET
Description	Get all rules of dredger detection.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_DredgerDetectionRuleList</i> Failed: <i>JSON_ResponseStatus</i>

Table 5-90 PUT /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/dredgerDetection/regions?format=json

Method	PUT
Description	Set all rules of dredger detection.
Query	format: determine the format of request or response message.
Request	<i>JSON_DredgerDetectionRuleList</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

- The <ID> in the URL refers to the channel No.
- The <SID> in the URL refers to the scene ID.

5.60 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>?format=json

Operations about specified ship detection scene.

Request URL Definition

Table 5-91 GET /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>?format=json

Method	GET
Description	Get the parameters of specified ship detection scene.

Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ShipsDetectionScene</i> Failed: <i>JSON_ResponseStatus</i>

Table 5-92 PUT /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>?format=json

Method	PUT
Description	Add a new scene, or set specified scene parameters.
Query	format: determine the format of request or response message.
Request	<i>JSON_ShipsDetectionScene</i>
Response	<i>JSON_ResponseStatus</i>

Table 5-93 DELETE /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>?format=json

Method	DELETE
Description	Delete the specified scene.
Query	format: determine the format of request or response message.
Request	<i>JSON_ShipsDetectionScene</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

- The <ID> in the URL refers to the channel No.
- The <SID> in the URL refers to the scene ID.
- You can add a new scene via URL: PUT /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>?format=json. When adding a new scene, you should input only the node **sid** (scene ID) and set other required nodes to NULL in the message *JSON_ShipsDetectionScene* . The optional nodes can not be applied.

5.61 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/goto?format=json

Switch scene for ship detection.

Request URL Definition

Table 5-94 PUT /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/goto?format=json

Method	PUT
Description	Switch scene for ship detection.

Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

- The <ID> in the URL refers to the channel No.
- The <SID> in the URL refers to the scene ID.

5.62 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipsDetectionCount?format=json

Get the ship statistics by scene.

Request URL Definition

Table 5-95 GET /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipsDetectionCount?format=json

Method	GET
Description	Get the ship statistics by scene.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ShipDetectionCount</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

- The <ID> in the URL refers to the channel No.
- The <SID> in the URL refers to the scene ID.

5.63 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipFlowDetection/regions/capabilities?format=json

Get the configuration capabilities of ship flow detection rules.

Request URL Definition**Table 5-96 GET /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipFlowDetection/
regions/capabilities?format=json**

Method	GET
Description	Get the configuration capabilities of ship flow detection rules.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ShipFlowDetectionRuleListCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

- The <ID> in the URL refers to the channel No.
- The <SID> in the URL refers to the scene ID.

**5.64 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/
shipFlowDetection/regions/<ID>?format=json**

Get or set a specified rule of ship flow detection.

Request URL Definition**Table 5-97 GET /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipFlowDetection/
regions/<ID>?format=json**

Method	GET
Description	Get a specified rule of ship flow detection.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ShipFlowDetectionRule</i> Failed: <i>JSON_ResponseStatus</i>

**Table 5-98 PUT /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipFlowDetection/
regions/<ID>?format=json**

Method	PUT
Description	Set a specified rule of ship flow detection.
Query	format: determine the format of request or response message.

Request	<i>JSON_ShipFlowDetectionRule</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

- The first <ID> in the URL refers to the channel No., the second <ID> in the URL refers to the rule ID
- The <SID> in the URL refers to the scene ID.

5.65 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipFlowDetection/regions?format=json

Get or set all rules of ship flow detection.

Request URL Definition

Table 5-99 GET /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipFlowDetection/regions?format=json

Method	GET
Description	Get all rules of ship flow detection.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ShipFlowDetectionRuleList</i> Failed: <i>JSON_ResponseStatus</i>

Table 5-100 PUT /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipFlowDetection/regions?format=json

Method	PUT
Description	Set all rules of ship flow detection.
Query	format: determine the format of request or response message.
Request	<i>JSON_ShipFlowDetectionRuleList</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

- The <ID> in the URL refers to the channel No.
- The <SID> in the URL refers to the scene ID.

5.66 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipsFunctionMode/capabilities?format=json

Get the configuration capabilities of scene function mode.

Request URL Definition

Table 5-101 GET /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipsFunctionMode/capabilities?format=json

Method	GET
Description	Get the configuration capabilities of scene function mode.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Cap_functionMode</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

- The <ID> in the URL refers to the channel No.
- The <SID> in the URL refers to the scene ID.

5.67 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipsFunctionMode?format=json

Get or set the scene function mode of ship detection.

Request URL Definition

Table 5-102 GET /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipsFunctionMode?format=json

Method	GET
Description	Get the scene function mode of ship detection.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_functionMode</i> Failed: <i>JSON_ResponseStatus</i>

Table 5-103 PUT /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/shipsFunctionMode?format=json

Method	PUT
Description	Set the scene function mode of ship detection.
Query	format : determine the format of request or response message.
Request	None.
Response	<i>JSON_functionMode</i>

Remarks

- The <ID> in the URL refers to the channel No.
- The <SID> in the URL refers to the scene ID.

5.68 /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/resetCount?format=json

Clear the ship statistics by scene.

Request URL Definition**Table 5-104 PUT /ISAPI/Thermal/channels/<ID>/shipsDetection/<SID>/resetCount?format=json**

Method	PUT
Description	Clear the ship statistics by scene.
Query	format : determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

- The <ID> in the URL refers to the channel No.
- The <SID> in the URL refers to the scene ID.

5.69 /ISAPI/Thermal/channels/<ID>/shipsDetection?format=json

Operations about all ship detection scenes.

Request URL Definition

Table 5-105 GET /ISAPI/Thermal/channels/<ID>/shipsDetection?format=json

Method	GET
Description	Get parameters of all ship detection scenes.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ShipsDetectionSceneList</i> Failed: <i>JSON_ResponseStatus</i>

Table 5-106 PUT /ISAPI/Thermal/channels/<ID>/shipsDetection?format=json

Method	PUT
Description	Add new scenes, or set the parameters of all scenes.
Query	format: determine the format of request or response message.
Request	<i>JSON_ShipsDetectionSceneList</i>
Response	<i>JSON_ResponseStatus</i>

Table 5-107 DELETE /ISAPI/Thermal/channels/<ID>/shipsDetection?format=json

Method	DELETE
Description	Delete all scenes.
Query	format: determine the format of request or response message.
Request	<i>JSON_ShipsDetectionSceneList</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URL refers to the channel No.

5.70 /ISAPI/Thermal/channels/<ID>/streamParam

Get or set the thermal stream parameters.

Request URL Definition

Table 5-108 GET /ISAPI/Thermal/channels/<ID>/streamParam

Method	GET
Description	Get the thermal stream parameters.

Query	None.
Request	None.
Response	<i>XML_ThermalStreamParam</i>

Table 5-109 PUT /ISAPI/Thermal/channels/<ID>/streamParam

Method	PUT
Description	Set the thermal stream parameters.
Query	None.
Request	<i>XML_ThermalStreamParam</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URL is the channel ID.

5.71 /ISAPI/Thermal/channels/<ID>/streamParam/capabilities

Get the thermal stream parameters capability.

Request URL Definition

Table 5-110 GET /ISAPI/Thermal/channels/<ID>/streamParam/capabilities

Method	GET
Description	Get the thermal stream parameters capability.
Query	None.
Request	None.
Response	<i>XML_ThermalStreamParamCap</i>

Remarks

The <ID> in the URL is the channel ID.

5.72 /ISAPI/Thermal/channels/<ID>/thermIntell

Operations about functional mutex configuration.

Request URL Definition**Table 5-111 GET /ISAPI/Thermal/channels/<ID>/thermIntell**

Method	GET
Description	Get the functional mutex configuration parameters.
Query	None.
Request	None.
Response	<i>XML_ThermIntell</i>

Table 5-112 PUT /ISAPI/Thermal/channels/<ID>/thermIntell

Method	PUT
Description	Get the functional mutex parameters.
Query	None.
Request	<i>XML_ThermIntell</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URL is the channel ID.

5.73 /ISAPI/Thermal/channels/<ID>/thermIntell/capabilities

Get the capability of functional mutex configuration.

Request URL Definition**Table 5-113 GET /ISAPI/Thermal/channels/<ID>/thermIntell/capabilities**

Method	GET
Description	Get the capability of functional mutex configuration.
Query	None.
Request	None.
Response	<i>XML_Cap_ThermIntell</i>

Remarks

The <ID> in the URL is the video input channel ID.

5.74 /ISAPI/Thermal/channels/<ID>/thermometry/<SID>

Operations about preset (scene) configurations of temperature measurement.

Request URI Definition

Table 5-114 GET /ISAPI/Thermal/channels/<ID>/thermometry/<SID>

Method	GET
Description	Get preset (scene) configurations of temperature measurement.
Query	None.
Request	None.
Response	<i>XML_ThermometryScene</i>

Remarks

The <ID> in the URI refers to the video input channel ID, and the <SID> in the URI is the scene ID (or preset No.).

5.75 /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/alarmRules

Operations about the rule configuration of temperature alarm or temperature difference alarm of a specific preset (scene).

Request URL Definition

Table 5-115 GET /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/alarmRules

Method	GET
Description	Get the rule parameters of temperature alarm or temperature difference alarm of a specific preset (scene).
Query	None.
Request	None.
Response	<i>XML_ThermometryAlarmRule</i>

Table 5-116 PUT /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/alarmRules

Method	PUT
Description	Set the rule parameters of temperature alarm or temperature difference alarm of a specific preset (scene).
Query	None.

Request	<i>XML_ThermometryAlarmRule</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URL is the channel ID, and the <SID> in the URL is the scene ID (or preset No.).

5.76 /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/alarmRules/capabilities

Get the configuration capability of temperature alarm or temperature difference alarm of a specific preset (scene).

Request URL Definition

Table 5-117 GET /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/alarmRules/capabilities

Method	GET
Description	Get the configuration capability of temperature alarm or temperature difference alarm of a specific preset (scene).
Query	None.
Request	None.
Response	<i>XML_Cap_ThermometryAlarmRule</i>

Remarks

The <ID> in the URL refers to the video input channel ID, and the <SID> in the URL is the scene ID (or preset No.).

5.77 /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/capabilities

Get the preset (scene) configuration capability of temperature measurement.

Request URI Definition

Table 5-118 GET /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/capabilities

Method	GET
Description	Get the preset (scene) configuration capability of temperature measurement.
Query	None.

Request	None.
Response	<i>XML_Cap_ThermometryScene</i>

Remarks

The <ID> in the URI refers to the video input channel ID, and the <SID> in the URI is the scene ID (or preset No.).

5.78 /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/regions

Operations about the temperature measurement points, lines, and frames configuration of all detection regions in the preset (scene).

Request URI Definition

Table 5-119 GET /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/regions

Method	GET
Description	Get the temperature measurement points, lines, and frames configuration of all detection regions in the preset (scene).
Query	None.
Request	None.
Response	<i>XML_ThermometryRegionList</i>

Table 5-120 PUT /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/regions

Method	PUT
Description	Set the temperature measurement points, lines, and frames of all detection regions in the preset (scene).
Query	None.
Request	<i>XML_ThermometryRegionList</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the video input channel ID, and the <SID> in the URI is the scene ID (or preset No.).

5.79 /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/regions/<ID>

Operations about the temperature measurement points, lines, and frames configuration of a specific detection region in the preset (scene).

Request URI Definition

Table 5-121 GET /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/regions/<ID>

Method	GET
Description	Get the thermometry points, lines, and frames parameters of a specific detection region in the preset (scene).
Query	None.
Request	None.
Response	<i>XML_ThermometryRegion</i>

Table 5-122 PUT /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/regions/<ID>

Method	PUT
Description	Set the temperature measurement points, lines, and frames of a specific detection region in the preset (scene)..
Query	None.
Request	<i>XML_ThermometryRegion</i>
Response	<i>XML_ResponseStatus</i>

Remarks

- The <SID> in the URI refers to scene ID (or preset No).
- The first <ID> in the URI refers to channel ID, and the second <ID> refers to detection region ID.

5.80 /ISAPI/Thermal/channels/<ID>/thermometry/basicParam

Operations about basic configuration of temperature measurement.

Request URL Definition

Table 5-123 GET /ISAPI/Thermal/channels/<ID>/thermometry/basicParam

Method	GET
Description	Get the basic parameters of temperature measurement.
Query	None.

Request	None.
Response	<i>XML_ThermometryBasicParam</i>

Table 5-124 PUT /ISAPI/Thermal/channels/<ID>/thermometry/basicParam

Method	PUT
Description	Set the basic parameters of temperature measurement.
Query	None.
Request	<i>XML_ThermometryBasicParam</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URL is the channel ID.

5.81 /ISAPI/Thermal/channels/<ID>/thermometry/basicParam/capabilities

Get the capability of basic configuration of temperature measurement.

Request URL Definition**Table 5-125 GET /ISAPI/Thermal/channels/<ID>/thermometry/basicParam/capabilities**

Method	GET
Description	Get the capability of basic configuration of temperature measurement.
Query	None.
Request	None.
Response	<i>XML_Cap_ThermometryBasicParam</i>

Remarks

The <ID> in the URL refers to the channel ID.

5.82 /ISAPI/Thermal/channels/<ID>/thermometry/jpegPicWithAppendData?format=json

Get JPEG picture with pixel-to-pixel temperature measurement data.

Request URL Definition

Table 5-126 GET /ISAPI/Thermal/channels/<ID>/thermometry/jpegPicWithAppendData?format=json

Method	GET
Description	Get JPEG picture with pixel-to-pixel temperature measurement data.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: Data in Form format Failed: <i>JSON_ResponseStatus</i>

Example

Returned JPEG Picture Data in Form Format

```
GET /ISAPI/Thermal/channels/<ID>/thermometry/jpegPicWithAppendData?format=json HTTP/1.1
```

```
Accept: text/html, application/xhtml+xml, */*
```

```
Accept-Language: en-US
```

```
User-Agent: Mozilla/5.0 (Windows NT 6.1; WOW64; Trident/7.0; rv:11.0) like Gecko
```

```
Accept-Encoding: gzip, deflate
```

```
Host: 10.17.133.46
```

```
DNT: 1
```

```
Cookie: language=zh; sdMarkMenu=8%3Avehicle; sdMarkTab_1_0=0%3AsettingBasic;
sdMarkTab_6_0=5%3AeventException; sdMarkTab_6_1=6%3AsmartLoiterDetection;
sdMarkTab_7_0=1%3AplanCapture; sdMarkTab_7_1=0%3AstorageManageHarddisk;
sdMarkTab_8=0%3AvehicleParam; WebSession=f81610c130711300cf30
```

```
HTTP/1.1 200 OK
```

```
MIME-Version: 1.0
```

```
Content-Type: multipart/form-data; boundary=boundary
```

```
--boundary
```

```
Content-Type: application/json; charset="UTF-8"
```

```
Content-Length:
```

```
{
  "jpegPictureWithAppendData":
  {
    "channel": 1,
    /*req, integer32, channel ID*/
    "jpegPicLen": 1,
    /*req, integer32, JPEG picture length*/
    "jpegPicWidth": 1,
    /*req, integer32, JPEG picture width*/
    "jpegPicHeight": 1,
    /*req, integer32, JPEG picture height*/
    "p2pDataLen": 1,
    /*req, integer32, length of pixel-to-pixel temperature measurement data*/
  }
}
```

```

"isFreezedata": false,
/*req, boolean, whether freezes data*/
"temperatureDataLength": 2,
/*req, integer32, temperature measurement data length (2 or 4)*/
"scale": 1,
/*opt, integer32, zoom in scale, it is valid when temperatureDataLength values "2"*/
"offset": 1,
/*opt, integer32, offset, it is valid when temperatureDataLength values "2"*/
}
}
--boundary
Content-Disposition: form-data;
Content-Type: image/jpeg
Content-Length:

Picture data
--boundary
Content-Disposition: form-data;
Content-Type: application/octet-stream
Content-Length:

Pixel-to-pixel temperature measurement data
--boundary--

```

5.83 /ISAPI/Thermal/channels/<ID>/thermometry/pixelToPixelParam

Get or set the pixel-to-pixel temperature measurement parameters.

Request URI Definition

Table 5-127 GET /ISAPI/Thermal/channels/<ID>/thermometry/pixelToPixelParam

Method	GET
Description	Get the pixel-to-pixel temperature measurement parameters.
Query	None.
Request	None.
Response	<i>XML_PixelToPixelParam</i>

Table 5-128 PUT /ISAPI/Thermal/channels/<ID>/thermometry/pixelToPixelParam

Method	PUT
Description	Set the pixel-to-pixel temperature measurement parameters.
Query	None.

Request	<i>XML_PixelToPixelParam</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to channel ID.

5.84 /ISAPI/Thermal/channels/<ID>/thermometry/pixelToPixelParam/capabilities

Get the configuration capability of pixel-to-pixel temperature measurement parameters.

Request URI Definition

Table 5-129 GET /ISAPI/Thermal/channels/<ID>/thermometry/pixelToPixelParam/capabilities

Method	GET
Description	Get the configuration capability of pixel-to-pixel temperature measurement parameters.
Query	None.
Request	None.
Response	<i>XML_PixelToPixelParamCap</i>

Remarks

The <ID> in the URL is the channel ID.

5.85 /ISAPI/Thermal/channels/<ID>/thermometry/realTimethermometry/rules/<RuleID>?format=json

Get real-time temperature measurement information of a single rule.

Request URI Definition

Table 5-130 GET /ISAPI/Thermal/channels/<ID>/thermometry/realTimethermometry/rules/<RuleID>?format=json

Method	GET
Description	Get real-time temperature measurement information of a single rule.
Query	format: determine the format of request or response message.

Request	None.
Response	Succeeded: <i>JSON_ThermometryUpload</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

- This URI is valid only in the export mode.
- The <ID> in the URI is defined as channel ID.
- The <RuleID> in the URI is defined as rule ID.

5.86 /ISAPI/Thermal/channels/<ID>/thermometry/realTimethermometry/rules?format=json

Get real-time temperature measurement information of all rules.

Request URI Definition

Table 5-131 GET /ISAPI/Thermal/channels/<ID>/thermometry/realTimethermometry/rules?format=json

Method	GET
Description	Get real-time temperature measurement information of all rules.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ThermometryUploadList</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

- This URI is available in both normal mode and expert mode.
- The <ID> in the URI is defined as channel ID.

5.87 /ISAPI/Thermal/channels/<ID>/thermometryMode

Operations about temperature measurement mode.

Request URI Definition

Table 5-132 GET /ISAPI/Thermal/channels/<ID>/thermometryMode

Method	GET
Description	Get the temperature measurement mode.

Query	None.
Request	None.
Response	<i>XML_ThermometryMode</i>

Table 5-133 PUT /ISAPI/Thermal/channels/<ID>/thermometryMode

Method	PUT
Description	Set the temperature measurement mode.
Query	None.
Request	<i>XML_ThermometryMode</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to channel ID.

5.88 /ISAPI/Thermal/channels/<ID>/thermometryMode/capabilities

Get the configuration capability of temperature measurement mode.

Request URI Definition**Table 5-134 GET /ISAPI/Thermal/channels/<ID>/thermometryMode/capabilities**

Method	GET
Description	Get the configuration capability of temperature measurement mode.
Query	None.
Request	None.
Response	<i>XML_Cap_ThermometryMode</i>

Remarks

The <ID> in the URI refers to video input channel ID.

5.89 /ISAPI/Thermal/channels/<ID>/burningPrevention

Get or set configuration parameters of burning prevention.

Request URL Definition

Table 5-135 GET /ISAPI/Thermal/channels/<ID>/burningPrevention

Method	GET
Description	Get configuration parameters of burning prevention.
Query	None.
Request	None.
Response	<i>XML_BurningPrevention</i>

Table 5-136 PUT /ISAPI/Thermal/channels/<ID>/burningPrevention

Method	PUT
Description	Set the burning prevention parameters.
Query	None.
Request	<i>XML_BurningPrevention</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URL is the channel ID.

5.90 /ISAPI/Thermal/channels/<ID>/burningPrevention/capabilities

Get burning prevention capabilities.

Request URL Definition

Table 5-137 GET /ISAPI/Thermal/channels/<ID>/burningPrevention/capabilities

Method	GET
Description	Get burning prevention capabilities.
Query	None.
Request	None.
Response	<i>XML_BurningPreventionCap</i>

5.91 /ISAPI/Thermal/channels/<ID>/clickToThermometry/initialization

Initialize clicking to detect temperature.

Request URL Definition**Table 5-138 PUT /ISAPI/Thermal/channels/<ID>/clickToThermometry/initialization**

Method	PUT
Description	Initialize clicking to detect temperature.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

5.92 /ISAPI/Thermal/channels/<ID>/clickToThermometry/rules/<ID>?format=json

Set the rules of temperature measurement by point.

Request URI Definition**Table 5-139 PUT /ISAPI/Thermal/channels/<ID>/clickToThermometry/rules/<ID>?format=json**

Method	PUT
Description	Set the rules of temperature measurement by point.
Query	format: determine the format of request or response message.
Request	<i>JSON_ClickToThermometryRule</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The first <ID> in the URI is the channel ID, the second <ID> in the URL is the rule ID.

5.93 /ISAPI/Thermal/channels/<ID>/temperatureCorrect/capabilities?format=json

Get the temperature calibration capability.

Request URI Definition**Table 5-140 GET /ISAPI/Thermal/channels/<ID>/temperatureCorrect/capabilities?format=json**

Method	GET
Description	Get the temperature calibration capability.

Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_TemperatureCorrectCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI refers to channel No.

5.94 /ISAPI/Thermal/channels/<ID>/temperatureCorrect?format=json

Get or set the temperature calibration configuration parameters

Request URI Definition

Table 5-141 GET /ISAPI/Thermal/channels/<ID>/temperatureCorrect?format=json

Method	GET
Description	Get the temperature calibration configuration parameters.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_TemperatureCorrect</i> Failed: <i>JSON_ResponseStatus</i>

Table 5-142 POST /ISAPI/Thermal/channels/<ID>/temperatureCorrect?format=json

Method	POST
Description	Set the temperature calibration parameters.
Query	format: determine the format of request or response message.
Request	<i>JSON_TemperatureCorrect</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI refers to channel No.

5.95 /ISAPI/Thermal/channels/<ID>/thermalDataConversion

For thermal data conversion.

Request URL Definition

Table 5-143 POST /ISAPI/Thermal/channels/<ID>/thermalDataConversion

Method	POST
Description	Thermal data conversion.
Query	None
Request	<i>XML_ThermalDataConversionDescription</i>
Response	<i>XML_ThermalDataConversionResult</i>

Remarks

The <ID> in the URL refers to the channel ID.

5.96 /ISAPI/Thermal/channels/<ID>/thermalDataConversion/capabilities

Get the capabilities of thermal data conversion.

Request URL Definition

Table 5-144 GET /ISAPI/Thermal/channels/<ID>/thermalDataConversion/capabilities

Method	GET
Description	Get the capabilities of thermal data conversion.
Query	None
Request	None.
Response	<i>XML_ThermalDataConversionCap</i>

Remarks

The <ID> in the URL refers to the channel ID.

5.97 /ISAPI/Thermal/channels/<ID>/thermometry/OffLineCapture?format=json

Device captures one picture offline.

Request URL Definition

Table 5-145 GET /ISAPI/Thermal/channels/<ID>/thermometry/OffLineCapture?format=json

Method	GET
Description	Device captures one picture offline.
Query	None.
Request	None.
Response	Succeeded: <i>JSON_thermometryOffLineCapture</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URL refers to the channel ID.

5.98 /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/rulesTemperatureInfo?format=json

Manually get the temperature information of temperature measurement rules in batch.

Request URI Definition

Table 5-146 GET /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/rulesTemperatureInfo?format=json

Method	GET
Description	Manually get the temperature information of temperature measurement rules in batch.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ThermometryRulesTemperatureInfoList</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI refers to channel ID.

5.99 /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/rulesTemperatureInfo/<ID>?format=json

Manually get the single temperature information of temperature measurement rules.

Request URI Definition

Table 5-147 GET /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/rulesTemperatureInfo/<ID>?format=json

Method	GET
Description	Manually get the single temperature information of temperature measurement rules.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ThermometryRulesTemperatureInfo</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The first <ID> in the URI is the channel No., the second <ID> in the URL is the rule ID.

5.100 /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/rulesTemperatureInfo/capabilities?format=json

Get capability of manually getting temperature information of temperature measurement rules.

Request URI Definition

Table 5-148 GET /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/rulesTemperatureInfo/capabilities?format=json

Method	GET
Description	Get capability of manually getting temperature information of temperature measurement rules.
Query	format : determine the format of request or response message.
Request	None
Response	Succeeded: <i>JSON_ThermometryRulesTemperatureInfoListCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI refers to channel No.

5.101 /ISAPI/Thermal/channels/<ID>/thermometry/basicParam/calibrationCoefficient

Get the calibration coefficient.

Request URL Definition

Table 5-149 POST /ISAPI/Thermal/channels/<ID>/thermometry/basicParam/calibrationCoefficient

Method	POST
Description	Get the calibration coefficient.
Query	None.
Request	<i>XML_CalibrationCoefficientDescription</i>
Response	<i>XML_CalibrationCoefficientResult</i>

5.102 /ISAPI/Thermal/Power

Get the camera battery parameters.

Request URL Definition

Table 5-150 GET /ISAPI/Thermal/Power

Method	GET
Description	Get the camera battery parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Power</i> Failed: <i>XML_ResponseStatus</i>

5.103 /ISAPI/Thermal/Power/capabilities

Get the configuration capability of camera battery.

Request URL Definition**Table 5-151 GET /ISAPI/Thermal/Power/capabilities**

Method	GET
Description	Get the configuration capability of camera battery.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_Power</i> Failed: <i>XML_ResponseStatus</i>

5.104 rtsp://<ipaddress>[:port]/ISAPI/Streaming/thermal/channels/<ID>/streamType/<Type>/trackID=

Upload the real-time thermal stream of specified channel via RTSP.

Request URL Definition**Table 5-152 SETUP rtsp://<ipaddress>[:port]/ISAPI/Streaming/thermal/channels/<ID>/streamType/<Type>/trackID=**

Method	SETUP
Description	Interact the session information.

Remarks

- The **<Type>** in the URL is defined as thermal stream data type as follows:
thermal_raw_data (the original outputted data); pixel-to-pixel_thermometry_data (temperature data after the original data has processed by algorithm); real-time_raw_data (real-time raw data)
- The trackID in the URL is defined as stream ID, and for thermal stream, the trackID=5.

5.105 rtsp://<ipaddress>[:port]/ISAPI/Streaming/thermal/channels/<ID>/streamType/<Type>

Upload the real-time thermal stream of specified channel via RTSP.

Request URL Definition

Table 5-153 DESCRIBE `rtsp://<ipaddress>[:port]/ISAPI/Streaming/thermal/channels/<ID>/streamType/<Type>`

Method	DESCRIBE
Description	Get the basic information of stream.

Table 5-154 PLAY `rtsp://<ipaddress>[:port]/ISAPI/Streaming/thermal/channels/<ID>/streamType/<Type>`

Method	PLAY
Description	Start stream transmission.

Table 5-155 PAUSE `rtsp://<ipaddress>[:port]/ISAPI/Streaming/thermal/channels/<ID>/streamType/<Type>`

Method	PAUSE
Description	Pause stream transmission.

Table 5-156 TEARDOWN `rtsp://<ipaddress>[:port]/ISAPI/Streaming/thermal/channels/<ID>/streamType/<Type>`

Method	TEARDOWN
Description	Stop stream transmission.

Remarks

The **<Type>** in the URL is defined as thermal stream data type as follows:
thermal_raw_data (the original outputted data); pixel-to-pixel_thermometry_data (temperature data after the original data has processed by algorithm); real-time_raw_data (real-time raw data)

5.106 /ISAPI/Event/notification/alertStream

Get the uploaded heartbeat or alarm/event information.

Request URI Definition

Table 5-157 GET `/ISAPI/Event/notification/alertStream`

Method	GET
Description	Get the heartbeat or uploaded alarm/event information.
Query	None.

Request	None.
Response	Option 1: <i>XML_EventNotificationAlert_AlarmEventInfo</i> or <i>XML_EventNotificationAlert_HeartbeatInfo</i> Option 2: <i>JSON_EventNotificationAlert_Alarm/EventInfo</i>  Note The messages here only show the format of alarm/event information to be uploaded. For details, refer to the corresponding alarm/event configuration chapters.

Remarks

- After calling this URI, a persistent connection is set up between the device and the platform, and the alarm or event information will be uploaded from device continuously once the alarm is triggered or event occurred.
- You can check if the XML response message is the heartbeat information according to the nodes **<eventType>** and **<eventState>**. If the values of these two node are "videoloss" and "inactive", respectively, the returned message is the heartbeat information.

5.107 /ISAPI/Event/notification/httpHosts

Get or set parameters of all HTTP listening servers, add a HTTP listening server, and delete all HTTP listening servers.

Request URI Definition

Table 5-158 GET /ISAPI/Event/notification/httpHosts

Method	GET
Description	Get parameters of all HTTP listening servers.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.
Request	None
Response	Succeeded: <i>XML_HttpHostNotificationList</i> Failed: <i>XML_ResponseStatus</i>

Table 5-159 PUT /ISAPI/Event/notification/httpHosts

Method	PUT
Description	Set parameters of all HTTP listening servers.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.
Request	<i>XML_HttpHostNotificationList</i>
Response	<i>XML_ResponseStatus</i>

Table 5-160 POST /ISAPI/Event/notification/httpHosts

Method	POST
Description	Add a HTTP listening server.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.
Request	<i>XML_HttpHostNotification</i>
Response	<i>XML_ResponseStatus</i>

Table 5-161 DELETE /ISAPI/Event/notification/httpHosts

Method	DELETE
Description	Delete all HTTP listening servers.
Query	None
Request	None
Response	<i>XML_ResponseStatus</i>

5.108 /ISAPI/Event/notification/httpHosts/<ID>

Get or set the parameters of a HTTP listening server, and delete a HTTP listening server.

Request URI Definition

Table 5-162 GET /ISAPI/Event/notification/httpHosts/<ID>

Method	GET
Description	Get the parameters of a HTTP listening server.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.
Request	None
Response	Succeeded: <i>XML_HttpHostNotification</i> Failed: <i>XML_ResponseStatus</i>

Table 5-163 PUT /ISAPI/Event/notification/httpHosts/<ID>

Method	PUT
Description	Set the parameters of a HTTP listening server.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.
Request	<i>XML_HttpHostNotification</i>
Response	<i>XML_ResponseStatus</i>

Table 5-164 DELETE /ISAPI/Event/notification/httpHosts/<ID>

Method	DELETE
Description	Delete a HTTP listening server.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.

Request	None
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HTTP listening server No.

5.109 /ISAPI/Event/notification/httpHosts/<ID>/test

Check whether the HTTP listening server is working normally.

Request URI Definition

Table 5-165 POST /ISAPI/Event/notification/httpHosts/<ID>/test

Method	POST
Description	Check whether the HTTP listening server is working normally.
Query	None
Request	<i>XML_HttpHostNotification</i>
Response	Succeeded: <i>XML_HttpHostTestResult</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HTTP listening server ID.

5.110 /ISAPI/Event/notification/httpHosts/capabilities

Get the configuration capabilities of all HTTP listening servers.

Request URI Definition

Table 5-166 GET /ISAPI/Event/notification/httpHosts/capabilities

Method	GET
Description	Get the configuration capabilities of all HTTP listening servers.
Query	None
Request	None
Response	Succeeded: <i>XML_HttpHostNotificationCap</i> Failed: <i>XML_ResponseStatus</i>

5.111 http://ipAddress:portNo/url

Listening sever sends alarm information to alarm center.

Request URL Definition

Table 5-167 POST http://ipAddress:portNo/url

Method	POST
Description	Listening sever sends alarm information to alarm center.
Query	None.
Request	None.
Response	Succeeded: <i>XML_EventNotificationAlert_AlarmEventInfo</i> or <i>JSON_EventNotificationAlert_Alarm/EventInfo</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The default port number in the URL is 80, so the URL without port No. is also valid.

Appendix A. Request and Response Messages

A.1 JSON_advanceParam

advanceParam message in JSON format

```
{
  "advanceParam":{
    "targetDetectionThreshold":,
    /*optional, integer type, target detection sensitivity threshold, ranges from 0 to 4, the default value is 3*/
    "targetGenerationSpeed":
    /*optional, integer type, target generation speed, ranges from 0 to 4, the default value is 3*/
  }
}
```

A.2 JSON_advanceParamCap

advanceParamCap message in JSON format

```
{
  "advanceParamCap":{
    "targetDetectionThreshold":{
    /*optional, integer type, target detection sensitivity threshold, ranges from 0 to 4, the default value is 3*/
      "@min":0,
      "@max":4,
      "def":3
    },
    "targetGenerationSpeed":{
    /*optional, integer type, target generation speed, ranges from 0 to 4, the default value is 3*/
      "@min":0,
      "@max":4,
      "def":3
    }
  }
}
```

A.3 JSON_basicParam

basicParam message in JSON format

```
{
  "basicParam":{
    "enabled":,
    /*required, boolean type, enable ship detection*/
    "horizontalHeight":,
    /*required, float type, device horizontal height, corrects to one decimal place*/
  }
}
```

```
"displayShipInfoOnStreamEnabled";
/*optional, boolean type, enable displaying ship information on stream*/
"backToSceneTime":
/*optional, integer type, back to scene time, ranges from 0 to 100, the default value is 15, unit: second*/
}
}
```

A.4 JSON_basicParamCap

basicParamCap message in JSON format

```
{
  "basicParamCap":{
    "enabled":{
      /*required, boolean type, enable ship detection*/
      "@opt":"true,false"
    },
    "horizontalHeight":{
      /*required, float type, device horizontal height, corrects to one decimal place*/
      "@min":0.0,
      "@max":100.0,
      "def":0.0
    },
    "displayShipInfoOnStreamEnabled":{
      /*optional, boolean type, enable displaying ship information on stream*/
      "@opt":"true,false"
    },
    "backToSceneTime":{
      /*optional, integer type, back to scene time, ranges from 0 to 100, the default value is 15, unit: second*/
      "@min":0,
      "@max":100,
      "def":15
    }
  }
}
```

A.5 JSON_Cap_functionMode

functionMode capability message in JSON format

```
{
  "functionMode":{
    /*required, string type, scene function mode of ship detection: shipFlowDetection-ship flow detection mode,
    dredgerDetection-dredger detection mode*/
    "@opt":"shipFlowDetection,dredgerDetection"
  }
}
```

A.6 JSON_ClickToThermometryRule

ClickToThermometryRule message in JSON format

```
{
  "ClickToThermometryRule":{
    "Point":{
      "positionX": "",
      /*required, float, point coordinates, normalized value, ranges from 0 to 1, corrects to three decimal places*/
      "positionY": ""
      /*required, float, point coordinates, normalized value, ranges from 0 to 1, corrects to three decimal places*/
    }
  }
}
```

A.7 JSON_DredgerDetectionRule

DredgerDetectionRule message in JSON format

```
{
  "DredgerDetectionRule":{
    "id":,
    /*required, integer type, rule ID*/
    "enabled":,
    /*required, boolean type, enable rule*/
    "overStayingAlarmTime":,
    /*optional, integer type, overstay alarm time, ranges from 60 to 3600, the default value is 180, unit: second*/
    "RegionCoordinatesList":[
      {
        /*optional, list of detection region coordinates*/
        "positionX":,
        /*required, integer type, X-coordinate*/
        "positionY":
        /*required, integer type, Y-coordinate*/
      }
    ],
    "TriggerLineList":[
      {
        "id":,
        /*required, integer type, trigger line ID*/
        "enabled":,
        /*required, boolean type, enable trigger line*/
        "MinimumTargetSize":{
          /*required, the minimum target size*/
          "width":,
          /*required, integer type, width, ranges from 0 to 100, the default value is 52*/
          "height":35
          /*required, integer type, height, ranges from 0 to 100, the default value is 35*/
        },

```

```
}  
]  
}  
}
```

A.8 JSON_DredgerDetectionRuleList

DredgerDetectionRuleList message in JSON format

```
{  
  "DredgerDetectionRuleList": [  
    {  
      "DredgerDetectionRule": {  
        "id":,  
        /*required, integer type, rule ID*/  
        "enabled":,  
        /*required, boolean type, enable rule*/  
        "overStayingAlarmTime":,  
        /*optional, integer type, overstay alarm time, ranges from 60 to 3600, the default value is 180, unit: second*/  
        "RegionCoordinatesList": [  
          {  
            /*optional, list of detection region coordinates*/  
            "positionX":,  
            /*required, integer type, X-coordinate*/  
            "positionY":  
            /*required, integer type, Y-coordinate*/  
          }  
        ],  
        "TriggerLineList": [  
          {  
            "id":,  
            /*required, integer type, trigger line ID*/  
            "enabled":,  
            /*required, boolean type, enable trigger line*/  
            "MinimumTargetSize": {  
              /*required, the minimum target size*/  
              "width":,  
              /*required, integer type, width, ranges from 0 to 100, the default value is 52*/  
              "height":  
              /*required, integer type, height, ranges from 0 to 100, the default value is 35*/  
            },  
          }  
        ]  
      }  
    }  
  ]  
}
```

A.9 JSON_DredgerDetectionRuleListCap

DredgerDetectionRuleListCap message in JSON format

```
{
  "DredgerDetectionRuleListCap":{
    "maxSize":5,
    /*required, integer type, the maximum number of supported dredger detection rules*/
    "DredgerDetectionRuleCap":{
      "enabled":{
        /*required, boolean type, enable rule*/
        "@opt":"true,false"
      },
      "overStayingAlarmTime":{
        /*optional, integer type, overstay alarm time, ranges from 60 to 3600, the default value is 180, unit: second*/
        "@min":60,
        "@max":3600,
        "@def":180
      },
      "RegionCoordinatesList":{
        /*optional, list of detection region coordinates*/
        "maxSize":5,
        /*required, integer type, the number of supported regions*/
        "positionX":{
          /*required, integer type, X-coordinate*/
          "@min":0,
          "@max":1000
        },
        "positionY":{
          /*required, integer type, Y-coordinate*/
          "@min":0,
          "@max":1000
        }
      },
      "TriggerLineList":{
        "maxSize":5,
        /*required, integer type, the number of supported trigger lines*/
        "enabled":{
          /*required, boolean type, enable trigger line*/
          "@opt":"true,false"
        },
        "MinimumTargetSize":{
          /*required, the minimum target size*/
          "width":{
            /*required, integer type, width, ranges from 0 to 100, the default value is 52*/
            "@min":0,
            "@max":100,
            "@def":52
          },
          "height":{
            /*required, integer type, height, ranges from 0 to 100, the default value is 35*/
```

```
    "@min":0,
    "@max":100,
    "@def":35
  },
}
}
}
}
```

A.10 JSON_EpidemicControlCap

JSON message about face thermography capability

```
{
  "EpidemicControlCap":{
    "reportType":{
      /*required, string, report type: "daily,weekly,monthly,yearly"*/
      "@opt":[
        "daily",
        "weekly",
        "monthly",
        "yearly"
      ]
    },
    "startTime":{
      /*required, string, start time*/
      "@min":0,
      "@max":0
    },
    "endTime":{
      /*required, string, end time*/
      "@min":0,
      "@max":0
    },
    "channel":{
      /*optional, array of strings, channel No., multiple channels can be separated by comma*/
      "@min":0,
      "@max":0
    }
  }
}
```

A.11 JSON_EventNotificationAlert_dredgerDetection

The information of dredger detection alarm is uploaded in the JSON format, see details below.

```
{
  "ipAddress":"","
```

```
/*required, alarm device IPV4 address, string type, the maximum length is 32*/
"ipv6Address":"","
/*optional, alarm device IPV6 address, string type, the maximum length is 128*/
"portNo":"","
/*optional, alarm device port No., integer32 type*/
"protocol":"","
/*optional, protocol type: HTTP or HTTPS, string type, the maximum length is 32*/
"macAddress":"","
/*optional, MAC address, string type, the maximum length is 32*/
"channelID",
/*optional, channel ID number, which triggers the alarm, integer32 type*/
"dateTime":"","
/*required, alarm triggered time, ISO8601 time format, string type, the maximum length is 32*/
"activePostCount",
/*required, the number of one alarm uploaded times, integer32 type*/
"eventType":"","
/*required, event type: dredgerDetection-dredger detection, string type, the maximum length is 128*/
"eventState":"","
/*required, event triggering status: active-triggered, inactive-not triggered (heartbeat data), string type, the maximum
length is 32*/
"eventDescription":"","
/*required, event description: dredger detection alarm)*/
"DetectionRegionList":[{
/*ship detection region list*/
  "DetectionRegionEntry":{
    "RegionCoordinatesList":[{
/*ship detection region*/
      "RegionCoordinates":{
        "positionX",
/*required, float type, X-coordinate*/
        "positionY":
/*required, float type, Y-coordinate*/
      }
    }],
    "ShipsDetection":{
      "shipsInfoList":[{
        "shipsInfo":{
          "id",
/*required, integer type, ship ID*/
          "shipsLength",
/*required, float type, ship length*/
          "shipsHeight",
/*required, float type, ship height*/
          "shipsWidth":
/*required, float type, ship width*/
        }
      }
    ]
  }
}],
"sId",
/*optional, integer type, scene ID*/
```

```
"sceneName":"","  
/*optional, string type, scene name*/  
}
```

A.12 JSON_EventNotificationAlert_TDAMsg

JSON message about temperature difference alarm details

Content-Type: multipart/form-data; boundary=MIME_boundary

--MIME_boundary

Content-Type: application/json

Content-Length:

```
{  
  "ipAddress":"","  
  /*required, string, IPv4 address of alarm device; the maximum string length is 32 bytes*/  
  "ipv6Address":"","  
  /*optional, string, IPv6 address of alarm device; the maximum string length is 128 bytes*/  
  "portNo":,  
  /*optional, integer32, device's communication port No.*/  
  "protocol":"","  
  /*optional, string, transport protocol type: HTTP, HTTPS, EHome; the maximum string length is 32 bytes*/  
  "macAddress":"","  
  /*optional, string, MAC address; the maximum string length is 32 bytes*/  
  "channelID":,  
  /*optional, integer32, channel No., which triggers alarm*/  
  "dateTime":"","  
  /*required, string, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00; the maximum  
string length is 32 bytes*/  
  "activePostCount":,  
  /*required, integer32, event occurred times*/  
  "eventType":"","  
  /*required, string, event types, here it should be set to "TDA"; the maximum string length is 128 bytes*/  
  "eventState":"","  
  /*required, string, event status (for persistent event): active, inactive; the maximum string length is 32 bytes*/  
  "eventDescription":"","  
  /*required, event description: Temperature Diff Alarm*/  
  "DetectionRegionList":{  
  /*region list of temperature difference alarm*/  
    "DetectionRegionEntry":{  
      "AlarmRuleList":{  
        "alarmID":,  
        /*required, integer, rule ID of temperature difference alarm*/  
        "RegionCoordinatesList":{  
  /*region of temperature difference alarm*/  
        "RegionCoordinates":{  
          "positionX":,  
          /*required, float, x-coordinate*/  
          "positionY":  
          /*required, float, y-coordinate*/
```

```
    }
  }}
},
"TDA":{
  "thermometryUnit": "",
  /*optional, string, temperature unit: "celsius, fahrenheit, kelvin"*/
  "ruleTemperatureDiff":,
  /*optional, float, configured temperature difference threshold, corrects to one decimal place*/
  "currTemperatureDiff":,
  /*optional, float, current temperature difference, corrects to one decimal place*/
  "ruleCalibType": "",
  /*optional, string, rule calibration type: "point,line,region"*/
  "alarmType": "",
  /*optional, string, alarm type: "MaxTemperature"-highest temperature, "MinTemperature"-lowest temperature,
  "AverageTemperature"-average temperature*/
  "alarmRule": "",
  /*optional, string, alarm rule: "greater"-higher than, "less"-lower than*/
  "toleranceTemperature":,
  /*optional, float, temperature tolerance, corrects to one decimal place*/
  "alarmFilteringTime":,
  /*optional, integer, alarm dwell time, unit: second, range:[1,200], the default value is 0s*/
  "AbsoluteHigh":{
  /*PTZ coordinates*/
    "elevation":,
    /*optional, float, corrects to two decimal place*/
    "azimuth":,
    /*optional, float, corrects to two decimal place*/
    "absoluteZoom":,
    /*optional, float, corrects to two decimal place*/
  },
  "presetNo":
  /*optional, int, preset No. 可0, integer 0型, 0置点号*/
  }
},
"channelName": "",
/*optional, string, channel name*/
"deviceId": "",
/*optional, string, device ID*/
"detectionPicturesNumber":,
/*optional, integer, the number of detected pictures*/
"visibleLightURL": "",
/*optional, string, visible picture URL*/
"thermalURL": "",
/*optional, string, thermal picture URL*/
"thermalInfoURL": "",
/*optional, string, thermal additional information URL*/
"URLCertificationType": ""
/*optional, string, authentication mode of picture URL: no (none authentication), digest (by digest, for local picture
URL returned by NVR or DVR; the maximum string length is 32 bytes*/
}
```

```
--MIME_boundary
Content-Disposition: form-data; name="F4F665D6A18E41308CE9934DCDDD1111"; filename="TDA.jpg";
Content-Type: image/jpeg
Content-Length: 5798
Content-ID: TDA
```

A.13 JSON_FaceThermReportSearchCond

JSON message about search condition of face thermography statistical report

```
{
  "reportType": "",
  /*required, string, report type: "daily,weekly,monthly,yearly"*/
  "startTime": "",
  /*required, string, start time*/
  "endTime": "",
  /*optional, string, end time*/
  "channel": "",
  /*optional, array of strings, channel No., multiple channels can be separated by comma*/
}
```

A.14 JSON_FaceThermReportSearchResult

JSON message about search result of face thermography statistical report

```
{
  "numOfMatches":,
  /*required, int, the number of returned records*/
  "matchList":[
    {
      "timeSpan":{
        /*time span for daily report is one hour, e.g., from 2020-02-20T02:00:00+08:00 to 2020-02-20T03:00:00+08:00; for
        weekly and monthly report is one day, e.g., from 2020-02-20T00:00:00+08:00 to 2020-02-21T00:00:00+08:00; for
        annual report is one month, e.g., from 2020-02-00T00:00:00+08:00 to 2020-03-00T00:00:00+08:00*/
        "startTime": "",
        /*required, string, start time*/
        "endTime": "",
        /*required, string, end time*/
      },
      "totalPersonCount":,
      /*required, int, total number of person, totalPersonCount=abnormalTempPersonCount +
      normalTempPersonCount*/
      "abnormalTempPersonCount":,
      /*required, int, total number of abnormal temperature person*/
      "normalTempPersonCount":,
      /*required, int, total number of normal temperature person*/
      "noMaskPersonCount":,
      /*required, int, total number of person who is not wearing mask*/
    }
  ]
}
```

```
]
}
```

A.15 JSON_fireDetection_AdvanceParam

JSON message about advanced parameters of fire detection

```
{
  "AdvanceParam": {
    "minPixel":
      /*optional, int, the minimum pixel of fire detection, range: [1,2]*/
  }
}
```

A.16 JSON_fireDetection_AdvanceParamCap

JSON message about capability of configuring advanced parameters of fire detection.

```
{
  "AdvanceParamCap": {
    "minPixel": {
      /*optional, int, the minimum pixel of fire detection, range: [1,2]*/
      "@min": 1,
      "@max": 2
    }
  }
}
```

A.17 JSON_functionMode

functionMode message in JSON format

```
{
  "functionMode":""
  /*required, string type, scene function mode of ship detection: shipFlowDetection-ship flow detection mode,
  dredgerDetection-dredger detection mode*/
}
```

A.18 JSON_ResponseStatus

JSON message about response status

```
{
  "requestURL":"","
  /*optional, string, request URL*/
}
```

```
"statusCode": ,
/*required, int, status code*/
"statusString": "",
/*required, string, status description*/
"subStatusCode": "",
/*required, string, sub status code*/
"errorCode": ,
/*optional, int, error code, which corresponds to subStatusCode, this field is required when statusCode is not 1. The
returned value is the transformed decimal number*/
"errorMsg": "",
/*optional, string, error details, this field is required when statusCode is not 1*/
}
```



Note

See **ISAPI Response Codes** for details about the status codes, sub status codes, error codes, and error descriptions.

A.19 JSON_ShipDetectionCount

ShipDetectionCount message in JSON format

```
{
  "ShipsDetectionCount":{
    "sid",
    /*required, integer type, scene ID*/
    "upShipsCount",
    /*required, integer type, number of upstream ships*/
    "downShipsCount",
    /*required, integer type, number of downstream*/
    "leftShipsCount",
    /*required, integer type, number of ships on the left*/
    "rightShipsCount",
    /*required, integer type, number of ships on the right*/
    "totalCount",
    /*required, integer type, total number of ships*/
    "beginTime":""
  }
}
```

A.20 JSON_shipsDetectionCap

shipsDetectionCap message in JSON format

```
{
  "shipsDetectionCap":{
    "isSupportBasicConfig",
    /*required, boolean type, whether supports basic parameters configuration of ship detection*/
  }
}
```

```
"isSupportAdvanceConfig":,
/*optional, boolean type, whether supports advanced parameters configuration of ship detection*/
"isSupportSceneConfig":,
/*optional, boolean type, whether supports scene parameters configuration of ship detection*/
"isSupportCaptureRatio":,
/*optional, boolean type, whether supports capture ratio configuration of ship detection*/
"isSupportFunctionMode":,
/*optional, boolean type, whether supports function mode configuration of ship detection*/
"isSupportShipsFlowDetectionRuleConfig":,
/*optional, boolean type, whether supports rule configuration of ship flow detection mode*/
"isSupportDredgerDetectionRuleConfig":,
/*optional, boolean type, whether supports rule configuration of dredger detection mode*/
"isSupportShipsDetectionCount":,
/*optional, boolean type, whether supports ship counting*/
"isSupportSceneCruise":,
/*optional, boolean type, whether supports scene auto-switch*/
"sceneRatioId":{
/*optional, string type, the ID of channel, which supports scene ratio*/
"@opt":""
},
"shipsDetectionSceneNum":
/*optional, integer type, the maximum number of supported scenes for ship detection*/
}
}
```

A.21 JSON_ShipsDetectionScene

ShipsDetectionScene message in JSON format

```
{
  "ShipsDetectionScene":{
    "sid":,
    /*required, integer type, scene ID*/
    "sceneName":"","
    /*optional, string type, scene name*/
    "sceneRatio":,
    /*optional, integer type, scene ratio (actual ratio of each channel). You can get the channel No. via URL: /ISAPI/
    Thermal/shipsDetection/capabilities?format=json*/
    "SceneNameOverlay":{
    /*optional, scene name on-screen display*/
    "enabled":,
    /*required, enable scene name OSD*/
    "normalizedScreenSize":{
    /*normalized size*/
    "normalizedScreenWidth":,
    /*optional, float type, normalized width*/
    "normalizedScreenHeight":
    /*optional, float type, normalized height*/
    },
    "positionX":,
```

```
/*optional, integer type, original point X-coordinate*/
  "positionY":
/*optional, integer type, original point Y-coordinate*/
  }
}
}
```

A.22 JSON_ShipsDetectionSceneCap

ShipsDetectionSceneCap message in JSON format

```
{
  "ShipsDetectionSceneCap":{
    "sid";
/*required, integer type, scene ID*/
    "sceneName":"","
/*optional, string type, scene name*/
    "sceneRatio":{
/*optional, integer type, scene ratio (actual ratio of each channel). You can get the channel No. via URL: /ISAPI/
Thermal/shipsDetection/capabilities?format=json*/
    "@min";
    "@max";
    "def";
  },
  "SceneNameOverlay":{
/*optional, scene name on-screen display*/
    "enabled":{
/*required, enable scene name OSD*/
    "@opt":"true,false"
    },
    "normalizedScreenSize":{
/*normalized size*/
    "normalizedScreenWidth":{
/*optional, float type, normalized width*/
    "@min":0,
    "@max":1,
    },
    "normalizedScreenHeight":{
/*optional, float type, normalized height*/
    "@min":0,
    "@max":1,
    }
  },
  "positionX":{
/*optional, integer type, original point X-coordinate*/
  "@min";
  "@max";
  },
  "positionY":{
/*optional, integer type, original point Y-coordinate*/
```

```
"@min";  
"@max";  
}  
}  
}  
}
```

A.23 JSON_ShipsDetectionSceneList

ShipsDetectionSceneList message in JSON format

```
{  
  "ShipsDetectionSceneList":[{"  
    "ShipsDetectionScene":{  
      "sid";  
      /*required, integer type, scene ID*/  
      "sceneName":"","  
      /*optional, string type, scene name*/  
      "sceneRatio";  
      /*optional, integer type, scene ratio (actual ratio of each channel). You can get the channel No. via URL: /ISAPI/  
Thermal/shipsDetection/capabilities?format=json*/  
      "SceneNameOverlay":{  
        /*optional, scene name on-screen display*/  
        "enabled";  
        /*required, enable scene name OSD*/  
        "normalizedScreenSize":{  
          /*normalized size*/  
          "normalizedScreenWidth";  
          /*optional, float type, normalized width*/  
          "normalizedScreenHeight":  
          /*optional, float type, normalized height*/  
          },  
          "positionX";  
          /*optional, integer type, original point X-coordinate*/  
          "positionY":  
          /*optional, integer type, original point Y-coordinate*/  
          }  
        }  
      }  
    }  
  ]  
}
```

A.24 JSON_ShipsDetectionCaptureRatio

ShipsDetectionCaptureRatio message in JSON format

```
{  
  "ShipsDetectionCaptureRatio":{  
    "centerCaptureEnabled";  
  }  
}
```

```
/*required, boolean type, enable center capture mode*/
  "captureRatio":
/*optional, integer type, capture ratio*/
  }
}
```

A.25 JSON_ShipsDetectionCaptureRatioCap

ShipsDetectionCaptureRatioCap message in JSON format

```
{
  "ShipsDetectionCaptureRatioCap":{
    "centerCaptureEnabled":"true,false",
/*required, boolean type, enable center capture mode*/
    "captureRatio":{
/*optional, integer type, capture ratio*/
      "@min":1,
      "@max":32
    }
  }
}
```

A.26 JSON_ShipsDetectionTraceListCap

ShipsDetectionTraceListCap message in JSON format

```
{
  "ShipsDetectionTraceListCap":{
    "maxSize":"","
/*required, integer type, the maximum number of supported scenes*/
    "ShipsDetectionTraceCap":{
      "functionMode":{
/*optional, string type, supported function mode*/
        "@opt":"shipFlowDetection"
      },
      "dwelltime":{
/*optional, integer type, scene dwell time, ranges from 30s to 3600s*/
        "@min":30,
        "@max":3600
      },
      "overStayingAlarmTime":{
/*optional, integer type, overstay alarm time, ranges from 60s to 3600s. The scene dwell time should be no less than overstay alarm time*/
        "@min":60,
        "@max":3600
      },
      "SceneNameOverlay":{
/*optional, scene name OSD*/
        "enabled":{
```

```
/*required, enable scene name OSD*/
  "opt": "true,false"
},
"normalizedScreenSize": {
/*normalized size*/
  "normalizedScreenWidth": {
/*optional, integer type, normalized width*/
    "@min": 0,
    "@max": 1
  },
  "normalizedScreenHeight": {
/*optional, integer type, normalized height*/
    "@min": 0,
    "@max": 1
  }
},
"positionX": {
/*optional, integer type, original point X-coordinate*/
  "@min":,
  "@max":
},
"positionY": {
/*optional, integer type, original point Y-coordiante*/
  "@min":,
  "@max":
}
}
}
}
```

A.27 JSON_ShipsDetectionSceneTraceList

ShipsDetectionSceneTraceList message in JSON format

```
{
  "ShipsDetectionSceneTraceList": [{
    "ShipsDetectionSceneTrace": {
      "sid":,
/*required, integer type, scene ID*/
      "sceneName": "",
/*required, string type, scene name*/
      "patrolId":,
/*optional, integer type, patrol sequence ID, it is the scene serial number in the patrol*/
      "functionMode": "",
/*optional, string type, function mode: shipFlowDetection-ship flow detection mode, dredgerDetection-dredger
detection mode*/
      "dwelltime":,
/*optional, integer type, scene dwell time, ranges from 30s to 3600s*/
      "overStayingAlarmTime":,
```

```
/*optional, integer type, ranges from 60s to 3600s. The scene dwell time should be no less than overstay alarm time*/
  "SceneNameOverlay":{
/*optional, scene name OSD*/
    "enabled":,
/*required, enable scene name OSD*/
    "ScreenSize":{
/*normalized size*/
      "ScreenWidth":1,
/*optional, integer type, normalized width*/
      "ScreenHeight":1
/*optional, integer type, normalized height*/
    },
    "positionX":1,
/*optional, integer type, original point X-coordinate*/
    "positionY":1
/*optional, integer type, original point Y-coordiante*/
  }
}
}}
```

A.28 JSON_ShipFlowDetectionRuleListCap

ShipFlowDetectionRuleListCap message in JSON format

```
{
  "ShipFlowDetectionRuleListCap":{
    "maxSize":5,
/*required, integer type, the maximum number of supported ship flow detection rules*/
    "ShipFlowDetectionRuleCap":{
      "enabled":{
/*required, boolean type, enable ship flow detection rule*/
        "@opt":"true,false"
      },
      "regionEnOrExAlarmEnabled":{
/*optional, boolean type, enable alarm of region entrance or region exit*/
        "@opt":"true,false"
      },
      "RegionCoordinatesList":{
/*optional, list of detection region coordinates*/
        "maxSize":5,
/*required, integer type, the maximum number of supported regions*/
        "positionX":{
/*required, integer type, X-coordinate*/
          "@min":0,
          "@max":1000
        },
        "positionY":{
/*required, integer type, Y-coordinate*/
          "@min":0,
```

```
"@max":1000
}
},
"TriggerLineList":{
  "maxSize":5,
  /*required, integer type, the number of supported trigger lines*/
  "enabled":{
    /*required, boolean type, enable trigger line*/
    "@opt":"true,false"
  },
  "MinimumTargetSize":{
    /*required, the minimum target size*/
    "width":{
      /*required, integer type, width, ranges from 0 to 100, the value is 52 by default*/
      "@min":0,
      "@max":100,
      "@def":52
    },
    "height":{
      /*required, integer type, height, ranges from 0 to 100, the value is 35 by default*/
      "@min":0,
      "@max":100,
      "@def":35
    },
  },
  "TriggerLineCoordinatesList":{
    "positionX":{
      /*required, integer type, X-coordinate*/
      "@min":0,
      "@max":1000
    },
    "positionY":{
      /*required, integer type, Y--coordinate*/
      "@min":0,
      "@max":1000
    }
  }
}
}
```

A.29 JSON_ShipFlowDetectionRuleList

ShipFlowDetectionRuleList message in JSON format

```
{
  "ShipFlowDetectionRuleList":[
    {
      "ShipFlowDetectionRule":{
```

```
"id";
/*required, integer type, rule ID*/
"enabled";
/*required, boolean type, enable the rule*/
"regionEnOrExAlarmEnabled";
/*optional, boolean type, enable alarm of region entrance or region exit*/
"RegionCoordinatesList":[
  {
/*optional, list of detection region coordinates*/
    "positionX";
/*required, integer type, X-coordiante*/
    "positionY";
/*required, integer type, Y-coordinate*/
  }
],
"TriggerLineList":[
  {
    "id";
/*required, integer type, trigger line ID*/
    "enabled";
/*required, boolean type, enable trigger line*/
    "MinimumTargetSize":{
/*required, the minimum target size*/
      "width";
/*required, integer type, width, ranges from 0 to 100, the value is 52 by default*/
      "height";
/*required, integer type, height, ranges from 0 to 100, the value is 35 by default*/
    },
    "TriggerLineCoordinatesList":[
      {
        "positionX";
/*required, integer type, X-coordinate*/
        "positionY";
/*required, integer type, Y-coordinate*/
      }
    ]
  }
]
}
```

A.30 JSON_ShipFlowDetectionRule

ShipFlowDetectionRule message in JSON format

```
{
  "ShipFlowDetectionRule":{
    "id";
```

```
/*required, integer type, rule ID*/
"enabled":,
/*required, boolean type, enable the rule*/
"regionEnOrExAlarmEnabled":,
/*optional, boolean type, enable alarm of region entrance or region exit*/
"RegionCoordinatesList":[
{
/*optional, list of detection region coordinates*/
"positionX":,
/*required, integer type, X-coordiante*/
"positionY":
/*required, integer type, Y-coordinate*/
}
],
"TriggerLineList":[
{
"id":,
/*required, integer type, trigger line ID*/
"enabled":,
/*required, boolean type, enable trigger line*/
"MinimumTargetSize":{
/*required, the minimum target size*/
"width":,
/*required, integer type, width, ranges from 0 to 100, the value is 52 by default*/
"height":
/*required, integer type, height, ranges from 0 to 100, the value is 35 by default*/
},
"TriggerLineCoordinatesList":[
{
"positionX":,
/*required, integer type, X-coordinate*/
"positionY":
/*required, integer type, Y-coordinate*/
}
]
}
]
}
}
```

A.31 JSON_thermometryOffLineCapture

thermometryOffLineCapture message in JSON format

```
{
"thermometryOffLineCapture":{
"channel": ,
/*required, integer32, channel ID*/
"thermometryOffLineCaptureDataLen": ,
/*optional, integer32, the size of offline captured picture*/
```

```
}  
}
```

A.32 JSON_ThermometryRulesTemperatureInfo

ThermometryRulesTemperatureInfo message in JSON format

```
{  
  "ThermometryRulesTemperatureInfo": {  
    "id": "",  
    /*required, integer32, rule ID*/  
    "maxTemperature": "",  
    /*required, float, the maximum temperature (corrects to one decimal place, unit: degree centigrade)*/  
    "minTemperature": "",  
    /*required, float, the minimum temperature (corrects to one decimal place, unit: degree centigrade)*/  
    "averageTemperature": "",  
    /*required, float, the average temperature (corrects to one decimal place, unit: degree centigrade)*/  
    "MaxTemperaturePoint": {  
      "positionX": "",  
      /*required, float, the coordinates of max. temperature position, normalized value, ranges from 0 to 1, accuracy: 0.001,  
      reference point refers to top left corner of picture*/  
      "positionY": ""  
    },  
    /*required, float, the coordinates of max. temperature position, normalized value, ranges from 0 to 1, accuracy: 0.001,  
    reference point refers to top left corner of picture*/  
    "MinTemperaturePoint": {  
      "positionX": "",  
      /*required, float, the coordinates of min. temperature position, normalized value, ranges from 0 to 1, accuracy: 0.001,  
      reference point refers to top left corner of picture*/  
      "positionY": ""  
    },  
    /*required, float, the coordinates of min. temperature position, normalized value, ranges from 0 to 1, accuracy: 0.001,  
    reference point refers to top left corner of picture*/  
    "isFreezedata": ""  
  },  
  /*optional, boolean, whether freezes data*/  
}
```

A.33 JSON_ThermometryRulesTemperatureInfoList

ThermometryRulesTemperatureInfoList message in JSON format

```
{  
  "ThermometryRulesTemperatureInfoList": {  
    "ThermometryRulesTemperatureInfo": [{  
      "id": "",  
      /*required, integer32, rule ID*/  
      "maxTemperature": "",  
      /*required, float, the maximum temperature (corrects to one decimal place, unit: degree centigrade)*/  
    }  
  ]  
}
```

```
"minTemperature": "",
/*required, float, the minimum temperature (corrects to one decimal place, unit: degree centigrade)*/
"averageTemperature": "",
/*required, float, the average temperature (corrects to one decimal place, unit: degree centigrade)*/
"MaxTemperaturePoint": {
  "positionX": "",
/*required, float, the coordinates of max. temperature position, normalized value, ranges from 0 to 1, accuracy: 0.001,
reference point refers to top left corner of picture*/
  "positionY": ""
/*required, float, the coordinates of max. temperature position, normalized value, ranges from 0 to 1, accuracy: 0.001,
reference point refers to top left corner of picture*/
},
"MinTemperaturePoint":{
  "positionX": "",
/* required, float, the coordinates of min. temperature position, normalized value, ranges from 0 to 1, accuracy:
0.001, reference point refers to top left corner of picture*/
  "positionY": ""
/* required, float, the coordinates of min. temperature position, normalized value, ranges from 0 to 1, accuracy:
0.001, reference point refers to top left corner of picture*/
}
"isFreezedata": ""
/*optional, boolean, whether freezes data*/
}}
}
}
```

A.34 JSON_ThermometryRulesTemperatureInfoListCap

ThermometryRulesTemperatureInfoListCap message in JSON format

```
{
  "ThermometryRulesTemperatureInfoListCap":{
    "ThermometryRulesTemperatureInfoList":{
      "ThermometryRulesTemperatureInfo": {
        "id":{
/*required, integer32, rule ID*/
          "@min": ,
          "@max":
        },
        "maxTemperature":{
/*required, float, the maximum temperature (corrects to one decimal place, unit: degree centigrade)*/
          "@min": ,
          "@max":
        },
        "minTemperature":{
/*required, float, the minimum temperature (corrects to one decimal place, unit: degree centigrade)*/
          "@min": ,
          "@max":
        },
        "averageTemperature":{
```

```
/*required, float, the average temperature (corrects to one decimal place, unit: degree centigrade)*/
  "@min": ,
  "@max":
},
"MaxTemperaturePoint":{
  "positionX":{
/*required, float, the coordinates of max. temperature position, normalized value, ranges from 0 to 1, accuracy: 0.001,
reference point refers to top left corner of picture*/
    "@min": ,
    "@max":
  },
  "positionY":{
/*required, float, the coordinates of max. temperature position, normalized value, ranges from 0 to 1, accuracy: 0.001,
reference point refers to top left corner of picture*/
    "@min": ,
    "@max":
  },
},
"MinTemperaturePoint":{
  "positionX":{
/* required, float, the coordinates of min. temperature position, normalized value, ranges from 0 to 1, accuracy:
0.001, reference point refers to top left corner of picture*/
    "@min": ,
    "@max":
  },
  "positionY":{
/* required, float, the coordinates of max. temperature position, normalized value, ranges from 0 to 1, accuracy:
0.001, reference point refers to top left corner of picture*/
    "@min": ,
    "@max":
  },
},
"isFreezedata": ""
/*optional, boolean, whether freezes data*/
}
}
}
```

A.35 JSON_ThermometryUpload

JSON message about real-time temperature measurement data

Content-Type: multipart/form-data; boundary=boundary

--boundary

Content-Type: application/json; charset="UTF-8"

Content-Length:

{

```
"ThermometryUpload":{
  "relativeTime": ,
  /*required,integer32, relative time*/
  "absTime": ,
  /*required,integer32, absolute time*/
  "ruleName": "",
  /*required,string, rule name*/
  "ruleID": ,
  /*required,integer32, rule ID, ranges from 1 to 21*/
  "ruleCalibType": ,
  /*required,integer32, rule calibration type: 0-point, 1-frame, 2-line*/
  "presetNo": ,
  /*required,integer32, preset number*/
  "PointThermCfg":{
    /*optional, point thermography information, it is valid when ruleCalibType values "0"*/
    "temperature":
    /*required,float, current temperature, corrects to one decimal place*/
    "Point": {
    /*optional, coordinates of point thermography*/
    "positionX":,
    /*required,float, X-coordinate, normalized value, ranges from 0 to 1, accuracy is 0.001, the origin of coordinates is
    upper-left corner*/
    "positionY":
    /*required,float, Y-coordinate, normalized value, ranges from 0 to 1, accuracy is 0.001, the origin of coordinates is
    upper-left corner*/
    }
    }
    "LinePolygonThermCfg":{
    /*optional, frame/line thermography, it is valid when ruleCalibType values "1" or "2"*/
    "MaxTemperature":
    /*required,float, max. temperature, corrects to one decimal place*/
    "MinTemperature":
    /*required,float, min. temperature, corrects to one decimal place*/
    "AverageTemperature":
    /*required,float, average temperature, corrects to one decimal place*/
    "TemperatureDiff":
    /*required,float, temperature difference, corrects to one decimal place*/
    "Region":[{
    /*optional, region coordinates*/
    "Point":{
    /*optional, point coordinate*/
    "positionX":,
    /*required,float, X-coordinate, normalized value, ranges from 0 to 1, accuracy is 0.001, the origin of coordinates is
    upper-left corner*/
    "positionY":
    /*required,float, Y-coordinate, normalized value, ranges from 0 to 1, accuracy is 0.001, the origin of coordinates is
    upper-left corner*/
    }
    }
    }
    "thermometryUnit": ,
    /*required,integer32, temperature measurement unit: 0-Celsius (°C), 1-Fahrenheit (°F), 2-Kelvin (K).*/
  }
```

```
"dataType": ,
/*required,integer32, data status: 0-measuring, 1-start, 2-end*/
"centerPointTemperature":
/*optional, float, central point temperature, it is not returned if not supported, corrects to one decimal place*/
"highestPointTemperature":
/*optional, float, temperature of highest point, it is not returned if not supported, corrects to one decimal place*/
"lowestPointTemperature":
/*optional, float, temperature of lowest point, it is not returned if not supported, corrects to one decimal place*/
"isFreezedata": ,
/*required,boolean, whether freezes data*/
"HighestPoint":{
/*optional, the coordinates of maximum temperature position, it is valid when ruleCalibType values "1" or "2"*/
"positionX":,
/*required, float, X-coordinate, normalized value, ranges from 0 to 1, accuracy is 0.001, the origin of coordinates is
upper-left corner*/
"positionY":
/*required, float, Y-coordinate, normalized value, ranges from 0 to 1, accuracy is 0.001, the origin of coordinates is
upper-left corner*/
},
"LowestPoint":{
/*optional, the coordinates of minimum temperature position, it is valid when ruleCalibType values "1" or "2"*/
"positionX":,
/*required, float, X-coordinate, normalized value, ranges from 0 to 1, accuracy is 0.001, the origin of coordinates is
upper-left corner*/
"positionY":
/*required, float, Y-coordinate, normalized value, ranges from 0 to 1, accuracy is 0.001, the origin of coordinates is
upper-left corner*/
},
"faceSnapThermometryEnabled": ,
/*optional, boolean, whether to enable uploading alarm with temperature information*/
"faceRect": {
/*optional, face thumbnail rectangle*/
"height": ,
/*optional, height, float*/
"width": ,
/*optional, width, float*/
"x": ,
/*optional, float, X-coordinate*/
"y":
/*optional, float, Y-coordinate*/
}
}
}
}
--boundary
```

A.36 JSON_ThermometryUploadList

JSON message about real-time temperature measurement data of all rules

```
Content-Type: multipart/form-data; boundary=boundary

--boundary
Content-Type: application/json; charset="UTF-8"
Content-Length:
{
  "ThermometryUploadList":
  {
    "ThermometryUpload":
/*real-time temperature measurement data, see JSON_ThermometryUpload for details*/
  }
}
--boundary
```

See Also

JSON_ThermometryUpload

A.37 JSON_TemperatureCorrect

JSON message about temperature calibration configuration parameters

```
{
  "TemperatureCorrect": {
    "enabled": ,
/*optional, boolean, whether to enable the function*/
    "streamOverlay": ,
/*optional, string, whether to display calibrated temperature on video*/
    "correctEnabled": ,
/*optional, string, whether to enable temperature calibration*/
    "emissivity": ,
/*optional, float, emissivity, range: [0.01,1.00]*/
    "distance": ,
/*optional, float, distance: range: [0.3,2]m*/
    "temperature": ,
/*optional, float, temperature, range: [30.0,50.0]°C*/
    "CentrePoint":{
/*optional, normalized coordinates of black body center point, from 0 to 1000*/
      "CalibratingCoordinates":{
        "positionX": ,
/*required, int, x-coordinate*/
        "positionY":
/*required, int, y-coordinate*/
      }
    },
    "correctTemperature":
/*optional, float, temperature calibration coefficient, range: [-99.0,-99.0]°C*/
  }
}
```

A.38 JSON_TemperatureCorrectCap

JSON message about temperature calibration capability

```
{
  "TemperatureCorrectCap":{
    "enabled":{
      /*optional, boolean, whether to enable the function*/
      "@opt":[true,false]
    },
    "streamOverlay":{
      /*optional, string, whether to display calibrated temperature on video*/
      "@opt":[true,false]
    },
    "correctEnabled":{
      /*optional, string, whether to enable temperature calibration*/
      "@opt":[true,false]
    },
    "emissivity":{
      /*optional, float, emissivity, range: [0.01,1.00]*/
      "@min":0.01,
      "@max":1
    },
    "distance":{
      /*optional, float, distance: range: [0.3,2]m*/
      "@min":0.3,
      "@max":2
    },
    "temperature":{
      /*optional, float, temperature, range: [30.0,50.0]°C*/
      "@min":30,
      "@max":50
    },
    "CentrePoint":{
      /*optional, normalized coordinates of black body center point, from 0 to 1000*/
      "CalibratingCoordinates":{
        "positionX": ,
        /*required, int, x-coordinate*/
        "positionY":
        /*required, int, y-coordinate*/
      }
    },
    "correctTemperature":{
      /*optional, float, temperature calibration coefficient, range: [-99.0,-99.0]°C*/
      "@min":-99,
      "@max":99
    }
  }
}
```

A.39 XML_BodyTemperatureCompensation

XML message about body temperature compensation parameters

```
<?xml version="1.0" encoding="utf-8"?>
<BodyTemperatureCompensation version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <type><!--required, xs:string, compensation type: "manual"-manual, "auto"-automatic--></type>
  <ManualParam><!--dependent, type="manual"-->
    <compensationValue><!--optional, xs:float, compensation value, range: [-10.0 10.0], unit: celsius degree--></
compensationValue>
    <smartCorrection><!--optional, xs:float, manual calibration, the range varies according to device type--></
smartCorrection>
    <environmentalTemperature>
      <!--dependent, xs:float, environment temperature, it is configurable when the value of type is "manual" and read
only when the value is of type is "auto"-->
    </environmentalTemperature>
    <temperatureCompensation><!--optional, xs:boolean, whether to enable body temperature compensation--></
temperatureCompensation>
    <environmentalTemperatureMode>
      <!--optional, xs:string, environment temperature mode: "auto"-automatic, "manual"-manual, "precision"-precise-->
    </environmentalTemperatureMode>
  </ManualParam>
  <AutoParam><!--dependent, type="auto"-->
    <compensationValue><!--optional, ro, xs:float, compensation value, range: [-10.0 10.0], unit: celsius degree--></
compensationValue>
    <smartCorrection><!--optional, xs:float, smart calibration, range: [-10.0 10.0], unit: celsius degree--></
smartCorrection>
    <environmentalTemperature><!--optional, ro, xs:float, environment temperature--></environmentalTemperature>
    <temperatureCompensation><!--optional, xs:boolean, whether to enable body temperature compensation--></
temperatureCompensation>
    <environmentalTemperatureMode>
      <!--optional, xs:string, environment temperature mode: "auto"-automatic, "manual"-manual, "precision"-precise-->
    </environmentalTemperatureMode>
  </AutoParam>
</BodyTemperatureCompensation>
```

A.40 XML_BodyTemperatureCompensationCap

XML message about capability of body temperature compensation

```
<?xml version="1.0" encoding="utf-8"?>
<BodyTemperatureCompensationCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <type opt="manual, auto"><!--required, xs:string, compensation type: manual, automatic--></type>
  <ManualParam><!--dependent, type="manual"-->
    <compensationValue min="-10.0" max="10.0"><!--optional, xs:float, compensation value, range: [-10.0 10.0], unit:
celsius degree--></compensationValue>
    <smartCorrection min="" max=""><!--optional, xs:float, manual calibration, the range varies according to device
type--></smartCorrection>
    <environmentalTemperature min="" max="">
```

```
<!--dependent, xs:float, environment temperature, it is configurable when the value of type is "manual" and read
only when the value is of type is "auto"-->
</environmentalTemperature>
<temperatureCompensation opt="true,false"><!--optional, xs:boolean, whether to enable body temperature
compensation--></temperatureCompensation>
<environmentalTemperatureMode opt="auto>manual,precision">
  <!--optional,xs:string, environment temperature mode: "auto"-automatic, "manual"-manual, "precision"-precise-->
</environmentalTemperatureMode>
</ManualParam>
<AutoParam><!--dep type="auto"-->
  <compensationValue min="-10.0" max="10.0"><!--optional, ro, xs:float, compensation value, range: [-10.0 10.0],
unit: celsius degree"--></compensationValue>
  <smartCorrection min="-10.0" max="10.0"><!--optional, xs:float, smart calibration, range: [-10.0 10.0], unit: celsius
degree--></smartCorrection>
  <environmentalTemperature><!--optional, ro, xs:float, environment temperature--></environmentalTemperature>
  <temperatureCompensation opt="true,false"><!--optional, xs:boolean, whether to enable body temperature
compensation--></temperatureCompensation>
  <environmentalTemperatureMode opt="auto>manual,precision">
    <!--optional, xs:string, environment temperature mode: "auto"-automatic, "manual"-manual, "precision"-precise-->
  </environmentalTemperatureMode>
</AutoParam>
</BodyTemperatureCompensationCap>
```

A.41 XML_BurningPrevention

BurningPrevention message in XML format.

```
<BurningPrevention version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs:boolean "true,false"--></enabled>
  <mode><!--opt, xs:string, mode: manual, automatic--></mode>
  <closedDuration><!--opt, xs:integer, close duration, unit: second--></closedDuration>
  <shutterStatus><!--opt, xs:string, shutter status: closed, open--></shutterStatus>
  <protectionMode>
    <!--opt, xs:string, protection mode: lensMovement-lens movement, shutterClose-close shutter-->
  </protectionMode>
  <burningRecoveryEnabled>
    <!--opt, xs:boolean, enable burning recovery or not-->
  </burningRecoveryEnabled>
  <movementDuration>
    <!--dep, xs:integer, duration of lens movement for burning protection, unit: minute-->
  </movementDuration>
</BurningPrevention>
```

Remarks

- When **protectionMode** is "lensMovement", the node **movementDuration** is valid.
- When **protectionMode** is "shutterClose", the node **closedDuration**, **shutterStatus**, and **burningRecoveryEnabled** are valid.
- When **mode** is "automatic", the shutter is open, and cannot be configured.

A.42 XML_BurningPreventionCap

BurningPreventionCap message in XML format

```
<BurningPreventionCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled opt="true,false"><!--req, xs:boolean--></enabled>
  <mode opt="manual,automatic"><!--opt, xs:string, mode: manual, automatic--></mode>
  <closedDuration min="5" max="60" default="10">
    <!--opt, xs:integer, closed status duration, unit: second-->
  </closedDuration>
  <shutterStatus opt="closed,open"><!--opt, xs:string, shutter status: closed, open--></shutterStatus>
  <protectionMode opt="lensMovement,shutterClose">
    <!--opt, xs:string, protection mode: lensMovement-lens movement, shutterClose-close shutter-->
  </protectionMode>
  <burningRecoveryEnabled opt="true,false">
    <!--opt, xs:boolean, enable burning recovery or not-->
  </burningRecoveryEnabled>
  <movementDuration min="" max="" default="">
    <!--opt, xs:integer, duration of lens movement for burning protection, unit: minute-->
  </movementDuration>
</BurningPreventionCap>
```

Remarks

- When **protectionMode** is "lensMovement", the node **movementDuration** is valid.
- When **protectionMode** is "shutterClose", the node **closedDuration**, **shutterStatus**, and **burningRecoveryEnabled** are valid.
- When **mode** is "automatic", the shutter is open, and cannot be configured.

A.43 XML_CalibrationCoefficientDescription

CalibrationCoefficientDescription message in XML format

```
<CalibrationCoefficientDescription version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enviroTemperature><!--opt, xs:float, environment temperature, ranges from -273.3 to 1000.0--></enviroTemperature>
</CalibrationCoefficientDescription>
```

A.44 XML_CalibrationCoefficientResult

CalibrationCoefficientResult in XML format

```
<CalibrationCoefficientResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <calibrationCoefficient><!--dep, xs:float, calibration coefficient, ranges from 0 to 30, corrects to two decimal places--></calibrationCoefficient>
</CalibrationCoefficientResult>
```

A.45 XML_Cap_DPC

DPC capability message in XML format

```
<DPC version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mode opt="correction,moveup,movedown,moveleft,moveright,cancelCorrect,allCorrection,save, start, exit,
display">
    <!--req, xs:string, it is invalid when dpcMode values "auto"-->
  </mode>
  <dpcMode opt="manual,auto" def="auto">
    <!--opt, xs:string, defective pixel correction mode-->
  </dpcMode>
  <normalizedScreenSize><!--opt, ro-->
    <normalizedScreenWidth><!--req, ro,xs:integer--></normalizedScreenWidth>
    <normalizedScreenHeight><!--req, ro,xs:integer--></normalizedScreenHeight>
  </normalizedScreenSize>
  <Point><!--opt-->
    <positionX><!--req,xs:integer;coordinate--></positionX>
    <positionY><!--req,xs:integer;coordinate--></positionY>
  </Point>
</DPC>
```

A.46 XML_Cap_FaceThermometry

XML message about temperature screening configuration capability

```
<FaceThermometry version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <faceThermometryEnabled opt="true,false">
    <!--required, xs:boolean, whether enables temperature screening: true-yes, false-no-->
  </faceThermometryEnabled>
  <thermometrShowEnabled opt="true,false">
    <!--required, xs:boolean, whether enables temperature OSD: true-yes, false-no-->
  </thermometrShowEnabled>
  <alarmEnabled opt="true,false">
    <!--required, xs:boolean, whether enables alarm subscription: true-yes, false-no-->
  </alarmEnabled>
  <alarmIntervalTime min="0.5" max="600" def="1">
    <!--dependent,endent, xs:float, unit:s, alarm interval time, which is valid in non-card mode, the interval is between
0.5s and 60s, corrects to one decimal place. The default time interval is 1 second-->
  </alarmIntervalTime>
  <normalizedScreenSize>
    <!--required, read-only; it is the multiples of normalized coordinates returned by device-->
    <normalizedScreenWidth>
      <!--required, read-only, xs:integer, normalized screen width-->
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--required, read-only, xs:integer, normalized screen height-->
    </normalizedScreenHeight>
  </normalizedScreenSize>
```

```
<FaceThermometryRegionList size="1">
  <!--optional, temperature screening rule list-->
  <ThermometryRegion>
    <!--optional, temperature screening rule-->
    <id min="1" max="40">
      <!--required, xs:integer, rule ID-->
    </id>
    <name min="0" max="32">
      <!--optional, xs:string, rule name-->
    </name>
    <sensitivity min="1" max="5" def="3">
      <!--required, xs:integer, sensitivity, normalized value, ranges from 1 to 5, default value:3-->
    </sensitivity>
    <PupilParam>
      <!--required, pupil parameters-->
      <pupilDistance min="42" max="1000">
        <!--required, xs:integer, pupil distance, normalized value, range: [0,1000], valid range: [42,1000]-->
      </pupilDistance>
      <Region>
        <!--required, pupil region coordinates, normalized value, range: [0,1000],-->
        <RegionCoordinatesList size="">
          <!--required-->
          <RegionCoordinates>
            <!--optional, pupil region coordinates-->
            <positionX>
              <!--required, xs:integer, X-coordinate-->
            </positionX>
            <positionY>
              <!--required, xs:integer, Y-coordinate-->
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </Region>
    </PupilParam>
    <MaxPupilParam><!--optional, the maximum pupil distance-->
    <pupilDistance min="10" max="625">
      <!--optional, xs:integer, pupil distance range, normalized value, range: [0,1000], valid range: [10,625], default
value: 625-->
    </pupilDistance>
    <Region>
      <RegionCoordinatesList size=""><!--req-->
      <RegionCoordinates><!--list, pupil region coordinates-->
      <positionX><!--required, xs:integer, coordinate, X-coordinate--></positionX>
      <positionY><!--required, xs:integer, coordinate, Y-coordinate--></positionY>
    </RegionCoordinates>
    </RegionCoordinatesList>
  </Region>
</MaxPupilParam>
</PupilParam>
<targetSpeed min="1" max="5" def="3">
  <!--required, xs:integer, target generating speed, ranges from 1 to 5, and default value is 3-->
</targetSpeed>
<alarmTemperature min="0.0" max="60.0">
```

```
<!--optional, xs:float, alarm triggered temperature, unit:°C, ranges from -20.0 to 60.0, and corrects to one decimal place-->
</alarmTemperature>
<type opt="region">
  <!--required, xs:string, rule type-->
</type>
<Region>
  <!--required, rule region coordinates-->
  <RegionCoordinatesList size="">
    <!--dependent-->
    <RegionCoordinates>
      <!--opt-->
      <positionX>
        <!--required, xs:integer;coordinate-->
      </positionX>
      <positionY>
        <!--required, xs:integer;coordinate-->
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</Region>
<RegionBoundary>
  <RegionCoordinatesList size="">
    <!--dependent-->
    <RegionCoordinates>
      <!--opt-->
      <positionX>
        <!--required, xs:integer;coordinate-->
      </positionX>
      <positionY>
        <!--required, xs:integer;coordinate-->
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</RegionBoundary>
<alarmRule opt="highestGreater,highestLess">
  <!--optional, xs:string, alarm rule: highestGreater-Max. temperature higher than, highestLess-Max. temperature lower than-->
</alarmRule>
<alert min="-100.0" max="1000.0"><!--optional, xs:float, pre-alarm threshold, unit: Celsius, corrects to one decimal place-->
</alert>
</ThermometryRegion>
</FaceThermometryRegionList>
<isSupportFaceThermDetectionInfo>
  <!--optional, xs:boolean, whether supports getting temperature screening result, corresponds to URL of /ISAPI/Thermal/channels/<ID>/faceThermometry/regions/<ID>/detectionInfo-->
</isSupportFaceThermDetectionInfo>
<imageQuality opt="high,medium,low">
  <!--optional, xs:string-->
</imageQuality>
<mode opt="barrierPassing, targeting">
```

```
<!--optional, xs:string, temperature screening mode: "barrierPassing"-barrier passing (transmit the custom in
airport), "targeting"-targeting (detect the person with high temperature)-->
</mode>
<faceSnapUploadEnabled opt="true,false">
  <!--optional, xs:boolean, whether to enable uploading captured face picture: true-enable, false-disable-->
</faceSnapUploadEnabled>
<maxTemperatureCoordinatesEnabled>
  <!--optional, xs:boolean, whether to enable displaying the maximum temperature position: true-enable, false-
disable-->
</maxTemperatureCoordinatesEnabled>
<faceRectShowEnabled opt="true,false">
  <!--optional, xs:boolean, whether to enable displaying a frame on the target person: true-enable, false:disable-->
</faceRectShowEnabled>
<faceTemperatureShowEnabled opt="true,false">
  <!--optional, xs:boolean, whether to enable displaying face temperature: true-enable, false:disable-->
</faceTemperatureShowEnabled>
</FaceThermometry>
```

A.47 XML_Cap_FireDetection

FireDetection capability message in XML format

```
<FireDetection version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <enabled><!--req, xs:boolean--></enabled>
  <sensitivity min="1" max="100"><!--req, xs:integer, detection sensitivity, which ranges from 1 to 10--></sensitivity>
  <fireComfirmTime min="0" max="120"><!--opt, xs:integer, ranges from 0 to 120--></fireComfirmTime>
  <fireRegionOverlay><!--opt, xs:boolean, ro, display fire frame on stream--></fireRegionOverlay>
  <fireFrameDis><!--opt, xs:boolean, ro, display fire frame--></fireFrameDis>
  <fireMaxTemp><!--opt, xs:boolean, ro--></fireMaxTemp>
  <fireMaxTempPosition><!--opt, xs:boolean, ro--></fireMaxTempPosition>
  <fireDistance><!--opt, xs:boolean, ro--></fireDistance>
  <detectionMode opt="multipleFarme,singleFarme">
    <!--opt, xs:string, fire detection mode: multipleFarme-multiple frame, singleFarme-single frame-->
  </detectionMode>
  <fireFocusMode opt="auto,cruise"><!--opt, xs:string--></fireFocusMode>
  <FireZoom>
    <zoomMode opt="auto, manual"><!--req, xs:string--></zoomMode>
    <zoomLevel min="1" max="100"><!--dep, xs:integer--></zoomLevel>
  </FireZoom>
  <AlarmStrategy>
    <strategyType opt="any,cooperate,multisystem,appointFire,appointSmoke"><!--req, xs:string--></strategyType>
  </AlarmStrategy>
  <SmokeDetection>
    <enabled><!--req, xs:boolean--></enabled>
    <sensitivity min="1" max="100"><!--opt, xs:integer, ranges from 1 to 100--></sensitivity>
    <patrolSensitivity min="1" max="100" default="50"><!--opt, xs:integer, sensitivity of patrol detection, ranges from 1
to 100--> </patrolSensitivity>
    <doubleCheckSensitivity min="1" max="100" default="50"><!--opt, xs:integer, sensitivity of double filtering, ranges
from 1 to 100--> </doubleCheckSensitivity>
    <displaySmokeInfoOnStreamEnabled opt="true,false"><!--opt, xs:boolean, overlay smoke information on stream-->
```

```
</displaySmokeInfoOnStreamEnabled>
</SmokeDetection>
<smokeFireEnabled><!--opt, xs:boolean--></smokeFireEnabled>
<ApplicationScene>
  <mode opt="forest-Fire_Prevention,strawBurning,high-building,Indoor/Perimeter"><!--opt, xs:string--></mode>
  <InstallationHeight min="1" max="500"><!--xs:integer, it is valid only when mode is "strawBurning", "high-
building", or "Indoor/Perimeter", ranges from 1 to 500, unit: m--></InstallationHeight>
</ApplicationScene>
<cancelRepeatedAlarmEnabled opt="true, false">
  <!--opt, xs:boolean, cancel repeated alarm, it is valid only when detectionMode is "multipleFarme"-->
</cancelRepeatedAlarmEnabled>
<fireManualWaitEnabled opt="true,false"><!--opt,xs:boolean,--></fireManualWaitEnabled>
<isSupportFireScanStart opt="true,false"><!--opt,xs:boolean, whether the fire continue scan command is supported,
related URI : /ISAPI/Thermal/channels/<ID>/fireScanStart--></isSupportFireScanStart>
<isSupportFireScanState opt="true,false"><!--opt,xs:boolean, whether the fire scan status command is supported, ,
related URI : /ISAPI/Thermal/channels/<ID>/fireScanState--></isSupportFireScanState>
<displayFireInfoOnStreamEnabled opt="true,false"><!--opt, xs:boolean, overlay fire source information on stream--
></displayFireInfoOnStreamEnabled>
<fireSourceDetection opt="dynamicFire, smokingMode"><!--opt, xs:string, fire source detection mode: dynamic fire
source, smoking--></fireSourceDetection>
<smokeAuxiliaryDetectionEnabled opt="true,false">
  <!--dep, xs:boolean, enable fire and smoke detection or not, it is valid only when detectionMode is
"multipleFarme"-->
</smokeAuxiliaryDetectionEnabled>
<verificationSensitivity min="1" max="100" default="50">
  <!--opt, xs:integer, sensitivity of double verification, ranges from 1 to 100-->
</verificationSensitivity>
<fireAlgorithmModel opt="patternRecognition, machineLearning">
  <!--opt, xs:string, fire detection algorithm mode: "patternRecognition"-pattern recognition, "machineLearning"-
machine learning-->
</fireAlgorithmModel>
<agriculturalMachineryFilterEnabled>
  <!--opt, xs:boolean, enable agricultural machinery filter-->
</agriculturalMachineryFilterEnabled>
<waterReflectionEnabled><!--opt, xs:boolean, enable water reflection--></waterReflectionEnabled>
<patrolSensitivity min="1" max="100" default="50">
  <!--opt, xs:integer, patrol sensitivity, only valid for fire detection, ranges from 1 to 100-->
</patrolSensitivity>
</FireDetection>
```

A.48 XML_Cap_Power

Power capability message in XML format

```
<?xml version="1.0" encoding="utf-8"?>
<Power version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <powerSwitch>
    <!--req, xs:boolean, switch on/off: "true"-switch on, wake up device, "false"-switch off, device will be in sleep
mode-->
  </powerSwitch>
```

```
<batteryPower min="1" max="100">
  <!--opt, xs:integer, battery percentage-->
</batteryPower>
</Power>
```

A.49 XML_Cap_ThermalBlackBody

ThermalBlackBody capability message in XML format.

```
<ThermalBlackBody version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <emissivity min="0.01" max="1" def="0.97">
    <!--required, xs:float, black body emissivity is between 0.01 and 1, which corrects to two decimal places. The default
value is 0.97-->
  </emissivity>
  <distance min="0.0" max="10.0" def="2.0">
    <!--required, xs: float, the distance between lens and black body, ranges from 0.0 m to 10.0 m, which corrects to
one decimal place. The default value is 2.0 m, unit:m-->
  </distance>
  <temperature min="30.0" max="50.0" def="35.0">
    <!--required, xs:float, black body temperature is between 30.0 °C and 50.0 °C, which corrects to one decimal place.
The default value is 35.0 °C, unit: °C-->
  </temperature>
  <CentrePoint><!--required, center position of black body. After clicking this position, the normalized coordinate
information (between 0 and 1000) will be applied-->
    <CalibratingCoordinates><!--dep-->
      <positionX><!--required, xs:integer; coordinate--></positionX>
      <positionY><!--required, xs:integer; coordinate--></positionY>
    </CalibratingCoordinates>
  </CentrePoint>
  <normalizedScreenSize><!--required, ro, read-only, it is the multiples of normalized coordinates returned by device-->
    <normalizedScreenWidth><!--required, ro,xs:integer--></normalizedScreenWidth>
    <normalizedScreenHeight><!--required, ro,xs:integer--></normalizedScreenHeight>
  </normalizedScreenSize>
  <enabled opt="true,false"><!--optional, xs:boolean, whether to enable black body--></enabled>
  <BlackBodyReigon><!--optional, black body detection area-->
    <type opt="point,region"><!--required, xs:string, "point, region", area type: "point", "region"--></type>
    <Point><!--dependent, point coordinate, it is valid when value of type is "point"-->
      <CalibratingCoordinates><!--dependent, point coordinate-->
        <positionX><!--required, xs:integer; coordinate--></positionX>
        <positionY><!--required, xs:integer; coordinate--></positionY>
      </CalibratingCoordinates>
    </Point>
    <Region><!--dependent, frame coordinate, it is valid when value of type is "region"-->
      <RegionCoordinatesList size="4"><!--dependent, list of region coordinates-->
        <RegionCoordinates><!--list, optional-->
          <positionX><!--required, xs:integer; X-coordinate--></positionX>
          <positionY><!--required, xs:integer; Y-coordinate--></positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </Region>
  </BlackBodyReigon>
</ThermalBlackBody>
```

```
</BlackBodyReigon>
</ThermalBlackBody>
```

A.50 XML_Cap_ThermIntell

ThermIntell capability message in XML format.

```
<ThermIntell version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:integer--></id>
  <intellType
opt="thermometryAndSmart,shipsDetection,fireDetection,pip,faceThermometry,thermometryAndSmokeFireDetection
,thermometryAndFireDetection,basicBehavior,thermometry" def="thermometryAndSmart">
  <!--req, xs:string,smart function resource configuration type: "thermometryAndSmart"-temperature measurement
+behavior analysis (default), "shipsDetection"-ship detection, "fireDetection"-fire source detection, "pip"-picture in
picture function, "faceThermometry"-temperature screening, "thermometryAndSmokeFireDetection"-temperature
measurement+smoke and fire source detection, "thermometryAndFireDetection"-temperature measurement+fire
source detection, "basicBehavior"-behavior analysis, "thermometry"-temperature measurement-->
  </intellType>
</ThermIntell>
```

A.51 XML_Cap_ThermalPip

ThermalPip capability message in XML format

```
<ThermalPip version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <!--req, capability of picture-in-picture configuration-->
  <enabled><!--req, xs:boolean, enable or not--></enabled>
  <pipMode opt="overlap,fusion, normal" def="normal">
    <!--req, xs:string, picture-in-picture mode: overlap-overlay mode, fusion-integration mode, normal-normal mode-->
  </pipMode>
  <overlapType opt="visibleOverlapThermal,thermalOverlapVisible">
    <!--req, xs:string, overlay type: visibleOverlapThermal-overlay visible light on thermal imaging picture,
thermalOverlapVisible-overlay thermal imaging on visible light-->
  </overlapType>
  <transparency min="0" max="100">
    <!--opt, xs: integer, transparency-->
  </transparency>
  <normalizedScreenSize><!--req, ro, screen with normalized coordinates-->
    <normalizedScreenWidth><!--req, ro,xs:integer--></normalizedScreenWidth>
    <normalizedScreenHeight><!--req, ro,xs:integer--></normalizedScreenHeight>
  </normalizedScreenSize>
  <PipRegion> <!--dep, picture-in-picture area-->
    <RegionCoordinatesList size ="4"><!--dep, coordinates list of picture-in-picture area-->
      <RegionCoordinates><!--opt, picture-in-picture area coordinate-->
        <positionX><!--req, xs:integer;x-coordinate--> </positionX>
        <positionY><!--req, xs:integer;y-coordinate--> </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </PipRegion>
```

```
<imageFusionRatio min="0" max="100" default="75">
  <!--dep, xs: integer, image fusion ratio, it is valid when pipMode is "fusion"-->
</imageFusionRatio>
<borderFusionRatio min="0" max="100" default="100">
  <!--dep, xs: integer, border fusion ratio, it is valid when pipMode is "fusion"-->
</borderFusionRatio>
<distance><!--dep, xs:float, fusion distance, it is valid when pipMode is "fusion", ranges from 0.1 m to 4.0 m--></
distance>
</ThermalPip>
```

A.52 XML_Cap_ThermometryAlarmRule

XML message about alarm rules parameters of temperature measurement preset

```
<ThermometryAlarmRule version="2.0">
  <ThermometryAlarmModeList size="">
    <!--req, alarm rules parameters of temperature measurement preset -->
    <ThermometryAlarmMode>
      <id min="" max="">
        <!--req, xs:inter, rule ID -->
      </id>
      <enabled opt="true,false">
        <!--req, xs:boolean, enable: false-no, true-yes-->
      </enabled>
      <name min="0" max="32">
        <!--req, xs:string, ro, rule name-->
      </name>
      <pointRule opt="averageGreater,averageLess" def="averageGreater">
        <!--req, xs:string, alarm temperature comparison mode of point thermography: averageGreater-average
temperature higher than (default), averageLess-average temperature lower than-->
      </pointRule>
      <lineRule opt="highestGreater,highestLess,lowestGreater,lowestLess, averageGreater,averageLess"
def="averageGreater">
        <!--req, xs:string, alarm temperature comparison mode of line thermography: highestGreater-Max. temperature
higher than, highestLess-Max. temperature lower than, lowestGreater-Min. temperature lower than, lowestLess-Min.
temperature lower than, averageGreater-average temperature higher than (default), averageLess-average
temperature lower than-->
      </lineRule>
      <regionRule
opt="highestGreater,highestLess,lowestGreater,lowestLess,averageGreater,averageLess,diffTempGreater,diffTempLess"
def="averageGreater">
        <!--req, xs:string, alarm temperature comparison of frame thermography: highestGreater-Max. temperature
higher than, highestLess-Max. temperature lower than, lowestGreater-Min. temperature lower than, lowestLess-Min.
temperature lower than, averageGreater-average temperature higher than (default), averageLess-average
temperature lower than, diffTempGreater-temperature difference higher than, diffTempLess-temperature difference
lower than-->
      </regionRule>
      <alert min="0" max="32">
        <!--req, xs: float, pre-alarm temperature-->
      </alert>
```

```
<alarm min="0" max="32">
  <!--req, xs: float, alarm temperature-->
</alarm>
<threshold min="0" max="32">
  <!--req, xs: float, threshold temperature-->
</threshold>
<isSupportAlertOutputIOPortList>
  <!--opt, xs:boolean, "true", if supports, this node will be returned and the value is "true", if not support, it will not
be returned-->
</isSupportAlertOutputIOPortList>
<isSupportAlarmOutputIOPortList>
  <!--opt, xs:boolean, "true", if supports, this node will be returned and the vlue is "true", if not support, it will not
be returned-->
</isSupportAlarmOutputIOPortList>
<alertFilteringTime min="0" max="200" default="0">
  <!--opt, xs:integer, temperature pre-alarm dwell time, unit: s-->
</alertFilteringTime>
<alarmFilteringTime min="0" max="200" default="0">
  <!--opt, xs:integer, temperature pre-alarm dwell time, unit:s-->
</alarmFilteringTime>
<visibleLightLinkageEnabled opt="true, false">
  <!--opt, xs:boolean, visible light linkage-->
</visibleLightLinkageEnabled>
<TemperatureSuddenChange>
  <mode opt="close, temperatureSuddenIncrease, temperatureSuddenDecrease">
    <!--req, xs:string, temperature sudden change mode-->
  </mode>
  <cycle opt="1, 5, 10, 30, 60"><!--opt, xs:integer, cycle period, unit: second--></cycle>
  <alert min="" max=""><!--opt, xs: float, pre-alarm threshold--></alert>
  <alarm min="" max=""><!--opt, xs: float, alarm threshold--></alarm>
</TemperatureSuddenChange>
</ThermometryAlarmMode>
</ThermometryAlarmModeList>
<TemperatureDifferenceComparisonList size="">
  <!--req, temperature difference configuration parameters of temperature measurement preset-->
<TemperatureDifferenceComparison>
  <id min="" max="">
    <!--req, xs:inter, rule ID-->
  </id>
  <enabled opt="true,false">
    <!--req, xs:boolean, enable: false-no, true-yes-->
  </enabled>
  <ruleID1 min="" max="">
    <!--req, xs:string, alarm point 1-->
  </ruleID1>
  <ruleID2 min="" max="">
    <!--req, xs:string, alarm point 2-->
  </ruleID2>
  <pointRule opt="averageGreater,averageLess" def="averageGreater">
    <!--req, xs:string, alarm temperature comparison modes of point thermography: averageGreater-average
temperature higher than (default), averageLess-average temperature lower than-->
  </pointRule>
```

```
<lineRule opt="highestGreater,highestLess,lowestGreater,lowestLess, averageGreater,averageLess"
def="averageGreater">
  <!--req, xs:string, alarm temperature comparison modes of line thermography: highestGreater-Max. temperature
higher than, highestLess-Max. temperature lower than, lowestGreater-Min. temperature higher than, lowestLess-Min.
temperature lower than, averageGreater-average temperature higher than (default), averageLess-average
temperature lower than -->
  </lineRule>
  <regionRule
opt="highestGreater,highestLess,lowestGreater,lowestLess,averageGreater,averageLess,diffTempGreater,diffTempLess"
def="averageGreater">
  <!--req, xs:string, alarm temperature comparison modes of frame thermography: highestGreater-Max.
temperature higher than, highestLess-Max. temperature lower than, lowestGreater-Min. temperature lower than,
lowestLess-Min. temperature lower than, averageGreater-average temperature higher than (Default), averageLess-
average temperature lower than, diffTempGreater-temperature difference higher than, diffTempLess-temperature
difference lower than-->
  </regionRule>
  <temperatureDifference min="0" max="32">
    <!--req, xs: float, temperature difference-->
  </temperatureDifference>
</TemperatureDifferenceComparison>
</TemperatureDifferenceComparisonList>
</ThermometryAlarmRule>
```

A.53 XML_Cap_ThermometryBasicParam

XML message about capability of temperature measurement basic parameters

```
<ThermometryBasicParam version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:integer, channel number--></id>
  <enabled opt="true,false"><!--req, xs:boolean, enable or not: false-no, true-yes--></enabled>
  <streamOverlay opt="true,false"><!--req, xs:boolean, whether displays temperature information on the stream: false-
no, true-yes--></streamOverlay>
  <pictureOverlay opt="true,false"><!--req, xs:boolean, whether displays temperature information on the captured
picture: false-no, true-yes--></pictureOverlay>
  <temperatureRange
opt="-20-150,0-550,0-650,-4-302,32-1022,32-1200,-20-650,-20-1500,automatic,-20-120,20-350,20-45">
    <!--req, xs:string, temperature range-->
  </temperatureRange>
  <temperatureUnit opt="degreeCentigrade,degreeFahrenheit,degreeKelvin">
    <!--req, xs:string, temperature unit: degreeCentigrade-Celsius(°C), degreeFahrenheit-Fahrenheit(°F), degreeKelvin-
Kelvin(K)-->
  </temperatureUnit>
  <temperatureCurve opt="close,transverseTemperatureTrend,longitudinalTemperatureTrend">
    <!--opt, xs:string, temperature curve modes: close-closed, transverseTemperatureTrend-transverse temperature
mode, longitudinalTemperatureTrend-longitudinal temperature mode-->
  </temperatureCurve>
  <fireImageMode opt="blackWhite,thermalProbe,fireGround">
    <!--opt, xs:string, fire detection modes: blackWhite-black and white mode, thermalProbe-thermal detection mode,
fireGroud-fire scene mode-->
  </fireImageMode>
```

```

<emissivity min="0.01" max="1.00" default="0.96">
  <!--opt, xs:float, the emissivity is between 0.01 and 1.00, which corrects to two decimal places. This parameter is
  used by the device (i.e., DS-2TF03-260V/GLT, DS-2TF03-167V/GLT) that does not support regular frame and preset-->
</emissivity>
<distanceUnit opt="meter,feet,centimeter">
  <!--req, xs:string-->
</distanceUnit>
<TemperatureColor>
  <!--opt, set the alarm information color for the temperature measurement-->
  <type opt="highTemperature,lowTemperature,rangeTemperature,heatPreservation">
    <!--req, xs:string, set temperature alarm types: highTemperature field-high temperature alarm, when the
    measured temperature is higher than the configured value of highTemperature field, the measured temperature will
    be marked by color; lowTemperature-low temperature alarm, when the measured temperature is lower than the
    configured value of lowTemperature field, the measured temperature will be marked by color; rangeTemperature-
    range temperature alarm, when the measured temperature is between the configured values of highTemperature
    field and lowTemperature field, the measured temperature will be marked by color; heatPreservation-insulation
    alarm, when the measured temperature is not between the configured values of highTemperature field and
    lowTemperature field, the measured temperature will be marked by color-->
  </type>
  <highTemperature min="-273" max="10000"><!--dep, xs:integer--></highTemperature>
  <lowTemperature min="-273" max="10000"><!--dep, xs:integer--></lowTemperature>
</TemperatureColor>
<enviroTemperature min="-273" max="10000"><!--opt, xs:integer, environment temperature, unit: °C--></
enviroTemperature>
<enviroHumidity min="0" max="100"><!--opt, xs:integer, environment humidity, unit:%--></enviroHumidity>
<correctionVolume min="-100" max="100" def="0"><!--opt, xs:integer, temperature correction--></
correctionVolume>
<specialPointThermType opt="centerPoint,highestPoint,lowestPoint">
  <!--req, xs:string, display the special point temperature, central point temperature, highest temperature, lowest
  temperature. Supports multiple selections-->
</specialPointThermType>
<distance min="0" max="50"><!--req, xs:integer; unit:m, the distance range is [0, 10000]--></distance>
<reflectiveEnable><!--req, xs:boolean, whether enables temperature reflection--></reflectiveEnable>
<reflectiveTemperature min="" max=""><!--opt, xs:float, reflective temperature, which corrects to one decimal place--
></reflectiveTemperature>
<alert min="-73.3" max="1000.0"><!--opt, xs: float, pre-alarm threshold--></alert>
<alarm min="-73.3" max="1000.0"><!--opt, xs: float, alarm threshold--></alarm>
<showTempStripEnable><!--opt, xs:boolean, whether enables displaying temperature bar--></showTempStripEnable>
<thermalOpticalTransmittance min="0.001" max="1.000" def="1.000">
  <!--opt, xs: float, optical transmissivity is between 0.001 and 1.000, which corrects to three decimal places. The
  default value is 1.000-->
</thermalOpticalTransmittance>
<externalOpticsWindowCorrection min="-40.0" max="80.0" def="20.0">
  <!--opt, xs: float, external optical temperature is between -40.0 °C and 80.0 °C. The default value is 20 °C-->
</externalOpticsWindowCorrection>
<isSupportAlertOutputIOPortList><!--opt, xs:boolean, "true", true indicates support, no return indicates not support--
></isSupportAlertOutputIOPortList>
<isSupportAlarmOutputIOPortList><!--opt, xs:boolean, "true", true indicates support, no return indicates not
support--></isSupportAlarmOutputIOPortList>
<alertFilteringTime min="0" max="200" default="0"><!--opt, xs:integer, temperature pre-alarm dwell time, unit:s--></
alertFilteringTime>
<alarmFilteringTime min="0" max="200" default="0"><!--opt, xs:integer, temperature alarm dwell time, unit:s--></

```

```
alarmFilteringTime>
  <displayMaxTemperatureEnabled opt="true,false"><!--opt, xs:boolean, whether displays the maximum temperature-->
</displayMaxTemperatureEnabled>
  <displayMinTemperatureEnabled opt="true,false"><!--opt, xs:boolean, whether displays the minimum temperature-->
</displayMinTemperatureEnabled>
  <displayAverageTemperatureEnabled opt="true,false"><!--opt, xs:boolean, whether displays the average
temperature--></displayAverageTemperatureEnabled>
  <thermometryInfoDisplayposition opt="rules_around,top_left_of_screen">
    <!--opt, xs:string, position of temperature measurement information overlay-->
  </thermometryInfoDisplayposition>
  <calibrationCoefficientEnabled opt="true,false"><!--opt, xs:boolean, whether enables calibration coefficient--></
calibrationCoefficientEnabled>
  <calibrationCoefficient min="0.00" max="30.00"><!--dep, xs:float, calibration coefficient, ranges from 0 to 30,
corrects to two decimal places--></calibrationCoefficient>
  <emissivityMode opt="rougher,rough,smooth,smoother,customsettings">
    <!--opt,xs:string, emissivity type: "rougher"-rougher 0.95, "rough"-rough 0.80, "smooth"-smooth 0.60, "smoother"-
smoother 0.30, "customsettings"-customized value, ranges from 0.01 to 1.00, the larger the value, the higher the
roughness-->
  </emissivityMode>
  <displayTemperatureInOpticalChannelEnabled opt="true,false">
    <!--opt,xs:boolean, display the temperature information of optical channel-->
  </displayTemperatureInOpticalChannelEnabled>
  <distanceMode opt="selfAdapt,fixed"><!--optional, xs:string, distance mode: "selfAdapt"-self-adaption, "fixed"-fixed
distance--></distanceMode>
  <faceTemperatureInfoUploadEnabled opt="true,false"><!--optional, xs:boolean, whether to enable uploading face
temperature information: true-enable, false-disable--></faceTemperatureInfoUploadEnabled>
</ThermometryBasicParam>
```

Remarks

The following nodes are not supported by the thermographic automation thermal camera (DS-2TA03-15SVI, DS2TA06-25SVI): **<TemperatureColor>**, **<specialPointThermType>**, and **<reflectiveEnable>**.

A.54 XML_Cap_ThermometryMode

XML message about temperature measurement mode capability

```
<ThermometryMode version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <!--req, capability sets of temperature measurement mode-->
  <mode opt="normal,expert" def="normal">
    <!--req, xs:string, temperature measurement: normal-normal mode, expert-expert mode-->
    <!--Normal mode: configure overall temperature measurement without distinguishing rule and preset. In normal
mode, the related configuration in the basic parameter configuration structure is valid. Expert mode: configure
temperature measurement configuration by rule and preset. In expert mode, the related configuration in the preset
configuration structure is valid-->
  </mode>
  <thermometryROIEnabled opt="true,false"><!--opt,xs:boolean, whether enables ROI temperature measurement--></
thermometryROIEnabled>
</ThermometryMode>
```

Remarks

The ROI temperature measurement and mode configuration is mutually exclusive, ROI temperature measurement is mainly applied to temperature measurement.

A.55 XML_Cap_ThermometryScene

ThermometryScene capability message in XML format.

```
<ThermometryScene version="2.0">
  <normalizedScreenSize>
    <!--req, ro, coordinate normalization-->
    <normalizedScreenWidth>
      <!--req, ro, xs:integer, normalized width-->
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, ro, xs:integer, normalized height-->
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <ThermometryRegionList size="">
    <ThermometryRegion>
      <id min="1" max="21">
        <!--req, xs:integer, channel number-->
      </id>
      <enabled opt="true,false">
        <!--req, xs:boolean, enable or not: false-No, true-Yes-->
      </enabled>
      <name min="0" max="32">
        <!--req, xs:string, rule name-->
      </name>
      <emissivity min="0.01" max="1.00">
        <!--req, xs:float, emissivity-->
      </emissivity>
      <distance min="0" max="50">
        <!--req, xs:integer, unit:m, distance-->
      </distance>
      <reflectiveEnable>
        <!--req, xs:boolean, enable reflective temperature? false-No, true-Yes-->
      </reflectiveEnable>
      <reflectiveTemperature min="" max="">
        <!--opt, xs:float, reflective temperature-->
      </reflectiveTemperature>
      <type opt="point,region,line">
        <!--req, xs:string, rule calibration type: point-point, region-frame, line-line-->
      </type>
      <Point>
        <!--dep, temperature measurement by point, display the average temperature-->
        <TempValue min="-273.0" max="10000.0">
          <!--dep, xs:float, ro-->
        </TempValue>
      </Point>
    </ThermometryRegion>
  </ThermometryRegionList>
</ThermometryScene>
```

```
<CalibratingCoordinates>
  <!--dep, point coordinates-->
  <positionX>
    <!--req, xs:integer;coordinate -->
  </positionX>
  <positionY>
    <!--req, xs:integer;coordinate -->
  </positionY>
</CalibratingCoordinates>
</Point>
<Region>
  <!--dep, temperature measurement by frame, display the maximum, minimum, average temperature and the
temperate difference-->
  <highestTempValue min="-273.0" max="10000.0">
    <!--dep, xs: float, ro, maximum temperature-->
  </highestTempValue>
  <lowestTempValue min="-273.0" max="10000.0">
    <!--dep, xs: float, ro, minimum temperature-->
  </lowestTempValue>
  <averageTempValue min="-273.0" max="10000.0">
    <!--dep, xs: float, ro, average temperature-->
  </averageTempValue>
  <diffTempValue min="-273.0" max="10000.0">
    <!--dep, xs: float, ro, temperature difference-->
  </diffTempValue>
  <RegionCoordinatesList size="">
    <!--dep, area coordinates list of the frame-->
    <RegionCoordinates>
      <!--opt-->
      <positionX>
        <!--req, xs:integer;coordinate -->
      </positionX>
      <positionY>
        <!--req, xs:integer;coordinate -->
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</Region>
<Line>
  <!--dep, temperature measurement by line, display the maximum and minimum temperature-->
  <highestTempValue min="-273.0" max="10000.0">
    <!--dep, xs: float, ro, maximum temperature-->
  </highestTempValue>
  <lowestTempValue min="-273.0" max="10000.0">
    <!--dep, xs: float, ro, minimum temperature-->
  </lowestTempValue>
  <RegionCoordinatesList size="">
    <!--dep, endpoint coordinates list of the line-->
    <RegionCoordinates><!--opt-->
      <positionX><!--req, xs:integer;coordinate--></positionX>
      <positionY><!--req, xs:integer;coordinate--></positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</Line>
```

```
</RegionCoordinatesList>
</Line>
<distanceUnit opt="meter,feet,centimeter"><!--opt, xs:string--></distanceUnit>
<emissivityMode opt="rougher,rough,smooth,smoother,customsettings">
  <!--opt,xs:string, emissivity type: "rougher"-rougher 0.95, "rough"-rough 0.80, "smooth"-smooth 0.60,
"smoother"-smoother 0.30, "customsettings"-customized value, ranges from 0.01 to 1.00, the larger the value, the
higher the roughness-->
</emissivityMode>
<RegionBoundary><!--opt, rule region boundary-->
  <RegionCoordinatesList size=""><!--dep-->
    <RegionCoordinates><!--opt-->
      <positionX><!--req, xs:integer; x-coordinate--></positionX>
      <positionY><!--req, xs:integer; y-coordinate--></positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</RegionBoundary>
</ThermometryRegion>
<maxPointNum><!--req, xs:integer, maximum number of point rules that can be configured--></maxPointNum>
<maxLineNum><!--req, xs:integer, maximum number of line rules that can be configured--></maxLineNum>
<maxRegionNum><!--req, xs:integer, maximum number of frame rules that can be configured--></maxRegionNum>
</ThermometryRegionList>
</ThermometryScene>
```

A.56 XML_CorrectionParam

CorrectionParam in XML format.

```
<xml version="1.0" encoding="utf-8">
< CorrectionParam version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <distance><!--req, distance, float, unit:m--></distance>
  <enviroTemperature><!--req, environment temperature, corrects to one decimal place, unit: °C--></
enviroTemperature>
  <emissivity><!--req, emissivity, float, corrects to two decimal places--></emissivity>
  <TemperaturePointList><!--req, temperature point information list-->
    <TemperaturePoint><!--req, temperature point information-->
      <id><!--req, temperature point number, integer--></id>
      <presetTemperature><!--req, predefined temperature, corrects to one decimal place, float, unit: °C--></
presetTemperature>
      <Coordinates> <!--req, temperature point coordinates-->
        <x><!--req, temperature point X-coordinate, float--></x>
        <y><!--req, temperature point Y-coordinate, float--></y>
      </Coordinates>
    </TemperaturePoint>
  </TemperaturePointList>
</CorrectionParam>
```

A.57 XML_CorrectionParamCap

CorrectionParamCap message in XML format.

```
<xml version="1.0" encoding="utf-8">
<CorrectionParamCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <distance min="0" max="10.0"><!--req, distance, float, unit:m--></distance>
  <enviroTemperature min="-273.0" max="100">
    <!--req, environment temperature, corrects to one decimal place, float, unit: °C-->
  </enviroTemperature>
  <emissivity min="0.01" max="1.00">
    <!--req, emissivity, float, corrects to two decimal places-->
  </emissivity>
  <TemperaturePointList size="4"><!--opt-->
    <TemperaturePoint>
      <!--opt, temperature point information-->
      <id min="0" max="4"><!--req, temperature point number, integer--></id>
      <presetTemperature min="-40.0" max="650.0">
        <!--req, predefined temperature, corrects to one decimal place, float, unit: °C-->
      </presetTemperature>
      <Coordinates>
        <!--opt, temperature point coordinates-->
        <x min="0" max="1"><!--req, temperature point X-coordinates, float--></x>
        <y min="0" max="1"><!--req, temperature point Y-coordinates, float--></y>
      </Coordinates>
    </TemperaturePoint>
  </TemperaturePointList>
</CorrectionParamCap>
```

A.58 XML_DeviceCap

XML message about device capability

```
<DeviceCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <SysCap><!--optional-->
    <isSupportDst><!--optional, xs: boolean, whether it supports daylight saving time--></isSupportDst>
    <NetworkCap><!--optional, xs: boolean, network capability-->
    <IOCap><!--optional, IO capability-->
    <SerialCap><!--optional, serial port capability-->
    <VideoCap><!--optional, video capability, see details in the message of XML_VideoCap-->
    <AudioCap><!--optional, audio capability-->
    <isSupportHoliday><!--optional, xs:boolean--></isSupportHoliday>
    <RebootConfigurationCap>
      <Genetec><!--optional, xs:boolean--></Genetec>
      <ONVIF><!--optional, xs:boolean--></ONVIF>
      <RTSP><!--optional, xs:boolean--></RTSP>
      <HTTP><!--optional, xs:boolean--></HTTP>
      <SADP>
        <ISDiscoveryMode><!--optional, xs:boolean--></ISDiscoveryMode>
```

```
<PcapMode><!--optional, xs:boolean--></PcapMode>
</SADP>
<IPCAddStatus><!--optional, xs:boolean--></IPCAddStatus>
</RebootConfigurationCap>
<isSupportExternalDevice><!--optional, xs:boolean--></isSupportExternalDevice>
<isSupportChangedUpload>
  <!--optional, xs: boolean, whether it supports uploading status changes-->
</isSupportChangedUpload>
<isSupportGettingWorkingStatus>
  <!--optional, xs:boolean, whether it supports getting device status-->
</isSupportGettingWorkingStatus>
<isSupportGettingChannelInfoByCondition>
  <!--optional, xs:boolean-->
</isSupportGettingChannelInfoByCondition>
<isSupportDiagnosedDataParameter>
  <!--optional, xs:boolean-->
</isSupportDiagnosedDataParameter>
<isSupportSimpleDevStatus>
  <!--optional, xs: boolean, whether it supports getting device working status-->
</isSupportSimpleDevStatus>
<isSupportFlexible>
  <!--optional, xs: boolean, whether it supports getting channel status by condition-->
</isSupportFlexible>
<isSupportPTZChannels>
  <!--optional, xs:boolean, whether it supports returning PTZ channel (which is different from the video channel)-->
</isSupportPTZChannels>
<isSupportSubscribeEvent>
  <!--optional, xs:boolean, whether it supports alarm or event subscription: "true,false"-->
</isSupportSubscribeEvent>
<isSupportDiagnosedData>
  <!--optional, xs:boolean, "true,false", whether it supports diagnosis data-->
</isSupportDiagnosedData>
<isSupportTimeCap>
  <!--optional, xs:boolean, whether it supports time capability-->
</isSupportTimeCap>
<isSupportThermalStreamData>
  <!--optional, xs:boolean, whether it supports uploading thermal stream data in real-time. If it is supported, the
returned value is "true"; otherwise, this node will not be returned-->
</isSupportThermalStreamData>
<isSupportPostUpdateFirmware>
  <!--optional,xs:boolean,"true,false", whether it supports upgrading the firmware-->
</isSupportPostUpdateFirmware>
<isSupportPostConfigData>
  <!--optional, xs:boolean,"true,false", whether it supports importing or exporting the configuration file-->
</isSupportPostConfigData>
<isSupportUserLock>
  <!--optional, xs:boolean,"true,false", whether it supports locking user-->
</isSupportUserLock>
<isSupportModuleLock><!--optional, xs:boolean, whether it supports locking the module: "true,false"--></
isSupportModuleLock>
<isSupportSoundCfg><!--optional, xs:boolean--></isSupportSoundCfg>
<isSupportMetadata>
```

```
<!--optional, xs:boolean, if it is supported, return "true", otherwise, this node will not be returned-->
</isSupportMetadata>
<isSupportShutdown><!--optional, xs:boolean, whether it supports shutdown configuration--></
isSupportShutdown>
<supportSmartOverlapChannles opt="1"/><!--optional, xs:boolean, whether it supports stream configuration of
smart events. If this function is supported, this node and the corresponding channel ID will be returned; otherwise,
this node will not be returned-->
</SysCap>
<voicetalkNums><!--optional, xs:integer, the number of two-way audio channels--></voicetalkNums>
<isSupportSnapshot><!--optional, xs:boolean, whether it supports capture: "true, false"--></isSupportSnapshot>
<SecurityCap/><!--optional, security capability-->
<EventCap/><!--optional, event capability-->
<ITCCap><!--opt--></ITCCap>
<ImageCap/><!--optional, image capability-->
<RacmCap/><!--optional, storage capability-->
<PTZCtrlCap>
<isSupportPatrols><!--optional, xs:boolean--></isSupportPatrols>
</PTZCtrlCap>
<SmartCap/><!--optional, intelligent capability-->
<isSupportEhome><!--optional, xs:boolean--></isSupportEhome>
<isSupportStreamingEncrypt><!--optional, xs:boolean--></isSupportStreamingEncrypt>
<TestCap>
<isSupportEmailTest><!--optional, xs:boolean--></isSupportEmailTest>
</TestCap>
<ThermalCap/><!--optional, temperature measurement capability-->
<WLAAlarmCap/><!--optional, wireless alarm capability-->
<SecurityCPCapabilities/><!--optional, security control panel capability-->
<isSupportGIS>
<!--optional, xs:boolean, whether it supports GIS capability-->
</isSupportGIS>
<isSupportCompass>
<!--optional, xs:boolean-->
</isSupportCompass>
<isSupportRoadInfoOverlays>
<!--optional, xs:boolean-->
</isSupportRoadInfoOverlays>
<isSupportFaceCaptureStatistics>
<!--optional, xs:boolean-->
</isSupportFaceCaptureStatistics>
<isSupportExternalDevice>
<!--optional, xs:boolean-->
</isSupportExternalDevice>
<isSupportElectronicsEnlarge>
<!--optional, xs:boolean, whether it supports digital zoom-->
</isSupportElectronicsEnlarge>
<isSupportRemoveStorage>
<!--optional, xs:boolean-->
</isSupportRemoveStorage>
<isSupportCloud>
<!--optional, xs:boolean-->
</isSupportCloud>
<isSupportRecordHost>
```

```
<!--optional, xs:boolean-->
</isSupportRecordHost>
<isSupportEagleEye>
  <!--optional, xs:boolean, whether it supports PanoVu series camera-->
</isSupportEagleEye>
<isSupportPanorama>
  <!--optional, xs:boolean, whether it supports panorama-->
</isSupportPanorama>
<isSupportFirmwareVersionInfo>
  <!--optional, xs:boolean, whether it supports displaying firmware version information-->
</isSupportFirmwareVersionInfo>
<isSupportExternalWirelessServer>
  <!--optional, xs: boolean-->
</isSupportExternalWirelessServer>
<isSupportSetupCalibration>
  <!--optional, xs:boolean, whether it supports setting calibration-->
</isSupportSetupCalibration>
<isSupportGetmutexFuncErrMsg>
  <!--optional, xs:boolean, whether it supports getting mutex information-->
</isSupportGetmutexFuncErrMsg>
<isSupportTokenAuthenticate><!--optional, xs:boolean--></isSupportTokenAuthenticate>
<isSupportStreamDualVCA><!--optional, xs:boolean--></isSupportStreamDualVCA>
<isSupportlaserSpotManual>
  <!--optional, boolean, whether it supports laser spot configuration-->
</isSupportlaserSpotManual>
<isSupportRTMP><!--optional, xs:boolean--></isSupportRTMP>
<isSupportTraffic><!--optional, xs:boolean--></isSupportTraffic>
<isSupportLaserSpotAdjustment>
  <!--optional, boolean, whether it supports adjusting laser spot size-->
</isSupportLaserSpotAdjustment>
<VideoIntercomCap/><!--optional, video intercom capability-->
<isSupportSafetyCabin>
  <!--optional, xs:boolean-->
</isSupportSafetyCabin>
<isSupportPEA>
  <!--optional, xs:boolean, whether it supports one-touch security control panel capability-->
</isSupportPEA>
<isSupportCurrentLock>
  <!--optional, xs:boolean, whether it supports locking current configuration-->
</isSupportCurrentLock>
<isSupportGuardAgainstTheft>
  <!--optional, xs:boolean, whether it supports device anti-theft configuration-->
</isSupportGuardAgainstTheft>
<isSupportPicInfoOverlap>
  <!--optional, xs:boolean, whether it supports picture information overlay-->
</isSupportPicInfoOverlap>
<isSupportPlay>
  <!--optional, xs: boolean, whether it supports live view: "true,false"-->
</isSupportPlay>
<isSupportPlayback>
  <!--optional, xs: boolean, whether it supports playback: "true,false"-->
</isSupportPlayback>
```

```
<UHFRFIDReader>
  <!--optional, supported capability of UHF RFID card reader-->
  <isSupportBasicInformation>
    <!--optional, xs:boolean, whether it supports basic parameters of UHF RFID card reader-->
  </isSupportBasicInformation>
  <isSupportHardDiskStorageTest>
    <!--optional, xs:boolean, whether it supports hard disk storage test of UHF RFID card reader-->
  </isSupportHardDiskStorageTest>
</UHFRFIDReader>
<isSupportIntelligentStructureAnalysis>
  <!--optional, xs:boolean, whether it supports structured VCA-->
</isSupportIntelligentStructureAnalysis>
<isSupportIntelligentAnalysisEngines>
  <!--optional, xs:boolean, whether it supports VCA engine configuration-->
</isSupportIntelligentAnalysisEngines>
<PreviewDisplayNum>
  <!--optional, xs:integer, the number of live view windows, which is the number of simultaneous live view windows
controlled by the device. Limited by the performance of DeepinMind series network video recorder, currently only live
view of a network camera is supported, and playback is not supported-->
</PreviewDisplayNum>
<isSupportBoard opt="true,false">
  <!--optional, xs:boolean, whether it supports protocol related to sub-board-->
</isSupportBoard>
<ResourceSwitch>
  <workMode opt="4KPreview,educationRecord">
    <!--req, xs:string, device working mode : "4KPreview"-4K live view mode, "educationRecord"-education recording
mode-->
  </workMode>
</ResourceSwitch>
<isSupportCustomStream><!--optional, xs:boolean--></isSupportCustomStream>
<isSupportTriggerCapCheck>
  <!--optional, xs:boolean, whether it supports verifying capability of alarm linkage actions-->
</isSupportTriggerCapCheck>
<isSupportActiveMulticast>
  <!--optional, xs: boolean, whether it supports active multicast-->
</isSupportActiveMulticast>
<isSupportChannelEventCap>
  <!--optional, xs:boolean, whether it supports getting event capability by channel-->
</isSupportChannelEventCap>
<isSupportPictureServer>
  <!-- opt, xs:boolean, whether it supports picture storage server-->
</isSupportPictureServer>
<isSupportVideoCompositeAlarm>
  <!--optional, xs:boolean, whether it supports video double check alarm-->
</isSupportVideoCompositeAlarm>
<isSupportSensorCalibrating>
  <!--optional, xs:boolean, whether it supports double sensor calibration-->
</isSupportSensorCalibrating>
<isSupportChannelEventListCap>
  <!--optional, xs:boolean, whether it supports getting event capability of all channels-->
</isSupportChannelEventListCap>
<VCAResourceChannelsCap>
```

```
<!--optional, whether it supports independently switching to another VCA resource by channel-->
<ChannelsList>
  <channelsID>
    <!--req, xs:integer, channel No. supported by the device-->
  </channelsID>
</ChannelsList>
</VCAResourceChannelsCap>
<SensorCap/><!--optional, intelligent cabinet capability-->
<isSupportSecurityCP/>
  <!--optional, xs:boolean, whether it supports the applications of security control panel: "true, false"-->
</isSupportSecurityCP>
<isSupportClientProxyWEB>
  <!--optional, xs:boolean, whether it supports the function that the client proxy passes through the remote web
configuration: "true"-->
</isSupportClientProxyWEB>
<WEBLocation>
  <!--optional, string type, web page location: "local"-local device, "remote"-remote location. If this node is not
returned, the web page will be in the local device by default-->
</WEBLocation>
<isSupportTime/>
  <!--optional, xs:boolean, "true, false", whether it supports time configuration-->
</isSupportTime>
<isSupportTimeZone/>
  <!--optional, xs:boolean, "true, false", whether it supports daylight saving time (DST) configuration-->
</isSupportTimeZone>
<isSupportMixedTargetDetection>
  <!--optional, xs:boolean, "true, false", whether it supports multi-target-type detection-->
</isSupportMixedTargetDetection>
<isSupportFaceContrastMode>
  <!--optional, xs:boolean, whether it supports face picture comparison mode-->
</isSupportFaceContrastMode>
<isSupportPictureCaptureComparision>
  <!--optional, xs:boolean, whether it supports face picture N:1 comparison between face pictures captured by the
camera and imported face pictures-->
</isSupportPictureCaptureComparision>
<isSupportGPSCalibratation>
  <!--optional, xs:boolean, whether it supports GPS calibration capability-->
</isSupportGPSCalibratation>
<isSupportChannelFullEventListCap>
  <!--optional, xs:boolean, whether it supports getting event list capability of all channels-->
</isSupportChannelFullEventListCap>
<isSupportAUXInfoCap>
  <!--optional, xs:boolean, whether it supports getting property capability of all channels-->
</isSupportAUXInfoCap>
<isSupportCalibrationFile>
  <!--optional, xs:boolean, whether it supports importing calibration file-->
</isSupportCalibrationFile>
<isSupportDisplayTrajectory>
  <!--optional, xs:boolean, whether it supports displaying trajectory-->
</isSupportDisplayTrajectory>
<maximumSuperPositionTime opt="5,10,20,30">
  <!--dep,xs:integer, the maximum time of trajectory displaying, unit: second, it is valid only when displaying
```

```
trajectory is supported-->
</maximumSuperPositionTime>
<isSupportUnitConfig>
  <!--optional, xs:boolean, whether it supports unit configuration-->
</isSupportUnitConfig>
<isSupportAutoMaintenance>
  <!--optional, xs:boolean, whether it supports automatic maintenance. When this node exists and values "true", it
indicates support-->
</isSupportAutoMaintenance>
<isSupportGetLinkSocketIP>
  <!--optional, xs: boolean, "true,false", whether it supports getting the SocketIP of current connection-->
</isSupportGetLinkSocketIP>
<isSupportIntelligentSearch>
  <!--optional, xs:boolean, whether it supports intelligent search-->
</isSupportIntelligentSearch>
<IOTCap><!--optional, xs:boolean, IoT device access capability-->
  <supportChannelNum>
    <!--req, xs:integer, number of supported channels of IoT device-->
  </supportChannelNum>
  <startChannelNo>
    <!--optional, xs:integer, initial channel ID, if this node is not inputted, it indicates that the initial channel ID is 1-->
  </startChannelNo>
  <isSupportlinkageChannelsSearch>
    <!--optional, boolean, returns "true" if support, returns "false" if not support-->
  </isSupportlinkageChannelsSearch>
</IOTCap>
<isSupportEncryption>
  <!--optional, xs: boolean, stream encryption capability-->
</isSupportEncryption>
<AIDEventSupport opt="abandonedObject, pedestrian, congestion, roadBlock, construction, trafficAccident,
fogDetection, wrongDirection, illegalParking, SSharpDriving, lowSpeed, dragRacing">
  <!--optional, xs:string, supported traffic incident type: "abandonedObject"-objects dropped down, "pedestrian"-
pedestrian, "congestion"-congestion, "roadBlock"-roadblock, "construction"-construction, "trafficAccident"-traffic
accident, "fogDetection"-fog, "wrongDirection"-wrong-way driving, "illegalParking"-illegal parking, "SSharpDriving"-
slalom driving, "lowSpeed"-driving in low speed, "dragRacing"-street racing-->
</AIDEventSupport>
<TFSEventSupport
opt="illegalParking ,wrongDirection,crossLane,laneChange,vehicleExist,turnRound,parallelParking,notKeepDistance,not
SlowZebraCrossing,overtakeRightSide,lowSpeed,dragRacing,changeLaneContinuously,SSharpDriving,largeVehicleOccup
yLine,jamCrossLine">
  <!--optional, xs:string, supported enforcement event type: "illegalParking"-illegal parking, "wrongDirection"-wrong-
way driving, "crossLane"-driving on the lane line, "laneChange"-illegal lane change, "vehicleExist"-motor vehicle on
non-motor vehicle lane, "turnRound"-illegal U-turn, "parallelParking"-parallel parking, "notKeepDistance"-not keeping
vehicle distance, "notSlowZebraCrossing"-not slowing down at zebra corssing, "overtakeRightSide"-overtaking on the
right, "lowSpeed"-driving in low speed, "dragRacing"-street racing, "changeLaneContinuously"-continuous lane
change, "SSharpDriving"-slalom driving, "largeVehicleOccupyLine"-lane occupation by large-sized vehicle,
"jamCrossLine"-queue jumping-->
</TFSEventSupport>
<isVehicleStatisticsSupport>
  <!--optional, xs: boolean, whether it supports setting parameters for traffic data collection-->
</isVehicleStatisticsSupport>
<isSupportIntersectionAnalysis>
```

```
<!--optional, xs: boolean, whether it supports intersection analysis-->
</isSupportIntersectionAnalysis>
<supportRemoteCtrl opt="up,down,left,right,enter,menu,num,power,esc,edit,F1,.prev,rec,play,stop,notSupport"/><!--
whether it supports remote control-->
<isSptDiagnosis>
  <!--optional, xs:boolean, whether it supports device diagnosis: "true", "false"-->
</isSptDiagnosis>
<isSptSerialLogCfg>
  <!--optional, xs:boolean, whether it supports configuring serial port log redirection: "true", "false"-->
</isSptSerialLogCfg>
<isSptFileExport>
  <!--optional, xs:boolean, whether it supports exporting files from the device: "true", "false"-->
</isSptFileExport>
<isSptCertificationStandard>
  <!--optional, xs:boolean, whether it supports configuring authentication standard for security control panel: "true",
"false"-->
</isSptCertificationStandard>
<isSptKeypadLock>
  <!--optional, xs:boolean, whether it supports locking keypad: "true", "false"-->
</isSptKeypadLock>
<MixedTargetDetection><!--optional, whether the device supports recognizing specific target among mixed targets-->
  <isSupportFaceRecognition><!--optional, xs:boolean, whether it supports face recognition--></
isSupportFaceRecognition>
  <isSupportHumanRecognition><!--optional, xs:boolean, whether it supports human body recognition--></
isSupportHumanRecognition>
  <isSupportVehicleRecognition><!--optional, xs:boolean, whether it supports vehicle recognition--></
isSupportVehicleRecognition>
</MixedTargetDetection>
<isSupportDiscoveryMode><!--optional, xs:boolean--></isSupportDiscoveryMode>
<streamEncryptionType>
  <!--dep, xs:string, stream encryption type: "RTP/TLS", "SRTP/UDP", "SRTP/MULTICAST". This node is valid when
<isSupportEncryption> is "true", and the device can support one or more stream encryption types-->
</streamEncryptionType>
<isSupportLms><!--optional, xs:boolean, whether it supports laser--></isSupportLms>
<isSupportLCDScreen><!--optional, xs:boolean, whether it supports LCD screen--></isSupportLCDScreen>
<isSupportBluetooth><!--optional, xs:boolean, whether it supports bluetooth--></isSupportBluetooth>
<isSupportAcsUpdate>
  <!--optional, whether it supports upgrading slave access control devices or peripheral modules: "true"-yes, this
node is not returned-no-->
</isSupportAcsUpdate>
<isSupportAccessControlCap>
  <!--optional, whether it supports access control capability: "true"-yes, this node is not returned-no-->
</isSupportAccessControlCap>
<isSupportIDCardInfoEvent><!--optional, whether it supports ID card swiping event: "true"-yes. This node will not be
returned if this function is not supported--></isSupportIDCardInfoEvent>
<OpenPlatformCap><!--optional, embedded open platform capability, refer to the message XML_OpenPlatformCap
for details-->
<isSupportInstallationAngleCalibration>
  <!--optional, xs:boolean, whether it supports installation angle calibration-->
</isSupportInstallationAngleCalibration>
<isSupportZeroBiasCalibration>
  <!--optional, xs:boolean, whether it supports zero bias calibration-->
```

```
</isSupportZeroBiasCalibration>
<isSupportDevStatus><!--optional, xs:boolean, whether device supports getting device status--></
isSupportDevStatus>
<isSupportRadar><!--optional, xs:boolean, whether it supports the security radar--></isSupportRadar>
<isSupportRadarChannels><!--optional, xs:boolean, whether it supports getting radar channels--></
isSupportRadarChannels>
<radarIPDForm><!--optional, xs:string, radar form: "single"-single radar, "double_diagonal"-two radars forming an
180° diagonal, "double_vertical"-two radars forming a 90° vertical angle--></radarIPDForm>
<isSupportRadarFieldDetection><!--optional, xs:boolean, whether it supports intrusion detection (radar)--></
isSupportRadarFieldDetection>
<isSupportRadarLineDetection><!--optional, xs:boolean, whether it supports line crossing detection (radar)--></
isSupportRadarLineDetection>
<mixedTargetDetectionWebNoDisplay><!--optional, xs:boolean, whether to enable not displaying multi-target-type
recognition--></mixedTargetDetectionWebNoDisplay>
<SHMCap><!--opt-->
<isSupportHighHDDTemperature><!--optional, xs:boolean, whether it supports HDD high temperature detection--></
isSupportHighHDDTemperature>
<isSupportLowHDDTemperature><!--optional, xs:boolean, whether it supports HDD low temperature detection--></
isSupportLowHDDTemperature>
<isSupportHDImpact><!--optional, xs:boolean, whether it supports HDD impact detection--></isSupportHDImpact>
<isSupportHDBadBlock><!--optional, xs:boolean, whether it supports HDD bad sector detection--></
isSupportHDBadBlock>
<isSupportSevereHDFailure><!--optional, xs:boolean, whether it supports HDD severe fault detection--></
isSupportSevereHDFailure>
</SHMCap>
<isSupportBVCorrect><!--optional, xs:boolean, whether it supports configuring camera correction parameters--></
isSupportBVCorrect>
<guideEventSupport opt="linkageCapture">
<!--optional,xs:string, events which support quick setup by instruction, "linkageCapture"-capture by linkage-->
</guideEventSupport>
<isSupportAutoSwitch><!--optional, xs:boolean, whether it supports auto switch--> true</isSupportAutoSwitch>
<isSupportDataPrealarm><!--optional,xs:boolean, whether it supports traffic pre-alarm event--></
isSupportDataPrealarm>
<supportGISEvent opt="AID,TPS,ANPR,mixedTargetDetection">
<!--optional, xs:string, event types that support GIS information access: AID (corresponding SDK event:
COMM_ALARM_AID_V41), TPS (corresponding SDK event: COMM_ALARM_TPS_REAL_TIME), ANPR (corresponding
SDK event: COMM_ITS_PLATE_RESULT), mixedTargetDetection-mixed targets detection-->
</supportGISEvent>
<isSupportIntelligentMode><!--optional, xs:boolean, whether it supports intelligent scene switch (related URI:/ISAPI/
System/IntelligentSceneSwitch?format=json)--></isSupportIntelligentMode>
<isSupportCertificateCaptureEvent><!--optional, xs:boolean, whether it supports certificate capture and comparison
events: true=yes. If this function is not supported, this node will not be returned--></
isSupportCertificateCaptureEvent>
<isSupportAlgorithmsInfo><!--optional, xs:boolean, whether it supports getting the algorithm library version
information: true=yes. If this function is not supported, this node will not be returned--></isSupportAlgorithmsInfo>
<isSupportVibrationDetection><!--optional, xs:boolean, whether it supports vibration detection--></
isSupportVibrationDetection>
<isSupportFaceTemperatureMeasurementEvent><!--optional, xs:boolean, whether it supports uploading face
thermography events (eventType: "FaceTemperatureMeasurementEvent")--></
isSupportFaceTemperatureMeasurementEvent>
<isSupportQRCodeEvent><!--optional, xs:boolean, whether it supports uploading QR code events (eventType:
"QRCodeEvent")--></isSupportQRCodeEvent>
```

```
<isSupportPersonArmingTrack><!--optional, xs:boolean, whether device supports person arming (related URI: /ISAPI/Intelligent/channels/<ID>/personArmingTrack/capabilities?format=json)--></isSupportPersonArmingTrack>
<isSupportManualPersonArmingTrack><!--optional, xs:boolean, whether device supports manual person arming (related URI: /ISAPI/Intelligent/channels/<ID>/manualPersonArmingTrack?format=json)--></isSupportManualPersonArmingTrack>
<isSupportGPSCalibrationMode><!--optional, xs:boolean, whether device supports GPS calibration (related URI: /ISAPI/System/GPSCalibration/channels/<ID>/mode?format=json)--></isSupportGPSCalibrationMode>
<isSupportGPSVerification><!--optional, xs:boolean, whether device supports GPS verification (related URI: /ISAPI/System/GPSVerification/channels/<ID>/points?format=json)--></isSupportGPSVerification>
<isSupportHBDLib><!--optional, xs:boolean, whether device supports human body picture library (related URI: /ISAPI/Intelligent/HBDLib/capabilities?format=json)--></isSupportHBDLib>
<isSupportFireEscapeDetection><!--optional, xs:boolean, whether the device supports fire engine access detection (related URI: /ISAPI/Intelligent/channels/<ID>/fireEscapeDetection/capabilities?format=json)--></isSupportFireEscapeDetection>
<isSupportTakingElevatorDetection><!--optional, xs:boolean, whether the device supports elevator detection (related URI: /ISAPI/Intelligent/channels/<ID>/takingElevatorDetection/capabilities?format=json)--></isSupportTakingElevatorDetection>
</DeviceCap>
```

A.59 XML_DPC

DPC message in XML format

```
<DPC version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mode>
    <!--req, xs:string,
opt="correction,moveup,movedown,moveleft,moveright,cancelCorrect,allCorrection,save,start,exit,display"-->
  </mode>
  <dpcMode><!--opt, xs:string, defective pixel correction mode, opt="manual,auto"--></dpcMode>
  <normalizedScreenSize><!--opt, ro-->
    <normalizedScreenWidth><!--req, ro,xs:integer--></normalizedScreenWidth>
    <normalizedScreenHeight><!--req, ro,xs:integer--></normalizedScreenHeight>
  </normalizedScreenSize>
  <Point><!--dep, it is valid when mode values "display"-->
    <positionX><!--req,xs:integer;coordinate--></positionX>
    <positionY><!--req,xs:integer;coordinate--></positionY>
  </Point>
</DPC>
```

A.60 XML_EventNotificationAlert_FireSmokeAlarmEg

The fire and smoke alarm information is uploaded in the XML format of EventNotificationAlert message, see details below.

```
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dep, xs:string--></ipAddress>
  <ipv6Address><!--dep, xs:string--></ipv6Address>
  <portNo><!--opt, xs:integer--></portNo>
```

```
<protocol><!--opt, xs:string,"HTTP,HTTPS"--></protocol>
<macAddress><!--opt, xs:string;MAC--></macAddress>
<channelID><!--dep, xs:string--></channelID>
<dateTime>
  <!--req, xs:datetime based on IOS8601, e.g., 2017-04-22T15:39:01+08:00-->
</dateTime>
<activePostCount><!--req, xs:integer--></activePostCount>
<eventType>
  <!--req, xs:string, "fireDetection"-->
</eventType>
<eventState>
  <!--req, xs:string, "active,inactive"-->
</eventState>
<eventDescription>
  <!--req, xs:string, "Fire and Smoke Alarm"-->
</eventDescription>
<DetectionRegionList>
  <DetectionRegionEntry>
    <regionID><!--required, integer, region ID--></regionID>
    <RegionCoordinatesList><!--opt, coordinates of detection region-->
      <RegionCoordinates>
        <positionX><!--req, xs:integer; X-coordinate--></positionX>
        <positionY><!--req, xs:integer; Y-coordinate--></positionY>
      </RegionCoordinates>
      <RegionCoordinates>
        <positionX><!--req, xs:integer; X-coordinate--></positionX>
        <positionY><!--req, xs:integer; Y-coordinate--></positionY>
      </RegionCoordinates>
      <RegionCoordinates>
        <positionX><!--req, xs:integer; X-coordinate--></positionX>
        <positionY><!--req, xs:integer; Y-coordinate--></positionY>
      </RegionCoordinates>
      <RegionCoordinates>
        <positionX><!--req, xs:integer; X-coordinate--></positionX>
        <positionY><!--req, xs:integer; Y-coordinate--></positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </DetectionRegionEntry>
  <FireDetection>
    <FireRegion><!--coordinates of fire source region-->
      <x><!--req, xs:integer; X-coordinate--></x>
      <y><!--req, xs:integer; Y-coordinate--></y>
      <width><!--region width--></width>
      <height><!--region height--></height>
    </FireRegion>
    <HighestPoint><!--coordinates of point with highest temperature-->
      <x><!--req, xs:integer; X-coordinate--></x>
      <y><!--req, xs:integer; Y-coordinate--></y>
    </HighestPoint>
    <temperatureUnit><!--required, xs: string, temperature unit--></temperatureUnit>
    <fireMaxTemperature><!--maximum temperature--></fireMaxTemperature>
    <targetDistance><!--target distance--></targetDistance>
    <AbsoluteHigh>
```

```
<elevation><!-- opt, xs:integer, between -900 and 2700--></elevation>
<azimuth><!--opt, xs:integer, between 0 and 3600--></azimuth>
<absoluteZoom><!--opt, xs:integer, between 0 and 1000--></absoluteZoom>
</AbsoluteHigh>
</FireDetection>
</DetectionRegionEntry>
</DetectionRegionList>
<channelName><!--opt,xs:string, channel name--></channelName>
<detectedPicturesNumber>
  <!--opt,xs:string, picture transmission format: "url,binary"-->
</detectedPicturesNumber>
</EventNotificationAlert>
```

Example

```
GET /ISAPI/Event/notification/alertStream HTTP/1.1
Authorization: Digest username="admin",realm="IP
Camera(C2183)",nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",uri="/ISAPI/Event/notification/
alertStream",cnonce="3d183a245b8729121ae4ca3d41b90f18",nc=00000001,qop="auth",response="f2e0728991bb03
1f83df557a8f185178"
Host: 10.6.165.192

HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=boundary

--boundary
Content-Type: application/xml; charset="UTF-8"
Content-Length: 1579

<EventNotificationAlert version="2.0" xmlns="http://www.hikvision.com/ver20/XMLSchema">
  <ipAddress>10.6.165.192</ipAddress>
  <portNo>80</portNo>
  <protocol>HTTP</protocol>
  <macAddress>a4:14:37:80:2b:04</macAddress>
  <channelID>2</channelID>
  <dateTime>2018-04-01T18:58:59+08:00</dateTime>
  <activePostCount>1</activePostCount>
  <eventType>fireDetection</eventType>
  <eventState>active</eventState>
  <eventDescription>fireDetection alarm</eventDescription>
  <DetectionRegionList>
    <DetectionRegionEntry>
      <regionID>1</regionID>
      <RegionCoordinatesList>
        <RegionCoordinates>
          <positionX>382</positionX>
          <positionY>364</positionY>
        </RegionCoordinates>
        <RegionCoordinates>
          <positionX>629</positionX>
          <positionY>364</positionY>
```

```

</RegionCoordinates>
<RegionCoordinates>
  <positionX>629</positionX>
  <positionY>641</positionY>
</RegionCoordinates>
<RegionCoordinates>
  <positionX>382</positionX>
  <positionY>611</positionY>
</RegionCoordinates>
</RegionCoordinatesList>
<FireDetection>
  <FireRegion>
    <x>382</x>
    <y>364</y>
    <width>247</width>
    <height>277</height>
  </FireRegion>
  <HighestPoint>
    <x>505</x>
    <y>489</y>
  </HighestPoint>
  <temperatureUnit>celsius</temperatureUnit>
  <fireMaxTemperature>88</fireMaxTemperature>
  <targetDistance>-1</targetDistance>
  <AbsoluteHigh>
    <elevation>26202</elevation>
    <azimuth>633</azimuth>
    <absoluteZoom>100</absoluteZoom>
  </AbsoluteHigh>
</FireDetection>
</DetectionRegionEntry>
</DetectionRegionList>
<channelName>Camera 02</channelName>
<detectionPicturesNumber>2</detectionPicturesNumber>
</EventNotificationAlert>
--boundary
Content-Disposition: form-data;
Content-Type: image/jpeg
Content-Length: 114648

.....JFIF.....`.....8....".....
.....}......!1A..Qa."q.2....#B...R..$3br.
....

```

A.61 XML_EventNotificationAlert_IntrusionAlarmMsg

XML Message about alarm details of intrusion detection

```

<?xml version="1.0" encoding="utf-8"?>
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">

```

```
<ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
<ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
<portNo><!--optional, xs:integer, device's communication port No.--></portNo>
<protocolType><!--optional, xs:string, transmission protocol type: "HTTP, HTTPS, EHome"--></protocolType>
<macAddress><!--optional, xs:string, MAC address--></macAddress>
<dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
<channelID><!--dependent, xs:string, channel No.--></channelID>
<dateTime>
  <!--required, xs:datetime, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00-->
</dateTime>
<activePostCount><!--required, xs:integer, event occurred times--></activePostCount>
<eventType><!--required, xs:string, event types, here it should be set to "fielddetection"--></eventType>
<eventState><!--required, xs:string, event status (for persistent event): "active, inactive"--></eventState>
<eventDescription><!--required, xs:string, event description--></eventDescription>
<deviceId><!--optional, xs:string, device ID--></deviceId>
<DetectionRegionList>
  <DetectionRegionEntry><!--list-->
    <regionID><!--required, xs:string, detection region ID--></regionID>
    <sensitivityLevel><!--optional, xs:integer, sensitivity level, which is between 0 and 100--></sensitivityLevel>
    <RegionCoordinatesList><!--optional, target region-->
      <RegionCoordinates><!--optional-->
        <positionX><!--required, xs:integer, x-coordinate--></positionX>
        <positionY><!--required, xs:integer, y-coordinate--></positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <detectionTarget><!--optional, xs:string, target type: "human, vehicle, others"--></detectionTarget>
    <TargetRect><!--optional-->
      <X><!--required, xs:float; x-coordinate of target position, value range: [0,1]--></X>
      <Y><!--required, xs:float; y-coordinate of target position, value range: [0,1]--></Y>
      <width><!--required, xs:float; target frame width, value range: [0,1]--></width>
      <height><!--req, xs:float; target frame height, value range: [0,1]--></height>
    </TargetRect>
  </DetectionRegionEntry>
</DetectionRegionList>
<channelName><!--optional, xs:string, channel name--></channelName>
<Extensions>
  <serialNumber><!--required, xs:string, serial No.--></serialNumber>
  <eventPush><!--required, xs:string, event or alarm notification contents--></eventPush>
</Extensions>
<detectionPictureTransType><!--optional, xs:string, picture transmission mode: "url" or "binary"--></
detectionPictureTransType>
<detectionPicturesNumber>
  <!--optional, xs:integer, number of alarm pictures received in arming mode. If there is no alarm picture, this node
will not be returned-->
</detectionPicturesNumber>
<bkgUrl><!--optional, xs:string, background picture URL--></bkgUrl>
<URLCertificationType>
  <!--optional, xs:string, authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication
(it is used for local picture URL returned by NVR or DVR)-->
</URLCertificationType>
<pId><!--optional, xs:string, picture ID--></pId>
<visibleLightURL>
```

```

<!--optional, xs:string, optical picture URL-->
</visibleLightURL>
<thermalURL>
<!--optional, xs:string, thermal picture URL-->
</thermalURL>
<targetAttrs><!--optional, transparent transmission field, the following fields are recommended-->
<![CDATA[{
/*optional, transparent transmission field, the following fields are recommended*/
"deviceId": "",
/*optional, string, device ID, the maximum size is 64 bytes*/
"deviceChannel": ,
/*optional, integer32, device channel ID*/
"deviceName": ""
/*optional, string, device name, the maximum size is 128 bytes*/
}]]>
</targetAttrs>
<TaskInfo><!--optional, task information submitted by Data Center products-->
<taskID><!--required, xs:string, task ID, the maximum size is 64 bytes--></taskID>
<algorithmType>
<!--optional, xs:integer, algorithm type: 1-"hightQualityVehicle" (high-quality vehicle),
2-"hightPerformanceVehicle" (high-performance vehicle), 3-"personStructModel" (person structural modeling),
4-"vehicleAndPersonStructModel" (person and vehicle structural modeling), 5-"videoFaceDetect" (video face
detection), 6-"videoFaceDetectModel" (video face detection and modeling), 7-"smart" (face detection (output frame
without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->
</algorithmType>
<taskName><!--optional, xs:string, task name--></taskName>
<algorithmID><!--optional, xs:string, algorithm package ID--></algorithmID>
<monitorPointID><!--optional, xs:string, camera No.--></monitorPointID>
<monitorPointName><!--optional, xs:string, camera name--></monitorPointName>
<streamType><!--optional, xs:string, stream type: "realtime,historyvideo,localvideo--></streamType>
<ruleID><!--optional, xs:string, rule ID, which is used to distinguish different tasks with the same name, and the
ruleID in the event uploading message indicates that the event is related to this task--></ruleID>
<ruleCustomName><!--required, xs:string, custom event name, which is included in the message when the alarm is
uploaded, and the maximum size is 128 bytes--></ruleCustomName>
<algorithmInfo><!--optional, algorithm version information, this field is required for secondary analysis-->
<algorithmId><!--optional, xs:string, algorithm package --></algorithmId>
<algoVendorId><!--optional, xs:string, manufacturer ID--></algoVendorId>
<algoVersionId><!--optional, xs:string, algorithm version ID--></algoVersionId>
<algoChipType><!--optional, xs:string, chip type--></algoChipType>
<algoModelVersion><!--optional, xs:string, model version--></algoModelVersion>
</algorithmInfo>
</TaskInfo>
<timeStamp><!--optional, xs:string, time of the alarm being triggered, ISO 8601 time format--></timeStamp>
<uid><!--optional, xs:string, unique ID of event upload, the maximum length is 64 bytes--></uid>
</EventNotificationAlert>

```

A.62 XML_EventNotificationAlert_LineCrossingAlarmMsg

XML Message about alarm details of line crossing detection

```
<?xml version="1.0" encoding="utf-8"?>
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
  <ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
  <portNo><!--optional, xs:integer, device's communication port No.--></portNo>
  <protocolType><!--optional, xs:string, transmission protocol type: "HTTP, HTTPS, EHome"--></protocolType>
  <macAddress><!--optional, xs:string, MAC address--></macAddress>
  <dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
  <channelID><!--dependent, xs:string, channel No.--></channelID>
  <dateTime>
    <!--required, xs:datetime, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00-->
  </dateTime>
  <activePostCount><!--required, xs:integer, event occurred times--></activePostCount>
  <eventType><!--required, xs:string, event types, here it should be set to "linedetection"--></eventType>
  <eventState><!--required, xs:string, event status (for persistent event): "active, inactive"--></eventState>
  <eventDescription><!--required, xs:string, event description--></eventDescription>
  <deviceId><!--optional, xs:string, device ID--></deviceId>
  <DetectionRegionList>
    <DetectionRegionEntry><!--list-->
      <regionID><!--required, xs:string, detection region ID--></regionID>
      <sensitivityLevel><!--optional, xs:integer, sensitivity level, which is between 0 and 100--></sensitivityLevel>
      <RegionCoordinatesList><!--optional, target region-->
        <RegionCoordinates><!--optional-->
          <positionX><!--required, xs:integer, x-coordinate--></positionX>
          <positionY><!--required, xs:integer, y-coordinate--></positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget><!--optional, xs:string, target type: "human, vehicle, others"--></detectionTarget>
      <TargetRect><!--optional-->
        <X><!--required, xs:float; x-coordinate of target position, value range: [0,1]--></X>
        <Y><!--required, xs:float; y-coordinate of target position, value range: [0,1]--></Y>
        <width><!--required, xs:float; target frame width, value range: [0,1]--></width>
        <height><!--req, xs:float; target frame height, value range: [0,1]--></height>
      </TargetRect>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName><!--optional, xs:string, channel name--></channelName>
  <Extensions>
    <serialNumber><!--required, xs:string, serial No.--></serialNumber>
    <eventPush><!--required, xs:string, event or alarm notification contents--></eventPush>
  </Extensions>
  <detectionPictureTransType><!--optional, xs:string, picture transmission mode: "url" or "binary"--></
detectionPictureTransType>
  <detectionPicturesNumber>
    <!--optional, xs:integer, number of alarm pictures received in arming mode. If there is no alarm picture, this node
will not be returned-->
  </detectionPicturesNumber>
  <bkgUrl><!--optional, xs:string, background picture URL--></bkgUrl>
  <URLCertificationType>
    <!--optional, xs:string, authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication
(it is used for local picture URL returned by NVR or DVR)-->
  </URLCertificationType>
```

```
<pId><!--optional, xs:string, picture ID--></pId>
<visibleLightURL>
  <!--optional, xs:string, optical picture URL-->
</visibleLightURL>
<thermalURL>
  <!--optional, xs:string, thermal picture URL-->
</thermalURL>
<targetAttrs><!--optional, transparent transmission field, the following fields are recommended-->
  <![CDATA[{
/*optional, transparent transmission field, the following fields are recommended*/
  "deviceId": "",
/*optional, string, device ID, the maximum size is 64 bytes*/
  "deviceChannel": ,
/*optional, integer32, device channel ID*/
  "deviceName": ""
/*optional, string, device name, the maximum size is 128 bytes*/
  }]]>
</targetAttrs>
<TaskInfo><!--optional, task information submitted by Data Center products-->
  <taskId><!--required, xs:string, task ID, the maximum size is 64 bytes--></taskId>
  <algorithmType>
    <!--optional, xs:integer, algorithm type: 1-"highQualityVehicle" (high-quality vehicle),
2-"highPerformanceVehicle" (high-performance vehicle), 3-"personStructModel" (person structural modeling),
4-"vehicleAndPersonStructModel" (person and vehicle structural modeling), 5-"videoFaceDetect" (video face
detection), 6-"videoFaceDetectModel" (video face detection and modeling), 7-"smart" (face detection (output frame
without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->
    </algorithmType>
    <taskName><!--optional, xs:string, task name--></taskName>
    <algorithmId><!--optional, xs:string, algorithm package ID--></algorithmId>
    <monitorPointId><!--optional, xs:string, camera No.--></monitorPointId>
    <monitorPointName><!--optional, xs:string, camera name--></monitorPointName>
    <streamType><!--optional, xs:string, stream type: "realtime,historyvideo,localvideo"--></streamType>
    <ruleId><!--optional, xs:string, rule ID, which is used to distinguish different tasks with the same name, and the
ruleId in the event uploading message indicates that the event is related to this task--></ruleId>
    <ruleCustomName><!--required, xs:string, custom event name, which is included in the message when the alarm is
uploaded, and the maximum size is 128 bytes--></ruleCustomName>
    <algorithmInfo><!--optional, algorithm version information, this field is required for secondary analysis-->
      <algorithmId><!--optional, xs:string, algorithm package --></algorithmId>
      <algoVendorId><!--optional, xs:string, manufacturer ID--></algoVendorId>
      <algoVersionId><!--optional, xs:string, algorithm version ID--></algoVersionId>
      <algoChipType><!--optional, xs:string, chip type--></algoChipType>
      <algoModelVersion><!--optional, xs:string, model version--></algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp><!--optional, xs:string, time of the alarm being triggered, ISO 8601 time format--></timeStamp>
  <uid><!--optional, xs:string, unique ID of event upload, the maximum length is 64 bytes--></uid>
</EventNotificationAlert>
```

Example

Sample Message of Line Crossing Detection Alarm in XML Format

```
<?xml version="1.0" encoding="utf-8"?>
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress>10.17.114.210</ipAddress>
  <protocolType>HTTP</protocolType>
  <macAddress>94:e1:ac:0e:dd:59</macAddress>
  <channelID>1</channelID>
  <dateTime>2018-03-23T14:40:56+08:00</dateTime>
  <activePostCount>20</activePostCount>
  <eventType>linedetection</eventType>
  <eventState>active</eventState>
  <eventDescription>linedetection alarm</eventDescription>
  <DetectionRegionList>
    <DetectionRegionEntry>
      <regionID>1</regionID>
      <sensitivityLevel>1</sensitivityLevel>
      <RegionCoordinatesList>
        <RegionCoordinates>
          <positionX>1</positionX>
          <positionY>1</positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>human</detectionTarget>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>Camera 01</channelName>
  <detectionPictureTransType>url</detectionPictureTransType>
  <detectionPicturesNumber>1</detectionPicturesNumber>
  <bkgUrl>http://10.16.98.38:8088/BFAF2EB16A878E2708B0A25D2466112Aundefined</bkgUrl>
  <URLCertificationType>digest</URLCertificationType>
</EventNotificationAlert>
```

A.63 XML_EventNotificationAlert_RegionEntranceAlarmMsg

XML Message about alarm details of region entrance detection

```
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
  <ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
  <portNo><!--optional, xs:integer, device's communication port No.--></portNo>
  <protocolType><!--optional, xs:string, transmission protocol type: "HTTP, HTTPS, EHome"--></protocolType>
  <macAddress><!--optional, xs:string, MAC address--></macAddress>
  <dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
  <channelID><!--dependent, xs:string, channel No.--></channelID>
  <dateTime>
    <!--required, xs:datetime, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00-->
  </dateTime>
  <activePostCount><!--required, xs:integer, event occurred times--></activePostCount>
  <eventType><!--required, xs:string, event types, here it should be set to "regionEntrance"--></eventType>
  <eventState><!--required, xs:string, event status (for persistent event): "active, inactive"--></eventState>
  <eventDescription><!--required, xs:string, event description--></eventDescription>
```

```
<deviceId><!--opt, xs:string, device ID--></deviceId>
<DetectionRegionList>
  <DetectionRegionEntry><!--list-->
    <regionID><!--required, xs:string, detection region ID--></regionID>
    <sensitivityLevel><!--optional, xs:integer, sensitivity level, which is between 0 and 100--></sensitivityLevel>
    <RegionCoordinatesList><!--optional, target region-->
      <RegionCoordinates><!--optional-->
        <positionX><!--required, xs:integer, x-coordinate--></positionX>
        <positionY><!--required, xs:integer, y-coordinate--></positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <detectionTarget><!--optional, xs:string, target type: "human, vehicle, others"--></detectionTarget>
    <TargetRect><!--optional-->
      <X><!--required, xs:float; x-coordinate of target position, value range: [0,1]--></X>
      <Y><!--required, xs:float; y-coordinate of target position, value range: [0,1]--></Y>
      <width><!--required, xs:float; target frame width, value range: [0,1]--></width>
      <height><!--req, xs:float; target frame height, value range: [0,1]--></height>
    </TargetRect>
  </DetectionRegionEntry>
</DetectionRegionList>
<channelName><!--optional, xs:string, channel name--></channelName>
<detectionPictureTransType><!--optional, xs:string, picture transmission mode: "url" or "binary"--></
detectionPictureTransType>
<detectionPicturesNumber>
  <!--optional, xs:integer, number of alarm pictures received in arming mode. If there is no alarm picture, this node
will not be returned-->
</detectionPicturesNumber>
<bkgUrl><!--optional, xs:string, background picture URL--></bkgUrl>
<URLCertificationType>
  <!--optional, xs:string, authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication
(it is used for local picture URL returned by NVR or DVR)-->
</URLCertificationType>
<pId><!--optional, xs:string, picture ID--></pId>
<visibleLightURL>
  <!--optional, xs:string, optical picture URL-->
</visibleLightURL>
<thermalURL>
  <!--optional, xs:string, thermal picture URL-->
</thermalURL>
<targetAttrs><!--optional, transparent transmission field, the following fields are recommended-->
  <![CDATA[{
/*optional, transparent transmission field, the following fields are recommended*/
  "deviceId": "",
/*optional, string, device ID, the maximum size is 64 bytes*/
  "deviceChannel": ,
/*optional, integer32, device channel ID*/
  "deviceName": ""
/*optional, string, device name, the maximum size is 128 bytes*/
}]]>
</targetAttrs>
<TaskInfo><!--optional, task information submitted by Data Center products-->
  <taskID><!--required, xs:string, task ID, the maximum size is 64 bytes--></taskID>
```

```
<algorithmType>
  <!--optional, xs:integer, algorithm type: 1-"hightQualityVehicle" (high-quality vehicle),
  2-"hightPerformanceVehicle" (high-performance vehicle), 3-"personStructModel" (person structural modeling),
  4-"vehicleAndPersonStructModel" (person and vehicle structural modeling), 5-"videoFaceDetect" (video face
  detection), 6-"videoFaceDetectModel" (video face detection and modeling), 7-"smart" (face detection (output frame
  without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->
</algorithmType>
<taskName><!--optional, xs:string, task name--></taskName>
<algorithmID><!--optional, xs:string, algorithm package ID--></algorithmID>
<monitorPointID><!--optional, xs:string, camera No.--></monitorPointID>
<monitorPointName><!--optional, xs:string, camera name--></monitorPointName>
<streamType><!--optional, xs:string, stream type: "realtime,historyvideo,localvideo"--></streamType>
<ruleID><!--optional, xs:string, rule ID, which is used to distinguish different tasks with the same name, and the
ruleID in the event uploading message indicates that the event is related to this task--></ruleID>
<ruleCustomName><!--required, xs:string, custom event name, which is included in the message when the alarm is
uploaded, and the maximum size is 128 bytes--></ruleCustomName>
<algorithmInfo><!--optional, algorithm version information, this field is required for secondary analysis-->
  <algorithmId><!--optional, xs:string, algorithm package --></algorithmId>
  <algoVendorId><!--optional, xs:string, manufacturer ID--></algoVendorId>
  <algoVersionId><!--optional, xs:string, algorithm version ID--></algoVersionId>
  <algoChipType><!--optional, xs:string, chip type--></algoChipType>
  <algoModelVersion><!--optional, xs:string, model version--></algoModelVersion>
</algorithmInfo>
</TaskInfo>
<timeStamp><!--optional, xs:string, time of the alarm being triggered, ISO 8601 time format--></timeStamp>
<uid><!--optional, xs:string, unique ID of event upload, the maximum length is 64 bytes--></uid>
</EventNotificationAlert>
```

A.64 XML_EventNotificationAlert_RegionExitingAlarmMsg

XML message about alarm details of region exiting detection

```
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
  <ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
  <portNo><!--optional, xs:integer, device's communication port No.--></portNo>
  <protocolType><!--optional, xs:string, transmission protocol type: "HTTP, HTTPS, EHome"--></protocolType>
  <macAddress><!--optional, xs:string, MAC address--></macAddress>
  <dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
  <channelID><!--dependent, xs:string, channel No.--></channelID>
  <dateTime>
    <!--required, xs:datetime, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00-->
  </dateTime>
  <activePostCount><!--required, xs:integer, event occurred times--></activePostCount>
  <eventType><!--required, xs:string, event types, here it should be set to "regionExiting"--></eventType>
  <eventState><!--required, xs:string, event status (for persistent event): "active, inactive"--></eventState>
  <eventDescription><!--required, xs:string, event description--></eventDescription>
  <deviceId><!--opt, xs: string; device ID--></deviceId>
  <DetectionRegionList>
    <DetectionRegionEntry><!--list-->
```

```
<regionID><!--required, xs:string, detection region ID--></regionID>
<sensitivityLevel><!--optional, xs:integer, sensitivity level, which is between 0 and 100--></sensitivityLevel>
<RegionCoordinatesList><!--optional, target region-->
  <RegionCoordinates><!--optional-->
    <positionX><!--required, xs:integer, x-coordinate--></positionX>
    <positionY><!--required, xs:integer, y-coordinate--></positionY>
  </RegionCoordinates>
</RegionCoordinatesList>
<detectionTarget><!--optional, xs:string, target type: "human, vehicle, others"--></detectionTarget>
<TargetRect><!--optional-->
  <X><!--required, xs:float; x-coordinate of target position, value range: [0,1]--></X>
  <Y><!--required, xs:float; y-coordinate of target position, value range: [0,1]--></Y>
  <width><!--required, xs:float; target frame width, value range: [0,1]--></width>
  <height><!--req, xs:float; target frame height, value range: [0,1]--></height>
</TargetRect>
</DetectionRegionEntry>
</DetectionRegionList>
<channelName><!--optional, xs:string, channel name--></channelName>
<detectionPictureTransType><!--optional, xs:string, picture transmission mode: "url" or "binary"--></
detectionPictureTransType>
<detectionPicturesNumber>
  <!--optional, xs:integer, number of alarm pictures received in arming mode. If there is no alarm picture, this node
will not be returned-->
</detectionPicturesNumber>
<bkgUrl><!--optional, xs:string, background picture URL--></bkgUrl>
<URLCertificationType>
  <!--optional, xs:string, authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication
(it is used for local picture URL returned by NVR or DVR)-->
</URLCertificationType>
<pId><!--optional, xs:string, picture ID--></pId>
<visibleLightURL>
  <!--optional, xs:string, optical picture URL-->
</visibleLightURL>
<thermalURL>
  <!--optional, xs:string, thermal picture URL-->
</thermalURL>
<targetAttrs><!--optional, transparent transmission field, the following fields are recommended-->
  <![CDATA[{
/*optional, transparent transmission field, the following fields are recommended*/
    "deviceId": "",
/*optional, string, device ID, the maximum size is 64 bytes*/
    "deviceChannel": ,
/*optional, integer32, device channel ID*/
    "deviceName": ""
/*optional, string, device name, the maximum size is 128 bytes*/
  }]]>
</targetAttrs>
<TaskInfo><!--optional, task information submitted by Data Center products-->
  <taskID><!--required, xs:string, task ID, the maximum size is 64 bytes--></taskID>
  <algorithmType>
    <!--optional, xs:integer, algorithm type: 1-"highQualityVehicle" (high-quality vehicle),
2-"highPerformanceVehicle" (high-performance vehicle), 3-"personStructModel" (person structural modeling),
```

4-"vehicleAndPersonStructModel" (person and vehicle structural modeling), 5-"videoFaceDetect" (video face detection), 6-"videoFaceDetectModel" (video face detection and modeling), 7-"smart" (face detection (output frame without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->
</algorithmType>
<taskName><!--optional, xs:string, task name--></taskName>
<algorithmID><!--optional, xs:string, algorithm package ID--></algorithmID>
<monitorPointID><!--optional, xs:string, camera No.--></monitorPointID>
<monitorPointName><!--optional, xs:string, camera name--></monitorPointName>
<streamType><!--optional, xs:string, stream type: "realtime,historyvideo,localvideo"--></streamType>
<ruleID><!--optional, xs:string, rule ID, which is used to distinguish different tasks with the same name, and the **ruleID** in the event uploading message indicates that the event is related to this task--></ruleID>
<ruleCustomName><!--required, xs:string, custom event name, which is included in the message when the alarm is uploaded, and the maximum size is 128 bytes--></ruleCustomName>
<algorithmInfo><!--optional, algorithm version information, this field is required for secondary analysis-->
<algorithmId><!--optional, xs:string, algorithm package --></algorithmId>
<algoVendorId><!--optional, xs:string, manufacturer ID--></algoVendorId>
<algoVersionId><!--optional, xs:string, algorithm version ID--></algoVersionId>
<algoChipType><!--optional, xs:string, chip type--></algoChipType>
<algoModelVersion><!--optional, xs:string, model version--></algoModelVersion>
</algorithmInfo>
</TaskInfo>
<timeStamp><!--optional, xs:string, time of the alarm being triggered, ISO 8601 time format--></timeStamp>
<uid><!--optional, xs:string, unique ID of event upload, the maximum length is 64 bytes--></uid>
</EventNotificationAlert>

A.65 XML_EventNotificationAlert_shipFlowDetection

The information of ship flow detection alarm is uploaded in the XML format of EventNotificationAlert message, see details below.

```
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dep, xs:string--></ipAddress>
  <ipv6Address><!--dep, xs:string--></ipv6Address>
  <portNo><!--opt, xs:integer--></portNo>
  <protocol><!--opt, xs:string,"HTTP,HTTPS"--></protocol>
  <macAddress><!--opt, xs:string;MAC--></macAddress>
  <channelID><!--dep, xs:string--></channelID>
  <dateTime><!--req, xs:datetime ios8601,eg: 2017-04-22T15:39:01+08:00--></dateTime>
  <activePostCount><!--req, xs:integer--></activePostCount>
  <eventType><!--req, xs:string, "shipsDetection"--></eventType>
  <eventState><!--req, xs:string,"active,inactive"--></eventState>
  <eventDescription><!--req, xs:string--></eventDescription>
  <DetectionRegionList\>
    <channelName><!--opt, xs:string--></channelName>
    <detectionPicturesNumber><!--opt, xs:integer--></detectionPicturesNumber>
    <sid><!--opt, xs:integer, scene ID--></sid>
    <sceneName><!--opt, xs:string, scene name--></sceneName>
  </EventNotificationAlert>
```

A.66 XML_EventNotificationAlert_SubscriptionHeartbeat

Heartbeat information message returned when subscribing alarm/event, and it is in XML format.

```
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <ipAddress>10.17.133.46</ipAddress>
  <portNo>80</portNo>
  <protocol>HTTP</protocol>
  <macAddress>44:19:b6:6d:24:85</macAddress>
  <channelID>1</channelID>
  <dateTime>2017-05-04T11:20:02+08:00</dateTime>
  <activePostCount>0</activePostCount>
  <eventType>heartBeat</eventType>
  <eventState>active</eventState>
  <eventDescription>heartBeat</eventDescription>
</EventNotificationAlert>
```

See Also

XML_EventNotificationAlert_AlarmEventInfo

A.67 XML_EventNotificationAlert_TemperatureAlarmMsg

XML message about temperature alarm details

```
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=boundary

--boundary
Content-Type: application/xml; charset="UTF-8"
Content-Length: 1645

<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
  <ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
  <portNo><!--optional, xs:integer, device's communication port No.--></portNo>
  <protocolType><!--optional, xs:string, transport protocol type: HTTP, HTTPS, EHome--></protocolType>
  <macAddress><!--optional, xs:string, MAC address--></macAddress>
  <dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
  <channelID><!--dependent, xs:string, channel No.--></channelID>
  <dateTime>
    <!--required, xs:datetime, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00-->
  </dateTime>
  <activePostCount><!--required, xs:integer, event occurred times--></activePostCount>
  <eventType><!--required, xs:string, event types, here it should be set to "TMA"--></eventType>
  <eventState><!--required, xs:string, event status (for persistent event): active, inactive--></eventState>
  <eventDescription><!--required, xs:string, event description--></eventDescription>
  <DetectionRegionList><!--dependent-->
```

```
<DetectionRegionEntry><!--required-->
  <regionID><!--required, xs:string--></regionID>
  <RegionCoordinatesList><!--optional-->
    <RegionCoordinates><!--optional-->
      <positionX><!--required, xs:integer; x-coordinate--></positionX>
      <positionY><!--required, xs:integer; y-coordinate--></positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <TMA><!--dependent-->
    <thermometryUnit><!--required, xs:string, temperature unit: "celsius, fahrenheit, kelvin"--></thermometryUnit>
  <ruleTemperature><!--required, xs:float, configured temperature threshold--></ruleTemperature>
  <currTemperature><!--required, xs:float, current temperature--></currTemperature>
  <ruleCalibType><!--required, xs:string, "point,line,region"--></ruleCalibType>
  <MaximumTemperaturePoint><!--dependent, it is valid when <ruleCalibType> is "line" or "region"-->
    <RegionCoordinates><!--optional-->
      <positionX><!--required, xs:integer; x-coordinate--></positionX>
      <positionY><!--required, xs:integer; y-coordinate--></positionY>
    </RegionCoordinates>
  </MaximumTemperaturePoint>
  <AbsoluteHigh>
    <elevation><!-- optional, xs:integer, between -900 and 2700--></elevation>
    <azimuth><!--optional, xs:integer, between 0 and 3600--></azimuth>
    <absoluteZoom><!--optional, xs:integer, between 0 and 1000--></absoluteZoom>
  </AbsoluteHigh>
  <presetNo><!--optional, xs:integer, preset No.--></presetNo>
</TMA>
</DetectionRegionEntry>
</DetectionRegionList>
<channelName><!--optional, xs:string, channel name--></channelName>
<deviceId><!--optional, xs:string, device ID--></deviceId>
<detectedPicturesNumber><!--optional, xs:integer--></detectedPicturesNumber>
<visibleLightURL><!--optional, xs:string, visible picture URL--></visibleLightURL>
<thermalURL><!--optional, xs:string, thermal picture URL--></thermalURL>
<thermalInfoURL><!--optional, xs:string, thermal additional information URL--></thermalInfoURL>
<URLCertificationType>
  <!--optional, xs:string, authentication mode of picture URL: no (none authentication), digest (by digest, for local
picture URL returned by NVR or DVR)-->
</URLCertificationType>
</EventNotificationAlert>

--boundary
Content-Disposition: form-data;
Content-Type: image/jpeg
Content-Length: 47432
<Picture data>
.....JFIF.....
...
--boundary--
```

A.68 XML_EventNotificationAlert_TMPAMsg

XML message about temperature measurement precautionary alarm

```
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=boundary

--boundary
Content-Type: application/xml; charset="UTF-8"
Content-Length: 1662

<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
  <ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
  <portNo><!--optional, xs:integer, device's communication port No.--></portNo>
  <protocolType><!--optional, xs:string, transport protocol type: HTTP, HTTPS, EHome--></protocolType>
  <macAddress><!--optional, xs:string, MAC address--></macAddress>
  <dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
  <channelID><!--dependent, xs:string, channel No.--></channelID>
  <dateTime>
    <!--required, xs:datetime, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00-->
  </dateTime>
  <activePostCount><!--required, xs:integer, event occurred times--></activePostCount>
  <eventType><!--required, xs:string, event types, here it should be set to "TMPA"--></eventType>
  <eventState><!--required, xs:string, event status (for persistent event): active, inactive--></eventState>
  <eventDescription><!--required, xs:string, event description--></eventDescription>
  <DetectionRegionList><!--dependent-->
    <DetectionRegionEntry><!--required-->
      <regionID><!--required, xs:string--></regionID>
      <RegionCoordinatesList><!--optional-->
        <RegionCoordinates><!--optional, the coordinates of detected area, in the temperature screening, it is the face
coordinates-->
          <positionX><!--required, xs:integer; x-coordinate--></positionX>
          <positionY><!--required, xs:integer; y-coordinate--></positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    <TMPA><!--temperature measurement precautionary information-->
      <thermometryUnit><!--required, xs:string,"celsius,fahrenheit,kelvin"--></thermometryUnit>
      <ruleTemperature><!--required, xs:float, configured temperature threshold--></ruleTemperature>
      <currTemperature><!--required, xs:float, current temperature--></currTemperature>
      <ruleCalibType><!--required, xs:string, rule calibration type: "point,line,region"--></ruleCalibType>
      <ruleType><!--optional, xs:string, rule type: "highest temp is higher than"-highest temperature is higher than,
"lowest temp is higher than"-lowest temperature is higher than, "average temp is higher than"-average temperature
is higher than, "temp diff is higher than"-temperature difference is higher than, "highest temp is lower than"-highest
temperature is lower than, "lowest temp is lower than"-lowest temperature is lower than, "average temp is lower
than"-average temperature is lower than, "temp diff is lower than"-temperature difference is lower than--></
ruleType>
      <MaximumTemperaturePoint><!--dependent, it is valid when <ruleCalibType> is "line" or "region"-->
      <RegionCoordinates><!--optional-->
        <positionX><!--required, xs:integer; x-coordinate--></positionX>
```

```
<positionY><!--required, xs:integer; y-coordinate--></positionY>
</RegionCoordinates>
</MaximumTemperaturePoint>
<AbsoluteHigh>
  <elevation><!-- optional, xs:integer, between -900 and 2700--></elevation>
<azimuth><!--optional, xs:integer, between 0 and 3600--></azimuth>
<absoluteZoom><!--optional, xs:integer, between 0 and 1000--></absoluteZoom>
</AbsoluteHigh>
<presetNo><!--optional, xs:integer, preset No.--></presetNo>
<alarmRuleTemperature><!--required, xs:float, pre-alarm rule temperature--></alarmRuleTemperature>
</TMPA>
</DetectionRegionEntry>
</DetectionRegionList>
<channelName><!--optional, xs:string, channel name--></channelName>
<deviceId><!--optional, xs:string, device ID--></deviceId>
<detectedPicturesNumber><!--optional, xs:integer, the number of pictures--></detectedPicturesNumber>
<visibleLightURL><!--optional, xs:string, visible picture URL, it is not encrypted--></visibleLightURL>
<thermalURL><!--optional, xs:string, thermal picture URL, it is not encrypted--></thermalURL>
<thermalInfoURL><!--optional, xs:string, thermal additional information URL, it is not encrypted--></thermalInfoURL>
<URLCertificationType><!--optional, xs:string, authentication mode of picture URL: no (none authentication), digest
(by digest, for local picture URL returned by NVR or DVR; the maximum string length is 32 bits--></
URLCertificationType>
</EventNotificationAlert>

--boundary
Content-Disposition: form-data;
Content-Type: image/jpeg
Content-Length: 49164
<Picture data>
.....JFIF.....
.....
--boundary--
```

A.69 XML_EventTrigger

Linkage parameter message in XML format

```
<EventTrigger version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, xs:string, ID--></id>
  <eventType>
    <!--required, xs:string, see details in the "Remarks" below-->
  </eventType>
  <eventDescription><!--optional, xs:string--></eventDescription>
  <inputIOPortID><!--dependent, xs:string, alarm input ID--></inputIOPortID>
  <dynInputIOPortID><!--dependent, xs:string, dynamic alarm input ID--></dynInputPortID>
  <videoInputChannelID>
    <!--dependent, xs:string, video input channel ID, it is valid when <eventType> is "VMD, videoloss, tamperdetection,
regionEntrance, regionExiting, loitering, group, rapidMove, parking, unattendedBaggage, attendedBaggage"-->
  </videoInputChannelID>
  <dynVideoInputChannelID><!--dependent, xs:string, dynamic video input channel ID--></dynVideoInputChannelID>
```

```
<intervalBetweenEvents><!--optional, xs:integer, event time interval, unit: second--></intervalBetweenEvents>
<WLSensorID><!--dependent, xs:string, ID--></WLSensorID>
<EventTriggerNotificationList/><!--optional, alarm/event linkage actions, see details in the message of
XML_EventTriggerNotificationList-->
</EventTrigger>
```

Remarks

The node **<eventType>** can be the following values: IO, VMD, videoloss, raidfailure, recordingfailure, badvideo, POS, analytics, fanfailure, overheat, tamperdetection, diskfull, diskerror, nicbroken, ipconflict, illaccess, videomismatch, resolutionmismatch, radifailure, PIR, WLSensor, spareException, poePowerException, heatmap, counting, linedetection, fielddetection, regionEntrance, regionExiting, loitering, group,rapidMove, parking, unattendedBaggage, attendedBaggage, HUMANATTRIBUTE, blacklist, whitelist, peopleDetection, allVehicleList, otherVehicleList, vehicledetection, storageDetection, shipsDetection, humanAttribute, faceContrast, blacklistFaceContrast, whitelistFaceContrast, faceSnap, faceLib, personDensityDetection, personQueueDetecton, mixedTargetDetection, HVTVehicleDetection, illegalParking, pedestrian, trafficAccident, construction, roadblock, abandonedObject, parallelParking, parkingState, congestion, intersectionAnalysis, heatMap, thermometry, shipsFlowDetection, dredgerDetection, reverseEntrance, luma, highHDTemperature, lowHDTemperature, hdImpact, hdBadBlock, SevereHDFailure, safetyHelmetDetection, vibrationDetection, HBDLib,TMPA,faceThermometry,noMaskDetection

See Also

XML_EventTriggerNotificationList

A.70 XML_EventTriggerList

EventTriggerList message in XML format

```
<EventTriggerList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <EventTrigger/><!--opt, see details in the message of XML_EventTrigger-->
</EventTriggerList>
```

See Also

XML_EventTrigger

Example

XML_EventTriggerList Message Example of Linkage Configurations of Multiple Alarms

```
<EventTriggerList version="2.0" xmlns="http://www.hikvision.com/ver20/XMLSchema">
  <EventTrigger>
    <id>VMD-1</id>
    <eventType>VMD</eventType>
    <eventDescription>VMD Event trigger Information</eventDescription>
    <videoInputChannelID>1</videoInputChannelID>
    <dynVideoInputChannelID>1</dynVideoInputChannelID>
    <EventTriggerNotificationList></EventTriggerNotificationList>
```

```
</EventTrigger>
<EventTrigger>
  <id>tamper-1</id>
  <eventType>tamperdetection</eventType>
  <eventDescription>shelteralarm Event trigger Information</eventDescription>
  <videoInputChannelID>1</videoInputChannelID>
  <dynVideoInputChannelID>1</dynVideoInputChannelID>
  <EventTriggerNotificationList></EventTriggerNotificationList>
</EventTrigger>
<EventTrigger>
  <id>diskfull</id>
  <eventType>diskfull</eventType>
  <eventDescription>exception Information</eventDescription>
  <videoInputChannelID>1</videoInputChannelID>
  <dynVideoInputChannelID>1</dynVideoInputChannelID>
  <EventTriggerNotificationList></EventTriggerNotificationList>
</EventTrigger>
<EventTrigger>
  <id>diskerror</id>
  <eventType>diskerror</eventType>
  <eventDescription>exception Information</eventDescription>
  <videoInputChannelID>1</videoInputChannelID>
  <dynVideoInputChannelID>1</dynVideoInputChannelID>
  <EventTriggerNotificationList>
    <EventTriggerNotification>
      <id>beep</id>
      <notificationMethod>beep</notificationMethod>
      <notificationRecurrence>beginning</notificationRecurrence>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
<EventTrigger>
  <id>nicbroken</id>
  <eventType>nicbroken</eventType>
  <eventDescription>exception Information</eventDescription>
  <videoInputChannelID>1</videoInputChannelID>
  <dynVideoInputChannelID>1</dynVideoInputChannelID>
  <EventTriggerNotificationList></EventTriggerNotificationList>
</EventTrigger>
<EventTrigger>
  <id>ipconflict</id>
  <eventType>ipconflict</eventType>
  <eventDescription>exception Information</eventDescription>
  <videoInputChannelID>1</videoInputChannelID>
  <dynVideoInputChannelID>1</dynVideoInputChannelID>
  <EventTriggerNotificationList></EventTriggerNotificationList>
</EventTrigger>
<EventTrigger>
  <id>illaccess</id>
  <eventType>illaccess</eventType>
  <eventDescription>exception Information</eventDescription>
  <videoInputChannelID>1</videoInputChannelID>
```

```
<dynVideoInputChannelID>1</dynVideoInputChannelID>
<EventTriggerNotificationList></EventTriggerNotificationList>
</EventTrigger>
<EventTrigger>
  <id>videomismatch</id>
  <eventType>videomismatch</eventType>
  <eventDescription>exception Information</eventDescription>
  <videoInputChannelID>1</videoInputChannelID>
  <dynVideoInputChannelID>1</dynVideoInputChannelID>
  <EventTriggerNotificationList>
    <EventTriggerNotification>
      <id>beep</id>
      <notificationMethod>beep</notificationMethod>
      <notificationRecurrence>beginning</notificationRecurrence>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
<EventTrigger>
  <id>badvideo</id>
  <eventType>badvideo</eventType>
  <eventDescription>exception Information</eventDescription>
  <videoInputChannelID>1</videoInputChannelID>
  <dynVideoInputChannelID>1</dynVideoInputChannelID>
  <EventTriggerNotificationList></EventTriggerNotificationList>
</EventTrigger>
<EventTrigger>
  <id>storageDetection-1</id>
  <eventType>storageDetection</eventType>
  <eventDescription>storageDetection Event trigger Information</eventDescription>
  <videoInputChannelID>1</videoInputChannelID>
  <dynVideoInputChannelID>1</dynVideoInputChannelID>
  <EventTriggerNotificationList></EventTriggerNotificationList>
</EventTrigger>
<EventTrigger>
  <id>illegalParking-1</id>
  <eventType>illegalParking</eventType>
  <eventDescription>illegalParking Event trigger Information</eventDescription>
  <videoInputChannelID>1</videoInputChannelID>
  <dynVideoInputChannelID>1</dynVideoInputChannelID>
  <EventTriggerNotificationList>
    <EventTriggerNotification>
      <id>center</id>
      <notificationMethod>center</notificationMethod>
      <notificationRecurrence>beginning</notificationRecurrence>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
<EventTrigger>
  <id>pedestrian-1</id>
  <eventType>pedestrian</eventType>
  <eventDescription>pedestrian Event trigger Information</eventDescription>
  <videoInputChannelID>1</videoInputChannelID>
```

```
<dynVideoInputChannelID>1</dynVideoInputChannelID>
<EventTriggerNotificationList>
  <EventTriggerNotification>
    <id>center</id>
    <notificationMethod>center</notificationMethod>
    <notificationRecurrence>beginning</notificationRecurrence>
  </EventTriggerNotification>
</EventTriggerNotificationList>
</EventTrigger>
<EventTrigger>
  <id>trafficAccident-1</id>
  <eventType>trafficAccident</eventType>
  <eventDescription>trafficAccident Event trigger Information</eventDescription>
  <videoInputChannelID>1</videoInputChannelID>
  <dynVideoInputChannelID>1</dynVideoInputChannelID>
  <EventTriggerNotificationList>
    <EventTriggerNotification>
      <id>center</id>
      <notificationMethod>center</notificationMethod>
      <notificationRecurrence>beginning</notificationRecurrence>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
<EventTrigger>
  <id>construction-1</id>
  <eventType>construction</eventType>
  <eventDescription>construction Event trigger Information</eventDescription>
  <videoInputChannelID>1</videoInputChannelID>
  <dynVideoInputChannelID>1</dynVideoInputChannelID>
  <EventTriggerNotificationList>
    <EventTriggerNotification>
      <id>center</id>
      <notificationMethod>center</notificationMethod>
      <notificationRecurrence>beginning</notificationRecurrence>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
<EventTrigger>
  <id>roadBlock-1</id>
  <eventType>roadBlock</eventType>
  <eventDescription>roadBlock Event trigger Information</eventDescription>
  <videoInputChannelID>1</videoInputChannelID>
  <dynVideoInputChannelID>1</dynVideoInputChannelID>
  <EventTriggerNotificationList>
    <EventTriggerNotification>
      <id>center</id>
      <notificationMethod>center</notificationMethod>
      <notificationRecurrence>beginning</notificationRecurrence>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
<EventTrigger>
```

```
<id>abandonedObject-1</id>
<eventType>abandonedObject</eventType>
<eventDescription>abandonedObject Event trigger Information</eventDescription>
<videoInputChannelID>1</videoInputChannelID>
<dynVideoInputChannelID>1</dynVideoInputChannelID>
<EventTriggerNotificationList>
  <EventTriggerNotification>
    <id>center</id>
    <notificationMethod>center</notificationMethod>
    <notificationRecurrence>beginning</notificationRecurrence>
  </EventTriggerNotification>
</EventTriggerNotificationList>
</EventTrigger>
<EventTrigger>
  <id>parallelParking-1</id>
  <eventType>parallelParking</eventType>
  <eventDescription>parallelParking Event trigger Information</eventDescription>
  <videoInputChannelID>1</videoInputChannelID>
  <dynVideoInputChannelID>1</dynVideoInputChannelID>
  <EventTriggerNotificationList>
    <EventTriggerNotification>
      <id>center</id>
      <notificationMethod>center</notificationMethod>
      <notificationRecurrence>beginning</notificationRecurrence>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
<EventTrigger>
  <id>trafficJam-1</id>
  <eventType>trafficJam</eventType>
  <eventDescription>trafficJam Event trigger Information</eventDescription>
  <videoInputChannelID>1</videoInputChannelID>
  <dynVideoInputChannelID>1</dynVideoInputChannelID>
  <EventTriggerNotificationList>
    <EventTriggerNotification>
      <id>center</id>
      <notificationMethod>center</notificationMethod>
      <notificationRecurrence>beginning</notificationRecurrence>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
</EventTriggerList>
```

A.71 XML_EventTriggerNotification

Event linkage notification message in XML format

```
<EventTriggerNotification><!--opt-->
  <id><!--required, xs:string, device ID--></id>
  <notificationMethod>
```

```
<!--required, xs:string, linkage actions, opt="email,IM,IO,sylog,HTTP,FTP,beep,ptz,record, monitorAlarm, center,
LightAudioAlarm,focus,trace,cloud,SMS,whiteLight,audio,whiteLight,faceContrast"-->
</notificationMethod>
<notificationRecurrence>
  <!--optional, xs:string, "beginning,beginningandend,recurring"-->
</notificationRecurrence>
<notificationInterval><!--dependent, xs:integer, unit: millisecond--></notificationInterval>
<outputIOPortID><!--dependent, xs:string, video output No., it is required only when notificationMethod is "IO"--></
outputIOPortID>
<dynOutputIOPortID><!--dependent, xs:string, dynamic video output No., it is required only when
notificationMethod is "IO"--></dynOutputIOPortID>
<videoInputID><!--dependent, xs:string, video input No., it is required only when notificationMethod is "record"--></
videoInputID>
<dynVideoInputID><!--dependent, xs:string, dynamic video input No., it is required only when notificationMethod is
"record"--></dynVideoInputID>
<ptzAction><!--dependent, it is required only when notificationMethod is "ptz"-->
  <ptzChannelID><!--required, xs:string, PTZ channel ID--></ptzChannelID>
  <actionName><!--required, xs:string, PTZ control type: "preset", "pattern", "patrol"--></actionName>
  <actionNum><!--dependent, xs:integer></actionNum>
</ptzAction>
<WhiteLightAction><!--dependent, white light linkage parameters, this node is valid when notificationMethod is
"whiteLight"-->
  <whiteLightDurationTime><!--required, xs:integer, white light flashing duration, it is between 1 and 60, unit:
second--></whiteLightDurationTime>
</WhiteLightAction>
<cellphoneNumber><!--dependent, xs:string, min="0" max="11",cellphone number--></cellphoneNumber-->
</EventTriggerNotification>
```

A.72 XML_EventTriggerNotificationList

EventTriggerNotificationList message in XML format

```
<EventTriggerNotificationList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <EventTriggerNotification/><!--opt, see details in the message of XML_EventTriggerNotification-->
</EventTriggerNotificationList>
```

See Also

XML_EventTriggerNotification

A.73 XML_EventTriggersCap

XML message about linkage capabilities of different alarm categories

```
<EventTriggersCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <DiskfullTriggerCap><!--optional, xs: EventTriggerCapType--></DiskfullTriggerCap>
  <DiskerrorTriggerCap><!--optional, xs: EventTriggerCapType--></DiskerrorTriggerCap>
  <NicbrokenTriggerCap><!--optional, xs: EventTriggerCapType--></NicbrokenTriggerCap>
  <IpconflictTriggerCap><!--optional, xs: EventTriggerCapType--></IpconflictTriggerCap>
```

```
<IllaccesTriggerCap><!--optional, xs: EventTriggerCapType--></IllaccesTriggerCap>
<BadvideoTriggerCap><!--optional, xs: EventTriggerCapType--></BadvideoTriggerCap>
<VideomismatchTriggerCap><!--optional, xs: EventTriggerCapType--></VideomismatchTriggerCap>
<IOTriggerCap><!--optional, xs: EventTriggerCapType--></IOTriggerCap>
<LineDetectTriggerCap><!--optional, xs: EventTriggerCapType--></LineDetectTriggerCap>
<RegionEntranceTriggerCap><!--optional, xs: EventTriggerCapType--></RegionEntranceTriggerCap>
<RegionExitingTriggerCap><!--optional, xs: EventTriggerCapType--></RegionExitingTriggerCap>
<LoiteringTriggerCap><!--optional, xs: EventTriggerCapType--></LoiteringTriggerCap>
<GroupDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></GroupDetectionTriggerCap>
<RapidMoveTriggerCap><!--optional, xs: EventTriggerCapType--></RapidMoveTriggerCap>
<ParkingTriggerCap><!--optional, xs: EventTriggerCapType--></ParkingTriggerCap>
<UnattendedBaggageTriggerCap><!--optional, xs: EventTriggerCapType--></UnattendedBaggageTriggerCap>
<AttendedBaggageTriggerCap><!--optional, xs: EventTriggerCapType--></AttendedBaggageTriggerCap>
<FireDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></FireDetectionTriggerCap>
<FireDetectionCap><!--optional, xs: EventTriggerCapType--></FireDetectionCap>
<StorageDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></StorageDetectionTriggerCap>
<ShipsDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></ShipsDetectionTriggerCap>
<ThermometryCap><!--optional, xs: EventTriggerCapType--></ThermometryCap>
<VandalProofTriggerCap><!--optional, xs: EventTriggerCapType--></VandalProofTriggerCap>
<BlackListTriggerCap><!--optional, xs: EventTriggerCapType, configuration capability of blacklist arming linkage--></
BlackListTriggerCap>
<WhiteListTriggerCap><!--optional, xs: EventTriggerCapType, configuration capability of whitelist arming linkage--></
WhiteListTriggerCap>
<AllVehicleListTriggerCap><!--optional,xs:EventTriggerCapType, configuration capability of other list arming linkage--
></AllVehicleListTriggerCap>
<OtherVehicleListTriggerCap><!--optional,xs:EventTriggerCapType--></OtherVehicleListTriggerCap>
<PeopleDetectionTriggerCap><!--optional,xs:EventTriggerCapType--></PeopleDetectionTriggerCap>
<PIRAAlarmCap><!--optional, xs: EventTriggerCapType--></PIRAAlarmCap>
<TamperDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></TamperDetectionTriggerCap>
<DefocusDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></DefocusDetectionTriggerCap>
<FaceDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></FaceDetectionTriggerCap>
<SceneChangeDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></SceneChangeDetectionTriggerCap>
<VandalProofAlarmCap><!--optional, xs: EventTriggerCapType--></VandalProofAlarmCap>
<JudgmentTriggerCap><!--optional, xs: EventTriggerCapType--></JudgmentTriggerCap>
<FightingTriggerCap><!--optional, xs: EventTriggerCapType--></FightingTriggerCap>
<RisingTriggerCap><!--optional, xs: EventTriggerCapType--></RisingTriggerCap>
<DozingTriggerCap><!--optional, xs: EventTriggerCapType--></DozingTriggerCap>
<CountingTriggerCap><!--optional, xs: EventTriggerCapType--></CountingTriggerCap>
<VideoLossTriggerCap><!--optional, xs: EventTriggerCapType--></VideoLossTriggerCap>
<HideTriggerCap><!--optional, xs:EventTriggerCapType--></HideTriggerCap>
<AlarmInTriggerCap><!--optional, xs: EventTriggerCapType--></AlarmInTriggerCap>
<VehicleDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></VehicleDetectionTriggerCap>
<AudioExceptionCap><!--optional, xs: EventTriggerCapType--></AudioExceptionCap>
<FiledDetectTriggerCap><!--optional, xs: EventTriggerCapType--></FiledDetectTriggerCap>
<MotionDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></MotionDetectionTriggerCap>
<TemperatureCap><!--optional, xs: EventTriggerCapType--></TemperatureCap>
<IntelligentTriggerCap><!--optional, xs: EventTriggerCapType--></IntelligentTriggerCap>
<FaceContrastTriggerCap><!--optional, xs: EventTriggerCapType, face picture comparison alarm linkage--></
FaceContrastTriggerCap>
<PersonDensityDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></PersonDensityDetectionTriggerCap>
<PersonQueueDetectionTriggerCap><!--optional, xs: EventTriggerCapType, queue management alarm linkage--></
PersonQueueDetectionTriggerCap>
```

```
<WhiteListFaceContrastTriggerCap><!--optional, xs: EventTriggerCapType--></WhiteListFaceContrastTriggerCap>
<HumanRecognitionTriggerCap><!--optional,xs: EventTriggerCapType--></HumanRecognitionTriggerCap>
<FaceSnapTriggerCap><!--optional, xs: EventTriggerCapType--></FaceSnapTriggerCap>
<isSupportWhiteLightAction>
  <!--dependent, xs: boolean, see details in EventTriggerCapType, it is valid when isSupportWhiteLight is "true"-->
</isSupportWhiteLightAction>
<isSupportAudioAction>
  <!--dependent, xs: boolean, see details in EventTriggerCapType, it is valid when isSupportBeep is "true"-->
</isSupportAudioAction>
<HFPDTriggerCap><!--optional, xs: EventTriggerCapType--></HFPDTriggerCap>
<MixedTargetDetectionCap><!--optional, xs: EventTriggerCapType--></MixedTargetDetectionCap>
<HVTVehicleDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></HVTVehicleDetectionTriggerCap>
<VCATriggerCap><!--optional, xs: EventTriggerCapType--></VCATriggerCap>
<PIRCap><!--optional, xs: EventTriggerCapType--></PIRCap>
<IllegalParkingTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports illegal parking detection--></
IllegalParkingTriggerCap>
<PedestrianTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports pedestrian detection--></
PedestrianTriggerCap>
<TrafficAccidentTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports traffic accident detection--></
TrafficAccidentTriggerCap>
<ConstructionTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports construction detection--></
ConstructionTriggerCap>
<RoadBlockTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports roadblock detection--></
RoadBlockTriggerCap>
<AbandonedObjectTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports objects dropped down
detection--></AbandonedObjectTriggerCap>
<ParallelParkingTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports parallel parking detection--></
ParallelParkingTriggerCap>
<ParkingStateTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports parking space status detection,
currently this node is not supported--></ParkingStateTriggerCap>
<CongestionTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports congestion detection--></
CongestionTriggerCap>
<IntersectionAnalysisCap><!--optional, xs: EventTriggerCapType, whether it supports intersection analysis--></
IntersectionAnalysisCap>
<ShipsFlowDetectionTriggerCap><!--optional,xs:EventTriggerCapType, ship flow detection--></
ShipsFlowDetectionTriggerCap>
<dredgerDetectionTriggerCap><!--optional,xs:EventTriggerCapType, dredger detection--></
dredgerDetectionTriggerCap>
<voltageInstableTriggerCap><!--optional,xs:EventTriggerCapType, supply voltage exception--></
voltageInstableTriggerCap>
<HighHDDTemperatureTriggerCap><!--optional, xs:EventTriggerCapType, HDD high temperature detection--></
HighHDDTemperatureTriggerCap>
<LowHDDTemperatureTriggerCap><!--optional, xs:EventTriggerCapType, HDD low temperature detection--></
LowHDDTemperatureTriggerCap>
<HDImpactTriggerCap><!--optional, xs:EventTriggerCapType, HDD impact detection--></HDImpactTriggerCap>
<HDBadBlockTriggerCap><!--optional, xs:EventTriggerCapType, HDD bad sector detection--></
HDBadBlockTriggerCap>
<SevereHDFailureTriggerCap><!--optional, xs:EventTriggerCapType, HDD severe fault detection--></
SevereHDFailureTriggerCap>
<HUMANATTRIBUTECap><!--optional, xs:EventTriggerCapType--></HUMANATTRIBUTECap>
<HumanAttributeTriggerCap><!--optional, xs:EventTriggerCapType, human body attribute--></
HumanAttributeTriggerCap>
```

```
<BlackListFaceContrastTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of blacklist face
comparison--></BlackListFaceContrastTriggerCap>
<FaceLibTriggerCap><!--optional, xs:EventTriggerCapType--></FaceLibTriggerCap>
<SafetyHelmetDetectionTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of hard hat
detection--></SafetyHelmetDetectionTriggerCap>
<VibrationDetectionTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of vibration detection--
></VibrationDetectionTriggerCap>
<RadarLineDetectionTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of radar line crossing
detection--></RadarLineDetectionTriggerCap>
<RadarFieldDetectionTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of radar intrusion
detection--></RadarFieldDetectionTriggerCap>
<HBDLibTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of human body picture library--></
HBDLibTriggerCap>
<FaceThermometryCap><!--optional, xs:EventTriggerCapType--></FaceThermometryCap>
<NoMaskDetectionTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of no wearing mask
detection--></NoMaskDetectionTriggerCap>
<TMPATriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of temperature measurement pre-
alarm--></TMPATriggerCap>
<FireEscapeDetectionTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of fire engine access
detection--></FireEscapeDetectionTriggerCap>
<TakingElevatorDetectionTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of elevator
detection--></TakingElevatorDetectionTriggerCap>
<RuleTriggerCap><!--optional, linkage capability of rule triggered alarm -->
  <isSupportCityManagement>
    <!--optional, xs:boolean, whether the city management supports setting linkage actions by area; if supports, the
value is true, otherwise, this node will not be returned-->
  </isSupportCityManagement>
</RuleTriggerCap>
</EventTriggersCap>
```

See Also

XML_EventTriggerCapType

A.74 XML_EventTriggerCapType

XML message about capability of alarm linkage action types

```
<EventTriggerCapType version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <isSupportCenter><!--optional, xs:boolean--></isSupportCenter>
  <isSupportRecord><!--optional, xs:boolean--></isSupportRecord>
  <isSupportMonitorAlarm><!--optional, xs:boolean--></isSupportMonitorAlarm>
  <isSupportBeep><!--optional, xs: boolean, whether it supports audible warning--></isSupportBeep>
  <isSupportIO><!--optional, xs:boolean--></isSupportIO>
  <isSupportFTP><!--optional, xs:boolean--></isSupportFTP>
  <isSupportEmail><!--optional, xs:boolean--></isSupEmail>
  <isSupportLightAudioAlarm><!--optional, xs:boolean--></isSupportLightAudioAlarm>
  <isSupportFocus><!--optional, xs:boolean--></isSupportFocus>
  <isSupportPTZ><!--optional, xs:boolean--></isSupportPTZ>
  <maxPresetActionNum>
    <!--dependent, xs:integer, it is valid only when <isSupportPTZ> is "true"-->
```

```

</maxPresetActionNum>
<maxPatrolActionNum>
  <!--dependent, xs:integer, it is valid only when <isSupportPTZ> is "true"-->
</maxPatrolActionNum>
<maxPatternActionNum>
  <!--dependent, xs:integer, it is valid only when <isSupportPTZ> is "true"-->
</maxPatternActionNum>
<isSupportTrack><!--optional, xs:boolean, whether it supports PTZ linked tracking--></isSupportTrack>
<isSupportWhiteLight>
  <!--optional, xs: boolean, whether it supports supplement light alarm linkage-->
</isSupportWhiteLight>
<isSupportCloud><!--optional, xs:boolean, whether it supports upload to the cloud--></isSupportCloud>
<targetNotificationInterval max="1000" min="0" default="30"><!--xs:integer, range: [0, 1000], the default value is 30,
unit: seconds, this node is valid for <MotionDetectionTriggerCap> and <TamperDetectionTriggerCap> and this node is
valid when <isSupportPTZ> is "true"--></targetNotificationInterval>
  <direction opt="both,forward,reverse"><!--xs:string, triggering direction, this node is valid for the node
<BlackListTriggerCap>, <WhiteListTriggerCap>, and <VehicleDetectionTriggerCap>--></direction>
  <presetDurationTime min="" max=""><!--dependent, xs:integer--></presetDurationTime>
  <isSupportSMS><!--optional, xs:boolean, whether to support SMS (Short Message Service)--></isSupportSMS>
  <maxCellphoneNum><!--dependent, xs:integer, the maximum number of cellphones, which is node is valid only
when <isSupportSMS> is "true"--></maxCellphoneNum>
  <isSupportOSD><!--optional, xs:boolean--></isSupportOSD>
  <isSupportAudio><!--optional, xs:boolean, whether it supports setting audio alarm independently. If this node is set
to "true", audio alarm and buzzer alarm can be linked separately, and the lineage method is audio--></isSupportAudio>
  <AudioAction><!--dependent, this node is valid when <isSupportBeep> is "true" or <isSupportAudio> is "true"-->
    <audioTypeList>
      <audioType><!--list-->
        <audioID><!--required, xs:integer, alarm sound type--></audioID>
        <audioDescription><!--required, xs:string, alarm sound description, it should correspond to the alarm sound type-->
      </audioType>
    </audioTypeList>
    <alarmTimes opt="0,1,2,3,4,5,6,7,8,9,255"><!--required, xs:integer, alarm times, it is between 0 and 9, 255-
continuous alarm, unit: time--></alarmTimes>
  </AudioAction>
  <isSupportSMS><!--optional, xs:boolean --></isSupportSMS>
  <maxCellphoneNum><!--dependent, if <isSupportSMS> is true, xs:integer--></maxCellphoneNum>
  <isNotSupportCenterModify><!--optional, xs:boolean, whether editing configuration parameters of the surveillance
center is not supported: "true"-yes (configuration parameters of the surveillance center cannot be edited), "false" or
this node is not returned-no (configuration parameters of the surveillance center can be edited)--></
isNotSupportCenterModify>
  <isSupportMessageConfig>
    <!--optional, xs:boolean, whether it supports SMS configuration, if supports, set cellphoneNumber to null-->
  </isSupportMessageConfig>
  <isSupportAnalogOutput><!--optional, xs:boolean, whether it supports IO output of linkage analog channel--></
isSupportAnalogOutput>
  <isSupportIOOutputUnify><!--optional, xs:boolean, whether it supports configuration of IO output--></
isSupportIOOutputUnify>
  <isSupportFaceContrast><!--optional, xs:boolean, whether it supports face picture comparison linkage--></
isSupportFaceContrast>
</EventTriggerCapType>

```

A.75 XML_FaceThermDetectionInfo

XML message about temperature screening result.

```
<FaceThermDetectionInfo version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id> <!--req, xs:integer, rule ID--> </id>
  <name> <!--req, xs:string, rule name--> </name>
  <time> <!--req, xs:ISO8601_time, detection time--> </time>
  <faceDetectionState>
    <!--req, xs:boolean, face detection status: false-no face detected, true-face detected-->
  </faceDetectionState>
  <alarmTemperature>
    <!--opt, xs:float, alarm triggered temperature, ranges from 20.0°C to 150.0°C-->
  </alarmTemperature>
  <visibleLightImageLen><!--opt, xs:integer, the size of visible light image binary data--></visibleLightImageLen>
  <faceImageLen>
    <!--opt, xs:integer, size of face thumbnail binary data, that is the size of Opaque Data(visibleLightImage), if no
Opaque Data(visibleLightImage), it values 0-->
  </faceImageLen>
  <FaceRect>
    <!--opt, face thumbnail coordinates-->
    <height><!--req, xs:float--></height>
    <width><!--req, xs:float--></width>
    <x><!--req, xs:float--></x>
    <y><!--req, xs:float--></y>
  </FaceRect>
  <minTemperature> <!--opt, xs:float, -20.0~150.0°C--> </minTemperature>
  <ruleTemperature> <!--opt, xs:float, 0.0°C~60.0°C--> </ruleTemperature>
  <averageTemperature> <!--opt, xs:float, -20.0~150.0°C--> </averageTemperature>
  <MinTemperaturePoint>
    <!--opt, the coordinates of lowest temperature position-->
    <x><!--req, xs:float--></x>
    <y><!--req, xs:float--></y>
  </MinTemperaturePoint>
  <MaxTemperaturePoint>
    <!--opt, the coordinates of highest temperature position-->
    <x><!--req, xs:float--></x>
    <y><!--req, xs:float--></y>
  </MaxTemperaturePoint>
  <alarmRule>
    <!--opt, xs:string, alarm rule: highestGreater-max. temperature higher than, highestLess-min. temperature lower
than-->
  </alarmRule>
  <highTemperatureTargetImageLen><!--opt, xs:integer--></ highTemperatureTargetImageLen>
  <HighTemperatureTargetRect>
    <!--opt, the coordiantes of high object thumbnail-->
    <height><!--req, xs:float--></height>
    <width><!--req, xs:float--></width>
    <x><!--req, xs:float--></x>
    <y><!--req, xs:float--></y>
```

```
</HighTemperatureTargetRect>  
</FaceThermDetectionInfo>
```

Example

Transmit Result Information in Form Format

```
Accept: text/html, application/xhtml+xml, */*  
Accept-Language: en-US  
User-Agent: Mozilla/5.0 (Windows NT 6.1; WOW64; Trident/7.0; rv:11.0) like Gecko  
Accept-Encoding: gzip, deflate  
Host: 10.17.133.46  
DNT: 1  
Connection: Keep-Alive  
Cookie: language=zh; sdMarkMenu=8%3Avehicle; sdMarkTab_1_0=0%3AsettingBasic;  
sdMarkTab_6_0=5%3AeventException; sdMarkTab_6_1=6%3AsmartLoiterDetection;  
sdMarkTab_7_0=1%3AplanCapture; sdMarkTab_7_1=0%3AstorageManageHarddisk;  
sdMarkTab_8=0%3AvehicleParam; WebSession=f81610c130711300cf30  
  
HTTP/1.1 200 OK  
MIME-Version: 1.0  
Content-Type: multipart/mixed; boundary=boundary  
  
--boundary  
Content-Type: application/xml; charset="UTF-8"  
Content-Length: 480  
  
<FaceThermDetectionInfo>  
--boundary  
Content-Disposition: form-data;  
Content-Type: image/jpeg  
Content-Length: 480  
  
Opaque Data(visibleLightImage)  
--boundary  
Content-Disposition: form-data;  
Content-Type: image/jpeg  
Content-Length: 480  
  
Opaque Data(facelImage)  
--boundary  
Content-Disposition: form-data;  
Content-Type: image/jpeg  
Content-Length: 480  
  
Opaque Data(highTemperatureTargetImage)// High temperature object thumbnail  
--boundary--
```

A.76 XML_FaceThermometry

XML message about temperature screening parameters

```
<FaceThermometry version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, xs:integer, channel No.--></id>
  <faceThermometryEnabled><!--required, xs:boolean, whether enables temperature screening: true=yes, false=no--></faceThermometryEnabled>
  <thermometrShowEnabled><!--required, xs:boolean, whether enables temperature OSD: true=yes, false=no--></thermometrShowEnabled>
  <alarmEnabled><!--required, xs:bool, enable alarm subscription, opt="true,false"--></alarmEnabled>
  <alarmIntervalTime min="0.5" max="600" def="1"><!--dep, xs:float, alarm interval time, which is valid in non-card mode, the interval is between 0.5 s and 60 s. The default time interval is 1s, unit:s--></alarmIntervalTime>
  <normalizedScreenSize><!--required, read-only, read-only, it is the multiples of normalized coordinates returned by device-->
    <normalizedScreenWidth><!--required, read-only, xs:integer--></normalizedScreenWidth>
    <normalizedScreenHeight><!--required, read-only, xs:integer--></normalizedScreenHeight>
  </normalizedScreenSize>
  <FaceThermometryRegionList/><!--optional, temperature screening rule list-->
  <imageQuality><!--optional, xs:string, image quality, including high, medium, and low, "high,medium,low"--></imageQuality>
  <mode><!--optional, xs:string, temperature screening mode: "barrierPassing"-barrier passing (transmit the custom in airport), "targeting"-targeting (detect the person with high temperature)--></mode>
  <faceSnapUploadEnabled><!--optional, xs:boolean, whether to enable uploading captured face picture: true-enable, false-disable--></faceSnapUploadEnabled>
  <maxTemperatureCoordinatesEnabled><!--optional, xs:boolean, whether to enable displaying the maximum temperature position: true-enable, false-disable--></maxTemperatureCoordinatesEnabled>
  <faceRectShowEnabled><!--optional, xs:boolean, whether to enable displaying a frame on the target person: true-enable, false:disable--></faceRectShowEnabled>
  <faceTemperatureShowEnabled><!--optional, xs:boolean, whether to enable displaying face temperature: true-enable, false:disable--></faceTemperatureShowEnabled>
</FaceThermometry>
```

A.77 XML_FaceThermometryRegionList

XML message about temperature screening rule list

```
<FaceThermometryRegionList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ThermometryRegion/><!--opt, single rule information of temperature screening-->
</FaceThermometryRegionList>
```

A.78 XML_FaceThermometryScheduleList

FaceThermometryScheduleList message in XML format.

```
<FaceThermometryScheduleList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <Schedule/><!--opt, see details in XML_Schedule-->
</FaceThermometryScheduleList>
```

See Also

XML_Schedule

A.79 XML_FireDetection

FireDetection message in XML format

```
<FireDetection version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <enabled><!--req, xs:boolean--></enabled>
  <sensitivity><!--req, xs:integer, ranges from 1 to 100--></sensitivity>
  <fireComfirmTime><!--opt, xs:integer, ranges from 0 to 120--></fireComfirmTime>
  <fireRegionOverlay><!--opt, xs:boolean--></fireRegionOverlay>
  <detectionMode><!--opt, xs:string, "multipleFarme, singleFarme"--></detectionMode>
  <fireFocusMode><!--opt, xs:string, "auto, cruise"--></fireFocusMode>
  <FireZoom>
    <zoomMode><!--req, xs:string, "auto, manual"--></zoomMode>
    <zoomLevel><!--dep, xs:integer, ranges from 1 to 100--></zoomLevel>
  </FireZoom>
  <AlarmStrategy>
    <strategyType><!--req, xs:string, "any, cooperate, multisystem, appointFire, appointSmoke"--></strategyType>
    <alarmType><!--dep, xs:string, "fire, smoke"--></alarmType>
  </AlarmStrategy>
  <SmokeDetection>
    <enabled><!--req, xs:boolean--></enabled>
    <sensitivity><!--opt, xs:integer, ranges from 1 to 100--></sensitivity>
    <patrolSensitivity><!--opt, xs:integer, sensitivity of patrol detection, ranges from 1 to 100--></patrolSensitivity>
    <doubleCheckSensitivity><!--opt, xs:integer, sensitivity of double filtering, ranges from 1 to 100--></
doubleCheckSensitivity>
    <displaySmokeInfoOnStreamEnabled><!--opt, xs:boolean, overlay smoke information on stream--></
displaySmokeInfoOnStreamEnabled>
  </SmokeDetection>
  <smokeFireEnabled><!--req, xs:boolean--></smokeFireEnabled>
  <ApplicationScene>
    <mode><!--opt, xs:string. "forest-Fire_Prevention, strawBurning, high-building, Indoor/Perimeter"--></mode>
    <InstallationHeight><!--xs:interger, it is valid only when mode is "strawBurning", "high-building", or "Indoor/
Perimeter", ranges from 1 to 500, unit: m--></InstallationHeight>
  </ApplicationScene>
  <cancelRepeatedAlarmEnabled>
    <!--opt, xs:boolean, cancel repeated alarm, it is valid only when detectionMode is "multipleFarme"-->
  </cancelRepeatedAlarmEnabled>
  <displayFireInfoOnStreamEnabled><!--opt, xs:boolean, overlay fire source information on stream--></
displayFireInfoOnStreamEnabled>
  <smokeAuxiliaryDetectionEnabled>
    <!--dep, xs:boolean, enable fire and smoke detection or not, it is valid only when detectionMode is
"multipleFarme"-->
  </smokeAuxiliaryDetectionEnabled>
  <verificationSensitivity>
    <!--opt, xs:integer, sensitivity of double verification, ranges from 1 to 100-->
  </verificationSensitivity>
  <fireAlgorithmMode>
    <!--opt, xs:string, fire detection algorithm mode: "patternRecognition"-pattern recognition, "machineLearning"-
machine learning-->
  </fireAlgorithmMode>
  <agriculturalMachineryFilterEnabled>
```

```
<!--opt, xs:boolean, enable agricultural machinery filter-->
</agriculturalMachineryFilterEnabled>
<waterReflectionEnabled><!--opt, xs:boolean, enable water reflection--></waterReflectionEnabled>
<patrolSensitivity>
  <!--opt, xs:integer, patrol sensitivity, only valid for fire detection, ranges from 1 to 100-->
</patrolSensitivity>
<fireManualWaitEnabled opt="true,false"><!--opt,xs:boolean,--></fireManualWaitEnabled>
<isSupportFireScanStart opt="true,false"><!--opt,xs:boolean, whether the fire continue scan command is supported,
related URI : /ISAPI/Thermal/channels/<ID>/fireScanStart--></isSupportFireScanStart>
  <isSupportFireScanState opt="true,false"><!--opt,xs:boolean, whether the fire scan status command is supported, ,
related URI : /ISAPI/Thermal/channels/<ID>/fireScanState--></isSupportFireScanState>
</FireDetection>
```

A.80 XML_HistoryTemperatureCap

HistoryTemperatureCap message in XML format

```
<HistoryTemperatureCap version="2.0" xmlns="http://www.hikvision.com/ver20/XMLSchema">
  <id min="1" max="40">
    <!--req, xs:integer, preset ID-->
  </id>
  <thermometryUnit opt = "celsius,fahrenheit,kelvin">
    <!--req, temperature measurement unit, xs:string:"celsius,fahrenheit,kelvin"-->
  </thermometryUnit>
  <ruleId min = "1" max = "10">
    <!--req, xs:integer -->
  </ruleId>
  <tempValue min="-40.0" max="1000.0">
    <!--req, xs: float ro-->
  </tempValue>
</HistoryTemperatureCap>
```

A.81 XML_HistoryTemperatureDescription

HistoryTemperatureDescription message in XML format

```
<HistoryTemperatureDescription version="2.0" xmlns="http://www.hikvision.com/ver20/XMLSchema">
  <id><!--req, xs:integer, preset ID--></id>
  <startTime><!--req,xs:time, ISO8601 time--></startTime>
  <RuleIdList><!--req-->
    <ruleId><!-- req ,xs:integer--></ruleId>
  </RuleIdList>
</HistoryTemperatureDescription>
```

A.82 XML_HistoryTemperatureResult

HistoryTemperatureResult message in XML format

```
<HistoryTemperatureResult version="2.0" xmlns="http://www.hikvision.com/ver20/XMLSchema">
  <thermometryUnit>
    <!--req, xs:string, "celsius, fahrenheit, kelvin"-->
  </thermometryUnit>
  <RuleInfoList><!--req-->
    <RuleInfo>
      <ruleId><!--req, xs:integer--></ruleId>
      <TempInfoList>
        <TempInfo>
          <time><!--req, xs:time, ISO8601 time--></time>
          <tempValue>
            <!--dep, xs:float, "-40.0 .. 1000.0", corrects to one decimal place, ro-->
          </tempValue>
        </TempInfo>
      </TempInfoList>
    </RuleInfo>
  </RuleInfoList>
</HistoryTemperatureResult>
```

A.83 XML_IntelliCap

XML message about intelligent capability

```
<IntelliCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <isFaceSupport><!--opt, xs:boolean, whether it supports face detection--></isFaceSupport>
  <isBehaviorSupport><!--opt, xs:boolean, whether it supports behavior analysis--></isBehaviorSupport>
  <isLineDetectionSupport><!--opt, xs:boolean, whether it supports line crossing detection--></
isLineDetectionSupport>
  <isFieldDetectionSupport><!--opt, xs:boolean, whether it supports intrusion detection--></isFieldDetectionSupport>
  <isRegionEntranceSupport><!--opt, xs:boolean, whether it supports region entrance detection--></
isRegionEntranceSupport>
  <isRegionExitingSupport><!--opt, xs:boolean, whether it supports region exiting detection--></
isRegionExitingSupport>
  <isLoiteringSupport><!--opt, xs:boolean, whether it supports loitering detection--></isLoiteringSupport>
  <isGroupSupport><!--opt, xs:boolean, whether it supports people gathering detection--></isGroupSupport>
  <isRapidMoveSupport><!--opt, xs:boolean, whether it supports fast moving detection--></isRapidMoveSupport>
  <isParkingSupport><!--opt, xs:boolean, whether it supports parking detection--></isParkingSupport>
  <isUnattendedBaggageSupport><!--opt, xs:boolean, whether it supports unattended baggage detection--></
isUnattendedBaggageSupport>
  <isAttendedBaggageSupport><!--opt, xs:boolean, whether it supports object removal detection--></
isAttendedBaggageSupport>
  <isTeacherSupport><!--opt, xs:boolean, whether it supports teacher's behavior detection--></isTeacherSupport>
  <isStudentSupport><!--opt, xs:boolean, whether it supports student's behavior detection--></isStudentSupport>
  <isCombinedSupport><!--opt, xs:boolean, whether it supports combined detection--></isCombinedSupport>
  <isTrafficSupport><!--opt, xs:boolean, whether it supports traffic detection--></isTrafficSupport>
```

```
<RestoreLib>
  <libName opt="studentsStoodUp,peopleCounting,shipDetection"><!--opt,xs:string,--></libName>
</RestoreLib>
<RestoreDefParamForbid>
  <!--opt, xs:boolean "true, false", forbid restoring algorithm library to default. When the node does not exist, it
indicates restoring to default is not supported. When the node exists and its value is "ture", it indicates support, when
the value is "false", it indicates not support-->
</RestoreDefParamForbid>
<RestoreAlgLibParam>
  <!--opt, the capabilities of restoring algorithm library parameters according to algorithm library name-->
  <libName opt="faceSnap,HMS,behaviorAnalysis"><!--opt, xs:string--></libName>
</RestoreAlgLibParam>
<isFaceCaptureStatisticsSupport><!--whether it supports face picture statistics--></isFaceCaptureStatisticsSupport>
<isSupportPersonQueueDetection><!--whether it supports queue management--></
isSupportPersonQueueDetection>
<isSupportIntersectionAnalysis>
  <!--opt, xs: boolean, whether it supports intersection analysis-->
</isSupportIntersectionAnalysis>
<mixedTargetDetectionWithoutAttribute><!--opt, xs: boolean--></mixedTargetDetectionWithoutAttribute>
<isSupportUploadFacePictureByForm><!--opt, xs:boolean, whether it supports uploading face pictures by form--></
isSupportUploadFacePictureByForm>
<isSupportUploadFacePictureByUrl><!--opt, xs: boolean, whether it supports uploading face pictures by URL--></
isSupportUploadFacePictureByUrl>
<isSupportUploadHumanPictureByForm><!--opt, xs:boolean, whether it supports uploading human pictures in form--
></isSupportUploadHumanPictureByForm>
<isSupportMisinfoFilterStatisticalMode><!--optioal, xs:boolean, whether it supports statistics of filtering face
comparison--></isSupportMisinfoFilterStatisticalMode>
<isSupportFaceScore><!--opt,xs:boolean, whether it supports face grading configuration (camera)--></
isSupportFaceScore>
<HumanRecognitionModeSearchCap>
  <searchTargetsNumMax><!--opt, xs:integer, maximum number of sample pictures that can be imported for
searching by picture--></searchTargetsNumMax>
  <HumanMode>
    <searchCond opt="age_group,gender,jacet_color,glass,bag,ride,unlimit"/><!--setting multiple search conditions is
supported-->
    <similarity min="0.0" max="100.0"/><!--opt, xs:float, similarity, range: [0.0,100.0]-->
  </HumanMode>
  <positive opt="true,false"/><!--opt, xs:boolean, whether it is false human body recognition alarm: "true"-yes,
"false"-no-->
  <eventType opt="unlimit,humanRecognition">
    <!--opt, xs:string, event type: "unlimit"-no limit, "humanRecognition"-human body detection alarm-->
  </eventType>
  <isSupportMultiChannelSearch>
    <!--opt, xs:boolean, whether it supports multi-channel search-->
  </isSupportMultiChannelSearch>
  <isSupportTotalSearchResult>
    <!--opt, xs:boolean, whether it supports limiting number of results that can be obtained after a single search-->
  </isSupportTotalSearchResult>
</HumanRecognitionModeSearchCap>
<VehicleRecognitionModeSearchCap>
  <searchTargetsNumMax><!--opt, xs:integer, maximum number of sample pictures that can be imported for
searching by picture--></searchTargetsNumMax>
```

```
<eventType opt="unlimit,vehicleBlackList,vehicleWhiteList"/><!--xs:string, event type: "unlimit"-no limit,
"vehicleBlackList"-vehicle blacklist, "vehicleWhiteList"-vehicle whitelist-->
<VehicleMode>
  <searchCond opt="licensePlate,vehicleLogo,vehicleSubLogoRecog,vehicleType,vehicleColor,unlimit"/>
  <similarity min="0.0" max="100.0"/><!--opt, xs:float, similarity, range: [0.0,100.0]-->
</VehicleMode>
<isSupportMultiChannelSearch>
  <!--opt, xs:boolean, whether it supports multi-channel search-->
</isSupportMultiChannelSearch>
<isSupportTotalSearchResult>
  <!--opt, xs:boolean, whether it supports limiting number of results that can be obtained after a single search-->
</isSupportTotalSearchResult>
</VehicleRecognitionModeSearchCap>
<FaceContrastAnalyzeCap><!--capability of manual comparison and analysis of face pictures-->
  <isSupportTrigger opt="false,true"><!--optional, xs:boolean--></isSupportTrigger>
  <SearchCondition><!--optional-->
    <startTime><!--required, xs:time, ISO8601 time--></startTime>
    <endTime><!--required, xs:time, ISO8601 time--></endTime>
  </SearchCondition>
</FaceContrastAnalyzeCap>
<FramesPeopleCountingCap><!--capability of people counting statistics in a single frame-->
  <Statistics>
    <SearchCondition>
      <startTime><!--required, xs:time,ISO8601 time--></startTime>
      <endTime><!--required, xs:time,ISO8601 time--></endTime>
    </SearchCondition>
  </Statistics>
  <MaskRegion><!--optional, shielded region-->
    <maxRegionNum><!--optional, xs:integer, number of regions--></maxRegionNum>
    <Region>
      <vertexNum min="3" max="10"><!--optional, xs:integer, number of region vertexes--></vertexNum>
    </Region>
  </MaskRegion>
</FramesPeopleCountingCap>
<FaceContrastPersonInfoExtend><!--configuration capability of face comparison tag-->
  <personInfoCap>
    <maxPersonInfo min="0" max="4">
      <!--opt, xs:integer, maximum number of person tags-->
    </maxPersonInfo>
  </personInfoCap>
  <personInfoFDlibCap>
    <maxPersonInfo min="0" max="4">
      <!--opt, xs:integer, maximum number of person tags-->
    </maxPersonInfo>
  </personInfoFDlibCap>
</FaceContrastPersonInfoExtend>
<isSupportSafetyHelmetDetection>
  <!-- opt, xs:boolean, whether it supports hard hat detection-->
</isSupportSafetyHelmetDetection>
<isSupportDistanceRecognition><!--optional, xs:boolean, whether supports distance measurement--></
isSupportDistanceRecognition>
<isSupportATM><!--optional, xs:boolean, ATM intelligent configuraiton, corresponds to URI: /ISAPI/Intelligent/ATM/
```

```
capabilities?format=json--></isSupportATM>
<isSupportFaceTemperature><!--optional, xs:boolean, whether the device supports face thermography, corresponds
to URI: /ISAPI/Intelligent/faceTemperature/capabilities?format=json--></isSupportFaceTemperature>
<isSupportFireEscapeDetectionSearch>
<!--optional, xs:boolean, whether the device supports searching for fire engine access detection events of multiple
channels (related URI: URI/ISAPI/Intelligent/fireEscapeDetection/search/capabilities?format=json)-->
</isSupportFireEscapeDetectionSearch>
<isSupportTakingElevatorDetectionSearch>
<!--optional, xs:boolean, whether the device supports searching for elevator detection events of multiple channels
(related URI: /ISAPI/Intelligent/takingElevatorDetection/search/capabilities?format=json)-->
</isSupportTakingElevatorDetectionSearch>
<isSupportCityManagementSearch>
<!--optional, xs:boolean, whether the device supports searching for city management events of multiple channels
(related URI: /ISAPI/Intelligent/cityManagement/search/capabilities?format=json)-->
</isSupportCityManagementSearch>
<isSupportSafetyHelmetDetectionSearch>
<!--optional, xs:boolean, whether the device supports searching for hard hat detection events of multiple channels
(related URI: /ISAPI/Intelligent/safetyHelmetDetection/search/capabilities?format=json)-->
</isSupportSafetyHelmetDetectionSearch>
</IntelliCap>
```

A.84 XML_PixelToPixelParam

XML message about parameters of pixel-to-pixel temperature measurement

```
<PixelToPixelParam version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:integer, channel No.--></id>
  <maxFrameRate>
    <!--opt, xs: integer, maximum frame rate-->
  </maxFrameRate>
  <reflectiveEnable>
    <!--opt, xs: boolean, enable reflection temperature or not-->
  </reflectiveEnable>
  <reflectiveTemperature>
    <!--opt, xs: float, reflection temperature-->
  </reflectiveTemperature>
  <emissivity>
    <!--opt, xs: float, emissivity-->
  </emissivity>
  <distance>
    <!--req, xs: integer, unit: m, distance-->
  </distance>
  <refreshInterval>
    <!--opt, xs: integer, the refresh interval of pixel-to-pixel temperature measurement-->
  </refreshInterval>
  <distanceUnit>
    <!--opt, xs:string, distance unit, opt="meter,feet,centimeter"-->
  </distanceUnit>
  <temperatureDataLength>
    <!--opt, xs:integer, temperature data length-->
```

```
</temperatureDataLength>
<JpegPictureWithAppendData>
  <!--opt, configuration of getting JPEG picture with pixel-to-pixel temperature measurement data-->
  <jpegPicEnabled>
    <!--opt, xs:boolean, whether returns JPEG picture-->
  </jpegPicEnabled>
</JpegPictureWithAppendData>
</PixelToPixelParam>
```

A.85 XML_PixelToPixelParamCap

XML message about capability of pixel-to-pixel temperature measurement

```
<PixelToPixelParamCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:integer, channel No.--></id>
  <maxFrameRate opt="">
    <!--opt, xs: integer, maximum frame rate-->
  </maxFrameRate>
  <reflectiveEnable opt="true false">
    <!--opt, xs: boolean, enable reflection temperature or not-->
  </reflectiveEnable>
  <reflectiveTemperature>
    <!--opt, xs: float, reflection temperature-->
  </reflectiveTemperature>
  <emissivity min="0.01" max="1.00">
    <!--opt, xs: float, emissivity-->
  </emissivity>
  <distance min="0" max="50">
    <!--req, xs: integer, unit: m, distance-->
  </distance>
  <refreshInterval min="1" max="500">
    <!--opt, xs: integer, the refresh interval of pixel-to-pixel temperature measurement-->
  </refreshInterval>
  <distanceUnit opt="meter,feet,centimeter">
    <!--opt, xs:string, distance unit-->
  </distanceUnit>
  <temperatureDataLength opt="2,4">
    <!--opt, xs:integer, temperature data length-->
  </temperatureDataLength>
  <JpegPictureWithAppendData>
    <!--opt, configuration of getting JPEG picture with pixel-to-pixel temperature measurement data-->
    <jpegPicEnabled opt="true,false">
      <!--opt, xs:boolean, whether returns JPEG picture-->
    </jpegPicEnabled>
  </JpegPictureWithAppendData>
</PixelToPixelParamCap>
```

A.86 XML_Power

Power message in XML format

```
<?xml version="1.0" encoding="utf-8"?>
<Power version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <powerSwitch>
    <!--req, xs:boolean, switch on/off: "true"-switch on, wake up device, "false"-switch off, device will be in sleep mode-->
  >
  </powerSwitch>
  <batteryPower><!--opt, ro, xs:integer, battery percentage, range: [1,100]--></batteryPower>
</Power>
```

A.87 XML_PTZPreset

XML message about preset parameters

```
<PTZPreset version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--required, xs:boolean, whether to enable preset configuration--></enabled>
  <id><!--required, xs:string, preset No.--></id>
  <presetName><!--required, xs:string, preset name--></presetName>
  <AbsoluteHigh>
    <elevation><!--optional, xs:integer, tilting parameter, the value is between -900 and 2700--></elevation>
    <azimuth><!--optional, xs:integer, panning parameter, the value is between 0 and 3600--></azimuth>
    <absoluteZoom><!--optional, xs:integer, zooming parameter, the value is between 1 and 1000--></absoluteZoom>
  </AbsoluteHigh>
</PTZPreset>
```

A.88 XML_PTZPresetList

XML message about preset information list

```
<PTZPresetList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <PTZPreset/><!--optional, see details in XML_PTZPreset-->
</PTZPresetList>
```

See Also

XML_PTZPreset

A.89 XML_RegionalExposure

RegionalExposure message in XML format

```
<RegionalExposure version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <StartPoint><!--from 0 to 1000-->
```

```
<positionX><!--req, xs:integer--></positionX>
<positionY><!--req, xs:integer--></positionY>
</StartPoint>
<EndPoint><!--from 0 to 1000-->
  <positionX><!--req, xs:integer--></positionX>
  <positionY><!--req, xs:integer--></positionY>
</EndPoint>
</RegionalExposure>
```

A.90 XML_ResponseStatus

XML message about response status

```
<ResponseStatus version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <requestURL>
    <!--required, read-only, xs:string, request URL-->
  </requestURL>
  <statusCode>
    <!--required, read-only, xs:integer, status code: 0,1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid
XML Format, 6-Invalid XML Content, 7-Reboot Required, 9-Additional Error-->
  </statusCode>
  <statusString>
    <!--required, read-only, xs:string, status description: OK, Device Busy, Device Error, Invalid Operation, Invalid XML
Format, Invalid XML Content, Reboot, Additional Error-->
  </statusString>
  <subStatusCode>
    <!--required, read-only, xs:string, describe the error reason in detail-->
  </subStatusCode>
</ResponseStatus>
```



Note

See **ISAPI Response Codes** for details about sub status codes and corresponding error codes.

A.91 XML_Schedule

Schedule message in XML format

```
<Schedule version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, xs:string, ID--></id>
  <eventType>
    <!--optional, xs:string, alarm/event types, see details in the "Remarks" below-->
  </eventType>
  <inputIOPortID><!--read-only, dependent, xs:string, alarm input No.--></inputIOPortID>
  <outputIOPortID><!--read-only, dependent, xs:string, alarm output No.--></inputIOPortID>
  <videoInputChannelID><!--read-only, dependent, xs:string, video input channel ID--></videoInputChannelID>
  <TimeBlockList size="8"><!--required-->
    <TimeBlock><!--list-->
      <dayOfWeek>
```

```
<!--optional, xs:integer, day of the week based on ISO8601, "1"=Monday, ...-->
</dayOfWeek>
<TimeRange><!--required-->
  <beginTime><!--required, xs:time, ISO8601 time--></beginTime>
  <endTime><!--required, xs:time, ISO8601 time--></endTime>
</TimeRange>
<CustomExtension>
  <vehicleDetectSceneID>
    <!--required, xs:interger-->
  </vehicleDetectSceneID>
</CustomExtension>
</TimeBlock>
</TimeBlockList>
<HolidayBlockList><!--optional-->
  <TimeBlock><!--list-->
    <TimeRange><!--required-->
      <beginTime><!--required, xs:time, ISO8601 time--></beginTime>
      <endTime><!--required, xs:time, ISO8601 time--></endTime>
    </TimeRange>
  </TimeBlock>
</HolidayBlockList>
</Schedule>
```

Remarks

The node **<eventType>** can be set to the following values: IO, VMD, videoloss, PIR, linedetection, fielddetection, audioexception, facedetection, regionEntrance, regionExiting, loitering, group, rapidMove, parking, unattendedBaggage, attendedBaggage, storageDetection, shipsDetection, HUMANATTRIBUTE, humanAttribute, faceContrast, faceSnap, faceLib, whiteListFaceContrast, personDensityDetection, personQueueDetection, mixedTargetDetection, fireDetection, illegalParking, pedestrian, trafficAccident, construction, roadblock, abandonedObject, parallelParking, parkingState, congestion, intersectionAnalysis, heatMap, reverseEntrance, vehicledetect, safetyHelmetDetection, vibrationDetection, TMPA, faceThermometry, HBDLib.

A.92 XML_SubscribeEvent

SubscribeEvent message in XML format

```
<SubscribeEvent version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema" >
  <heartbeat>
    <!--optional, xs:integer, heartbeat interval, unit: second, the default value is 30s-->
  </heartbeat>
  <eventMode>
    <!--required, xs:string, "all"-upload all alarms/events, "list"-upload specified alarm/event-->
  </eventMode>
  <EventList>
    <Event><!--uploading mode of specified alarm/event, this node exists only when eventMode is "list"-->
      <type>
        <!--required, xs:string, alarm/event types, which are obtained from the capability, refer to Alarm/Event Types for Subscription for its values-->
```

```
</type>
<minorAlarm>
  <!--opt, xs:string, minor alarm type: "0x400,0x401,0x402,0x403", see details in Access Control Event Type. This
node is required when type is "AccessControllerEvent"-->
</minorAlarm>
<minorException>
  <!--opt, xs:string, minor exception type: "0x400,0x401,0x402,0x403", see details in Access Control Event Type.
This node is required when type is "AccessControllerEvent"-->
</minorException>
<minorOperation>
  <!--opt, xs:string, minor operation type: "0x400,0x401,0x402,0x403", see details in Access Control Event Type.
This node is required when type is "AccessControllerEvent"-->
</minorOperation>
<minorEvent>
  <!--opt, xs:string, minor event type: "0x01,0x02,0x03,0x04", see details in Access Control Event Type. This node is
required when type is "AccessControllerEvent"-->
</minorEvent>
<pictureURLType>
  <!--opt, xs:string, alarm picture format: "binary"-binary, "localURL"-device local URL, "cloudStorageURL"-cloud
storage URL-->
</pictureURLType>
</Event>
</EventList>
<channels>
  <!--optional, xs:string, event linked channel information, and multiple channels can be linked, each channel is
separated by comma, e.g., "1,2,3,4..."-->
</channels>
<channels>
  <!--optional, xs:string, specify channels (each channel is separated by comma, e.g., "1,2,3,4...") to be armed, this
node does not exist if you want to arm all channels, and if this node exists, the sub node <channels> in the node
<Event> is invalid-->
</channels>
<identityKey max="64"/>
<!--opt, xs: string, interaction command of subscription, supports subscribing comparison results of face picture
library (importing with this command), the maximum length is 64-->
</SubscribeEvent>
```

A.93 XML_SubscribeEventCap

SubscribeEventCap capability message in XML format

```
<SubscribeEventCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <heartbeat min="" max="" />
  <!--optional, heartbeat time interval, unit: second-->
  <format opt="xml,json"/><!--req, supported message format-->
  <channelMode opt="all,list" />
  <!--required, channel subscription mode: "all"-subscribe events/alarms of all channels, "list"-subscribe events/
alarms of specific channels-->
  <eventMode opt="all,list" />
  <!--required, event subscription mode: "all"-subscribe all event types (must be supported), "list"-subscribe specific
```

event types, if "list" is returned, "all" will also be returned-->

<!--if both the **channelMode** and **eventMode** returns "all", it indicates that the device does not support subscribing event/alarm by event type or channel-->

<EventList><!--required, dependent, upload mode of specified alarms/events, it is valid only when **eventMode** is "list"-->

<Event><!--required-->

<type><!--required, xs:string, refer to *Supported Alarm/Event Types* for details--></type>

<minorAlarm opt="0x400,0x401,0x402,0x403">

<!--opt, xs:string, minor alarm type, see details in Access Control Event Type. This node is required when **type** is "AccessControllerEvent"-->

</minorAlarm>

<minorException opt="0x400,0x401,0x402,0x403">

<!--opt, xs:string, minor exception type, see details in Access Control Event Type. This node is required when **type** is "AccessControllerEvent"-->

</minorException>

<minorOperation opt="0x400,0x401,0x402,0x403">

<!--opt, xs:string, minor operation type, see details in Access Control Event Type. This node is required when **type** is "AccessControllerEvent"-->

</minorOperation>

<minorEvent opt="0x01,0x02,0x03,0x04">

<!--opt, xs:string, minor event type, see details in Access Control Event Type. This node is required when **type** is "AccessControllerEvent"-->

</minorEvent>

<pictureURLType opt="binary,localURL,cloudStorageURL" def=""/>

<!--opt, xs:string, alarm picture format: "binary"-binary, "localURL"-device local URL, "cloudStorageURL"-cloud storage URL, and the **def** is followed by the default format-->

</Event>

</EventList>

<pictureURLType opt="binary,localURL,cloudStorageURL" def=""/>

<!--opt, xs:string, alarm picture format: "binary"-binary picture, "localURL"-device local URL, "cloudStorageURL"-cloud storage URL. This node is the method of uploading all pictures related to the event. If this node is applied, **<pictureURLType>** in **<Event>** is invalid; otherwise, pictures will be uploaded using the default method returned by the device capability. For front-end devices, the default method is uploading binary pictures; for back-end devices, the default method is by device local URL-->

<ChangedUploadSub><!--message subscription-->

<interval/><!--opt, xs:integer, lifecycle of arming GUID, the default value is 5 minutes, unit: second. The device will generate new GUID for the arming connection after it is disconnected for the set lifecycle-->

<StatusSub>

<all/><!-- opt, xs:boolean, whether to subscribe all events-->

<channel/><!--opt, xs:boolean, whether to subscribe channel status. This node is not required when **<all>** is "true"-->

<hd/><!--opt, xs:boolean, whether to subscribe disk status. This node is not required when **<all>** is "true"-->

<capability/><!--opt, xs:boolean, whether to subscribe capability change status. This node is not required when **<all>** is "true"-->

</StatusSub>

</ChangedUploadSub>

<identityKey max="64"/>

<!--opt, xs: string, interaction command of subscription, supports subscribing comparison results of face picture library (importing with this command), the maximum length is 64-->

</SubscribeEventCap>

A.94 XML_SubscribeEventResponse

SubscribeEventResponse message in XML format

```
<SubscribeEventResponse>
  <id><!--req, xs:integer, subscription ID--></id>
  <FailedEventList>
    <!--opt, list of subscription failed events. When subscription failed, it should be returned, and the upper layer can
    check whether all event/alarm subscriptions are succeeded via the existence of node FailedEventList-->
    <Event>
      <type>
        <!--req, xs:string, refer to Supported Alarm/Event Types for details-->
      </type>
      <minorAlarm>
        <!--opt, xs:string, minor alarm type: "0x400,0x401,0x402,0x403", see details in Access Control Event Type. This
        node is required when type is "AccessControllerEvent"-->
      </minorAlarm>
      <minorException>
        <!--opt, xs:string, minor exception type: "0x400,0x401,0x402,0x403", see details in Access Control Event Type.
        This node is required when type is "AccessControllerEvent"-->
      </minorException>
      <minorOperation>
        <!--opt, xs:string, minor operation type: "0x400,0x401,0x402,0x403", see details in Access Control Event Type.
        This node is required when type is "AccessControllerEvent"-->
      </minorOperation>
      <minorEvent>
        <!--opt, xs:string, minor event type: "0x01,0x02,0x03,0x04", see details in Access Control Event Type. This node is
        required when type is "AccessControllerEvent"-->
      </minorEvent>
      <pictureURLType>
        <!--opt,xs:string, opt="binary,localURL,cloudStorageURL", alarm picture transmission mode: "binary"-binary,
        "localURL"-device local URL, "cloudStorageURL"-cloud storage URL-->
      </pictureURLType>
      <channels>
        <!--opt, xs:string, "1,2,3,4...", event related channel ID, supports multiple channels, and the channel ID is
        separated by commas-->
      </channels>
      <subStatusCode>
        <!--req, string, subscription failure error code-->
      </subStatusCode>
    </Event>
  </FailedEventList>
</SubscribeEventResponse>
```

A.95 XML_ThermalBlackBody

ThermalBlackBody message in XML format.

```
<ThermalBlackBody version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <emissivity><!--req, xs:float, black body emissivity is between 0.01 and 1, which corrects to two decimal places. The
  default value is 0.97 --></emissivity>
  <distance><!--req, xs: float, the distance between lens and black body, ranges from 0.0 m to 10.0 m, which corrects
  to one decimal place. The default value is 2.0 m, unit:m--></distance>
  <temperature><!--req, xs:float, black body temperature is between 30.0 °C and 50.0 °C, which corrects to one
  decimal place. The default value is 35.0 °C, unit: °C--></temperature>
  <CentrePoint><!--req, center position of black body. After clicking this position, the normalized coordinate
  information (between 0 and 1000) will be applied-->
    <CalibratingCoordinates><!--dep-->
      <positionX><!--req, xs:integer; coordinate--></positionX>
      <positionY><!--req, xs:integer; coordinate--></positionY>
    </CalibratingCoordinates>
  </CentrePoint>
  <normalizedScreenSize><!--req, ro, read-only, it is the multiples of normalized coordinates returned by device-->
    <normalizedScreenWidth><!--req, ro,xs:integer--></normalizedScreenWidth>
    <normalizedScreenHeight><!--req, ro,xs:integer--></normalizedScreenHeight>
  </normalizedScreenSize>
  <enabled opt="true,false"><!--optional, xs:boolean, whether to enable black body--></enabled>
  <BlackBodyReigon><!--optional, black body detection area-->
    <type opt="point,region"><!--required, xs:string, "point, region", area type: "point", "region"--></type>
    <Point><!--dependent, point coordinate, it is valid when value of type is "point"-->
      <CalibratingCoordinates><!--dependent, point coordinate-->
        <positionX><!--required, xs:integer; coordinate--></positionX>
        <positionY><!--required, xs:integer; coordinate--></positionY>
      </CalibratingCoordinates>
    </Point>
    <Region><!--dependent, frame coordinate, it is valid when value of type is "region"-->
      <RegionCoordinatesList size="4"><!--dependent, list of region coordinates-->
        <RegionCoordinates><!--list, optional-->
          <positionX><!--required, xs:integer; X-coordinate--></positionX>
          <positionY><!--required, xs:integer; Y-coordinate--></positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </Region>
  </BlackBodyReigon>
</ThermalBlackBody>
```

A.96 XML_ThermalCap

XML message about thermal capability

```
<ThermalCap version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <isSupportFireDetection><!--optional, xs:boolean, whether supports fire detection--></isSupportFireDetection>
  <isSupportThermometry><!--optional, xs:boolean, whether supports temperature measurement--></
isSupportThermometry>
  <isSupportRealtimeThermometry><!--optional, xs:boolean, whether supports uploading real-time temperature
  measurement data--></isSupportRealtimeThermometry>
  <isFireFocusZoomSupport><!--optional, xs:boolean, whether supports visible light lens zooming--></
isFireFocusZoomSupport>
```

```
<isSupportManualRanging>
  <!--optional, xs:boolean, this node will be returned if one or more channels of device supports this function, see
detailed channel capability in the response information"-->
</isSupportManualRanging>
<isSupportPower><!--optional, xs:boolean, whether supports power on/off capability--></isSupportPower>
<isSupportRealtimeTempHumi><!--optional, xs:boolean, whether supports real-time detection of temperature and
humidity--></isSupportRealtimeTempHumi>
<ManualThermCap>
  <manualThermRuleNum>
    <!--optional, xs:integer, the max. number of supported rules for manual temperature measurement. If this node is
not returned, it indicates manual temperature measurement is not supported-->
  </manualThermRuleNum>
</ManualThermCap>
<isSupportManualThermBasic>
  <!--optional, xs:boolean, whether supports basic configuration of manual temperature measurement-->
</isSupportManualThermBasic>
<isSupportFireShieldMask><!--optional, xs:boolean--></isSupportFireShieldMask>
<isSupportSmokeShieldMask><!--optional, xs:boolean--></isSupportSmokeShieldMask>
<isSupportThermometryMode>
  <!--optional, xs:boolean, whether the device supports the configuration of temperature measurement mode-->
</isSupportThermometryMode>
<isSupportThermalPip>
  <!--optional, xs:boolean, whether the device supports the PIP configuration-->
</isSupportThermalPip>
<isSupportThermalIntelRuleDisplay><!--optional, xs:boolean, whether supports VCA rule configuration--></
isSupportThermalIntelRuleDisplay>
<AlgVersionInfo><!--opt, whether supports getting the version information of thermal algorithms library-->
  <thermometryAlgName min = "1" max = "128">
    <!--read-only, xs:string, version information of temperature measurement algorithms library-->
  </thermometryAlgName>
  <shipsAlgName min = "1" max = "128"><!--read-only, xs:string, version name of ship detection algorithms library--
></shipsAlgName>
</AlgVersionInfo>
<isSupportFaceThermometry><!--optional, xs:boolean, whether supports temperature screening configuration--></
isSupportFaceThermometry>
<isSupportThermalBlackBody><!--optional, xs:boolean, whether supports black body configuration--></
isSupportThermalBlackBody>
<isSupportThermalStreamParam><!--optional, xs:boolean, whether supports stream configuration--></
isSupportThermalStreamParam>
<isSupportBodyTemperatureCompensation>
  <!--optional, xs:boolean, whether supports temperature compensation configuration-->
</isSupportBodyTemperatureCompensation>
<isSupportTemperatureCorrection><!--optional, xs:boolean, whether device supports temperature measurement
correction--></isSupportTemperatureCorrection>
<isSupportClickToThermometry><!--optional, xs:boolean, whether device supports clicking to detect temperature--
></isSupportClickToThermometry>
<isSupportThermometryHistorySearch><!--optional, xs:boolean--></isSupportThermometryHistorySearch>
<isSupportBurningPrevention><!--optional, xs:boolean, whether device supports burning prevention--></
isSupportBurningPrevention>
<isSupportTemperatureCollection><!--optional, xs:boolean, whether device supports temperature ANR--></
isSupportTemperatureCollection>
<isSupportJpegPicWithAppendData>
```

```
<!--optional, xs:boolean, whether device supports getting JPEG picture with pixel-to-pixel temperature
measurement data. If supports, it is returned and values true, if not support, it is not returned-->
</isSupportJpegPicWithAppendData>
<isSupportRealTimethermometryForHTTP>
  <!--optional, xs:boolean, whether device supports real-time temperature measurement. If supports, it is returned
and its value is true, if not support, it is not returned-->
</isSupportRealTimethermometryForHTTP>
<isSupportShipsDetectionWithScene>
  <!--optional, xs:boolean, whether device supports ship detection by scene, this node and isSupportShipsDetection
in XML_SmartCap are mutually exclusive-->
</isSupportShipsDetectionWithScene>
<isSupportthermometryOffLineCapture>
  <!--optional, xs:boolean, whether device supports offline capture. If supports, this node returned and its value is
true; if not, it is not returned-->
</isSupportthermometryOffLineCapture>
<isSupportThermalTemperatureCorrect>
  <!--optional, xs:boolean, whether device supports temperature calibration (related URI: /ISAPI/Thermal/channels/
<ID>/temperatureCorrect?format=json)-->
</isSupportThermalTemperatureCorrect>
<isSupportFaceSnapThermometry>
  <!--optional, xs:boolean, whether device supports uploading captured face picture with temperature information:
true-support, no return-not support-->
</isSupportFaceSnapThermometry>
</ThermalCap>
```

Remarks

When getting thermal product capabilities, **isSupportShipsDetectionWithScene** has a higher priority than **isSupportShipsDetection**. That is, firstly check if the node **isSupportShipsDetectionWithScene** exists and its value is "true", that indicates ship detection according to scene is supported, otherwise, check the if the node **isSupportShipsDetection** exists.

A.97 XML_ThermalDataConversionCap

ThermalDataConversionCap message in XML format

```
<ThermalDataConversionCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <conversionMethod opt="convertGrayscaleToTemperature,convertTemperatureToGrayscale">
    <!--req, xs:string, conversion method-->
  </conversionMethod>
  <temperature min="",max=""><!--opt, xs:float, temperature, accurates to one decimal place--></temperature>
  <grayscale min="",max=""><!--opt, xs:integer, grayscale--></grayscale>
</ThermalDataConversionCap>
```

A.98 XML_ThermalDataConversionDescription

ThermalDataConversionDescription in XML format

```
<ThermalDataConversionDescription version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <conversionMethod>
    <!--req, xs:string, conversion method, opt="convertGrayscaleToTemperature,convertTemperatureToGrayscale"-->
  </conversionMethod>
  <temperature>
    <!--dep, it is valid when conversionMethod is "convertTemperatureToGrayscale", xs:float, temperature, corrects to
one decimal place-->
  </temperature>
  <grayscale>
    <!--dep, it is valid when conversionMethod is "convertGrayscaleToTemperature", xs:integer, grayscale-->
  </grayscale>
</ThermalDataConversionDescription>
```

A.99 XML_ThermalDataConversionResult

ThermalDataConversionResult message in XML format

```
<ThermalDataConversionResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <conversionMethod>
    <!--req, xs:string, conversion method, opt="convertGrayscaleToTemperature,convertTemperatureToGrayscale"-->
  </conversionMethod>
  <temperature>
    <!--dep, it is valid when conversionMethod is "convertGrayscaleToTemperature", xs:float, temperature, corrects to
one decimal place-->
  </temperature>
  <grayscale>
    <!--dep, it is valid when conversionMethod is "convertTemperatureToGrayscale", xs:integer, grayscale-->
  </grayscale>
</ThermalDataConversionResult>
```

A.100 XML_ThermalPip

ThermalPip message in XML format

```
<ThermalPip version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs:boolean, enable or not--></enabled>
  <pipMode>
    <!--req, xs:string, picture-in-picture mode: overlap-overlay mode, fusion-integration mode, normal-normal mode-->
  </pipMode>
  <overlapType>
    <!--req, xs:string, overlay type: visibleOverlapThermal-overlay visible light on thermal imaging picture,
thermalOverlapVisible-overlay thermal imaging on visible light-->
  </overlapType>
  <transparency><!--opt, xs: integer, transparency--></transparency>
  <normalizedScreenSize><!--req, ro, screen with normalized coordinates-->
    <normalizedScreenWidth><!--req, ro,xs:integer--></normalizedScreenWidth>
    <normalizedScreenHeight> <!--req, ro,xs:integer--></normalizedScreenHeight>
  </normalizedScreenSize>
```

```
<PipRegion><!--dep, picture-in-picture area-->
  <RegionCoordinatesList><!--dep, coordinates list of picture-in-picture area-->
    <RegionCoordinates><!--opt, picture-in-picture area coordinate-->
      <positionX><!--req, xs:integer;x-coordinate--></positionX>
      <positionY><!--req, xs:integer;y-coordinate--></positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</PipRegion>
<imageFusionRatio>
  <!--dep, xs: integer, image fusion ratio, it is valid when pipMode is "fusion", ranges from 0 to 100, default value: 75-->
</imageFusionRatio>
<borderFusionRatio>
  <!--dep, xs: integer, border fusion ratio, it is valid when pipMode is "fusion", ranges from 0 to 100, default value: 100-->
</borderFusionRatio>
<distance><!--dep, xs:float, fusion distance, it is valid when pipMode is "fusion", ranges from 0.1 m to 4.0 m--></distance>
</ThermalPip>
```

A.101 XML_ThermalStreamParam

ThermalStreamParam message in XML format.

```
<ThermalStreamParam version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <videoCodingType>
    <!--opt, xs:string, coding type, opt="thermal_raw_data, pixel-to-pixel_thermometry_data, real-time_raw_data"-->
  </videoCodingType>
</ThermalStreamParam>
```

A.102 XML_ThermalStreamParamCap

ThermalStreamParamCap message in XML format.

```
<ThermalStreamParamCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <videoCodingType opt="thermal_raw_data, pixel-to-pixel_thermometry_data, real-time_raw_data"
def="thermal_raw_data">
    <!--opt, xs:string, coding type: "thermal_raw_data"-original data, "pixel-to-pixel_thermometry_data"-pixel-to-pixel
temperature measurement data, "real-time_raw_data"-real-time raw data-->
  </videoCodingType>
</ThermalStreamParamCap>
```

A.103 XML_ThermIntell

ThermIntell message in XML format.

```
<ThermIntell version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, xs:integer--></id>
  <intellType>
    <!--required, xs:string, intelligent function types, each two types are mutually exclusive, "thermometryAndSmart"-
    temperature measurement+behavior analysis (default), "shipsDetection"-ship detection, "fireDetection"-fire source
    detection, "pip"-picture in picture function, "faceThermometry"-temperature screening,
    "thermometryAndSmokeFireDetection"-temperature measurement+smoke and fire source detection,
    "thermometryAndFireDetection"-temperature measurement+fire source detection, "basicBehavior"-behavior
    analysis, "thermometry"-temperature measurement-->
  </intellType>
</ThermIntell>
```

A.104 XML_ThermometryAlarmRule

ThermometryAlarmRule message in XML format.

```
<ThermometryAlarmRule version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ThermometryAlarmModeList size="">
    <ThermometryAlarmMode>
      <id min="" max=""><!--req, xs:inter--></id>
      <enabled><!--req, xs:boolean--></enabled>
      <name><!--req, xs:string ro--></name>
      <rule opt="highestGreater, highestLess, lowestGreater, lowestLess, averageGreater, averageLess, diffTempGreater,
diffTempLess">
        <!--req, xs:string,-->
      </rule>
      <alert><!--req, xs: float--></alert>
      <alarm><!--req, xs: float--></alarm>
      <threshold><!--req, xs: float--></threshold>
      <AlertOutputIOPortList><!--opt, temperature pre-alarm I/O port list-->
        <OutputIOPort>
          <portID><!--req, xs:string--></portID>
          <enabled><!--req, xs:bool, "true,false"--></enabled>
        </OutputIOPort>
      </AlertOutputIOPortList>
      <AlarmOutputIOPortList><!--opt, temperature alarm I/O port list-->
        <OutputIOPort>
          <portID><!--req, xs:string--></portID>
          <enabled><!--req, xs:bool, "true,false"--></enabled>
        </OutputIOPort>
      </AlarmOutputIOPortList>
      <alertFilteringTime><!-- opt ,xs:integer, temperature pre-alarm dwell time, unit: second--></alertFilteringTime>
      <alarmFilteringTime><!-- opt ,xs:integer, temperature alarm dwell time, unit: second--></alarmFilteringTime>
    </ThermometryAlarmMode>
  </ThermometryAlarmModeList>
  <TemperatureDifferenceComparisonList size="">
    <TemperatureDifferenceComparison>
      <id min="" max=""><!--req, xs:inter--></id>
      <enabled><!--req, xs:boolean--></enabled>
      <ruleID1 min="" max=""><!--req, xs:string--></ruleID1>
```

```
<ruleID2 min="" max=""><!--req, xs:string--></ruleID2>
<rule opt="highestGreater, highestLess, lowestGreater, lowestLess, averageGreater, averageLess, diffTempGreater,
diffTempLess">
  <!--req, xs:string,-->
</rule>
<temperatureDifference><!--req, xs: float--></temperatureDifference>
</TemperatureDifferenceComparison>
</TemperatureDifferenceComparisonList>
</ThermometryAlarmRule>
```

A.105 XML_ThermometryBasicParam

XML message about temperature measurement basic parameters

```
<ThermometryBasicParam version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, xs:integer, channel number--></id>
  <enabled><!--required, xs:boolean, enable or not: false-no, true-yes--></enabled>
  <streamOverlay><!--required, xs:boolean, whether displays temperature information on the stream: false-no, true-yes--></streamOverlay>
  <pictureOverlay><!--required, xs:boolean, whether displays temperature information on the captured picture: false-no, true-yes--></pictureOverlay>
  <temperatureRange>
    <!--required, xs:string, temperature range,
    "-20-150,0-550,0-650,-4-302,32-1022,32-1200,-20-650,-20-1500,automatic,-20-120,20-350,20-45"-->
  </temperatureRange>
  <temperatureUnit>
    <!--required, xs:string, temperature unit: degreeCentigrade-Celsius (°C), degreeFahrenheit-Fahrenheit (°F),
    degreeKelvin-Kelvin(K)-->
  </temperatureUnit>
  <temperatureCurve>
    <!--optional, xs:string, temperature curve modes: close-closed, transverseTemperatureTrend-transverse
    temperature mode, longitudinalTemperatureTrend-longitudinal temperature mode-->
  </temperatureCurve>
  <fireImageMode>
    <!--optional, xs:string, fire detection modes: blackWhite-black and white mode, thermalProbe-thermal detection
    mode, fireGroud-fire scene mode-->
  </fireImageMode>
  <emissivity>
    <!--optional, xs:float, the emissivity is between 0.01 and 1.00, which corrects to two decimal places. This parameter
    is used by the device (i.e., DS-2TF03-260V/GLT, DS-2TF03-167V/GLT) that does not support regular frame and preset-->
  </emissivity>
  <distanceUnit>
    <!--required, xs:string, opt="meter,feet,centimeter"--></distanceUnit>
  <TemperatureColor>
    <!--optional, set the alarm information color for the temperature measurement-->
  <type>
    <!-- required, xs:string,"highTemperature, lowTemperature, rangeTemperature, heatPreservation", set temperature
    alarm types: highTemperature field-high temperature alarm, when the measured temperature is higher than the
    configured value of highTemperature field, the measured temperature will be marked by color; lowTemperature-low
    temperature alarm, when the measured temperature is lower than the configured value of lowTemperature field, the
```

measured temperature will be marked by color; rangeTemperature-range temperature alarm, when the measured temperature is between the configured values of highTemperature field and lowTemperature field, the measured temperature will be marked by color; heatPreservation-insulation alarm, when the measured temperature is not between the configured values of highTemperature field and lowTemperature field, the measured temperature will be marked by color-->

```
</type>
<highTemperature><!--dep, xs:integer--></highTemperature>
<lowTemperature><!--dep, xs:integer--></lowTemperature>
</TemperatureColor>
<enviroTemperature><!--optional, xs:integer, environment temperature, unit: °C--></enviroTemperature>
<enviroHumidity><!--optional, xs:integer, environment humidity, unit:%--></enviroHumidity>
<correctionVolume><!--optional, xs:integer, temperature correction--></correctionVolume>
<specialPointThermType>
  <!--required, xs:string,"centerPoint, highestPoint, lowestPoint", display the special point temperature, central point
temperature, highest temperature, lowest temperature. Supports multiple selections-->
</specialPointThermType>
<distance><!--required, xs:integer; unit:m, the distance range is [0, 10000]--></distance>
<reflectiveEnable><!--required, xs:boolean, whether enables temperature reflection--></reflectiveEnable>
<reflectiveTemperature><!--optional, xs:float, reflective temperature, which corrects to one decimal place--></
reflectiveTemperature>
<alert><!--optional, xs: float, pre-alarm threshold--></alert>
<alarm><!--optional, xs: float, alarm threshold--></alarm>
<showTempStripEnable><!--optional, xs:boolean, whether enables displaying temperature bar--></
showTempStripEnable>
<thermalOpticalTransmittance><!--optional, xs: float, optical transmissivity is between 0.001 and 1.000, which
corrects to three decimal places. The default value is 1.000--></thermalOpticalTransmittance>
<externalOpticsWindowCorrection><!--optional, xs: float, external optical temperature is between -40.0 °C and
80.0 °C. The default value is 20 °C--></externalOpticsWindowCorrection>
<AlertOutputIOPortList><!--optional, list of temperature pre-alarm output ports-->
  <OutputIOPort>
    <portID><!--required, xs:string --></portID>
    <enabled><!--required, xs:boolean, "true,false"--></enabled>
  </OutputIOPort>
</AlertOutputIOPortList>
<AlarmOutputIOPortList><!--optional, list of temperature alarm output ports-->
  <OutputIOPort>
    <portID><!--required, xs:string--></portID>
    <enabled><!--required, xs:boolean, "true,false"--></enabled>
  </OutputIOPort>
</AlarmOutputIOPortList>
<alertFilteringTime><!--optional, xs:integer, temperature pre-alarm dwell time, unit:s--></alertFilteringTime>
<alarmFilteringTime><!--optional, xs:integer, temperature alarm dwell time, unit:s--></alarmFilteringTime>
<displayMaxTemperatureEnabled><!--optional, xs:boolean, whether displays the maximum temperature--></
displayMaxTemperatureEnabled>
<displayMinTemperatureEnabled><!--optional, xs:boolean, whether displays the minimum temperature--></
displayMinTemperatureEnabled>
<displayAverageTemperatureEnabled><!--optional, xs:boolean, whether displays the average temperature--></
displayAverageTemperatureEnabled>
<thermometryInfoDisplayposition><!--optional, xs:string, position of temperature measurement information overlay,
"rules_around,top_left_of_screen"--></thermometryInfoDisplayposition>
<calibrationCoefficientEnabled><!--optional, xs:boolean, whether enables calibration coefficient--></
calibrationCoefficientEnabled>
```

```
<calibrationCoefficient><!--dep, xs:float, calibration coefficient, ranges from 0 to 30, corrects to two decimal places--></calibrationCoefficient>
<emissivityMode>
  <!--optional, xs:string, emissivity type: "rougner"-rougner 0.95, "rough"-rough 0.80, "smooth"-smooth 0.60,
  "smoother"-smoother 0.30, "customsettings"-customized setting, values from 0.01 to 1.00, the larger the value, the
  higher the roughness-->
</emissivityMode>
<displayTemperatureInOpticalChannelEnabled>
  <!--optional, xs:boolean, display the temperature information of optical channel-->
</displayTemperatureInOpticalChannelEnabled>
<distanceMode><!--optional, xs:string, distance mode: "selfAdapt"-self-adaption, "fixed"-fixed distance--></
distanceMode>
<faceTemperatureInfoUploadEnabled>
  <!--optional, xs:boolean, whether to enable uploading face temperature information: true-enable, false-disable-->
</faceTemperatureInfoUploadEnabled>
</ThermometryBasicParam>
```

Remarks

The following nodes are not supported by the thermographic automation thermal camera (DS-2TA03-15SVI, DS2TA06-25SVI): **<TemperatureColor>**, **<specialPointThermType>**, and **<reflectiveEnable>**.

A.106 XML_ThermometryMode

XML message about temperature measurement mode information

```
<ThermometryMode version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mode><!--required, xs:string, temperature measurement: "normal"-normal mode (default), "expert"-expert mode-->
</mode>
  <thermometryROIEnabled><!--optional, xs:boolean, whether to enable ROI temperature measurement, "true"-yes,
  "false"-no--></thermometryROIEnabled>
</ThermometryMode>
```

Remarks

The ROI temperature measurement and mode configuration is mutually exclusive, ROI temperature measurement is mainly applied to temperature measurement.

A.107 XML_ThermometryRegion

ThermometryRegion message in XML format

```
<ThermometryRegion version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:integer--></id>
  <enabled><!--req, xs:boolean--></enabled>
  <name><!--req, xs:string--></name>
  <emissivity><!--req, xs:float, corrects to two decimal places--></emissivity>
  <distance><!--req, xs:integer--></distance>
```

```
<reflectiveEnable><!--req, xs:boolean--></reflectiveEnable>
<reflectiveTemperature><!--opt, xs:float, reflective temperature, corrects to one decimal place--></
reflectiveTemperature>
<sensitivity><!--req, xs:integer, sensitivity, normalized value, from 1 to 5--></sensitivity>
<PupilParam><!--req, pupil parameters-->
  <Region><!--req, pupil region coordinates, normalized value, from 0 to 1000-->
    <RegionCoordinatesList> <!--dep-->
      <RegionCoordinates> <!--opt -->
        <positionX><!-- req, xs:integer;coordinate--></positionX>
        <positionY><!-- req, xs:integer;coordinate--></positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </Region>
</PupilParam>
<MaxPupilParam><!--optional, the maximum pupil distance-->
  <pupilDistance min="10" max="625">
    <!--optional, xs:integer, pupil distance range, normalized value, range: [0,1000], valid range: [10,625], default value:
625-->
    </pupilDistance>
    <Region>
      <RegionCoordinatesList size=""><!--req-->
        <RegionCoordinates><!--list, pupil region coordinates-->
          <positionX><!--required, xs:integer, coordinate, X-coordinate--></positionX>
          <positionY><!--required, xs:integer, coordinate, Y-coordinate--></positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </Region>
  </MaxPupilParam>
  <targetSpeed><!--req, xs:integer--></targetSpeed>
  <alarmTemperature><!--opt, xs:float, unit:°C, alarm triggered temperature, corrects to one decimal place, ranges
from 0.0°C to 60.0°C--></alarmTemperature>
  <type><!--opt, xs:string, "point,region,line"--></type>
  <Point>
    <TempValue><!--dep, xs:float "-40.0...1000.0" ro--></TempValue>
    <CalibratingCoordinates><!--dep-->
      <positionX><!--req, xs:integer;coordinate--></positionX>
      <positionY><!--req, xs:integer;coordinate--></positionY>
    </CalibratingCoordinates>
  </Point>
  <Region>
    <highestTempValue><!--dep, xs: float "-273.0...10000.0" ro--></highestTempValue>
    <lowestTempValue><!--dep, xs: float "-273.0...10000.0" ro--></lowestTempValue>
    <averageTempValue><!--dep, xs: float "-273.0...10000.0" ro--></averageTempValue>
    <diffTempValue><!--dep, xs: float "-273.0...10000.0" ro--></diffTempValue>
    <RegionCoordinatesList><!--dep-->
      <RegionCoordinates><!--opt-->
        <positionX><!--req, xs:integer;coordinate--></positionX>
        <positionY><!--req, xs:integer;coordinate--></positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </Region>
  <distanceUnit><!--opt, xs:string, opt="meter,feet,centimeter"--></distanceUnit>
```

```
<emissivityMode>
  <!--opt,xs:string, emissivity type: "rougher"-rougher 0.95, "rough"-rough 0.80, "smooth"-smooth 0.60, "smoother"-
  smoother 0.30, "customsettings"-customized setting, values from 0.01 to 1.00, the larger the value, the higher the
  roughness-->
</emissivityMode>
<alarmRule><!--opt, xs:string, alarm rule, "highestGreater,highestLess"--></alarmRule>
</ThermometryRegion>
```

A.108 XML_ThermometryRegionList

XML message about temperature measurement region list

```
<ThermometryRegionList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ThermometryRegion/><!--see details in XML_ThermometryRegion-->
</ThermometryRegionList>
```

See Also

XML_ThermometryRegion

A.109 XML_ThermometryScene

XML message about information of temperature measurement scene

```
<ThermometryScene version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, xs:integer--></id>
  <normalizedScreenSize><!--required, ro-->
    <normalizedScreenWidth><!--required, ro, xs:integer--></normalizedScreenWidth>
    <normalizedScreenHeight><!--required, ro, xs:integer--></normalizedScreenHeight>
  </normalizedScreenSize>
  <ThermometryRegionList/><!--see details in XML_ThermometryRegionList-->
</ThermometryScene>
```

See Also

XML_ThermometryRegionList

A.110 XML_ThermometryScheduleList

XML message about temperature measurement schedules

```
<ThermometryScheduleList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <Schedule/><!--opt, see details in XML_Schedule-->
</ThermometryScheduleList>
```

See Also

XML_Schedule

A.111 XML_TMPAScheduleList

XML message about temperature measurement pre-alarm arming schedules

```
<?xml version="1.0" encoding="utf-8"?>
<TMPAScheduleList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <Schedule/><!--optional, see XML_Schedule for details-->
</TMPAScheduleList>
```

See Also

XML_Schedule

A.112 JSON_EventNotificationAlert_Alarm/EventInfo

EventNotificationAlert message with alarm or event information in JSON format.

```
{
  "ipAddress": "",
  /*required, device IPv4 address , string, the maximum size is 32 bytes*/
  "ipv6Address": "",
  /*optional, device IPv6 address, string, the maximum size is 128 bytes*/
  "portNo": ,
  /*optional, device port No., integer32*/
  "protocol": "",
  /*optional, protocol type, "HTTP, HTTPS", string, the maximum size is 32 bytes*/
  "macAddress": "",
  /*optional, MAC address, string, the maximum size is 32 bytes, e.g., 01:17:24:45:D9:F4*/
  "channelID": "",
  /*optional, device channel No., integer32*/
  "dateTime": "",
  /*optional, string, alarm/event triggered or occurred time based on ISO8601, the maximum size is 32 bytes, e.g.,
  2009-11-14T15:27Z*/
  "activePostCount": "",
  /*required, alarm/event frequency, integer32*/
  "eventType": "",
  /*required, alarm/event type, "captureResult, faceCapture,...", string, the maximum size is 128 bytes*/
  "eventState": "",
  /*required, string, the maximum size is 32 bytes, durative alarm/event status: "active"-valid, "inactive"-invalid*/
  "eventDescription": "",
  /*required, event description, string, the maximum size is 128 bytes*/
  "deviceId": "",
  /*string type, device ID*/
  "uuid": "",
  /*string type, event UUID, which is used to uniquely identify an event, the standard UUID format is xxxxxxxx-xxxx-xxxx-
  xxxx-xxxxxxxxxxxx*/
  ...
  /*optional, for different alarm/event types, the nodes are different, see the message examples in different
```

```
applications*/  
}
```

A.113 XML_EventNotificationAlert_AlarmEventInfo

EventNotificationAlert message with alarm/event information in XML format.

```
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">  
  <ipAddress><!--dep, xs:string, device IPv4 address--></ipAddress>  
  <ipv6Address><!--dep, xs:string, device IPv6 address--></ipv6Address>  
  <portNo><!--opt, xs:integer, device port number--></portNo>  
  <protocol><!--opt, xs:string, protocol type for uploading alarm/event information, "HTTP,HTTPS"--></protocol>  
  <macAddress><!--opt, xs:string, MAC address--></macAddress>  
  <channelID><!--dep, xs:string, device channel No., starts from 1--></channelID>  
  <dateTime><!--req, alarm/event triggered or occurred time, format: 2017-07-19T10:06:41+08:00--></dateTime>  
  <activePostCount><!--req, xs:integer, alarm/event frequency, starts from 1--></activePostCount>  
  <eventType><!--req, xs:string, alarm/event type, "peopleCounting, ANPR,..."--></eventType>  
  <eventState>  
    <!--req, xs:string, durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is  
detected,  
    the alarm/event information will be uploaded continuously unit the status is set to "inactive"-->  
  </eventState>  
  <eventDescription><!--req, xs:string, alarm/event description--></eventDescription>  
  <...><!--opt, for different alarm/event types, the nodes are different, see the message examples in different  
applications--></...>  
</EventNotificationAlert>
```

A.114 XML_EventNotificationAlert_HeartbeatInfo

EventNotificationAlert message with heartbeat information (when there is no alarm is triggered) in XML format

```
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">  
  <ipv6Address><!--dep, xs:string, device IPv6 address--></ipv6Address>  
  <portNo><!--opt, xs:integer, device port number--></portNo>  
  <protocol><!--opt, xs:string, protocol type for uploading alarm/event information, "HTTP,HTTPS"--></protocol>  
  <macAddress><!--opt, xs:string, MAC address--></macAddress>  
  <channelID><!--dep, xs:string, device channel No., starts from 1--></channelID>  
  <dateTime><!--req, heartbeat uploaded time, format: 2017-07-19T10:06:41+08:00--></dateTime>  
  <activePostCount><!--req, xs:integer, heartbeat frequency, starts from 1--></activePostCount>  
  <eventType><!--req, xs:string, for heartbeat, it is "videoloss"--></eventType>  
  <eventState>  
    <!--req, xs:string, for heartbeat, it is "inactive"-->  
  </eventState>  
  <eventDescription><!--req, xs: string, description--></eventDescription>  
</EventNotificationAlert>
```

Remarks

- For network camera or network speed dome with the version 5.5.0 and lower, the heartbeat frequency is 300 ms per heartbeat.
- For network camera or network speed dome with the version 5.5.0 and higher, the heartbeat frequency is 10 s per heartbeat. If no heartbeat received for continuous 30 s, it indicates that the heartbeat is timed out.

Example

Message Example of Heartbeat

```
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <ipAddress>10.17.133.46</ipAddress>
  <portNo>80</portNo>
  <protocol>HTTP</protocol>
  <macAddress>44:19:b6:6d:24:85</macAddress>
  <channelID>1</channelID>
  <dateTime>2017-05-04T11:20:02+08:00</dateTime>
  <activePostCount>0</activePostCount>
  <eventType>videoloss</eventType>
  <eventState>inactive</eventState>
  <eventDescription>videoloss alarm</eventDescription>
</EventNotificationAlert>
```

A.115 XML_HttpHostNotification

XML message about parameters of a HTTP listening server

```
<HttpHostNotification version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, xs:string, ID--></id>
  <url><!--required, xs:string, the absolute path, e.g., http://<ipAddress>:<portNo>/<uri--></url>
  <protocolType><!--required, xs:string, "HTTP,HTTPS,EHome"--></protocolType>
  <parameterFormatType><!--required, xs:string, alarm/event information format, "XML,JSON"--></
parameterFormatType>
  <addressingFormatType><!--required, xs:string, "ipaddress,hostname"--></addressingFormatType>
  <hostName><!--dependent, xs:string--></hostName>
  <ipAddress><!--dependent, xs:string--></ipAddress>
  <ipv6Address><!--dependent, xs:string--></ipv6Address>
  <portNo><!--optional, xs:integer--></portNo>
  <userName><!--dependent, xs:string--></userName>
  <password><!--dependent, xs:string--></password>
  <httpAuthenticationMethod><!--required, xs:string, "MD5digest,none"--></httpAuthenticationMethod>
  <ANPR><!--optional-->
    <detectionUpLoadPicturesType>
      <!--optional, xs:string, types of alarm picture to be uploaded: "all, licensePlatePicture, detectionPicture"-->
    </detectionUpLoadPicturesType>
  </ANPR>
  <eventType optional="AID,TFS,TPS"><!--required, xs:string--></eventType>
  <uploadImagesDataType>
    <!--optional, xs:string, "URL", "binary" (default), for cloud storage, only "URL" is supported-->
```

```
</uploadImagesDataType>
<eventMode><!--optional, xs:string, "all,list"--></eventMode>
<EventList><!--dependent, it is valid only when eventMode is "list"-->
  <Event><!--required-->
    <type><!--required, xs:string--></type>
  </Event>
</EventList>
<channels><!--optional, xs:string, "1,2,3,4..."--></channels>
<SubscribeEvent/><!--optional, event subscription parameters, see details in the message of XML_SubscribeEvent-->
</HttpHostNotification>
```

A.116 XML_HttpHostNotificationCap

XML message about capability of HTTP listening server

```
<?xml version="1.0" encoding="utf-8"?>
<HttpHostNotificationCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <hostNumber>2</hostNumber>
  <urlLen max=""/>
  <protocolType opt="HTTP,HTTPS"/>
  <parameterFormatType opt="XML,querystring,JSON"/>
  <addressingFormatType opt="ipaddress,hostname"/>
  <ipAddress opt="ipv4,ipv6"/>
  <portNo min="" max=""/>
  <userNameLen min="" max=""/>
  <passwordLen min="" max=""/>
  <httpAuthenticationMethod opt="MD5digest,none"/>
  <Extensions>
    <intervalBetweenEvents min="" max=""/>
  </Extensions>
  <uploadImagesDataType opt="URL,binary"/>
  <ANPR><!--optional-->
    <detectionUpLoadPicturesType opt="all,licensePlatePicture,detectionPicture..."/><!--optional, xs:string, types of
alarm pictures to be uploaded-->
    <alarmHttpPushProtocol opt="baseline,custom"/>
  </ANPR>
  <httpBroken opt="true,false" def="true" ><!--optional, xs:boolean, whether to enable global ANR: true, false--></
httpBroken>
  <SubscribeEventCap>
    <heartbeat min="" max=""/><!--optional, heartbeat time interval, unit: second-->
    <channelMode opt="all,list"/><!--required, all-subscribe events of all channels, list-subscribe event by channel-->
    <eventMode opt="all,list"/><!--required, event subscription mode: all-subscribe all events of all channels, list-
subscribe events by type, channel, and target-->
    <!--if the values of the two nodes channelMode and eventMode are both "all", it indicates that the device does not
support subscribe events by type and channel-->
    <EventList><!--dependent, alarm uploading mode, this node is valid only when eventMode is "list"-->
      <Event><!--required-->
        <type><!--required, xs:string, event types--></type>
        <pictureURLType opt="binary,localURL,cloudStorageURL" def=""/>
        <!--optional, xs:string, transmission format of alarm picture: "binary"-picture binary data, "localURL"-picture URL
```

```
from local device, "cloudStorageURL"-picture URL from cloud storage-->
  </Event>
</EventList>
<pictureURLType opt="binary,localURL,cloudStorageURL" def=""/>
  <!--optional, xs:string, transmission format of all alarm pictures: "binary"-picture binary data (default for camera),
  "localURL"-picture URL from local device (default for NVR/DVR), "cloudStorageURL"-picture URL from cloud storage;
  this node is in highest priority-->
  <ChangedUploadSub>
    <interval/><!--optional, xs:integer, the life cycle of arming GUID, unit: second, the default life cycle is 5 minutes; if
    the reconnection is not started during the life cycle, a new GUID will be generated-->
    <StatusSub>
      <all/><!--optional, xs:boolean, whether to subscribe all-->
      <channel/><!--optional, xs:boolean, subscribe channel status, this node is not required when the node all is "true"-->
    >
      <hd/><!--optional, xs:boolean, subscribe the HDD status, this node is not required when the node all is "true"-->
      <capability/><!--optional, xs:boolean, subscribe the capability changed status, this node is not required when the
      node all is "true"-->
    </StatusSub>
  </ChangedUploadSub>
</SubscribeEventCap>
</HttpHostNotificationCap>
```

A.117 XML_HttpHostNotificationList

HttpHostNotificationList message in XML format

```
<HttpHostNotificationList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <HttpHostNotification>
    <id><!--req, xs:string, ID--></id>
    <url><!--req, xs:string--></url>
    <protocolType><!--req, xs:string, "HTTP,HTTPS"--></protocolType>
    <parameterFormatType><!--req, xs:string, alarm/event information format, "XML,JSON"--></parameterFormatType>
    <addressingFormatType><!--req, xs:string, "ipaddress,hostname"--></addressingFormatType>
    <hostName><!--dep, xs:string--></hostName>
    <ipAddress><!--dep, xs:string--></ipAddress>
    <ipv6Address><!--dep, xs:string--></ipv6Address>
    <portNo><!--opt, xs:integer--></portNo>
    <userName><!--dep, xs:string--></userName>
    <password><!--dep, xs:string--></password>
    <httpAuthenticationMethod><!--req, xs:string, "MD5digest,none"--></httpAuthenticationMethod>
    <uploadImagesDataType>
      <!--opt, xs:string, "URL", "binary" (default), for cloud storage, only "URL" is supported-->
    </uploadImagesDataType>
    <eventMode><!--opt, xs:string, "all,list"--></eventMode>
    <EventList><!--dep, it is valid only when eventMode is "list"-->
      <Event><!--req-->
        <type><!--req, xs:string--></type>
      </Event>
    </EventList>
    <channels><!--opt, xs:string, "1,2,3,4..."--></channels>
```

```
</HttpHostNotification>
</HttpHostNotificationList>
```

Example

HttpHostNotificationList Message Example

```
<HttpHostNotificationList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <HttpHostNotification>
    <id>1</id>
    <url></url>
    <protocolType>HTTP</protocolType>
    <parameterFormatType>XML</parameterFormatType>
    <addressingFormatType>ipaddress</addressingFormatType>
    <ipAddress>0.0.0.0</ipAddress>
    <portNo>80</portNo>
    <userName></userName>
    <httpAuthenticationMethod>none</httpAuthenticationMethod>
  </HttpHostNotification>
</HttpHostNotificationList>
```

A.118 XML_HttpHostTestResult

HttpHostTestResult message in XML format.

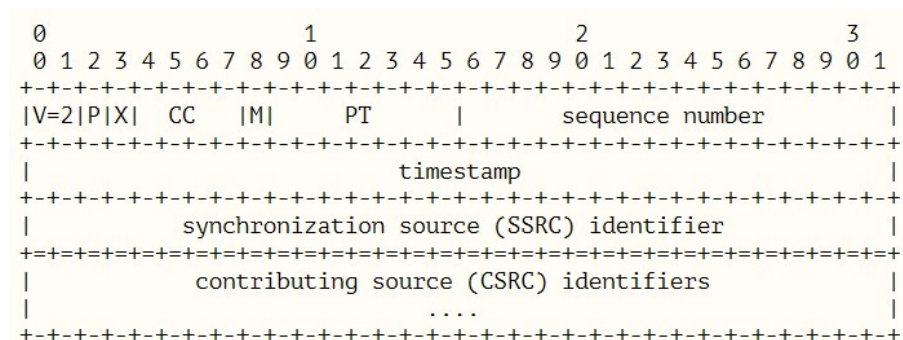
```
<HttpHostTestResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <errorDescription>
    <!--req, xs:string-->
  </errorDescription>
</HttpHostTestResult>
```

Appendix B. Appendixes

B.1 RTP Packet Format

The Real-time Transport Protocol (RTP) provides end-to-end network transport functions for transmitting real-time data, such as audio, video, or simulation data, over multicast or unicast network services. The RTP packet consists of RTP header (standard and extended header) and RTP payload.

RTP Header (Standard)



V

2-bit RTP version No.; currently, it is 2.

P

1-bit padding flag. If its value is 1, the message end will be padded with a or multiple 8-bit array(s), but the padded array(s) is (are) not a part of payload.

X

Extension flag, set to 0 and reserved for further extension requirement.

CC

4-bit CSRC counter, which indicates the number of CSRC identifiers.

M

1-bit maker flag, when there is a lot of stream data, the data will be packaging into multiple subpackets to be sent, for subpacket, the value of maker flag is 0, and for end packet, the value of maker flag is 1.

PT

7-bit payload type, which is obtained via DESCRIBE operation.

sequence number

16-bit serial No., which identifies the sent RTP message. The No. starts from 0 in HELIX server and will plus 1 when sending RTP message for once. The No. of video and audio message is counted separately.

When the lower-layer adopts UDP, or the network condition is bad, the serial No. can be used to check the packet loss; when the network jitter occurred, the serial No. can be used to reorder the data.

Timestamp: 32 bits

32-bit timestamp, which indicates the sampling time of the first 8-bit array in the RTP message. The receiver can calculate the delay or delay jitter, and control according to the timestamp.

SSRC

32-bit synchronization source identifier, which is randomly selected, and the identifier value of two synchronization sources in same RTP session cannot be same.

CSRC

32-bit contributing source identifier, up to 16 identifiers are supported, its value is between 0 and 15, and each CSRC identifier identifies all contributing sources in the payload of the RTP message.

RTP Payload

The payload consists of binary data, which transmits the detailed information and useful content.

B.2 ISAPI Response Codes

The response codes returned during ISAPI integration is based on the status codes of HTTP protocol. 7 kinds of status codes are predefined, including 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid Message Format), 6 (Invalid Message Content), and 7 (Reboot Required). Each kind of status code contains multiple sub status codes, and the response codes are in a one-to-one correspondence with the sub status codes.

StatusCode=1

SubStatusCode	Error Code	Description
ok	0x1	Operation completed.
riskPassword	0x10000002	Risky password.
armProcess	0x10000005	Arming process.

StatusCode=2

Sub Status Code	Error Code	Description
noMemory	0x20000001	Insufficient memory.
serviceUnavailable	0x20000002	The service is not available.
upgrading	0x20000003	Upgrading.
deviceBusy	0x20000004	The device is busy or no response.
reConnectIpc	0x20000005	The video server is reconnected.
transferUpgradePackageFailed	0x20000006	Transmitting device upgrade data failed.
startUpgradeFailed	0x20000007	Starting upgrading device failed.
getUpgradeProcessfailed.	0x20000008	Getting upgrade status failed.
certificateExist	0x2000000B	The Authentication certificate already exists.

StatusCode=3

Sub Status Code	Error Code	Description
deviceError	0x30000001	Hardware error.
badFlash	0x30000002	Flash operation error.
28181Uninitialized	0x30000003	The 28181 configuration is not initialized.
socketConnectError	0x30000005	Connecting to socket failed.
receiveError	0x30000007	Receive response message failed.
deletePictureError	0x3000000A	Deleting picture failed.
pictureSizeExceedLimit	0x3000000C	Too large picture size.
clearCacheError	0x3000000D	Clearing cache failed.
updateDatabasError	0x3000000F	Updating database failed.
searchDatabaseError	0x30000010	Searching in the database failed.

Sub Status Code	Error Code	Description
writeDatabaseError	0x30000011	Writing to database failed.
deleteDatabaseError	0x30000012	Deleting database element failed.
searchDatabaseElementError	0x30000013	Getting number of database elements failed.
cloudAutoUpgradeException	0x30000016	Downloading upgrade packet from cloud and upgrading failed.
HBPEException	0x30001000	HBP exception.
UDEPEException	0x30001001	UDEP exception
elasticSearchException	0x30001002	Elastic exception.
kafkaException	0x30001003	Kafka exception.
HBaseException	0x30001004	Hbase exception.
sparkException	0x30001005	Spark exception.
yarnException	0x30001006	Yarn exception.
cacheException	0x30001007	Cache exception.
trafficException	0x30001008	Monitoring point big data server exception.
faceException	0x30001009	Human face big data server exception.
SSDFileSystemsIsError	0x30001013	SSD file system error (Error occurs when it is non-Ext4 file system)
insufficientSSDCapacityForFPD	0x30001014	Insufficient SSD space for person frequency detection
wifiException	0x3000100A	Wi-Fi big data server exception
structException	0x3000100D	Video parameters structure server exception.
calibrationTimeout	0x30002051	Calibration timed out.
captureTimeout	0x30006000	Data collection timed out.
lowScore	0x30006001	Low quality of collected data.
uploadingFailed	0x30007004	Uploading failed.

StatusCode=4

Sub Status Code	Error Code	Description
notSupport	0x40000001	Not supported.
lowPrivilege	0x40000002	No permission.
badAuthorization	0x40000003	Authentication failed.
methodNotAllowed	0x40000004	Invalid HTTP method.
notSetHdiskRedund	0x40000005	Setting spare HDD failed.
invalidOperation	0x40000006	Invalid operation.
hdFormatFail	0x40002049	Formatting HDD failed.
notActivated	0x40000007	Inactivated.
hasActivated	0x40000008	Activated.
certificateAlreadyExist	0x40000009	The certificate already exists.
operateFailed	0x4000000F	Operation failed.
USBNotExist	0x40000010	USB device is not connected.
upgradePackageMorethan2GB	0x40001000	Up to 2GB upgrade package is allowed to be uploaded.
IDNotExist	0x40001001	The ID does not exist.
synchronizationError	0x40001003	Synchronization failed.
synchronizing	0x40001004	Synchronizing.
importError	0x40001005	Importing failed.
importing	0x40001006	Importing.
fileAlreadyExists	0x40001007	The file already exists.
invalidID	0x40001008	Invalid ID.
backupnodeNotAllowedLog	0x40001009	Accessing to backup node is not allowed.
exportingError	0x4000100A	Exporting failed.
exporting	0x4000100B	Exporting.
exportEnded	0x4000100C	Exporting stopped.
exported	0x4000100D	Exported.
IPOccupied	0x4000100E	The IP address is already occupied.

Sub Status Code	Error Code	Description
IDAAlreadyExists	0x4000100F	The ID already exists.
exportItemsExceedLimit	0x40001010	No more items can be exported.
noFiles	0x40001011	The file does not exist.
beingExportedByAnotherUser	0x40001012	Being exported by others.
needReAuthentication	0x40001013	Authentication is needed after upgrade.
unitAddNotOnline	0x40001015	The added data analysis server is offline.
unitControl	0x40001016	The data analysis server is already added.
analysis unitFull	0x40001017	No more data analysis server can be added.
unitIDError	0x40001018	The data analysis server ID does not exist.
unitExit	0x40001019	The data analysis server already exists in the list.
unitSearch	0x4000101A	Searching data analysis server in the list failed.
unitNotOnline	0x4000101B	The data analysis server is offline.
unitInfoError	0x4000101C	Getting data analysis server information failed.
unitGetNodeInfoError	0x4000101D	Getting node information failed.
unitGetNetworkInfoError	0x4000101E	Getting the network information of data analysis server failed
unitSetNetworkInfoError	0x4000101F	Setting the network information of data analysis server failed
setSmartNodeInfoError	0x40001020	Setting node information failed.
setUnitNetworkInfoError	0x40001021	Setting data analysis server network information failed.
unitRestartCloseError	0x40001022	Rebooting or shutting down data analysis server failed.
virtualIPnotAllowed	0x40001023	Adding virtual IP address is not allowed.
unitInstalled	0x40001024	The data analysis server is already installed.
badSubnetMask	0x40001025	Invalid subnet mask.
uintVersionMismatched	0x40001026	Data analysis server version mismatches.

Sub Status Code	Error Code	Description
deviceMModelMismatched	0x40001027	Adding failed. Device model mismatches.
unitAddNotSelf	0x40001028	Adding peripherals is not allowed.
noValidUnit	0x40001029	No valid data analysis server.
unitNameDuplicate	0x4000102A	Duplicated data analysis server name.
deleteUnitFirst	0x4000102B	Delete the added data analysis server of the node first.
getLocalInfoFailed	0x4000102C	Getting the server information failed.
getClientAddedNodeFailed	0x4000102D	Getting the added node information of data analysis server failed.
taskExit	0x4000102E	The task already exists.
taskInitError	0x4000102F	Initializing task failed.
taskSubmitError	0x40001030	Submitting task failed.
taskDelError	0x40001031	Deleting task failed.
taskPauseError	0x40001032	Pausing task failed.
taskContinueError	0x40001033	Starting task failed.
taskSeverNoCfg	0x40001035	Full-text search server is not configured.
taskPicSeverNoCfg	0x40001036	The picture server is not configured.
taskStreamError	0x40001037	Streaming information exception.
taskRecSDK	0x40001038	History recording is not supported.
taskCasaError	0x4000103A	Cascading is not supported.
taskVCARuleError	0x4000103B	Invalid VCA rule.
taskNoRun	0x4000103C	The task is not executed.
unitLinksNoStorageNode	0x4000103D	No node is linked with the data analysis server. Configure the node first.
searchFailed	0x4000103E	Searching video files failed.
searchNull	0x4000103F	No video clip.
userScheOffline	0x40001040	The task scheduler service is offline.
updateTypeUnmatched	0x40001041	The upgrade package type mismatches.

Sub Status Code	Error Code	Description
userExist	0x40001043	The user already exists.
userCannotDelAdmin	0x40001044	The administrator cannot be deleted.
userInexistence	0x40001045	The user name does not exist.
userCannotCreatAdmin	0x40001046	The administrator cannot be created.
monitorCamExceed	0x40001048	Up to 3000 cameras can be added.
monitorCunitOverLimit	0x40001049	Adding failed. Up to 5 lower-levels are supported by the control center.
monitorReginOverLimit	0x4000104A	Adding failed. Up to 5 lower-levels are supported by the area.
monitorArming	0x4000104B	The camera is already armed. Disarm the camera and try again.
monitorSyncCfgNotSet	0x4000104C	The system parameters are not configured.
monitorFdSyncing	0x4000104E	Synchronizing. Try again after completing the synchronization.
monitorParseFailed	0x4000104F	Parsing camera information failed.
monitorCreatRootFailed	0x40001050	Creating resource node failed.
deleteArmingInfo	0x40001051	The camera is already . Disarm the camera and try again.
cannotModify	0x40001052	Editing is not allowed. Select again.
cannotDel	0x40001053	Deletion is not allowed. Select again.
deviceExist	0x40001054	The device already exists.
IPErrorConnectFailed	0x40001056	Connection failed. Check the network port.
cannotAdd	0x40001057	Only the capture cameras can be added.
serverExist	0x40001058	The server already exists.
fullTextParamError	0x40001059	Incorrect full-text search parameters.
storParamError	0x4000105A	Incorrect storage server parameters.
picServerFull	0x4000105B	The storage space of picture storage server is full.

Sub Status Code	Error Code	Description
NTPUnconnect	0x4000105C	Connecting to NTP server failed. Check the parameters.
storSerConnectFailed	0x4000105D	Connecting to storage server failed. Check the network port.
storSerLoginFailed	0x4000105E	Logging in to storage server failed. Check the user name and password.
searchSerConnectFailed	0x4000105F	Connecting to full-text search server failed. Check the network port.
searchSerLoginFailed	0x40001060	Logging in to full-text search server failed. Check the user name and password.
kafkaConnectFailed	0x40001061	Connecting to Kafka failed. Check the network port.
mgmtConnectFailed	0x40001062	Connecting to system failed. Check the network port.
mgmtLoginFailed	0x40001063	Logging in to system failed. Check the user name and password.
TDAConnectFailed	0x40001064	Connecting to traffic data access server failed. Checking the server status.
86sdkConnectFailed	0x40001065	Connecting to listening port of iVMS-8600 System failed. Check the parameters.
nameExist	0x40001066	Duplicated server name.
batchProcessFailed	0x40001067	Processing in batch failed.
IDNotExist	0x40001068	The server ID does not exist.
serviceNumberReachesLimit	0x40001069	No more service can be added.
invalidServiceType.	0x4000106A	Invalid service type.
clusterGetInfo	0x4000106B	Getting cluster group information failed.
clusterDelNode	0x4000106C	Deletion node failed.
clusterAddNode	0x4000106D	Adding node failed.
clusterInstalling	0x4000106E	Creating cluster...Do not operate.
clusterUninstall	0x4000106F	Reseting cluster...Do not operate.
clusterInstall	0x40001070	Creating cluster failed.

Sub Status Code	Error Code	Description
clusterIpError	0x40001071	Invalid IP address of task scheduler server.
clusterNotSameSeg	0x40001072	The master node and slave node must be in the same network segment.
clusterVirIpError	0x40001073	Automatically getting virtual IP address failed. Enter manually.
clusterNodeUnadd	0x40001074	The specified master(slave) node is not added.
clusterNodeOffline	0x40001075	The task scheduler server is offline.
nodeNotCurrentIP	0x40001076	The analysis node of the current IP address is required when adding master and slave nodes.
addNodeNetFailed	0x40001077	Adding node failed. The network disconnected.
needTwoMgmtNode	0x40001078	Two management nodes are required when adding master and slave nodes.
ipConflict	0x40001079	The virtual IP address and data analysis server's IP address conflicted.
ipUsed	0x4000107A	The virtual IP address has been occupied.
cloudAlalyseOnline	0x4000107B	The cloud analytic server is online.
virIP&mainIPnotSame NetSegment	0x4000107C	The virtual IP address is not in the same network segment with the IP address of master/slave node.
getNodeDispatchInfoFailed	0x4000107D	Getting node scheduler information failed.
unableModifyManagementNetworkIP	0x4000107E	Editing management network interface failed. The analysis board is in the cluster.
notSpecifyVirtualIP	0x4000107F	Virtual IP address should be specified for master and slave cluster.
armingFull	0x40001080	No more device can be armed.
armingNoFind	0x40001081	The arming information does not exist.
disArming	0x40001082	Disarming failed.
getArmingError	0x40001084	Getting arming information failed.
refreshArmingError	0x40001085	Refreshing arming information failed.
ArmingPlateSame	0x40001086	The license plate number is repeatedly armed.
ArmingParseXLSError	0x40001087	Parsing arming information file failed.

Sub Status Code	Error Code	Description
ArmingTimeError	0x40001088	Invalid arming time period.
ArmingSearchTimeError	0x40001089	Invalid search time period.
armingRelationshipReachesLimit	0x4000108A	No more relation can be created.
duplicateArmingName	0x4000108B	The relation name already exists.
noMoreArmingListAdded	0x4000108C	No more blacklist library can be armed.
noMoreCamerasAdded	0x4000108D	No more camera can be armed.
noMoreArmingListAddedWithCamera	0x4000108E	No more library can be linked to the camera.
noMoreArmingPeriodAdded	0x4000108F	No more time period can be added to the arming schedule.
armingPeriodsOverlapped	0x40001090	The time periods in the arming schedule are overlapped.
noArmingAlarmInfo	0x40001091	The alarm information does not exist.
armingAlarmUnRead	0x40001092	Getting number of unread alarms failed.
getArmingAlarmError	0x40001093	Getting alarm information failed.
searchByPictureTimedOut	0x40001094	Searching picture by picture timeout. Search again.
comparisonTimeRangeError	0x40001095	Comparison time period error.
selectMonitorNumberUpperLimit	0x40001096	No more monitoring point ID can be filtered.
noMoreComparisonTasksAdded	0x40001097	No more comparison task can be executed at the same time.
GetComparisonResultFailed	0x40001098	Getting comparison result failed.
comparisonTypeError	0x40001099	Comparison type error.
comparisonUnfinished	0x4000109A	The comparison is not completed.
facePictureModelInvalid	0x4000109B	Invalid face model.

Sub Status Code	Error Code	Description
duplicateLibraryName.	0x4000109C	The library name already exists.
noRecord	0x4000109D	No record found.
countingRecordsFailed.	0x4000109E	Calculate the number of records failed.
getHumanFaceFrameFailed	0x4000109F	Getting face thumbnail from the picture failed.
modelingFailed.	0x400010A0	Modeling face according to picture URL failed.
1V1FacePictureComparisonFailed	0x400010A1	Comparison 1 VS 1 face picture failed.
libraryArmed	0x400010A2	The blacklist library is armed.
licenseExceedLimit	0x400010A3	Dongle limited.
licenseExpired	0x400010A4	Dongle expired.
licenseDisabled	0x400010A5	Unavailable dongle.
licenseNotExist	0x400010A6	The dongle does not exist.
SessionExpired	0x400010A7	Session expired .
beyondConcurrentLimit	0x400010A8	Out of concurrent limit.
stopSync	0x400010A9	Synchronization stopped.
getProgressFailed	0x400010AA	Getting progress failed.
uploadExtraCaps	0x400010AB	No more files can be uploaded.
timeRangeError	0x400010AC	Time period error.
dataPortNotConnected	0x400010AD	The data port is not connected.
addClusterNodeFailed	0x400010AE	Adding to the cluster failed. The device is already added to other cluster.
taskNotExist	0x400010AF	The task does not exist.
taskQueryFailed	0x400010B0	Searching task failed.
modifyTimeRuleFailed	0x400010B2	The task already exists. Editing time rule is not allowed.
modifySmartRuleFailed	0x400010B3	The task already exists. Editing VAC rule is not allowed.
queryHistoryVideoFailed	0x400010B4	Searching history video failed.

Sub Status Code	Error Code	Description
addDeviceFailed	0x400010B5	Adding device failed.
addVideoFailed	0x400010B6	Adding video files failed.
deleteAllVideoFailed	0x400010B7	Deleting all video files failed.
createVideoIndexFailed	0x400010B8	Indexing video files failed.
videoCheckTypeFailed	0x400010B9	Verifying video files types failed.
configStructuredAddressFailed	0x400010BA	Configuring IP address of structured server failed.
configPictureServerAddressFailed	0x400010BB	Configuring IP address of picture stored server failed.
storageServiceIPNotExist	0x400010BD	The storage server IP address does not exist.
syncBackupDatabaseFailed	0x400010BE	Synchronizing slave database failed. Try again.
syncBackupNTPTimeFailed	0x400010BF	Synchronizing NTP time of slave server failed.
clusterNotSelectLoopbackAddress	0x400010C0	Loopback address is not supported by the master or slave cluster.
addFaceRecordFailed	0x400010C1	Adding face record failed.
deleteFaceRecordFailed	0x400010C2	Deleting face record failed.
modifyFaceRecordFailed	0x400010C3	Editing face record failed.
queryFaceRecordFailed	0x400010C4	Searching face record failed.
faceDetectFailed	0x400010C5	Detecting face failed.
libraryNotExist	0x400010C6	The library does not exist.
blackListQueryExporting	0x400010C7	Exporting matched blacklists.
blackListQueryExported	0x400010C8	The matched blacklists are exported.
blackListQueryStopExporting	0x400010C9	Exporting matched blacklists is stopped.

Sub Status Code	Error Code	Description
blackListAlarmQueryExporting	0x400010CA	Exporting matched blacklist alarms.
blackListAlarmQueryExported	0x400010CB	The matched blacklists alarms are exported.
blackListAlarmQueryStopExporting	0x400010CC	Exporting matched blacklist alarms is stopped.
getBigDataCloudAnalysisFailed	0x400010CD	Getting big data cloud analytic information failed.
setBigDataCloudAnalysisFailed	0x400010CE	Configuring big data cloud analytic failed.
submitMapSearchFailed	0x400010CF	Submitting search by picture task failed.
controlRelationshipNotExist	0x400010D0	The relation does not exist.
getHistoryAlarmInfoFailed	0x400010D1	Getting history alarm information failed.
getFlowReportFailed	0x400010D2	Getting people counting report failed.
addGuardFailed	0x400010D3	Adding arming configuration failed.
deleteGuardFailed	0x400010D4	Deleting arming configuration failed.
modifyGuardFailed	0x400010D5	Editing arming configuration failed.
queryGuardFailed	0x400010D6	Searching arming configurations failed.
uploadUserSuperCaps	0x400010D7	No more user information can be uploaded.
bigDataServerConnectFailed	0x400010D8	Connecting to big data server failed.
microVideoCloudRequestInfoBuildFailed	0x400010D9	Adding response information of micro video cloud failed.
microVideoCloudResponseInfoBuildFailed	0x400010DA	Parsing response information of micro video cloud failed.
transcodingServerRequestInfoBuildFailed	0x400010DB	Adding response information of transcoding server failed.
transcodingServerResponseInfoParseFailed	0x400010DC	Parsing response information of transcoding server failed.

Sub Status Code	Error Code	Description
transcodingServerOffline	0x400010DD	Transcoding server is offline.
microVideoCloudOffline	0x400010DE	Micro video cloud is offline.
UPSServerOffline	0x400010DF	UPS monitor server is offline.
statisticReportRequestInfoBuildFailed	0x400010E0	Adding response information of statistics report failed.
statisticReportResponseInfoParseFailed	0x400010E1	Parsing response information of statistics report failed.
DisplayConfigInfoBuildFailed	0x400010E2	Adding display configuration information failed.
DisplayConfigInfoParseFailed	0x400010E3	Parsing display configuration information failed.
DisplayConfigInfoSaveFailed	0x400010E4	Saving display configuration information failed.
notSupportDisplayConfigType	0x400010E5	The display configuration type is not supported.
passError	0x400010E7	Incorrect password.
upgradePackageLarge	0x400010EB	Too large upgrade package.
sessionUserReachesLimit	0x400010EC	No more user can log in via session.
ISO8601TimeFormatError	0x400010ED	Invalid ISO8601 time format.
clusterDissolutionFailed	0x400010EE	Deleting cluster failed.
getServiceNodeInfoFailed	0x400010EF	Getting service node information failed.
getUPSInfoFailed	0x400010F0	Getting UPS configuration information failed.
getDataStatisticsReportFailed	0x400010F1	Getting data statistic report failed.
getDisplayConfigInfoFailed	0x400010F2	Getting display configuration failed.

Sub Status Code	Error Code	Description
namingAnalysisBoardNotAllowed	0x400010F3	Renaming analysis board is not allowed.
onlyDrawRegionsOfConvexPolygon	0x400010F4	Only drawing convex polygon area is supported.
bigDataServerResponseInfoParseFailed	0x400010F5	Parsing response message of big data service failed.
bigDataServerReturnFailed	0x400010F6	No response is returned by big data service.
microVideoReturnFailed	0x400010F7	No response is returned by micro video cloud service.
transcodingServerReturnFailed	0x400010F8	No response is returned by transcoding service.
UPSServerReturnFailed	0x400010F9	No response is returned by UPS monitoring service.
forwardingServerReturnFailed	0x400010FA	No response is returned by forwarding service.
storageServerReturnFailed	0x400010FB	No response is returned by storage service.
cloudAnalysisServerReturnFailed	0x400010FC	No response is returned by cloud analytic service.
modelEmpty	0x400010FD	No model is obtained.
mainAndBackupNodeCannotModifyManagementNetworkInterfaceIP	0x400010FE	Editing the management interface IP address of master node and backup node is not allowed.
IDTooLong	0x400010FF	The ID is too long.
pictureCheckFailed	0x40001100	Detecting picture failed.
pictureModelingFailed	0x40001101	Modeling picture failed.
setCloudAnalysisDefaultProvinceFailed	0x40001102	Setting default province of cloud analytic service failed.
InspectionAreasNumberExceedLimit	0x40001103	No more detection regions can be added.
picturePixelsTooLarge	0x40001105	The picture resolution is too high.
picturePixelsTooSmall	0x40001106	The picture resolution is too low.

Sub Status Code	Error Code	Description
storageServiceIPEmpty	0x40001107	The storage server IP address is required.
bigDataServerRequestInfoBuildFail	0x40001108	Creating request message of big data service failed.
analysisTimedOut	0x40001109	Analysis time out.
high-performanceModeDisabled.	0x4000110A	Please enable high-performance mode.
configuringUPSMonitoringServerTimedOut	0x4000110B	Configuring the UPS monitoring server time out. Check IP address.
cloudAnalysisRequestInformationBuildFailed	0x4000110C	Creating request message of cloud analytic service failed.
cloudAnalysisResponseInformationParseFailed	0x4000110D	Parsing response message of cloud analytic service failed.
allCloudAnalysisInterfaceFailed	0x4000110E	Calling API for cloud analytic service failed.
cloudAnalysisModelCompareFailed	0x4000110F	Model comparison of cloud analytic service failed.
cloudAnalysisFacePictureQualityRatingFailed	0x40001110	Getting face quality grading of cloud analytic service failed.
cloudAnalysisExtractFeaturePointsFailed	0x40001111	Extracting feature of cloud analytic service failed.
cloudAnalysisExtractPropertyFailed	0x40001112	Extracting property of cloud analytic service failed.
getAddedNodeInformationFailed	0x40001113	Getting the added nodes information of data analysis server failed.
noMoreAnalysisUnitsAdded	0x40001114	No more data analysis servers can be added.
detectionAreaInvalid	0x40001115	Invalid detection region.
shieldAreaInvalid	0x40001116	Invalid shield region.
noMoreShieldAreasAdded	0x40001117	No more shield region can be drawn.
onlyAreaOfRectangleShapeAllowed	0x40001118	Only drawing rectangle is allowed in detection area.

Sub Status Code	Error Code	Description
numberReachedLLimit	0x40001119	Number reached the limit.
wait1~3MinutesGetIPAfterSetupDHCP	0x4000111A	Wait 1 to 3 minutes to get IP address after configuring DHCP.
plannedTimeMustbeHalfAnHour	0x4000111B	Schedule must be half an hour.
oneDeviceCannotBuildCluster	0x4000111C	Creating master and backup cluster requires at least two devices.
updatePackageFileNotUploaded	0x4000111E	Upgrade package is not uploaded.
highPerformanceTasksNotSupportDrawingDetectionRegions	0x4000111F	Drawing detection area is not allowed under high-performance mode.
controlCenterIDDoesNotExist	0x40001120	The control center ID does not exist.
regionIDDoesNotExist	0x40001121	The area ID does not exist.
licensePlateFormatError	0x40001122	Invalid license plate format.
managementNodeDoesNotSupportThisOperation	0x40001123	The operation is not supported.
searchByPictureResourceNotConfiged	0x40001124	The conditions for searching picture by picture are not configured.
videoFileEncapsulationFormatNotSupported	0x40001125	The video container format is not supported.
videoPackageFailure	0x40001126	Converting video container format failed.
videoCodingFormatNotSupported	0x40001127	Video coding format is not supported.
monitorOfDeviceArmingdeleteArmingInfo	0x40001129	The camera is armed. Disarm it and try again.
getVideoSourceTypeFailed	0x4000112A	Getting video source type failed.
smartRulesBuildFailed	0x4000112B	Creating VAC rule failed.
smartRulesParseFailed	0x4000112C	Parsing VAC rule failed.

Sub Status Code	Error Code	Description
timeRulesBuildFailed	0x4000112D	Creating time rule failed.
timeRulesParseFailed	0x4000112E	Parsing time rule failed.
monitoInfoInvalid	0x4000112F	Invalid camera information.
addingFailedVersionMismatches	0x40001130	Adding failed. The device version mismatches.
theInformationReturnedAfterCloudAnalysisIsEmpty	0x40001131	No response is returned by the cloud analytic service.
selectingIpAddressOfHostAndSpareNodeFailedCheckTheStatus	0x40001132	Setting IP address for master node and backup node failed. Check the node status.
theSearchIdDoesNotExist	0x40001133	The search ID does not exist.
theSynchronizationIdDoesNotExist	0x40001134	The synchronization ID does not exist.
theUserIdDoesNotExist	0x40001136	The user ID does not exist.
theIndexCodeDoesNotExist	0x40001138	The index code does not exist.
theControlCenterIdDoesNotExist	0x40001139	The control center ID does not exist.
theAreaIdDoesNotExist	0x4000113A	The area ID does not exist.
theArmingLinkageIdDoesNotExist	0x4000113C	The arming relationship ID does not exist.
theListLibraryIdDoesNotExist	0x4000113D	The list library ID does not exist.
invalidCityCode	0x4000113E	Invalid city code.
synchronizingThePasswordOfSpareServerFailed	0x4000113F	Synchronizing backup system password failed.
editingStreamingTypesNotSupported	0x40001140	Editing streaming type is not supported.

Sub Status Code	Error Code	Description
switchingScheduledTaskToTemporaryTaskIsNotSupported	0x40001141	Switching scheduled task to temporary task is not supported.
switchingTemporaryTaskToScheduledTaskIsNotSupported	0x40001142	Switching temporary task to scheduled task is not supported.
theTaskIsNotDispatchedOrItIsUpdating	0x40001143	The task is not dispatched or is updating.
thisTaskDoesNotExist	0x40001144	This task does not exist in the cloud analytic service.
duplicatedSchedule	0x40001145	Schedule period cannot be overlapped.
continuousScheduleWithSameAlgorithmTypeShouldBeMerged	0x40001146	The continuous schedule periods with same algorithm type should be merged.
invalidStreamingTimeRange	0x40001147	Invalid streaming time period.
invalidListLibraryType	0x40001148	Invalid list library type.
theNumberOfMatchedResultsShouldBeLargerThan0	0x40001149	The number of search results should be larger than 0.
invalidValueRangeOfSimilarity	0x4000114A	Invalid similarity range.
invalidSortingType	0x4000114B	Invalid sorting type.
noMoreListLibraryCanBeLinkedToTheDevice	0x4000114C	No more lists can be added to one device.
InvalidRecipientAddressFormat	0x4000114D	Invalid address format of result receiver.
creatingClusterFailedTheDongleIsNotPluggedIn	0x4000114E	Insert the dongle before creating cluster.
theURLIsTooLong	0x4000114F	No schedule configured for the task.
noScheduleIsConfiguredForTheTask	0x40001150	No schedule configured for the task.
theDongleIsExpired	0x40001151	Dongle has expired.

Sub Status Code	Error Code	Description
dongleException	0x40001152	Dongle exception.
invalidKey	0x40001153	Invalid authorization service key.
decryptionFailed	0x40001154	Decrypting authorization service failed.
encryptionFailed	0x40001155	Encrypting authorization service failed.
AuthorizeServiceResponseError	0x40001156	Authorization service response exception.
incorrectParameter	0x40001157	Authorization service parameters error.
operationFailed	0x40001158	Operating authorization service error.
noAnalysisResourceOrNoDataInTheListLibrary	0x40001159	No cloud analytic resources or no data in the list library.
calculationException	0x4000115A	Calculation exception.
allocatingList	0x4000115B	Allocating list.
thisOperationIsNotSupportedByTheCloudAnalytics	0x4000115C	This operation is not supported by the cloud analytic service.
theCloudAnalyticsIsInterrupted	0x4000115D	The operation of cloud analytic service is interrupted.
theServiceIsNotReady	0x4000115E	The service is not ready.
searchingForExternalApiFailed	0x4000115F	Searching external interfaces failed.
noOnlineNode	0x40001160	No node is online.
noNodeAllocated	0x40001161	No allocated node.
noMatchedList	0x40001162	No matched list.
allocatingFailedTooManyFacePictureLists	0x40001163	Allocation failed. Too many lists of big data service.
searchIsNotCompletedSearchAgain	0x40001164	Current searching is not completed. Search again.
allocatingListIsNotCompleted	0x40001165	Allocating list is not completed.
searchingForCloudAnalyticsResultsFailed	0x40001166	Searching cloud analytic service overtime.

Sub Status Code	Error Code	Description
noDataOfTheCurrentLibraryFound	0x40001167	No data in the current library. Make sure there is data in the Hbase.
noFacePictureLibraryIsArmed	0x40001168	No face picture library is armed for big data service.
noAvailableDataSlicingVersionInformationArmedFirstAndSliceTheData	0x40001169	Invalid standard version information.
duplicatedOperationDataSlicingIsExecuting	0x4000116A	Slicing failed. Duplicated operation.
slicingDataFailedNoArmedFacePictureLibrary	0x4000116B	Slicing failed. No arming information in the face big data.
GenerateBenchmarkFileFailedSlicingAgain	0x4000116C	Generating sliced file failed. Slice again.
NonprimaryNodesProhibitedFromSlicingData	0x4000116D	Slicing is not allowed by the backup node.
NoReadyNodeToClusterServers	0x4000116E	Creating the cluster failed. No ready node.
NodeManagementServicesOffline	0x4000116F	The node management server is offline.
theCamera(s)OfTheControlCenterAreAlreadyArmed.DisarmThemFirst	0x40001170	Some cameras in control center are already armed. Disarm them and try again.
theCamera(s)OfTheAreaAreAlreadyArmed.DisarmThemFirst	0x40001171	Some cameras in this area are already armed. Disarm them and try again.
configuringHigh-frequencyPeopleDetectionFailed	0x40001172	Configuring high frequency people detection failed.
searchingForHigh-frequencyPeopleDetectionLogsFailed.	0x40001173	Searching detection event logs of high-frequency people detection failed.
gettingDetailsOfSearchHigh-	0x40001174	Getting the search result details of high frequency alarms failed.

Sub Status Code	Error Code	Description
frequencyPeopleDetectionLogsFailed.		
theArmedCamerasAlreadyExistInTheControlCenter	0x40001175	Some cameras in control center are already armed.
disarmingFailedTheCamerasNotArmed	0x40001177	Disarming failed. The camera is not armed.
noDataReturned	0x40001178	No response is returned by the big data service.
preallocFailure	0x40001179	Pre-allocating algorithm resource failed.
overDogLimit	0x4000117A	Configuration failed. No more resources can be pre-allocated.
analysisServicesDoNotSupport	0x4000117B	Not supported.
commandAndDispatchServiceError	0x4000117C	Scheduling service of cloud analytic service error.
engineModuleError	0x4000117D	Engine module of cloud analytic service error.
streamingServiceError	0x4000117E	Streaming component of cloud analytic service error.
faceAnalysisModuleError	0x4000117F	Face analysis module of cloud analytic service error.
vehicleAnalysisModuleError	0x40001180	Vehicle pictures analytic module of cloud analytic service error.
videoStructuralAnalysisModuleError	0x40001181	Video structuring module of cloud analytic service error.
postprocessingModuleError	0x40001182	Post-processing module of cloud analytic service error.
frequentlyAppearedPersonAlarmsAlreadyConfiguredForListLibrary	0x40001183	High frequency alarm is already armed for blacklist library.
creatingListLibraryFailed	0x40001184	Creating list library failed.
invalidIdentityKeyOfListLibrary	0x40001185	Invalid identity key of list library.

Sub Status Code	Error Code	Description
noMoreDevicesCanBeArmed	0x40001186	No more camera can be added.
settingAlgorithmTypeForDeviceFailed	0x40001187	Allocating task resource failed.
gettingHighFrequencyPersonDetectionAlarmInformationFailed	0x40001188	Setting high frequency alarm failed.
invalidSearchConfiton	0x40001189	Invalid result.
theTasksNotCompleted	0x4000118B	The task is not completed.
resourceOverRemainLimit	0x4000118C	No more resource can be pre-allocated.
frequentlyAppearedPersonAlarmsAlreadyConfiguredForTheCameraDisarmFirstAndTryAgain	0x4000118D	The high frequency alarm of this camera is configured. Delete the arming information and try again.
switchtimedifflesslimit	0x4000123b	Time difference between power on and off should be less than 10 minutes.
noMorePeopleNumChangeRulesAdded	0x4000128A	Maximum number of people number changing rules reached.
noMoreViolentMotionRulesAdded	0x4000128D	Maximum number of violent motion rules reached.
noMoreLeavePositionRulesAdded	0x4000128E	Maximum number of leaving position rules reached.
SMRDiskNotSupportRaid	0x40001291	SMR disk does not support RAID.
OnlySupportHikAndCustomProtocol	0x400012A3	IPv6 camera can only be added via Device Network SDK or custom protocols.
vehicleEnginesNoResource	0x400012A6	Insufficient vehicle engine resources.
noMoreRunningRulesAdded	0x400012A9	Maximum number of running rules reached.

Sub Status Code	Error Code	Description
noMoreGroupRulesAdded	0x400012AA	Maximum number of people gathering rules reached.
noMoreFailDownRulesAdded	0x400012AB	Maximum number of people falling down rules reached.
noMorePlayCellphoneRulesAdded	0x400012AC	Maximum number of playing cellphone rules reached.
ruleEventTypeDuplicate	0x400012C8	Event type duplicated.
noMoreRetentionRulesAdded	0x400015AD	Maximum number of people retention rules reached.
noMoreSleepOnDutyRulesAdded	0x400015AE	Maximum number of sleeping on duty rules reached.
polygonNotAllowedCrossing	0x400015C2	Polygons are not allowed to cross.
alreadyRunning	0x40002026	The application program is running.
notRunning	0x40002027	The application program is stopped.
packNotFound	0x40002028	The software packet does not exist.
alreadyExist	0x40002029	The application program already exists.
noMemory	0x4000202A	Insufficient memory.
invalidLicense	0x4000202B	Invalid License.
noClientCertificate	0x40002036	The client certificate is not installed.
noCACertificate	0x40002037	The CA certificate is not installed.
authenticationFailed	0x40002038	Authenticating certificate failed. Check the certificate.
clientCertificateExpired	0x40002039	The client certificate is expired.
HDMIResolutionIllegal	0x40002040	The HDMI video resolution cannot be larger than that of main and sub stream.
clientCertificateRevocation	0x4000203A	The client certificate is revoked.
CACertificateExpired	0x4000203B	The CA certificate is expired.
CACertificateRevocation	0x4000203C	The CA certificate is revoked.

Sub Status Code	Error Code	Description
connectFail	0x4000203D	Connection failed.
loginNumExceedLimit	0x4000203F	No more user can log in.
formattingFailed	0x40002056	Formatting HDD failed.
encryptedFormattingFailed	0x40002057	Formatting encrypted HDD failed.
wrongPassword	0x40002058	Verifying password of SD card failed. Incorrect password.
audiolsPlayingPleaseWait	0x40002067	Audio is playing. Please wait.
twoWayAudioInProgressPleaseWait	0x40002068	Two-way audio in progress. Please wait.
calibrationPointNumFull	0x40002069	The maximum number of calibration points reached.
completeTheLevelCalibrationFirst	0x4000206A	The level calibration is not set.
completeTheRadarCameraCalibrationFirst	0x4000206B	The radar-camera calibration is not set.
pointsOnStraightLine	0x4000209C	Calibrating failed. The calibration points cannot be one the same line.
TValueLessThanOrEqualZero	0x4000209D	Calibration failed. The T value of the calibration points should be larger than 0.
HBDLibNumOverLimit	0x40002092	The number of human body picture libraries reaches the upper limit
theShieldRegionError	0x40002093	Saving failed. The shielded area should be the ground area where the shielded object is located.
theDetectionAreaError	0x40002094	Saving failed. The detection area should only cover the ground area.
invalidLaneLine	0x40002096	Saving failed. Invalid lane line.
noDetectionArea	0x400050df	No detection area
armingFailed	0x40008000	Arming failed.
disarmingFailed	0x40008001	Disarming failed.
clearAlarmFailed	0x40008002	Clearing alarm failed.

Sub Status Code	Error Code	Description
bypassFailed	0x40008003	Bypass failed.
bypassRecoverFailed	0x40008004	Bypass recovery failed.
outputsOpenFailed	0x40008005	Opening relay failed.
outputsCloseFailed	0x40008006	Closing relay failed.
registerTimeOut	0x40008007	Registering timed out.
registerFailed	0x40008008	Registering failed.
addedByOtherHost	0x40008009	The peripheral is already added by other security control panel.
alreadyAdded	0x4000800A	The peripheral is already added.
armedStatus	0x4000800B	The partition is armed.
bypassStatus	0x4000800C	Bypassed.
zoneNotSupport	0x4000800D	This operation is not supported by the zone.
zoneFault	0x4000800E	The zone is in fault status.
pwdConflict	0x4000800F	Password conflicted.
audioTestEntryFailed	0x40008010	Enabling audio test mode failed.
audioTestRecoveryFailed	0x40008011	Disabling audio test mode failed.
addCardMode	0x40008012	Adding card mode.
searchMode	0x40008013	Search mode.
addRemoterMode	0x40008014	Adding keyfob mode.
registerMode	0x40008015	Registration mode.
exDevNotExist	0x40008016	The peripheral does not exist.
theNumberOfExDevLimited	0x40008017	No peripheral can be added.
sirenConfigFailed	0x40008018	Setting siren failed.
chanCannotRepeatedBinded	0x40008019	This channel is already linked by the zone.
masterSlavesEnable	0x4000802c	The master-slave relationship has taken effect, the slave radar does not support this operation.
forceTrackNotEnabled	0x4000802d	Mandatory tracking is disabled.

Sub Status Code	Error Code	Description
isNotSupportZoneConfigByLocalArea	0x4000802e	This area does not support the zone type.
alarmLineCross	0x4000802f	Trigger lines are overlapped.
zoneDrawingOutOfRange	0x40008030	The drawn zone is out of detection range.
alarmLineDrawingOutOfRange	0x40008031	The drawn alarm trigger line is out of detection range.
hasTargetInWarningArea	0x40008032	The warning zone already contains targets. Whether to enable mandatory arming?
radarModuleConnectFail	0x40008033	Radar module communication failed.
importCfgFilePasswordErr	0x40008034	Incorrect password for importing configuration files.
inProgramMode	0x4000801B	The keypad is in programming mode.
inPaceTest	0x4000801C	In pacing mode.
arming	0x4000801D	Arming.
subscribeTypeErr	0x4000a016	This metadata type is not supported to be subscribed.
startAppFail	/	Starting running application program failed.
yuvconflict	/	The raw video stream conflicted.
overMaxAppNum	/	No more application program can be uploaded.
noFlash	/	Insufficient flash.
noFlash	/	The platform mismatches.

StatusCode=5

Sub Status Code	Error Code	Description
badXmlFormat	0x50000001	Invalid XML format.

StatusCode=6

Sub Status Code	Error Code	Description
badParameters	0x60000001	Invalid parameter.
badHostAddress	0x60000002	Invalid host IP address.

Sub Status Code	Error Code	Description
badXmlContent	0x60000003	Invalid XML content.
badIPv4Address	0x60000004	Invalid IPv4 address.
badIPv6Address	0x60000005	Invalid IPv6 address.
conflictIPv4Address	0x60000006	IPv4 address conflicted.
conflictIPv6Address	0x60000007	IPv6 address conflicted.
badDomainName	0x60000008	Invalid domain name.
connectSreverFail	0x60000009	Connecting to server failed.
conflictDomainName	0x6000000A	Domain name conflicted.
badPort	0x6000000B	Port number conflicted.
portError	0x6000000C	Port error.
exportErrorData	0x6000000D	Importing data failed.
badNetMask	0x6000000E	Invalid sub-net mask.
badVersion	0x6000000F	Version mismatches.
badDevType	0x60000010	Device type mismatches.
badLanguage	0x60000011	Language mismatches.
incorrentUserNameOrPasswor d	0x60000012	Incorrect user name or password.
invalidStoragePoolOfCloudServ er	0x60000013	Invalid storage pool. The storage pool is not configured or incorrect ID.
noFreeSpaceOfStoragePool	0x60000014	Storage pool is full.
riskPassword	0x60000015	Risky password.
UnSupportCapture	0x60000016	Capturing in 4096*2160 or 3072*2048 resolution is not supported when H.264+ is enabled.
userPwdLenUnder8	0x60000023	At least two kinds of characters, including digits, letters, and symbols, should be contained in the password.
userPwdNameSame	0x60000025	Duplicated password.

Sub Status Code	Error Code	Description
userPwdNameMirror	0x60000026	The password cannot be the reverse order of user name.
beyondARGSRangeLimit	0x60000027	The parameter value is out of limit.
DetectionLineOutofDetectionRegion	0x60000085	The rule line is out of region.
DetectionRegionError	0x60000086	Rule region error. Make sure the rule region is convex polygon.
DetectionRegionOutOfCountingRegion	0x60000087	The rule region must be marked as red frame.
PedalAreaError	0x60000088	The pedal area must be in the rule region.
DetectionAreaABError	0x60000089	The detection region A and B must be in the a rule frame.
ABRegionCannotIntersect	0x6000008a	Region A and B cannot be overlapped.
customHBPIDError	0x6000008b	Incorrect ID of custom human body picture library
customHBPIDRepeat	0x6000008c	Duplicated ID of custom human body picture library
dataVersionsInHBDLibMismatches	0x6000008d	Database versions mismatches of human body picture library
invalidHBPID	0x6000008e	Invalid human body picture PID
invalidHBDID	0x6000008f	Invalid ID of human body picture library
humanLibraryError	0x60000090	Error of human body picture library
humanLibraryNumError	0x60000091	No more human body picture library can be added
humanImagesNumError	0x60000092	No more human body picture can be added
noHumanInThePicture	0x60000093	Modeling failed, no human body in the picture

Sub Status Code	Error Code	Description
analysisEnginesNoResourceError	0x60001000	No analysis engine.
analysisEnginesUsageExcced	0x60001001	The engine usage is overloaded.
PicAnalysisNoResourceError	0x60001002	No analysis engine provided for picture secondary recognition.
analysisEnginesLoadingError	0x60001003	Initializing analysis engine.
analysisEnginesAbnormaError	0x60001004	Analysis engine exception.
analysisEnginesFacelibImporting	0x60001005	Importing pictures to face picture library. Failed to edit analysis engine parameters.
analysisEnginesAssociatedChannel	0x60001006	The analysis engine is linked to channel.
smdEncodingNoResource	0x60001007	Insufficient motion detection encoding resources.
smdDecodingNoResource	0x60001008	Insufficient motion detection decoding resources.
diskError	0x60001009	HDD error.
diskFull	0x6000100a	HDD full.
facelibDataProcessing	0x6000100b	Handling face picture library data.
capturePackageFailed	0x6000100c	Capturing packet failed.
capturePackageProcessing	0x6000100d	Capturing packet.
noSupportWithPlaybackAbstract	0x6000100e	This function is not supported. Playback by video synopsis is enabled.
insufficientNetworkBandwidth	0x6000100f	Insufficient network bandwidth.
tapeLibNeedStopArchive	0x60001010	Stop the filing operation of tape library first.
identityKeyError	0x60001011	Incorrect interaction command.
identityKeyMissing	0x60001012	The interaction command is lost.

Sub Status Code	Error Code	Description
noSupportWithPersonDensityDetect	0x60001013	This function is not supported. The people density detection is enabled.
ipcResolutionOverflow	0x60001014	The configured resolution of network camera is invalid.
ipcBitrateOverflow	0x60001015	The configured bit rate of network camera is invalid.
tooGreatTimeDifference	0x60001016	Too large time difference between device and server.
noSupportWithPlayback	0x60001017	This function is not supported. Playback is enabled.
channelNoSupportWithSMD	0x60001018	This function is not supported. Motion detection is enabled.
channelNoSupportWithFD	0x60001019	This function is not supported. Face capture is enabled.
illegalPhoneNumber	0x6000101a	Invalid phone number.
illegalCertificateNumber	0x6000101b	Invalid certificate No.
linkedCameraOutLimit	0x6000101c	Connecting camera timed out.
achieveMaxChannelLimit	0x6000101e	No more channels are allowed.
humanMisInfoFilterEnabledChannelNumError	0x6000101f	No more channels are allowed to enable preventing false alarm.
humanEnginesNoResource	0x60001020	Insufficient human body analysis engine resources.
taskNumberOverflow	0x60001021	No more tasks can be added.
collisionTimeOverflow	0x60001022	No more comparison duration can be configured.
invalidTaskID	0x60001023	Invalid task ID.
eventNotSupport	0x60001024	Event subscription is not supported.
invalidEZVIZSecretKey	0x60001034	Invalid verification code for Hik-Connect.
needDoubleVerification	0x60001042	Double verification required

Sub Status Code	Error Code	Description
noDoubleVerificationUser	0x60001043	No double verification user
timeSpanNumOverLimit	0x60001044	Max. number of time buckets reached
channelNumOverLimit	0x60001045	Max. number of channels reached
noSearchIDResource	0x60001046	Insufficient searchID resources
noSupportDeleteStrangerLib	0x60001051	Deleting stranger library is not supported
noSupportCreateStrangerLib	0x60001052	Creating stranger library is not supported
behaviorAnalysisRuleInfoError	0x60001053	Behavior analysis rule parameters error.
safetyHelmetParamError	0x60001054	Hard hat parameters error.
OneChannelOnlyCanBindOneEngine	0x60001077	No more engines can be bound.
TransitionUseEmmc	0x60002000	Starting device failed. The EMMC is overused.
AdaptiveStreamNotEnabled	0x60002001	The stream self-adaptive function is not enabled.
AdaptiveStreamAndVariableBitrateEnabled	0x60002002	Stream self-adaptive and variable bitrate function cannot be enabled at the same time.
noSafetyHelmetRegion	0x60002023	The hard hat detection area is not configured (if users save their settings without configuring the arming area, they should be prompted to configure one).
unclosedSafetyHelmet	0x60002024	The hard hat detection is enabled (If users save their settings after deleting the arming area, they should be prompted to disable hard hat detection first and then delete the arming area).

Sub Status Code	Error Code	Description
width/ heightRatioOfPictureError	0x6000202C	The width/height ratio of the uploaded picture should be in the range from 1:2 to 2:1.
PTZNotInitialized	0x6000202E	PTZ is not initialized.
PTZSelfChecking	0x6000202F	PTZ is self-checking.
PTZLocked	0x60002030	PTZ is locked.
deployExceedMax	0x60006018	The arming connections exceed the maximum number.
detectorTypeMismatch	0x60008000	The detector type mismatched.
nameExist	0x60008001	The name already exists.
uploadImageSizeError	0x60008016	The size of the uploaded picture is larger than 5 MB.
laneAndRegionOverlap	/	The lanes are overlapped.
unitConfigurationNotInEffect	/	Invalid unit parameter.
ruleAndShieldingMaskConflict	/	The line-rule region overlaps with the shielded area.
wholeRuleInShieldingMask	/	There are complete temperature measurement rules in the shielded area.
LogDiskNotSetReadOnlyInGroupMode	0x60001100	The log HDD in the HDD group cannot be set to read-only.
LogDiskNotSetRedundancyInGroupMode	0x60001101	The log HDD in the HDD group cannot be set to redundancy.

StatusCode=7

SubStatusCode	Error Code	Description
rebootRequired	0x70000001	Reboot to take effect.

